4th Applied Active Inference Symposium (Part 2, Nov 14, 2024) ~ Full Upload

> **Source**: YouTube **Playlist**: Uncategorized **Duration**: 12:24:38 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

hello welcome back it's the fourth applied active inference Symposium day two part two on November 14th 2024 we're kicking off this session in the darkness of the preon on the west coast appropriately with Maron Hussein Zade bagani ml Dawn brain inthe dark design principles for the neuromatic inference under the free energy principle so thank you Maron for joining looking forward to your presentation thank you very much it's a pleasure to be here um so today I'm going to be talking about uh the design principles for neuromatic inference under the free energy principle namely the methods that are inspired by the brain um in order to model human perception um just um want to preface this by saying that this is a collaboration with my colleague Simon orbas from Manu University and Professor friston from UCL and this work has recently been accepted at the neuro AI workshop at neurs 2024 so let's just get started the content is going to be as follows we're going to be talking about first the motivation behind this work why we got started Ed with this idea anyway to to introduce the design principles particularly for these models and then I'm going to do a quick refresher on the math side just so we all get comfortable with the you know the upcoming derivations that I'm going to be uh expanding on which covers the math behind free the free energy principle particularly a variational free energy then I'll do a brief discussion on what free energy principle is I'm presume every already knows by now because we we' had so many talks uh since uh in the beginning of the Symposium but I'm going to do a quick run at that then we'll be talking about the beian brain hypothesis that furnishes the foundation for the idea of modeling human perception as a kind of inference namely beian inference next we dive into dejos between the generative model and a generative process what do we mean by each one of them knowing the distinction is crucial for understanding the free energy principle because Fe lies on this um notion that there is a boundary that separates an entity from uh the environment and separating uh basically my internal state from the external State beyond my marov blanket next we'll talk about variational free energy and how it's driven how it's derived basically and then we jump into the design principles in terms of the experiments not the things that you need to consider the things that you need to Define and how you should go about doing that in order to model human perception as a kind of online inference problem we'll go through some final thoughts some lessons learned and then I'll introduce some of the resources for those of you who are interested in uh you know pursuing free energy principle more seriously and delving into the theory and the Practical side of it and then we're going to wrap it up with uh a demonstration um using the GitHub repository that uh that is publicly available by um by by the way so the motivation is this so you said deep learning has revolutionized artificial intelligence um because it has automated the process of feature extraction from raw data however it actually faces a lot of problems for example lack of outof distribution generalizability catastrophic forgetting and poor interpretability it is known as a black box basically and the catastrophic forgetting is basically when the new knowledge overwrites the previous knowledge the weights are getting overwritten so while the network learns the concept of a cat it might momentarily forget the concept of a dog because the paths are shared between different Concepts that are to be learned you see in contrast to deep learning biological neural networks in your brain and a my brain they do not suffer from these issues and this inspires the AI researchers to actually explore this interesting and intriguing area called The neuromatic Deep learning that is all about replicating the brain mechanisms namely the neuronal message passing and belief updating within the AI models purely sprayed by our best knowledge about how the brain works um a foundational theory for this approach is the free energy principle um f which unfortunately despite its potential is often considered too complex to understand and to implement an AI simply because one would need a myriad of disciplines um under their belt knowledge about disciplines under their belt to be able to understand the free Andre principle namely you need to know things for example about dynamical systems random processes you need to know about basian mechanics some Neuroscience some maybe computational neuroscience and so on and so forth so as a as a result it's not it's an intimidating topic let's say for anybody who wants to just get started with it so we seek to demystify FP and basically provide a comprehensive framework for Designing neuromatic models within well where we are actually trying to mimic humanlike perception capabilities um so here we present a road map for implementing these models and by the way a pytorch code repository for applying FP in a predictive coding network is going to be introduced here the link is here that that's a code it's publicly available and also as I said the paper will soon be available um by the neuro AI Workshop um and so hopefully you guys will have a chance to take a look at that as well now let's get started with the ref refresher of the math so first foremost the idea of a PDF so it it's an acronym for a probability density function which is a function that describes the likelihood of a continuous random variable taking on a particular value it has two conditions it has to be positive so for a given parameter theta0 for example the mean of a gaan let's say for any input X it's always positive and second we have this normalizer uh sort of condition that if you integrate a PDF across the support space of of the input Nam the all possible values of X you will inevitably must receive get to the actual value of one what this means is the probability of X the input getting any value in in a in any range within its support is always one which means that X will always fall in in a range in that in its own accepted range of value so that's 100% so you need these two if you want to have a PDF next is the idea of an expectation I just want to remind you that the reason I'm going through this math is because later on when I introduced the steps of deriving um you know variational free energy these concepts are absolutely crucial so hopefully um I'm not going too fast but um yeah just bear that in mind that these are very important and for those of you who are you know math Savvy and these are two Basics my apologies now the idea of expectation is actually the weighted average of all the possible values that the variable consume weighted by their probability so you have the in the first line um the expected value of x over the probability distribution the PDF PX so you as you notice we're integrating all possible values of X multiplied by or weighted by their corresponding probability and it's important to note that you don't you're not limited to you know calculating the um the expected value with respect to a variable but you can also do this exactly the same thing with a function so the expected value of function is again the same thing but you are summing up a weight you're basically calculating a weighted average of a function where the weights are the probability of the input to that function which is X now some key Pro probability components here now we have this idea of a joint probability um so a joint distribution between two variable X X and Y you can divide it up between like like some conditional term which is the probability of x given y multipli by the prob probability of Y in the first place or you can go the other way around the probability of Y given x times the probability of X you notice that if X and Y are independent this conditional term reduces to probability of X or this conditional term would reduce the probability of Y so either way on both sides you're going to have the probability of x given y um if X and Y are indeed independent next if you want to move from a joint distribution to a marginalized distribution where you're actually marginalizing one of the variables out this is how you can do it so in this case if you integrate P of XY across all possible possible values of X in other words like you literally break this down and you know basically write it down in in its you know um constituent terms if you do this integration you're inevit what basically what you're doing is you're getting root of x and then you end up with a term with a probability distribution that is only a function of m y and that's what you call a marginalized probability distribution or PDF again this is key where will this come in handy it's going to come in handy when when I'm going to be talking about the basium brain hypothesis and base Rule and so on and so forth so keep these two equations in mind please now properties of logarithms so this is basically just the definition of a log so log of a with base b equals c the definition says that it just means a equals B to the power C right so for example the log of 8 base 2 equals 3 because 2 to^ 3 is 8 so the log of a multiplication is some of the logs of individual terms the log of a division is the subtraction of individual log terms that the log of denominator minus the log of the denominator um the log of a fraction is literally minus the log of the swapped fraction literally swap b and a together from like you know you end up with just a negative of the of the new log term um the power law the log of B to the power C so basically C drops down as a coefficient so you end up with this term C Times log of B with base a log of a with base a if the value and the base are both the same you end up with value one and by extension if you have this term C drops down as a coefficient and this term equals one so you end up with C log of one with any base is always equal to zero why because a whatever it is to the power zero is one and last but not least this one is again very important Ln of a is basically just a logarithm a logarithm with you know where the Bas is natural number e or it's called the natural log and E is basically the oiler Oiler number so you got log of a with base e it's called just the natural log so when in in Python we you say numpy do log is basically the natural log now the free energy principle it is a theoretical framework in Neuroscience it's been developed by Carl friston so basically things that persist over time especially living systems look as if they are minimizing a certain quantity it's called free energy it is as if they're uh sort of maintaining this boundary that is called a mark of blanket to persist in the face of random fluctuations in the uh in the environment um and this really connects well with the beian brain hypothesis you as you will see shortly so the brain from the perspective of beian brain hypothesis it's actually in the game of inferring the hidden causes behind its Sensations so namely if you have if you let's say y to denote the sensations and X as the hidden State and then this inference basically means that given my observation why I want to estimate the probability distribution over X right it's basically called a posterior probability which means that after having seen an observation um what is causing in the world beyond my markof blanket this particular sensation or observation inevitably this requires having a generative model GM of the world and this actually connects really well with Helm hold's um concept of unconscious inference and it is defined like that perception is kind of unconscious inference which means that we are in the game of you know inferring the hidden cause behind our our Sensations for example right now you're looking at the Monitor and the photons but the light is hitting the your sensory epithelia on your eyes those excitations those those hits those are your sensation and when you realize that I'm I'm watching this presentation at at at active inference Institute um and this guy is talking his name is mean this is the slide of a of a face Helm hols pictures here these are the inferences that your brain is making given that particular Sensation that is hitting the the neurons on the surface of your eyes right so it's estimating this probability distribution about what's happening beyond your markup blanket now let's talk about the idea of generative model versus generative process this is actually very very important on the right hand side you see this IDE this generative model as a box and it has some internal States let's call them hidden States this represents the neuronal activities in your brain um the excitabilities um uh different you know the the voltage the voltage level let's say of your neurons that are keep changing all the time they fluctuate all the time and these guys I basic basically holding information about what's happening beyond your Markov blanket right remember what's happening Beyond Your Mark of blanket is out of your reach you have absolutely no way of observing exactly what's happening right and here we have the generative process where xstar denotes the actual things that are happening in the world which are completely out of reach to you now if you imagine here you have the face of a person let's say c Ste job is standing in front of me but this is no like I'm I'm not going to see the truth what I'm going to see is some sort of a um a map some sort of a noisy transformation of the truth xstar right let's call it why this is what I'm going to be experiencing this is what's going to hit and uh induce excitation at my sensory epithelia on my eyes and then if you want to look at like um basically dig deeper into the idea of perception as um unconscious inference the G the game is this your brain the generative model in your brain predicts a sensation like I'm expecting this sensation to be felt or received at my sensory epithelia on my Markov blanket and then the actual sensation why is received so the difference between the actual sensation and my prediction is called the prediction error so using that prediction error I will update my jour model about the world namely I will update my belief about what's happening in the world so this is where my internal States will change to reflect a better image of the unknown World beyond my marov blanket or the parameters of my generative model are going to change um to again do a better estimate of what's happening in the world so that I will have a better prediction about the sensations that that world is going to cause right so generative process again is not directly observable and the internal uh States X this is very important carry information about the invisible external States xar beyond my mark of blanket this is where this connects to the IDE the discipline of information Theory this is a very important link by the way so the generative model what do we mean by that mathematically speaking now in a dynamical world a generative model has belief about the Dynamics of the world what it means is how fast is the world changing for example or how is it changing now if the world is at a given State X right what does its Dynamics look like for example it's velocity or acceleration or even higher temporal derivatives are like jerk basically all sorts of rates of changes the rates of change about the world that's what what I mean by Dynamics so in a generative model your generative model will have an understanding has to have a belief about how the world is changing beyond the smart C blanket so in other words if I'm in the game of estimating the Dynamics of the world it means I'm chasing a moving Target what it means is at any given time when I want to infer what's happening in the world by the time I've made that inference the world has already moved so that basically makes inference uh a tricky task especially if I'm in a in a you know highly dynamical World it makes it more challenging the second thing that in a generative model we should have is the generative model has a belief about how the world causes its Sensations namely if the world is at any given state that by my estimation beyond my mark of blanket what is the most likely sensation why that I will be experiencing right so these two are the key components of a generative model so like mathematically this is how you show it x dot is literally the velocity of uh the world how fast it's changing mathematically it's just DX over DT and then you have some function of the states X and causes new and Theta the parameters of my generative model plus some random fluctuation Omega X so that equation is called a state Dynamics equation but what it basically is telling you is that the Dynamics of the world is driven by X by the position of the world basically and these are all my belief about how the world is changing DX over DT the velocity of the world and technically speaking f is called the flow of the state Dynamics it's the deterministic part of the state Dynamics Omega X random fluctuation is what encodes your un certainty about your belief about how the world is changing yeah the second equation uh basically connects to the second concept that I introduced here it's basically telling you that my belief about my sensation about like what's the most likely um Sensation that I'm going to be receiving is encoded in this equation where my observation is equal to the G of the function of again hidden States the causes new and the parameters some parameters plus some random fluctuation Omega y it's called the observation model again Omega y encodes your uncertainty about your belief about the most likely Sensation that you will be experiencing a sensation that is caused by the world you notice that we have this these again your hidden St here x and then uh the cause is new now if you're wondering so you know what the thetas are these are just the parameters of the functional form of FNG G but if you're wondering what causes are this is something that I'm not going to be focusing in this um particular talk but just imagine that if you have a hierarchical generative model these causes are just a subset of the Hidden States these are like the hidden states that do not have Dynamics but they are the tools that different levels in the hierarchy use to be able to talk to each other so in interlayer communication is done using these news or the causes these are the ones that basically Drive the Dynamics you notice that that's why we have x. equals f of x and new they they drive they drive the Dynamics without them the layers cannot talk to each other basically now if you want to like turn this into a belief-based system where you have probabilities in play you you just need to assume a gush and distribution over your random fluctuation namely with mean mean zero and some Precision uh over your random variable X and your observation y just my you know Open brackets Precision is just inverse uncertainty um and close brackets that's just a notion in statistics Now using this if you literally replace um you know this um notion of a Galan distribution into your original equation you end up with this probabilistic sort of um uh you know notion that the probability of x dot given all of these three elements is distributed according to a normal distribution where again the Precision is the same as the Precision of your random fluctuation but you notice that because we're summing up f with this guy over here the the mean of that gussan distribution becomes this F term same story for the your belief about your why your your observations and that is why I'm saying that f of x and new and Theta is the expected your expected um Dynamics about X do because it's literally the expected value of the gussian distribution same s here G of X new and Theta is your expectation about your observation your sensation again it is the sort of expected value of this gussan distribution the value that is most likely to occur so that's generative model two things we know the Dynamics of the world we have beliefs let's say we don't know it we have beliefs about the Dynamics of the world and we have belief about uh basically the most likely Sensation that we will be experiencing if such and such was the State of Affair of the world let's say x was the State of Affairs in world now we talked about inference let's talk about Bas rule really quickly now here this is base Rule and the components of Base rule are as follows we have the likelihood term which is the probability of my observation given uh the state of the world we have a prior belief which means that before observing why this is my belief about what the world looks like we have the EV model evidence which is probability of data and these two together is called basically the the generative model again like this is in line with what what we discussed here this is basically your prior belief about the Dynamics and this is literally the mapping between X and Y which is your likelihood term so I'm not saying anything new here these two together are your generative model and this is your model evidence this is a devil right this is intractable you notice that in a real life scenario we will not be able to calculate this and this term on the left hand side is called posterior probability it's your belief about the distribution over hidden state of the world X and when you map your generative model using base rule to uh this posterior distribution we call this model inversion right so if whenever you hear model inversion in the context of FP that's what we mean by that so this evidence again going back to our idea of marginalizing a variable out if you again just remind yourself that this is how you marginalize the hidden State H or X out you just end up with probability of let's say sensation why you notice that this integration could get really nasty really fast which means that if you have you you could have basically any number of hidden States X and for each X you would have a range of possible values so this thing could turn into a potentially like potentially in infinite number of nested integrals basically it can it can become intractable really really quickly so what this means is we will not be able to calculate this posterior probability um exactly so what should we do the answer is variational inference so because U you you see probability of data your model evidence is intractable but at the same time ironically it's the true measure of how good your generative model is um which means that we call this also marginal likelihood right so if it is high it means whatever sensation you're experiencing right now under your model of the world under your generative model of the world the probability of that particular sensation is very high it means under your model of the world everything that you're sensing you already expected to to sense it so it means you have such a strong and uh you know and precise belief and precise yeah belief about the Dynamics of the world and how the sensations are generated from those uh those dynamics that whatever you're receiving is not surprising that that's why model evidence is such a strong metric for evaluating how good a generative model is but unfortunately it's intractable so can we find a bound to this namely a lower bound and the answer is yes I'm going to introduce the idea of a evidence lower bound and the next slide so by doing this I'm turning the problem on its head so I'm turning a converting let's say a an intractable exact calculation of the posterior and I'm replacing that with a tractable approximation um well basically a tractable approximate estimation of the posterior and that's called variational inference and I'll tell you what why we call this variational by the way because the objective function that we're going to be minimizing to do that is VAR free energy is actually a function of a function it's a functional so where wherever you're dealing with functionals and optimizing functionals U that's where the calculus of variation comes into play that's what what it means by variational inference now on the right hand side I'm just going to put some reminders the expected value and the this particular particular law of logarithms and in the middle this is called Jenson inequality just keep keep them keep an eye on this one it just says that the expected value of a log of something is always a lower bound to the log of the expected value of that thing right so this is going to come in handy so let's study the this uh amazing concept of evidence lower bound remember P of Y is intractable we want to find the bound to it so this is log of probability of data um this is how we mark maraliz X out and you can multiply and divide by this by by by something right whatever it is so here QX is we call this surrogate the surrogate posterior over the hidden State this is going to be a representative or your approximation to the true unknown posterior belief right so let's call it QX and it's by the way it's really a key factor in variational inference and Fe so keep that in mind this Q um um variable this Q function so moving on you you do a little bit of play here so this this sort of cube moves to the denominator and you notice that this is the definition of expectation with respect to this particular distribution Q right just getting the definition from uh from here and according to jens's inequality um this term has a lower B and this is the lower bound to it so the expected of of this term the expected value of this term over this probability distribution you have the joint here and you have the surrogate distribution here and you see so this right hand side again remind yourself this is your log model evidence and we found the lower bound to it using jens's inequality right so the lower bound is found this guy is your evidence lower bound um and that that's a key finding that by maximizing this evidence lower band you're actually maximizing uh your your log model evidence and you're getting closer and closer to the true posterior now elbow and vfe just want to make sure that we're clear on this we have elbow here um but VF or variational inference is just negative of your elbow so if you multiply NE minus one by both sides of this you end up with this term always bigger than minus log probability of data this is called surprisal and this is called variational fre energy that's why we say VF is always an upper bound to surprise so maximizing elbow or minimizing VF they they both mean the same thing and um this is basically called variational free energy and that's why we call this a functional because this thing is a function of a of another function and it's called surprisal or surprise and just shows you in a plot that here you've got variational free energy that's always an upper bound to the true unknown surprisal or by extension you got your elbow here that's always a lower bound to the true um log model evidence so VF has multiple incarnations it comes in different form by playing with the math so if this is your this is your vfe just expand you can just expand the joint and the denominator and then literally use the log rule of the you know uh the division turned into the subtraction and then you can distribute the expectation across the two different terms and this um this basically results into like just basically swapping these two terms uh by each other you end up with this equation right and these terms have names so this one actually this expected value between your belief about the the hidden State and the true unknown posterior this expectation is called a a a KL Divergence it measures how different your belief about X QX is from the true unknown posterior that's why it's called an approximate error approximation error and the second term is your surprise right um so um if I want to like show you like what this actually means visually this is really interesting so this is your surprisal right and on top of surprisal you're adding this approx imation error this is this sort of length over here and then this constitutes your free energy what this means is like free energy is always an upper bound on surprisal and the error the that Gap is your approximation error is how far your belief about X is from the true posterior so by tuning your generative model your variational free energy goes down and when it goes down this Gap is reducing so which means that your Q gets closer and closer to the true true posterior which means a better and better perception about the world and this doesn't end here your free energy will get closer and closer to the surprisal um basically by extension to the model evidence which is the metric that we're going to be use using for you know comparing different models with one another a second incarnation of vfe is again we're going to expand the denominator again into this joint so previously was P of x given y times probity of y now we go pro y given x times P of X and again Distributing um the log and then we distribute the uh you know the expectation and then just you know uh swapping the terms again you have this scale Divergence over here and you have this negative negative expectation now this thing over here is called accuracy it means how accurate am I under my belief of the world the expect the expect value of observing why under my estimation of where the world is X so the higher this is it means the more accurate your belief about the world is that is that is basically causing your observations but at the same time this term over here is called complexity it means that if you have some prior belief about the world you you will not be able to just move your queue wherever you want it is it has to stay close to your prior belief over here so this is basically a reg regularizer if you will um and the third Incarnation is um basically again here this is the definition of vfe again you know the log rule um so we end up with these two terms so this guy over here is called expected sorry it's called expected internal energy and this guy is your entropy so this the expected value of log of QX we call this entropy um how do we minimize raal free energy we have to maximize our entropy and we have to maximize our internal energy now this term makes sense that we need to maximize it but what does this one mean it means that your Q will stay as smooth as possible as simple as possible this connects really well with aom's Razer if you're from machine learning background that your the best model is the one that provides the most accurate account of the data but at the same time the simplest explanation for that data this Simplicity is enforced by this entropy term it it states smooth so this is very interesting so these were just three incarnations of variational free energy and the goal of inverting degenerative model is basically we the GM strives to minimize this vfe um this achieves two things I hinted this uh just a little bit previously but it means that the actual uh value of VF will approach the intractable minus log probability of data this is going to be used for model comparison and the surrogate distribution will approach the unknown posterior belief remember that was the goal of um you know perception in the first place and once you reach that basically minimize your variational free energy you can use the actual value of free energy as the metric as a proxy for the unknown model evidence to compare different generative models together right so the model with the lowest rational free energy is going to be the best model yeah I think I've explained that one now just to let you know that the true form of the posterior is unknown and you see the the the terms like the the probability distribution The Joint distribution here um it is not guaranteed that um you know they're analytically simple you know to to derive and and you know work with because of that um there is this idea that we can actually use llas approximation to break down each one of the distributions and basically approximate them using a tailor expansion um particularly up to the third term which means that I'm providing a quadratic approximation using tailor expansion for each one of the components um within variational fee energy um unfortunately I don't have the time to go over the detailed derivation but suffice it to say that the result of you know s approximating all the components is that the variational free energy simplifies the this equation for those of you who are interested to know the details I encourage you to look at the primer on variational Leo paper which is an amazing paper um it goes through all the details of derivation so basically free energy um reduces to log of the joint distribution and you have the posterior um covariance and then yeah n is basically the number of parameters and the states that you have and so on and so forth now predictive coding this this is basically where it gets even more interesting it is a basically a hierarchical generative model that we're going to be using to model human perception and the goal is to invert this hierarchical model so it's a powerful framework to describe how the cortex accumulates evidence from noisy stimuli um it's an application of the free energy principle it focuses on how uh this literally fantastic organ can perform probabilistic inference about the most likely hidden cause that has caused the sensory signal um it assumes that the brain has a generative model of the world through which it constantly infers the hidden states of the world and it can also learn the parameters while inferring the hidden State Out World so model inversion happens by minimizing variational free energy again even though it's hierarchical but the concept is stays the same so in predictive coding we have this concept of top- down predictions where predictions go from inside out about your Sensations right and then the prediction error is calculated like what I described before and then this these errors if I want to be more precise the Precision weighted prediction errors are sent back from outside in uh or they call it but up sort of Direction so that the the estimations of the world the estimation of the posterior beliefs about the world are corrected basically that's what model inversion that's when model inversion happens when the posterior estimates get adjusted and it looks like something like this we have this hierarchical model that we have the same story of State Dynamics and observation model but here we have this happening at each level and then you notice that these news these causes are the ones that link layer I to I minus one that's the important part that without these news without these causes the layers cannot talk to each other you notice that it is a hierarchical dynamical system uh basically um now how do we design an experiment um for you know for for for this kind of perception neuromatic modeling using the free energy principle I just want to give you the sort of ingredients here you the designer you first have to Define your generative process that's X that's the true uh hidden stat I should have maybe said xar yeah xstar would have been a better notion then you will generate the observations this is some sort of some transformation of the X's you can add noise to it you can transform it the way you want but that's going to be the observations that's going to hit the Markov blanket this these are what this uh this is basically where the sensory state observed and receive those Sensations you have to Define your generative model this this part is the tricky part right the fs and G's in your dynamical system that's where uh everything gets really tricky how would you define them if it's hierarchical how many layers should it have basically these are the big questions you have to define the evaluation metric that how would you define different generative models um with each other and then you uh you have to design the online State inference Loop now this is by the way just for this case for this presentation is just about State inference and then you have to compare different genal models this is where beian model comparison comes into play now here um you basically use a dynamical system here I'm just using a lotka voltera process that you know captures the Dynamics between uh Hunter and and the prey together it's a two-dimensional sort of world if you will so by solving the trajectories by solving the differential equations you get to this sort of trajectory so you have 1,000 data points here each which is two-dimensional so this is the unknown states of the world right this is the unknown dynamics of the world to gener the process again it's unknown it's beyond the mark of blanket again so this is the figure that we saw before now you can assume that you got the brain here and you've got uh the generative process xstar on the left hand side the observations why are B basically a noisy version of the same hidden states that you notice that this this is really noisy here right so the brain has to receive that and then infer the true States here right so for a one layer predictive coding Network for these particular experiments you have just s a simple sort of generative model just one layer and then uh I'm sort of comparing two generative models here where the flow of the of the state Dynamics are different on the left hand side you have something that we call just it's just just a pullback attractor uh we have this 2 by two Matrix here and we have this sort of point of Attraction which assumes that everything collapses into this world just just to be clear it means that my generative model of the world assumes that the world Beyond The Mark of blanket is attracted to a particular Center called saigh right that's what I mean by believing something about the world whereas the second flow the second generative model is a sinusoidal sort of belief it means that I believe there is something periodic about the world beyond my Markov blanket and the G's I I just kept them as identity which means that whatever new is is going to be my prediction about my sensation right so I kept the G's identical now the evaluation metric this is very important as the inference is happening online for every observation you have to make an inference um and then predict the sensation and then calculate the prediction error and then eventually calculate your variational free energy because it's happening online you can actually literally accumulate your free energy variational free energy across the entire inference period so at the end you will end up with one scalar value it's called the time integral of of your variational free energy they call it free action right it's just a fancy way of summing up variational free energy across the entire inference inference period in basically it's time integral and it is interesting it's an upper bound to the accumulated surprise so variational free energy is an upper bound to surprise and um free action is an is an upper bound to the accumulat surprise because free action is in and out itself an accumulation of variational free energy and this is you have to also then Define the online State inference Loop uh I don't want to go into too much detail here but um basically for each observation you will make predictions and you calculate the prediction errors you calculate your free energy you calculate the gradient of free energy with respect to your estimations of the posterior that's the key part and then this is where you have to update your beliefs about the world and it's you notice you notice that it's a differential equation because the way you're going to update your um your belief about the world it's continuous it happens in continuous time it means between each two observations Yi and Yi + one I have to integrate over this update rule to move from my previous belief about the world mux to a new mux at the next time step and that's why the integration happens here and just so you know this equation is basically saying that the amount by which I'm going to change my belief about the world mux is first of all the gradient of free energy with respect to mux right that's typical gradient descent but the second term it serves as a momentum it means that if I'm going to at my belief about where the world is I'm going to consider the velocity of the world as well right if I'm going to estimate the velocity of the world I'm going to consider the acceleration of the world as well it's it's similar to the idea of momentum in optimization algorithms this is the bit that helps you you know catch up with the chasing Target that is the world that I talked about in the beginning and then by you integrate this differential equation to update your belief about the world and then the next observation happens so this continues for all the observations and you're accumulating your free energy as time goes by and the result will be your ve your free action so the result is this so this is the first generative model with the pullback attractor um so this is the uh infert state right and this is the this is interesting this is the accumulated free action right um very very interesting you notice that free action um is having these regular jumps it means periodically I'm getting surprised about my Sensations remember that if you note here the this is where these nonlinearities happen it means that my simple pullback attractor fails to capture those Dynamics so it gets surprise but interestingly where things are linear and simple I'm not getting much surprise so it's almost constant here so I'm good at capturing simple Dynamics but not highly not the ones that are nonlinear whereas the sinusoidal one you notice that we are capturing the magnitudes much better we have a steady increase in free action we don't get surpris for some Dynamics and less surpris for other

Dynamics we are always equally surprised for everything which is interesting that like here we are getting surprised consistently for everything but here we are not getting much surpris for these particular Dynamics so one is good Capt good in capturing linear Dynamics periodic Dynamics the other one is better in capturing simpler linear Dynamics which is interesting and notice that the reaction here is much lower than the freeaction here so basically this means that this model is much better now looking at the Gen the generative power of these uh GMS so this is the G function if you remember G of X and mu that's the prediction of your one layer predictive coding Network about the sensation that's the bit that gets compared with the true sensation and the prediction error is calculated remember that observations just like hiden say are two-dimensional so this is the first dimension of why the true sensation and why had the predicted sensation you notice that they're almost close and also with the second um Dimension they it's not doing a bad job right but for the second generative model again it's doing a much better job in capturing the magnitudes of this of the system again you notice that it's not so good in generating these bits whereas this the first model is better in generating predic more accurate predictions of the of the first St now these are the results the pullback attractor and trigonometric the sinos soidal um sort of flow you notice that the freeaction of the second model is lower it's better and MSE loss which basically comp uh compares the estimated X and the true X the inferred X and true X together is also slightly lower so free action shows that the second model is good and this is where beia model comparison uh comes into play where we use base factor for comparison it basically says that we take the the model evidence for each one of the models and we divide them now because these are intractable we're using free action as a proxy and when you put the values of free action you get to this value because it's higher than one it basically means that you have to pick the set model because it has a lower free action it doesn't get as surprised as the first one and at some point of caution free action versus MSE msse as a measure of accuracy should never and I repeat never be consulted on its own because it ignores complexity so for example what if the freeaction of model one and model two the model one had a lower freeaction but what if the msse of the first model was actually higher than M than the second model which one would you choose again you have to choose model one because this has higher power of generalizability to the Unseen data so it has less chances of overfitting to the data we call that free action again as complexity minus accuracy so what if we remove complexity again we will overfit but that was important some final thoughts you can definitely learn the parameters and also estimate uncertainty about the world so these are the things that are not covered in this presentation but you can do that this is where the separation of time scales um will come into play the Dem algorithm does exactly that this is dynamic expectation maximization I encourage you guys to take a look at that algorithm it's you using that you can solve the triple estimation problem now can I go beyond x dot yes you can that's where you jump into the world of the generalized coordinates of motion where you're not only using X estimating x dot but you're also estimating higher temporal derivatives of uh of the world you have a a system of couple differential equations rather than having just two equations which is very very interesting actually um also remember that active inference is not the same as predictive coding um particularly in the philosophy um you know um communities I've heard that people use active vence when they're talking about predictive coding remember that in predictive coding we are simply doing um making sense of the world there is no concept of planning or action in play um the ml Community unfortunately um does not work with the Dynamics of the world and AI uh um basically is going to the opposite direction instead of uh using smart data we're just going bigger and bigger data so we would like a world that chat GPT would also ask you a question and becomes curious about you to receive information that it selects basically so where should you start with with the free energy principle these are the your sources that I strongly recommend you some videos on YouTube hopefully you can later on pause this video and go through these links some papers that I'm recommending here and the last one the two books that I believe the red one the predictive minus for philosophy community and the first one goes through the math extensively it's a beautiful uh book yeah the uh and the videos I I I recommend that you guys start with the videos in the first place and I would like to just acknowledge our funding from the merury and also my colleagues Simon bbas and professor priston and also acknowledge the folks at UCL UC UCD and monach University have always supported uh uh our work and here's the link to the the code that everybody can just take a look and run the read me file is self-explanatory unfortunately I ran out of time and I won't be able to show a demo but yeah that's it so thank you Maron that was great and and uh we have 10 minutes if you want to show anything or I can ask some questions but but we have till five minutes after if you want to show anything or even just scanning the repo could be cool okay sure sure sure that's great thank you for that but just wanted to comment your your educational material and the YouTube channel are great and and they are they're an excellent supplement to the textbook and to the math learning I appreciate that yes um absolutely uh so yeah I also have a YouTube channel mlon feel free to take a look I'm actually reading the active inference book on that channel so hopefully it would be of interest to you guys and this is the repository that is publicly available um the read me file is pretty self-explanatory and this is just a one layer predictive coding Network for the task of inference and modeling human perception as a task of inference these are the same plus that you've seen um so basically all the functions are here right everything you need is here so the main function is the one that you're going to be dealing with and the yaml file is where you actually choose the parameters uh of your experiment just to show you in action what it looks like so if you actually clone this repository on your machine you just have to go to the yaml file and just a quick scan of the readme file but basically U using this is where you define your generative process for example you can say I want it to be a lotka voltera process and this is how I'm going to be picking the DT and the total time uh to actually solve the trajectories of my generative process you can Define your generative model here you can Define the Dynamics to be like a pullback attractor or a trigonometric one the likelihood for example is identity here these are the dimensions of your X and Y two and one you might wonder why it's two for X is because you have x and x dot together where the world is and how fast the world is changing and this is just Why in your generative model you can increase them uh that's where you jump into the world of generalized coordinates of motion but this is beyond the scope of my talk today again this these bits are beyond the the scope of our talk today this is where uh you basically C talk about the correlation between the noise process across different levels of the generalized coordinates so the noise on the velocity how does it correlate with the noise process on the acceleration your belief about the acceleration of the world so how do they interact with one another that's where these um values come into play and then this is the kind of noise that you will use to alter the true states of the world and generate your observations for example you can use a colored noise here you define the mean and uh basically the standard deviation of noise and this is we're using the sort of Kernel here that convolves with the white noise to actually generate a more smooth noise the reason being that um the biological systems are believed to be uh working with uh you know to so all all the signals that you work with in biological and organic systems are believed to have been generated by some sort of Dynamics that's that's why we keep them as smooth noises infinitely differentiable noises rather than just white noise and uh if you actually run this so this is basically a pullback attractor this is the first uh sort of generative model that we worked in so you just run this main function so we have 1,000 observations noisy observations right and immediately this happens so for every 10 observations we are accumulating free action you notice that free action is ever increasing because it just accumulates free energy and on the right hand side you have VF or variational free energy um again if you look at the definitions inside uh the functions folder of how we calculate or compute variation of free energy you notice that um it's a far more simp simplified version of what you would see about energy is because everything is under llas approximation and you end up with far more simpler terms and the final result is reaction is this much and basically your final msse loss is0 53 and if you want to see the final output you just have to go into the results folder and these are the folders that are generated just so that you have two dimensions in x one dimension in y lot voltera is a generative process for example do of model follows a pullback attractor scheme and then your G is just identity this is your free action and then that is your Ms Loss and if you open it all the plots that I showed you in the presentation are saved as PDF here just to give you an idea like this is your X this is your true invisible uh states of the world right and if you want to look at exat this is the inferred states of the world and these are for example the the noisy observations of the world yeah so I hope that the repository would be um of help and of interest to all of you thank you very much great what kinds of problems or settings do you feel that this could be adapted to or useful for um I would say first foremost this could be used for um basically it's it's all about any kind of problem that's about inference you have some noisy observations that you want to infer the hidden state but in terms of adaptability you can build on that by introducing parameter learning as well as uncertainty estimation so our goal is to develop this into a triple estimation sort of package predictive coding Network and add more layers to it so to have a hierarchical predictive coding uh capable of solving the triple estimation problem yeah that's the ultimate thing that you can actually adapt this cool um what are your uh what are you excited about for ML Dawn or for your learning or research in next year um I'm very much excited in embedding these kinds of calculations into the world of AI because as I said AI is moving in my opinion and my humble opinion is moving to the towards a wrong direction instead of bombarding models with all sorts of data that the model should be able to choose its data so I'm very excited in you know developing this framework into a hierarchical predictive coding Network and then sort of compare the performance of such a model with other traditional AI based generative models and basically show its superiority um that I believe it has in different scenarios and ultimately this model is going to be used for in my grant proposal in a project I'm working on is going to be used for modeling hallucinations as a kind of false inference about the world um and it's going to be used as a proxy to for us to understand what happens in uh Neuropsychiatric disorders when people actually suffer from hallucinations in the brain so this predictive coding network is going to be a brain inspired AI model for us to understand what goes wrong in in the skull of a human being when they suffer from hallucinations about the world so that's what I'm really really excited about and really hoping to achieve cool it's like inspired by the brain for the brain absolutely thank youon great great presentation and good luck with the continuation of the work appreciate it thank you for having me okay till next time e e e hello we're back and with Demitri BV and the RX and fur developers in community so thank you for letting us peek and join into your meeting really appreciative for all the collaboration and learning that we've had with RX infer this year and looking forward to updates uh thanks for he us yes so yeah usually we have update meetings about the development for and fur but since we have uh now active in Symposium and we also have a new PhD students in our lab decided to make it a bit special and today we so brief I will briefly introduce arer again for people who are not familiar maybe bit and then we will discuss actual recent developments of it and it's just a basically an open discussion right you can you can stop me at any moment with some questions I unfort don't see YouTube so if there are some questions on YouTube just stop me and and I'll discuss um so okay so arxon fur uh is a library for scalable probabilistic program right uh and but what what is a probalistic programming by itself uh and let's let's imine we want to build an age like a drone right and we want to build a program for it and write some code and usually what we do is we take some data and uh we feed it into our algorithm and we get the result for example we find the pth right but in reality it's much more complicated because we don't really have a precise data prise measurements of our sensors or maybe we have some wind outside so our formula is not really correct in this conditions and basically we don't have exact values but we have some uncertainty around our parameters and how do we actually compute the result given the uncertainty around around the parameters that's the first question uh but more interesting question is actually uh the backwards question so given the observations of our path for for our agent what were the conditions under which it has been operated right and usually it's also it's cing around some uncertainties uh and in the in the context of active inference uh we can go even uh further so we can say Okay given the desired observations like given what we want to see uh what actions we actually should take given that we we don't also know the the conditions under which we operate so this is what essentially probabilistic programming is about uh and it's it's very challenging but good news is that we have the solution for it we already have it we we we know how to solve uh and this is this is the formula that solves everything essentially and this is the base rule uh and it looks pretty simple right it's just like one multiplication one uh not a big deal but it turns out that uh so this formula is is really

really challeng is is extremely challenging to apply in real situation uh and why is that well okay we we can I can I can demonstrate why so if we have a simple program uh with not a lot of parameters this this formula indeed is quite simple so you can actually solve it on the piece of paper uh and you will get the exact result uh and it's all fine but in real situations uh we actually have much more parameters and this formula uh it explodes pretty quickly and as soon as you add more parameters to your model uh you realize that it's no longer feasible to solve it on a piece of paper uh actually you need to write like an entire article about how to solve it for like some specific case uh but we we we want even more right so we want to not just have parameters but we want to have some constraints in our system uh like for example physical constraints like maybe we have an engine power constraint and like we have some trees around in the environment so when we deal with such a complex system the the math behind is just uh you cannot even comprehend it at this point right and actually in my opinion this is a very big problem in so active community in general is that map is pretty difficult um to to read uh and to like to double check and and to implement right so uh usually In Articles we have a lot of formulas and they they work in theory but as soon as we start implementing it uh we make mistakes uh our programs they have bucks and then uh it it just takes a lot of time to iterate through that uh but ideally we we don't even want to to think about it right we we want to design an agent and all of this math under good is known so this Bas uh and what we want to do is we want to have as many components as we want we want to press a button uh and we want to compile with the maash we want test works and just deploy it right uh and it should result in a robust systems that makes reasonable actions under uncertainty and unpredictable unpredict environment and if you think that's too much to ask I would uh actually draw an analogy with uh just regular programming so we now talking about probalistic programming uh but regular programming also started um or has the same path so in the beginning uh like a bunch of high level professionals phds they were designing the very very first algorithms uh and ifos error prone they they spend months designing a very simple program spend months designing computer uh but nowaday uh every yeah student or even in schools you can just download Python and write simple programs you don't you don't need to think about how computers work to do you just express your ideas in in some sort of high level uh programming language uh and compiler does the have Lifting for you so essentially what we want to achieve here is to have a language for uh solving probabilistic programming right and you don't have to think about math you only want to think about your application you want to design your drone or maybe it's a Buton cleaner or maybe something even more complex uh any questions so far maybe from YouTube not yet thank you though okay so uh okay so this is kind of the set right but uh how do we uh approach it how how can we approach it uh and in a sense there are a lot of similar projects uh in in various programming languages as well and they try to solve the same problem in a slightly different way right so and given that even in this slide there are like eight Solutions but in reality there are more why do we need a new one why why did we decide to write a firm where there are so many solutions already developing well um the idea here is that um for um autonomous agents we have um a pretty unique set of requirements so if we want to operate in real world uh we want to have a very big program because a program of a real world is is quite complicated has a lot of parameters so it should scale uh it also should be low power because our autonomous agents they usually operate on battery and you don't really want to uh have an empty battery soon right so it should it should preserve energy in the most efficient way uh it should it should be able to adapt to changes in the environment because the the real environment is chaotic uh it may change maybe you didn't account or something your model so you should be able to change the model fly should work in real time should be um yeah as efficient as possible in terms of performance and it also must be robust uh so we we basically are not allowed to stop our competion should run and should react circumstances uh and this is what a recent fa is essentially about and what it tries to achieve uh and in order to achieve this uh it combines this three main ideas um also in the literature right Factor graphs variational inference and reactive message passing so this these are the three key elements uh to and we basically try to understand here if if we combine them can we can we have these properties that I mentioned um and so to to talk a little bit about each of them so a factor graph is essentially a graph representation of a problemistic model uh and a good thing about Factor graphs is that they are explainable and they're not a black box right uh so we can in we we can clearly see the structure of the model and dependencies between different parameters or hidden States and also observations uh and it's also modular local we can reuse certain pieces and we can write specialized computations for certain pieces um so in AR Ander we have a special language to write such programs so have this model macro that take textual description of a puristic model uh and basically creates a factor graph under the so a user I should not create those graphs uh by hand and this is alterate uh and we can also create like sub module sub models and we can reuse them uh in the bigger context and as I mentioned uh we can for example do some local optimizations in our model because we clearly know the structure of it um so variation inference I assume that most of you may be familiar but for those of you who are not that's a basically an optimization procedure that approximate a base R so uh based rule in many situations is very challenging to solve directly uh so ation inference is is a very clever technique to trade off accuracy and maybe solve the task a bit easier at the cost of accuracy yeah and also it's it's faster because uh you don't have to solve the original problem exactly um so the formula actually looks a bit more complex than base rule itself um but yeah the math is sometimes strange and in practice it's actually easier to to solve this problem that the orinal uh so instead of solving the base rule directly we approximate the result with a simpler set on distributions from exponential fames for example which makes it simp and we have also language for for constraints for the variation in right so sometimes we want to make a mean field over our hidden States and maybe for some specific state B the ation so a user also can specify different constraints different speci um and last thing in our recipe is reactive message passing which is a central part of the the project so and each message carries a small piece of information required to solve this very complex math behind uh the inference procedure uh and um essentially in traditional message passing algorithms they require fixed Global shutle which has a lot of downsides and in AR recent fur we allow messages to flow asynchronously and each not node in the graph simply reacts on changes uh in neighborhood and Rec computes the result loly uh and it's more robust because if part of the system doesn't work we simply stop react on it uh but uh also it in principle supports variable update rates for different variables for example uh um orio signals they may update at a higher rate than V uh do we have any questions so far yeah I I can ask one now and you could go for it or address it later asked what has been the biggest yeah Arun asked what has been the biggest challenge building RX for so far uh oh uh that's a very good question so the biggest challenge um I I may say that uh the as as I mentioned already the math behind all of it is quite complex uh but the implementation side is also complex so in order to build such a system you need an expert in both uh you need to understand both math and a high level uh programming to make it efficient so the the the combination of both of these elements um is I would say unique uh and without understanding one or another it's it's it's basically impossible to be to build such a system any other questions okay um okay so uh basically to write the problemistic program we need to specify our model or a program if you would like uh we need to specify our variational distribution uh and we also need to specify how do we compute posteriors uh so we can Define models in our uh package we can Define variational distribution and for the minimization we use free energy minimization rational free energy minimization is just a single call to infer function that combines both model and constraints uh and runs the algorithm uh and it supports both static data sets but uh as I mentioned the whole idea about ariser reactivity and it also accepts reactive streams of data sets from external sources uh and this is basically we believe that these three key elements uh they do have properties which I mentioned earlier and they do allow us to um Implement all of those uh and AR andur uh yeah is free it's available on GitHub we have a website we have a documentation uh we also presented various conferences and also on YouTube um so yeah you you are we we invite you to check it out essentially and see yeah what it can do for you uh yeah so if we have have any other questions right now we can no because we just had the questions but again yeah free free to um stop me at any moment free so okay but the question is doesn't actually work uh so I had this example of uh the Drone uh and basically if you want to implement drone and there is we Define the Dynamics of the Drone how for example different parameters should influence its position and velocity and speed and then we Define a Model N ver uh and then basically we create an environment for it uh and see how the Drone would infer the action given its current state and the goal right so in this experiment uh our drone is instructed to fly to the Red Dot uh and it actively tries to infer uh its current position and the trajectory that it needs to take to reach the goal and and also on the bottom side of the screen you see that we change the parameters of the world and also the parameters of the Drone uh reactively and yeah the system still uh operates uh and actually to be honest it was much more difficult to create the environment for this drone that the Drone itself so first simplified this Stu but the environment was was still a challenge but for that we also work on a solution so which is called Aris environments which essentially would also hide all this complexity of building environments for uh different agent and we also we have a a set of other examples uh both for static data sets or preactive data sets uh yeah just feel free to check out uh our documentation uh and yeah we have reproduced many interesting bent inference tasks give a good performance and scalability so we we are say not solving all problems in this world uh but the set of um possibilities is is really uh other questions yeah I I'll ask a few from the chat so Discovery block wrote for the Drone would RX and fur be a able to manage actual image input to infer the position or velocity of the Drone instead of assuming the observation corresponds to real values with random error uh okay it's a good question not out of the box uh it depends on your model so in principle it's possible to design a model that would process images but uh we didn't try that so uh but AR Fur by itself is indifferent on on the input right so the in the the problem with images as input is that they're pretty high dimensional uh and since arer works on distributions this this image must be processed as a distribution so this might be quite challenging in practice but that's a direction to go with the composition of these systems so the Drone kernel could be designed with an app model taking in beliefs about position or about other features and then hypothetically like an RX infer or alternative approach to image processing could be brought in and concatenated and used and then the the kernel belief would remain the same uh exactly yes you you're correct okay and one more question from Frasier what is the largest remaining theoretical difficulty with this approach real time inference Universal inference um okay it's a good question so uh the limitations we we will talk actually a little bit about it in the current developments but I can already hint that indeed Universal inference is quite difficult because uh in order to execute the inference procedure we need to compute messages uh and we we can we can can compute them in many situations but in many situations it is still um way too challenging and we are actively working on that uh but yeah message computation is one of the main challenging components still awesome and one last question Bert and Arun both asked have you considered hierarchical or nested models or any of the renormalization type models we've been exploring uh great question yes we did and actually we last year we reimplemented completely the the language that we use for models to support hierarchical models and this is one of the directions that we are approaching currently in Bice laab as well uh to support to support more complex High models uh and also recent developments in active imprints literature yes questions no not yeah these are good questions but we can continue uh so um what about our current development um I can so I will talk about two and then I will give a stage to vter uh so we have an external C operation with active INF Institute about graph visualization that P the textual description of the model and actually display the graph so we have a data structure for the graph but we don't have a function that would create nice PNG out of it like picture uh and we have an ongoing project to um for the universal inference essentially for uh for to support more models so for uh graph visualization as I mentioned it's an external collaboration uh and it goes really well so we have an open pool request uh and for example for this simple model for a simple Point model pointos model I can generate this nice graph that will will show the dependencies between variables in this model uh and I can also generate uh more complex models or graphs for more structures uh and uh if uh yeah so and we have also Fraser and uh Chris joined our meeting so if you Fraser or Chris you want to talk something uh here you can yeah you can do that basically yeah I'll just can anyone can everyone hear me is that uh is this coming from yes yes uh yeah just really quickly so I'm I'm Fraser I'm the guy on the top right bubble there so along with Chris and Kus EST hoisen we have a project effectively at the Active inference Institute as Demitri said in collaboration with the bias lab to to visualize these models to be able to get some handle on on what they look like and ultimately to be able to even debug where certain messages are and and you know where issues are with respect to actual message computations um there's a there's a page essentially at the um I I could show where this is but um there's a very a publicly available page where you can kind of see our progress track you know what's happening and even reach out to to to join us and collaborate what where you will so it's very much open for um for collaborators and and participation absolutely Chris is there anything you'd like to say nope I think you covered it all thank you excellent back to you Demetri uh thank you uh Chris uh or or if you don't want to add anything we can just continue please continue okay that's uh great thanks uh and thanks for your effort by the way this is a really cool stuff so I was playing with with to prepare for the presentation uh and it Essen just works yeah so there are some yeah I suggested some improvement uh but yeah I think it's it's all good even in the current stage and also uh for example other um probabilistic programming libraries uh in Python for example they do similar approach uh in their modeling and also dis uh yeah how they display models um Okay so um so the next thing that I want to talk about is um yeah Universal inference essentially and we already had a question about it so suppose we have this model that essentially tells us that we have observations why that are coming from a gaan and the mean of this gaan is modeled as a beta distribution and by Beta is because uh the variable that is modeled as a beta distribution is constraint to be from 0 to one uh so basically I say here that my observations are constrained from 0 to one uh and I also say that the Precision of my observations is an exponent of R and R is also from Z to one so it's essentially it's a pretty simple model uh at a but the reality is that this is an extremely challenging model for uh message passing uh because the messages that are uh yeah that we need to compute are not easy to compute um so and the problem is that that the result of those computation is essentially a I say a bad distribution um so it has a it has a very strange form that is not parametric or cannot be easily described as a parametric distribution so we can also always has described those distributions as a set of samples but drawing a set of samples is uh costly uh and U it's essentially very difficult to make it real time so the idea here is that okay we we want a nice distribution us parametric and it's usually from exponential family and the question is here okay how how to find this best approx IM uh and essentially aren fur supports algorithms or approximations that would allow you to specify how you would find how you like to find those approximations and one of the recent uh algorith is a uh projections so this is is a very very new feature so what we can do is we can specify or instruct fur to use the projection algorithm to Beta or to other distributions from exponential family uh and it also works around the terministic noes or just for individual variables uh and if we combine these specifications so it's a little bit more complex than the original example but given this specification we can solve uh inference in this model um and the result is pretty good so I here I just display uh yeah generated some data set and I tried to infer the parameters back so as I mentioned in the very very beginning we have a set of observations and we want to know the parameters with which those observations were generated uh and you can see that uh the infert or projected distributions they are extremely close to the the ones that I used to actually generate data set uh and it scales pretty well comparing to other probabilistic programming libraries um so here on the left side bottom left side which US sampling and it takes up to three times or yeah real execution time to do this task and on the bottom part we have a automatic differentiation variational inference algorithm which is faster but uh we we basically uh because it's a new feature we still try to improve the performance and the performance is not uh ideal at the moment and we already know uh yeah how to improve it essentially we have a PhD student that works on this project and it goes really well and like around the week ago we had yet a better uh performance which is not reflected on this slide but um since it's like just last week we didn't have really time to invest properly uh any questions so far what stand n um not it's something like no uturn sampler uh and uturn is a is a yeah term that they use to describe their algorithm uh the yeah it stands for no urn s it's it's practically as far as I understand one of the best sampers right now to draw from an unnormalized distribution and those timings by the way uh so uh we Implement recent par Jula and those timings are from a different PR Library also written in jul so these timings are from tubing um okay so we have also we had already question about General influence models and here I will give stage to vter now hello after my name is R I'm also a PC student in bias lab and I'm going to have a short uh discussion about uh generalized inference in discret models so in active inference we really like work with pomdps partially observable Mark of decision processes and the simple pomdp or the vanilla PDP uh you can see on the slide so a joint distribution Over States observations and transition matrices and we see um that we have transition uh distributions both for the observation model and the state transition distribution okay you click so both this uh this observation distribution and this transition distribution they're both categorical transition distributions but they're slightly different right because the the observation model just depends on one categorical distribution or categorical variable only the state whereas the transition model depends on both the state and the control right so if we draw the factor graph representations of both these uh these these nodes you can see that they're they're extremely similar but they're slightly different now if we want to go towards this generalized inference there's two things we can do the first is we just implement this new node this uh two categoricals in one categorical out uh and we can just be done with it and uh we we could do control in in discrete environments but it it wouldn't be very general and since these transition distributions are very similar their functional forms are very similar the inference rules are also very similar so what I am investigating right now is how we can generalize this inference in these discrete models such that we can have an arbitrary number of categoricals coming in and transition to a categorical variable coming out so this would give us a general contingency tables or general uh discrete categorical transition distributions uh in RX f um there are some problems with this because this B Matrix um has a rank which is proportional to the amount of categorical interfaces so for the observation model we just have a transition Matrix but if we want to do discrete control then we already have a transition tensor which Maps States and controls to new States currently in ARX fer we only support these rank one or rank two dis lay distributions so this is something that we're working on right now to generalize this uh this inference so what we are currently doing is together with Rafael we are working to replace the rank one rank two Matrix DeRay implementation in arer with the general arbitrary rank uh tensor implementation um the the working name for now is a multi- durish lab we're very much open to suggestions um I am currently deriving rules for this General transition node so T already in this dissertation has the rules for the rank one and rank two uh 10 uh factors and matrices and I generalize them to arbitrary Rank tensors and then I will make this new transition node which takes an arbitrary number of interfaces so what can we do with this with this we can do control in discrete models so uh you can see on this slide that we have an agent which is navigating a maze and this is the type of stuff that will be possible with the generalized inference in discrete models because we will have categorical States coming in with categorical controls coming in um and a categorical State coming out of this so we would be able to go towards building um agents in discrete environments with discrete controls so we can go ahead and solve on the piece as soon as we're done with this uh thanks Al that uh and here it is here's the the entire presentation that hope gives an idea of what arer project is about uh and what we are currently trying to develop uh and yeah maybe we were a bit too fast but we have a plenty time for plenty of time for discussions do I have more questions online or rer if you want to ask your question directly one but he's also in the com yeah yeah just uh I was interested uh I'm relative I'm very very interested I did mathematics and computer science for my actual degree very interested in the the mathematics of the generalized inference and and node conr and all of that to what to what extent does message passing per se the the process of message passing um to what extent does that does that influence issues around non-conjugate inference is it effectively kind of only dependent on the content of the messages or does does the actual process of message passing have some kind of non-trivial effect on the on on the resulting inference I'm just curious about that so uh the problem with non conjug non- conjugate inference in message passing uh lies in the fact that in order to compute a message we need to compute an integral uh and in general computing an integral is not possible on a computer I'm talking about indefinite integrals that has no boundaries so for an integral that has a boundary from Like A to B say you not from A to B you can yeah indeed decompose it in to Regions uh and then solve this um integral numerically but for unbounded integrals that take from minus infinity to Infinity basically it is impossible right because we have finite memory on computers uh and you can always create a counter example where it doesn't work uh and it's even worse in um in multi-dimensional spaces so soal Cur curs dimensionality so and this is why people are coming with u many clever methods to take those integrals via sampling so sampling is essentially an algorithm to take an integral so with non conjugate inference uh uh so let me phrase the other way around so with conjugate inference those integrals they can actually be simplified to um an exact formula so you don't really need to solve the original integral you solve a much simpler problem because you can rearrange some terms and something just canceled out and you get the exact answer the non conjugate iner you cannot do that in general so you need to actually solve this integral and here is why you need those approximation methods like sampling or projection which essentially um try to solve this problem but at the cost of accuracy because we cannot afford um yeah infinite memory in our computer awesome I see so if I just just try and so effectively boils down to level ways to compute integrals in in ways that you know it's not going to take too long okay exactly exactly and you you also have to remember about uh dimensionality of your problem so uh for for example as I mentioned one of the recent developments that we just uh included in a re F would allow you to project a high dimensional gaussian distributions uh because before um so a gaan distribution can be described with its mean and with its ciance metrics uh so which means that for high dimensional G in distribution The cence Matrix is quite large so it grows uh and we recently added a simplified version of a gaou and that doesn't store the entire ciance Matrix but only the diagonal of it and only the same element on the entire diagonal so the number of parameters is essentially linear to the dimensionality and it scales much much better and it allows you to yet again solve bigger problems um yeah in in general in general this um this is a quite difficult problem I I'll ask a couple questions from live chat that's okay okay okay erun asked is rx and fur being used in production right now uh well we actually have been contacted in the past yes but I'm not I'm not aware of uh uh yeah any company actively using it we we uh actually we have a spin out from uh Technical University of that we yeah we founded a startup uh from bab Group which is called lazy Dynamics which uh the the entire idea is to commercialize aren fur to make it a professional toolbox with a really good support I mean it's it's always uh going to be free for research uh but uh yeah lazy Dynamics will uh create uh ice Tools around fur to make a process of uh creating models easier making process of deploying those models easier debugging models easier uh but essentially RX infer is free and perhaps someone is already using it in production but we may not know cool great okay next question from Dean tickles he wrote do your rules self-organize or do they remain constant does the rule recognize its own limits of applicability and adjust accordingly I'm just wondering if these rules can morph into heris sixs uh sorry you were glitching a bit can you repeat question yeah do your rules self-organize or do they remain constant does the rule recognize its own limits of applicability and adjust accordingly I'm just wondering if these rules can morph into heris sixs uh no like message passing if the questions about message passing update rules they work independently uh and this is also one of the main ideas of this approach is to not have any sort of global variables uh and uh yeah the graph is essentially each each message is unaware of any other message or computations such that we can run them asynchronously uh and do not have any sort of shling mechanism uh that would yeah decide how the messages should be computed or how they should cooperate awesome okay next question from he wrote is it possible to add a deep learning encoder and train it using messages Yeah we actually have I think we have an example in documentation where we use a neural network uh and we also use a normalizing flows inside theen fur so yeah that's definitely we also have an example where we use differential equations uh to infer parameters of differential equations maybe questions from students about the approach you know okay okay what what do you all usually do in the meetings or like how does it go and how are you going to be taking things in the coming months well we usually yeah we we already discussed the recent developments and already showed that yeah we we do exactly that we go through the current developments and we've tried to yeah see what we can do next so uh maybe here if someone in the chat would just post what for example they want to do right like what kind of applications they try to solve with active inference that would help us also to change our agenda because we have some certain things in our agenda so we want to continue with graph visualizations know that for example in the current version uh we don't have constraints visualization but constraints are pretty important uh for the inference so this is the next step in our agenda to also Implement con Str visualization uh we also have non- conjugate posterior computation in our agenda we have generalized inference in our agenda but for example people can also comment on their uses or they can also for example open AR recent for GitHub repository and open discussions right so maybe not right now uh maybe if someone watches this in the recording or maybe someone who is online but still uh don't have a concrete uh example you can always go to GitHub uh open discussions uh and just drop your ideas there and we can discuss and maybe we uh adjust uh yeah our goals of the project together and we are also looking for collaborators right so again Fraser and PR were super helpful thanks uh but if more people want to collaborate it's a uh it's an open project you can always just join uh the organization on GitHub uh and drop your ideas and collaborate awesome also I wanted to highlight kass's examples at learnable Loop and also the documentation that you all maintain and just looking back at our meeting notes from the beginning of the year and uh looking at the version history of RX and fur it's something really new it's really been one of the highlights of the year and seeing how it's all developing and a coin toss I mean never has a such a simple seemingly simple statistical situation been such an amazing learning opening and it it uh it's just exciting to see how the examples start to flow in and then those are useful for Learning and they're also useful for training code model models and the the the Paradigm difference with RX and fur which I think you very quickly move through with the app model declaration and reactive message passing is is really distinct from the script-like flow of essentially all other otive implementations so it's just really exciting to see and and also that this isn't just about action perception modeling we o see in the examples more conventional statistical models so that helps bring continuity between the joint distributional modeling modeling of active inference and all of these more standard statistical problems uh I also no just uh because I stopped presenting I had the chat YouTube and I see a question what's the best way to get started with aring uh and I think the best way to get started is to go to the doation that is to the page is called getting started uh but yeah so basically to start with the documentation uh that goes through the process of installing the library running first models uh and then um we have also a big collection of examples as I already mentioned maybe some of the some of the applications you can already find here uh we have we have also examples about active inference but not only uh we so This Global parameters mation uses neural network under the good so with lstm driven Dynamics oh something something is broken live stream issues we need to open the issue about but um but yeah this this example uses B neural network to train parameters and then lat on use it and we have also external examples from kobus we have also series of tutorials uh on AR and fur on YouTube from Toco to J uh from Theory point of view I really like the book that is called uh model based machine learning by John wi by John win so this is also a very good uh starting point it also explains the concept of yeah Factor graphs stochastic noes deterministic nodes the Run inference but also it it uh it explains the concept of uh running inference essentially yeah I have a question actually M so the the projects of uh um no conjugate posterior computation m to me sound simar to variational inflence so how is that great so uh variational inference uh is is a very very good question so variational inference uh indeed reframes the original problem as an optimization problem and you can solve it as a black box indeed so you you parameterize your distribution you take uh yeah gradients and you do a simple optimization problem and actually it does already solve many problem but uh in in arer case we actually do not treat models as black boxes we decompose those computations in a series of smaller computations uh which in a sense helps us because we don't need to run this big Global optimization procedure but uh in the result the result of this decomposition so this individual computations messages they still may be difficult uh and this projection algorithm is indeed uh yeah sort of a variational inference but at a smaller scale okay yeah but uh what what you maybe referred I can also refer to this ADI algorithm that I also showed in my slides automatic differ variational inference which yeah takes the entire model as a black box just just differentiates through it and yeah it doesn't care about contri thank you for this any last short comments from any of the RX and f team okay thank you again for for letting us kind of wander into your lab meeting and really looking forward to the collaborations that will continue okay yeah thank you thank you again for inviting us uh and yeah we are looking forward all right till next time bye hello we're back with Samuel ner Jonathan Larson and Peter Wade discussing active inference JL staying with the active inference plus Julia theme thank you all for coming on and sharing this work so looking forward to it [Music] hi everyone it is a pleasure to be here so like Daniel said will present um a new package for doing active influence simulations but also for applying it to data in Julia and we will leave time for questions at some moments during the presentation and then sometime in the end additionally um with three people because been working on this together so we'll alternate a little bit with the setting all right let's rock so first H it's nice just to mention who we are so we're all at o University I'm a PhD candidate in s and Jonathan are students who have been working on developing this package with me and we all come from computational cognitive science specifically it's called cognitive modeling you Model Behavior with competition models and of course within that specifically predicted presing models including active influence models and that's why we made this package and we use this in the context of competition Psychiatry for clinical questions and also more broadly for just modeling behavior and this package is part of a somewhat larger um process or project of developing a software for these kinds of research questions in Julia and I'll start with just giving a brush up on what cognitive modeling is in general so some members of the audience might be familiar with this but others might not um just to set the stage for an introduction to active inference you all have seen this before so we'll go through it briefly but it's nice to have this FS then we'll get to the meat of the matter which is of course how to do this specifically in code with active influence. ja and both simulating behavior and also fing into data and finally if there's time we have some applications and some future work that we'll be happy to spend some time we might skip that as well all right cognitive modeling or computational cognitive modeling um comes out of psychology C to science where the main question in those Fields is to find out what are the mental mechanisms that makes us do the things we do and so the classic perception action Loop we've depicted here people maybe be familiar with that too it's from the our activ book where you have some agent which could be a person but it could actually also be an animal or even an organoid or even a simulated um thing system that receives inputs from its environment and takes actions that changes its environment and we are here interested in essentially and I don't know if you can see my cursor um but essentially the relation between those observations and the actions that are created so basically the part um after the observation error ends and before the action error start and um traditionally in Psychology and in cognitive science this been done so um you take people out of natural situations and controlled laboratory situations where you measure well you you control both of these so you decide what they observe and you measure what they do and so the kind of classical example of this is have a computer screen you show people some image of something maybe you asked them to classify them or predict what's going to happen and you measure what they do and then you have a measurement of inputs and measurements of outputs and you then create some theory that will make some prediction as to how they relate and of course this is a big field you might have theories of personality types where you predict that actions will change faster depending on inputs there's theories of reinforcement learning Etc and so importantly here we're using already Natural Science approaches experimental approaches with quantitative analyses but still with purely conceptual theories so the theories are not formal as they often are in let's say physics we have mathematical Theory and the trick is often to go from some conceptual Theory to a quantitative prediction that you can then test and specifically give me a moment here sorry about that um specifically as many will know there's been a what's called a replication crisis in Psychology and cognitive science where the whole issue is that the experiments don't replicate and that's partly due to statistical methods issues but also partly due to problems stemming from the fact that our theories are not um clearly formulated so one sentence that's saying that there is social learning might mean many different things and so um a recent attempt to address them is to move into form theories just like in physics so computational models of this again relation between observations and action and this is inspired by physics so in physics people are no strangers to having things they can observe in our case it's the behavior but in physics it can be let's say radiation from the Sun and then trying to make inference about something you can't absorb in now case the mind and this cases perhaps the molecules or the of the Sun and so the methodology is exactly the same create mathematical equations that will describe how actions evolve over time given observations in the environment and test them on how will they describe empirical collected uh actions observations and here I just put on one of the very classic cognitive models which is the scholar Wagner where you have some very very simple equations that describe how actions depend on the observations in the previous time step with some learning and a classic learning dat and that's just to say that the many of the first Cog models were very simple in a few equations you would kind of describe these Rel I think primarily in three Fields is this approach used where you have some person or some system and you want to find formal models of its behavior and that's in computation Psychiatry where you want to use it to understand understand psychiatric conditions mathematical psychology which traditionally has been broader it's essentially the same questions but a different community and different types of models used but also broader just behavior in general not specifically Psychiatry and competitional Neuroscience where you want to relate these um models to for example neurobiological processes that could Implement and they're also overlay like overlap with artificial intelligence and Eng engineering and so on where have a all right so I think a primary use case especially in our field of active influence is theoretical simulations where you just take some phenomenon let's say learning or let's say learn helplessness you create a simulation that can do that uh create that kind of behavior and you suggest this as a proposed mechanism of how that behavior comes about you can use this for one agent doing one type of behavior or you can use it for many and then you're doing agent based models uh specifically cognitive agent based models where you rely on actual cognitive theories to name but you can also apply to behavioral data and here you're often trying to either find out what mechanisms provide actually empirically observed data you might want to distinguish let's say clinical populations from each other and what's called computational fingerprinting you might also use results here to inform future theortical simulations so you can have a BMS where every agent has parameter values that you found an experiment and you can also use this to design future experiment where you use simulation to make sure that your experimental design will give you as good results as possible and finally you can also apply these approaches in machine learning and engineering and here of course you can make robots that use cognitive models that people might also use in the way that in general technology and our knowledge of biological systems have always worked with each other so in the same way you might be inspired in how to build a plane by looking at a bird you might also be inspired in how to build an image recognizer by looking at a human being our packages specifically Target the application to behavioral data and so they can do thetical simulations and also in principle be used for engineering purposes but we've optimized them for this situation where you have participants you put them in an experiment you get behavioral data and you use those models to understand that dat and that's um what distinguishes it from some of the other software tools that are present and so PP now has this capability also and but otherwise many of you and there is in the metlab SPM software also some ways to apply to data but otherwise the focus has often been on theoretical simulations or on machine learning and if you look at packages like RX INF fur which is incredibly powerful and well made um that also at least to my understanding focuses more on the machine learning engineering part where the ultimate goal is to use these models to build tools that can do things um or maybe for simulations but not to specifically appli to be he all right um and I'll just give a sense of what the workflow here looks like because if you come from theoretical simulations or engineering this might seem really well to designing hypothesized models for how people work you then check these models before applying them to anything but whether they are worth applying so you do what's called parameter recovery and model recovery where you first simulate Behavior with some parameters you then estimate those parameters and you see if you can actually estimate the real parameters if you can't recover parameters that you know are true you can't trust this in when appli to real data and also prior and posterior predictive checks where you just check that your model can actually generate the kind of data that you might see in an experiment and if it can't it's also not where the pling be then you fited to data which means uh finding out what parameters of my model what they must have been to generate the data that I did observe So In classical models it would be learning rates we ask oh this person adapted this fast so his learning rate must be this this in the context of active influence it's usually um parameters of the generative model the a matrix and B matrices for example that we want to estimate and finally we can also estimate how well the model fits our given data because some models might describe some data better than others we might want to use this to select we use this to relate parameter estimates to some measure that's across participant often this is clinical population so we might say our learning rates or our amri parameters different between clinical populations but it might also be relations to let's say age or psychop pharmacological interventions ET we can also relate it to within participant things let's say physiological correlates and the the could pupil dation Etc but the primary use case of course is Neuro Imaging where you say oh according to my model he now this participant now has a high prediction error do I see a difference in brain activity at this moment compared to when has low prediction ER this often to find out what the neurobiological implementation of the computation that we think people do and finally um you might compare models and ask questions like which of these hypothesized models are the ones that best describe the data and therefore perhaps the actual mechanism within people so you could compare an active inflence and re enforcement learning model or let's say two active models with different gen inside finally it's worth mentioning that cing this to data there are many different approaches um broadly speaking their evasion and frequencies to methods they have different advantages and disadvantages we usually use Bean approaches and that's a long discussion of why that's nice they propagate uncertainty they're good we don't have so much data but I w't go to that just to say that there are also other ways to fit models to data and within basion methods uh because you can't do exact base to fit them there are also approximations one classical approximation is variational methods where you choose some arbitrary distribution we call it q and you make it be as similar as possible to the true posterior belief about the so the optimal parameter you do this by minimizing variational re energy and as it happens that's also what active INF agents use another in cognitive modeling widely used method is sampling approaches like Mar of chain M there are also others where essentially you try out many different parameter values and for each of them you calculate the likelihood of that value you do this in a smart way but it still takes a long time and you get a full estimate of the probabilities of different Prim values they have different claims to being useful for different um use cases so sampling is good um for parameter estimates but they take a long time they might not work varation base has some claims to being a better estimate of quality of fit or model evidence there's a whole few where people are still writing and if anyone's interested we can take questions on that but importantly the truth we make here can do either of these two finally it's just worth mentioning that we as research such as use ban methods to understand people's behavior but within active inflence we also theorize that people use Bion methods to understand the world it's nice to keep those separate so we might use a sampling approximation to Bent inflence to fit our models to data but still believe that people use variational B to understand their environment and so in the following when we talk about generative models we're not talking about our model of people we're talking about people's models of the world and again do ask questions in that that's one of the tricky points sometimes when we're doing this what's called metab Model good there are many paradigms within cognitive modeling mathematical psychology have many many ways of modeling often specific behavior tasks there are these old cognitive architecture models act R and sore are biggest ones where people also trying to do unified models of the behavior in general which rely on very different assumptions than active inference and often we're not aware of them perhaps we should be so I'll just bring them up here they are kind of raw old school computational list models different modules interacting there's of course reinforcement learning models where you assume that par people do actions get rewards from their environment then try to find actions that maximize those rewards there many others like deep learning models and but of course today most relevantly there are ban mind models you think that F's Minds use ban inflence of some sort to understand the environment there are many schools of basent Mind models might look at just ten B for a different School than ours but active inference specifically belongs to a variational Bas mind model we use variational approximate basion inference as a model of how people work and this has close ties to predictive coding and processing so broadly speaking we might call this rational basion predictive processing models they can be just a perception like gion filters or plenty of models out there or they can include action as well and when they include action it's called active difference so variational asion predictive processing models of perception and action is active and and now we'll give an introduction to that but we time for one question before we move on are there any questions sure I'll ask one maybe you'll revisit it later to Arun asked what's the biggest challenge you faced building this package that's a long and Technical discussion I think the main thing is that in Julia there are many ways of um doing what's called the auto differentiation which is a technical thing that you need to do the the parameter estimation and there's just a whole ecosystem of packages doing that and we've had to become very familiar with those to find the ones that work the best and and that's not an easy task the part of the package is to hide that from the user so they don't have to engag with that I don't know if you agree with thats probably yeah we can revisit it okay all right one more question just while we're while we're paused Frasier asked is there anything in active inference JL that is particularly suited to compositional agents or collectives kind of leading you into your paper I know so I return to that in the end if we have time but and basically you can use the same method that we use to make inferences about a given human participant in an experiment on instead of a human participant you can use it on a simulated Collective and treat that Collective as if it that is also doing some kind of clation or active entrance at least which is of course fundamental to active inference in general we expect a cell to do active inference but also person and also Collective and so if we treat a simulated Collective as let's say a foreign system that's doing something we don't know what is we can fit a model to that in exactly the same way we do to a person and make inferences about let's say the parameters of that simulated Collective gu model of the envir and in the end of time I'll just show like results from that paperwork we do exactly this just demonstrating it on example that's an exciting question obviously all right all right I will just briefly uh touch on the software ecosystem um that we build the package in um so firstly why jya uh it's fast and it's very intu meaning that if you come from python or madb you can carry a lot of the same intuition um over into Julia and then it has a community that is growing quite fast of people doing data science machine learning and Caron modelings we're doing then Julia is natively autod differentiable which means that you don't need to have a back end in another language to do auto differentiation uh which makes it very fast and Julia for this there's a very uh nice pack package called touring. JL uh which is just a package for probalistic programming in general and it can do both uh sampling methods such as mcmc or no u-turn sampler um and variational inference uh as well then in Julia there's also a sister package to active inference which takes different kinds of cognitive models and make it very easy to fit data and to those kind of models uh by kind of like wrapping touring um and that's where kind of like our active inf. JL package comes into um play as this is a cognitive model that through action models can be used uh to touring to fit data quite easily and we'll show this later in the presentation as well then we'll just briefly touch on some of the core elements of active inference uh with regards to perception action and learning so for the perception part this is uh the minimization of the quantity variational free energy where we through this uh minimization we can infer States optimally based on observations and there's different kinds of variational inference or sorry different kinds of algorithms um that we can use to do this uh I'll later go into the specific algorithm that we're using uh in this package then we have action and active inference which is Guided by the minimization of a quantity called expected free energy and this can have have a lot of different kind of Expressions if you look at the ones that we show here in the slide we have one that's Information Gain versus pragmatic value which is basically where you get get the expore exploit behavior for free in active inference which is really nice um and then we also use the expected free energy to calculate posterior probability over policies uh where policies are actions so this is kind of like the quantity that we use to kind of like guide the action selection uh we're also going to go into that a bit later then we have learning inactive inference this is uh specifically for categorical distribution beliefs so there's other ways of learning for continuous um cases however here we take a for categorical distributions uh for this we use derish lay priors over the generative model parameters this is basically D lays they have concentration parameters that we can then add counts to and this is the update equation that we show here where we have an Omega which is the forgetting value times current concentration parameter and then you add the new here is the learning rate and then you then add the kai which is the data observed and in this sense you just add counts to the concentration parameters and then through normalizing it uh it becomes your new update generative model parameter um we will come back to how that's implemented in the package as well and then a quick refresher on partially observable Market decision processes so we just want to make sure uh to say that this is one type of generative models uh in active inference there's other types of generative models as well and uh pmps they're not uh active inference specific um in the sense you can also use it in other Frameworks um and then this uh what we have done in this package is a discrete stat Spas uh implementation of pmdp you can have continued stat space implementation of pmpd uh pmds the movian element of it though is that the time steps uh they are discreet and not continuous um so we'll just briefly go over conceptually uh the different parameters in the pmdp and for this uh we have created sort of like this little um toy environment uh to explain this and if you can see this thing here where you kind of like if you imagine that it's a dial uh where we can spin it in different ways uh and then the goal is to get state three on top in the little rectangle um so then you have three action you can spin counterclockwise clockwise or you can just make it stay in place which you would do if you have uh state three on top for example um but we'll start with the observation State likelihood model basically what this encodes is the agent belief on how States generate observations so it's a categorical distribution where each column uh has to sum up to one so in this state in this case as we have a identity Matrix uh this indicates that the agent can't deterministically infer uh the state based on the observation so if the agent observes observation one then it will infer that it is in state one and this can be more or less entropic in the sense we could also have a completely uniform uh column where we just have 3.3 for all of these and then the agent wouldn't be able to infert the states based on the uh based on the observation all right so this was the observation State like model then we're going to go to the state transition likelihood model and this is basically how the agents believes that the stakes change over time um so in this case I've taken the example of doing it for the action of going counterclockwise because this is usually dependent um on actions so here we see that for example if the agent is in state one and it does the action spin counterclockwise then it believes that the next state it's going to be in uh is going to be state two and the same if it's in state two then it believes that it's going to be in state three um so that's the state likelihood uh transition model then we have the prior preference and so this is basically where we kind of like get to encode uh kind of like the preferences or the goals of the agent Um this can be done both Over States and observations but in our case we have done it for observations and this is the case in our package as well so here for example for observation one we have encoded a dislike uh for this observation which is coupled to the state one and then for observation two we have neither dislike nor preference and then for observation three which is uh coupled um to state three which is kind of like the goal that we want we have set a preference and this is important uh as this kind of encodes the pragmatic value um which is then part of the calculation of the expected free energy as you can see here below um yes and that's the C parameters then we'll continue to the D parameters which are basically just the agent's initial belief at time one uh which state is it going to be in and here we just encoded to be certain that it's going to be in state one then we have the prior over policies he'll just uh quickly introduce the notion of policies where policies are collections of actions that can have a specific L link determined by the policy link which is basically how long in the future how many time steps in the future uh should the agent uh plan so for this case we only have one time step in the future which is why we only have the three actions here and this is also sometimes described as the agent's kind of habits meaning that it has a preference for certain actions in this case we have given it a uniform distribution meaning that the agent doesn't really have any preference over uh any policies all right so now that we have the generative model down uh I'm going to show you how to impl this in our package um and we're going to do this um by providing a worked example of a teamas environment where the agent is going to learn the reward probabilities in the arms and for this I'll just quickly have to introduce the notion of factors and modalities so factors are basically different kinds of States so in this case it would be we have location States and we have reward condition states where reward condition states are where the left arm the right arm is going to provide a reward then we have modalities um modalities are basically different kinds of observations um so here we have a location observation we have a reward observation and we have a c observation and I'll go into these uh when we get to Define them uh in the generative model to start using our package you can just add it from the Julia registry using the Julia package manage manager uh as shown here um basically for this code you can run through it uh by yourself and it should be able to run um then we initialize the package and we also import the environments module which is important uh for the tmas environment that we're using as an example here and then we initialize the tmas environment and here we just said uh 95% probability of providing reward in the reward condition arm so now we can create uh the actual generative model uh the pmdp generative model and you can do this manually by yourself um but then you have to make sure that the generative model parameters are typed in the correct manner uh which is shown here where it have to be vectors of rays for a and vectors to vectors for c and d and Vector of floats for E uh it's very similar to object arrays in numpy however we have created a helper function that makes this uh a lot easier and so I'll just quickly go through the arguments that the helper function uh takes so the first argument um is um the number of states in each factor um so remember we had two factors we had the location State factor and the reward condition Factor where we had in the location Factor we had four states we had the sense the Q location and the two right arms so that's four and then we have two reward condition States reward condition right and reward condition left um so we take this as a vector and give it uh to this function as the first argument um then the second argument is the number of observations in each modality so we have uh four we have three different modalities in this case where we have four observations in the first modality the location modality we have the location sensor q and the arms then we have uh the second modality which is the reward modality it has three observations it has the observation of no reward reward and loss um then we have the last observation modality which is the cure observation modality these we also take as a vector and give to the function as the second argument the third argument is a number of controls for each of these factors so we have two factors the location factor and the reward condition Factor um we only want the agent to be able to control the location factor so it can move freely between the locations but in the reward condition State Factor it shouldn't be able to control what arm should be the reward arm and so therefore we just give it a one in this case then we have the policy link which I have described before it's how many steps in the future do the agent plan and then we have uh the fifth and last argument which is the initial fill which basically says like when we have when we have made the structure of these turn to model parameters uh what should be the initial filling of these and here it takes both uh it takes zeros which we have chosen in this instance and it also takes the argument random which is nice for testing and uniform uh which is the default u in this case good so now we just uh having created uh this helper or these generative model parameters uh we just need to fill them with the appropriate values so we'll start out uh with the observation State likelihood model and remember this is the one that encodes the agent beliefs of how States generates observ and then infer States based on the observations it receives so in this case we have the location OB observation which is not as interesting to us um basically it's an identity Matrix meaning that the agent just infers the correct States because we have encoded to do so um and then we have two matrices here because we have for each of the location states which is one Matrix and then we have two uh matrices one for each reward condition State Factor um if you go to the second modality which is the reward uh modality this is the one that is interesting to us because if we look at the First Column and the last column this is the Q and center location we see that it expects that these are going to provide the observation of no reward however for the second and third column which are the right and left arms uh we see that in the second uh row that it has a 50% chance uh that it believes that this state is going to produce a reward observation and then if we go to the third row we see that it has a 50% chance of gener ating this or the agent believe that there's a 50% chance of generating this observation so in this sense the agent doesn't know which of the arms is going to provide the reward observation and this is the modality that we want the agent to learn we wanted to approximate the actual reward uh probabilities in the teamm environment then we have um the Q observation modality which basically just tells the agent that only when it's in the Q location does it uh correctly get an observation that informs it on the reward condition state of the environment um and then we're going to add uh a prior over a and this is where the learning part comes in um the prior over a um is the D distribution and the way that we create this is basically just that we copy uh the structure of a and then we add a scaling uh concentration parameter which basically says that these counts in the DeRay distribution if they are very high like if we give a high scaling concentration parameter each uh new data observation is not going to matter as much if we set the scaling concentration parameter low then each data observation we have is going to going to matter a bit more and then it's important to say that and this is a bit clunky but H this is now going to define the a matrix so the uh the prior over a is going to define the a matrix meaning that the a matrix that we just made is not going to be exactly used however we used it as a template to create the prior over the a um great so now we have the observation state State likelihood model now we need to encode the state transition likelihood model and this is uh the matrices for the first uh Factor the location factor and basically we just note here that we have rows that are filled with ones meaning that from wherever the agent is for that action it goes to that place that it wants to go to um and then we have the second uh factor which is the uncontrollable one um which is the reward condition fact and basically this identity Matrix just tells the agent or the agent believes that the reward condition State doesn't change uh over time that it stays the same good so now uh we come to an interesting parameter which is the prior preference over observations so we're we have a vector here for each of the modalities however the one that we are interested in is the reward observation so in this case the first index is the index of the observation no reward and we said no preference or dislike for this the second index is the uh observation of reward which we said to a preference for that it wants to reward and then we have the third index which is the uh loss observation which we set a dislike for yes and then we continue to the D uh parameters uh where we have a vector for each of the factors we have the location factor um where we basically tell it that it's going to start the first index is the standard location that we're just telling it that it it's going to start in the center location and then for the second factor which is the reward condition Factor we just have a un uniform distribution mean that the agent does not know what the reward condition uh is going to be good then we have the last of these parameters uh which is the prior over policies and we're just going to hear provide it with a uniform U prior so this means that as the E parameters are the habits uh it's not going to have uh any habits in this case we see here that we have 16 different policies because we have a policy length of two and we have four actions so the total number of combinations of these two actions is going to be 16 um and then as said we just give it a uniform distribution over these um policies good so now we're almost ready uh to initialize the acit inference agent before we do that we have to specify some settings um and I'll just briefly go into these where what they do uh the policy length we have talked about the use utility uh basically means should we include the C parameter preferences that we just said and said this the true then we have the use States info gain which basically says that in the calculation of the expected free energies which is going to guide action um should we include the expected Information Gain and then we have the used parameter info gain uh setting which basically says that when we calculate the expected free energy should we include the expected learning or updating of the generative model then we have action selection which which can take two values either stochastic or deterministic um I'll go into what they do when we come to the function that actually uses the setting then we have modalities to learn um which basically specifies what modality do we want the agent to learn um in our case we wanted to learn the reward observation modality so we specify this by giving it two we also have factors to learn uh this is only relevant when we're doing B and D Matrix learning so it's not going to be relevant in our case then we have the fix Point iteration number of iterations and the fix Point iteration uh tolerance which is basically saying that um when should the inference algorithm stop so you can set it to either as we do here after 10 iterations or when the difference between iterations are equal to or lower between this threshold yes the last thing that we need to Define to initialize the agent is the parameters here I'll just briefly go over what they do the gamma is an inverse temperature function on the expected free energies uh then we have the alpha which is in this temperature on the action uh selection and then we have the learning and forgetting rate for the priors over a b and d uh parameters which are the ones that we currently have learning for the ones that we are interested in at the moment though is the prior over a uh which is the learning rate and forgetting rate for this which we're going to set to its default value of one and the the values here are all the default values good so now we can initialize our active inference uh agent by providing it with the Genera model and the settings and these parameters and we're almost ready for the perception and action Loop in itself and before we do that uh we just have to specify how many iterations does this Loop need to contain and then we reset the Tas which just gives the agent like an initial observation and this is the perception action learning uh loop and it has five functions and I'm going to go through them and just briefly explain how they relate to active inference um so the infast states function is the one that actually does uh the minimization of the variational free action uh the variational free energy um and this is the perception part of active inference um here in our package we use an algorithm called fixo iteration also known as coordinate Ascent variational inference um and here we also just uh we provided a picture of an observation where you can see that we have these three observations the location observation reward observation um and the Q observation then we have the infer policies uh this is the one that actually calculates the expected free energies and is affected by uh the settings that we talked about before and the gamma and the E parameters it Returns the expected free energies and it also returns uh the posterior uh over policies which is and it's this posterior probability that we are going to be sampling from when we are sampling actions um so basically this function sample action uh just samples from this posterior probability over policies um and it is affected by the action selection where it could be either stochastic or deterministic if it's stochastic it just samples from the probability distribution but if it's deterministic it always takes the policy with the highest probability then we have the update a uh which is where the learning occurs and this is where we basically add the derish lay counts uh to the prior over a and then it updates the a by normalizing the concentration parameters then we have the step teas which is basically just the agent interacting with the environment and then it provides uh the observation uh for the next iteration uh in the perception action loaring Loop um so this is the the loop um and this is the one that's uh is also going to be used uh in the model fitting part um and we are I think we're taking some questions now one question one question one question yes okay I'll I'll ask one question from Tintin in the chat they wrote hello thanks all for this great work have you implemented basian model reduction or incremental model construction SL structure learning with active inference JL rather than manual construction Haven no that's the so the package is just now ready for deployment and there's many ways to extend it which we work on but also I mean if people are interested in collaboration to develop the package will also be one so this is exciting for two reasons like in two ways by the way um basically model structure learning it can either be model structure learning from a perspective of the agent learning the structure of the environment but it can also be model structure learning from the perspective of us researchers wanting to find out what kind of structure a given agent has a en and and both of these purposes are something we we would like I I think a useful um kind of connective tissue question where would you put active inference JL in relationship to what pmdp does in Python and what RX Ander does in Julia so PVP was originally um a port of the SPN functions into python brilliant work by Conor and colleagues where they make it modular and easy to use but made for theoretical simulations only and I think this summer they just finished Jack's reimplementation of P which means now you can also use it to fit to behavior but it's not designed wasn't primarily designed for that and um it was also not made for engineering purposes python is too slow although I'm sure you could in principle and since then has come the C++ package right um which is much faster and I think but hard to use because it's in C although there are rers in different languages that would be useful for machine learning RX in fer implements active inflence and many other things on b style cograph which is a very efficient and Powerful way of thinking B inflence and variation B inference and that is optimized much more for engineering you want to build something that does something using active inference go C++ or um or RX infer and if you want to apply to behavioral data then go PP or active .jl and there are some tools for this also in metl but I think our package and mdp are have made to make this easier and as you might notice we use much of the syntax from PP so we heavily inspired and in the next part you'll see how to use it to apply the package to behavor does that answer those questions cool yeah perfect definitely the functions look great and your your team's understanding and Clarity around the problem setup taking that all the way through the helper functions for the major CES that's where so much of the friction is so this makes it really clear it's a great Open Source Point it's also where a kind of future development will be also there so the API can be improved for sure and we will work on it also to make it because I think one of the main challenges active influence has faced as a tool for car modeling is that it's hard to use and hard to understand and that's why much of the mathematical psychology world for example just leaves it to say we have a two or three equation model why would we use your beam of of a monster machine instead but just as it has been the case with many other methods making tools that make it easy to use and conceptualizations that make it easy to understand is the kind of thing that will make these models um feasible and therefore used more yeah that's that's making me think like RX and fur is the car factory this is the bike and active inference is the wheel you can think of it yeah we need a logo yeah all right let's do the the last important section and then see how much time okay and because we're short in time I will need to skip through a few of the sections but I hope it's going to be all right yeah so I think Peter already mentioned that we the sort of environment of packages in Julia and one of them is uh action models which is sort of a sister package uh for us which mediates the data fitting process for us right so here we are essentially going in the opposite direction where our inputs and the actions of our let's say experimental subjects uh are known but now we are interested in the hidden or latent uh parameters of the subjective um generative model um so there a sort of core component of action models is the action model function which which depending on our on our model can be an arbitrary function which takes a single input which in our case is a single observation it updates the states of the agent and then it sort of quantifies the probabilities of actions so we end up with a probability distribution from which we can sample uh our actions right and so in our case we have this nice function called action qmdp um which is our active inference action model and takes these two arguments which is our initialized aif object with the parameters of generative model settings Etc and it takes the takes the single observation and then the out it outputs the action distribution so in this particular case we have two action distributions one for each state Factor right and then another core component of action models is an agent so it sort of wraps our action model function and our um active inference um object in more more uh abstract structure which we then use will be will be using for pitting and for um simulation as well and it also stores parameters States uh history Etc right now I will just allow myself to skip through some of the convenience functions so I can just go straight to the data fitting with it's all right okay so um this is the I would assume the most interesting interesting part of this so how to actually fit active inference uh models to data so um all all of our fitting functions kind of operate on touring and we wrap all of the necessary Turing operations in our convenience functions for fitting so let's say for example we want to fit a single subject and we want to recover a single parameter is sort of a minimal simple example so we start by specifying the prior so here let's say we want to recover our action Precision parameter Alpha so we set the prior for it let's say a gan with a mean of six and sigma 2 and set it in a dictionary and then in case we want to put more parameters at once we just add them to the same dictionary and in their corresponding prior distributions and so now we can use the create model function which will instantiate our probalistic model on our data and it takes these uh inputs which is our initialized agents Tak are priors and are collected observations and actions which we collected let's say from from a behavioral experiment and in in this case uh we're just using some data from simulated teamas so here we um our actions um is a matrix which um which has rows corresponding to time steps or let's say trials and columns corresponding to State factors and similarly in observations uh we have columns corresponding to uh observation ities right and then we'll have our model object instantiated which we can easily fit in a single line using our fit model function which will run the sampler with appropriate defaults and then we can like obviously people would want to pass some some more specific settings into the sampler so here I'm passing for example uh the number of iterations per chain and I'm running four chains and then I'm using the distributed module to run the chains um in parallel right and then we after the sampling is finished then we will end up with the results object which which contains our chains and then we can work further with this right so here I'm just going to quickly go through some of our other convenience functions for example the rename chains which will rename the names of the parameters in our chains so something more more intuitive and we can extract quantities from our chains so we can just simply display the results of of our chain so we can get the summary statistics some information about

about the conversions of our chains quants Etc right or then we can just use the sort of native jully uploading functions we can pass our our chains to it uh to take a look at chain traces or posterior distributions or we can uh what we can do is we can sample from our prior so here I'm taking a th samples from the prior and I'm I'm using our plot parameters function to plot um the the posterior distribution against um against the prior distribution um for our um synthetic participant let's say okay and this is a part is going to be mostly interesting I think for people in computational Psychiatry and cognitive modeling and it's the case where we want to fit uh full theat set so in this is let's say that we have some data with 10 participants and let's assume that we have two clinical groups um in our data it can be I don't know um control and um whatever psychotic or it can be young um old um right and we can simply pass the whole data set with our collected um data let's say from from our behavioral experiment to say same create model function so instantiates the the probabilistic model um on the all data set and then an additional things that we will need to tell the function is which column to use to differentiate between individual subjects and which columns uh contain our observations and which columns contain our actions and again we have our model instantiated on the wall data set and we can just pass it on to to our fit model function right and again this following is sort of the same procedures before we can play around with our recovered chains here we have the the recovered values of alpha for for each participant and I'm using um Native Julia functions to to plot the chain traces and again we can let's say sample from the prior and quot the posterior against prior distributions for each participant right and then sort of keeping also with with this default procedure in cognitive modeling now let's say that we have our estimates and we want to see where there is actually a difference right in in our two groups right so here I'm just extracting the quantities from the chain and I'm getting the estimates which in this case are medians and we can see here that there are indeed um two groups present in our data right and then after let's say that we run some statistics on our estimates then we might be also interested uh to see whether there's any correlation with other states of our agent and let's say some physiological data here I'm plotting the state action prediction error Reas why people want to correlate let's say with brain data or pupilometry for example and uh for people who might want to use variational inference for uh various reasons over over sampling the good news is that touring can do variational inference too so here I just sort of remade the tutorial which you can find on the official touring um web page where I'm in the beginning just initializing the variational inference uh algorithm with a sort of default number of samples and a default number of uh iterations and importantly I'm setting the automatic differentiation back end to reverse mode automatic differentiation and then in the vi function which runs the variational inference algorithm I'm just reusing our probabilistic model uh which we have already created previously right so this is this is very nice and what we end up with is our variational posterior which is a type of distribution so we can sample from it or work further with right and just to rather quickly wrap up so um we Cy working on the documentation uh we know it's important so um it should be done in the next few weeks let's say uh until then you can either reach out or uh you can refer to action models documentation which is uh really nice to cre okay and yeah just briefly just to summarize what we need to do in our in our um next potential future uh so we need to optimize the package is fast but there are still a lot of things that need to be improved then the idea was to for example uh be able to fit all parameters because for now we we're only able to to recover the the sort of hyper parameters or what's been called the non-matrix parameters but in principle one can also one should be also able to fit an entire let's say a matrix or C Matrix which would be really nice and then okay we can um extend the parameter learning to all other uh parameters of the of the generative model and that idea was to introduce a sort of custom parameterizations where a user could introduce a sort temperature parameter on on on on an a matrix and this could be also in principle then recovered using data fitting from empirical data um and lastly I'm just going to say that um for our fitting we are using revers mod automatic differentiation which is fast but we are also planning to start experimenting with other uh engines which are currently under development in Chula also which includes enzyme and moon cake which might be potentially uh faster and then on the side of action models um Peter has been working for some time also on the possibility of adding uh linear regressions so the idea would be that we could fit the data we could recover our estimates and we could run linear regressions with our with our estimates or with with other um other variables that we might have in one model yes which is statistically up to yes okay and the last thing you're going to hear from me is that uh when you decide that you want to start experimenting with our package and you encounter any issue or do you have any specific feature in mind that you would like us to implement please feel free to your open issue uh any kind of feedback will be greatly um appreciated thank you and we're out of time so I'll I'll just briefly mentioned this point that you can treat a whole group of Agents as one agent and do c modeling art it's a separate paper so I'm sure there'll be time to talk about that at another time to reach out of that's interesting as well um yes I think that's it thanks s this is a huge Advance it's really awesome brings a lot of features a lot closer to people's hands and minds and and and it's exciting that that we'll continue developing and learning about it thank you okay thank you till next hello welcome back we're here with shanana Dobson discussing descent conditions on hyper Fantasia a new model of self-information looking forward to this presentation followed by questions at the end so thank you Shanna super looking forward to this hi Daniel great thank you for the invitation I'm excited to present these new ideas on hyper Fantasia so I hope this relates with someone and thank you all for listening so a small outline I'll be discussing the Fantasia spectrum and its two extremes a Fantasia and Hyper Fantasia and then we're going to discuss the what I'm calling the memory Continuum then we're going to review what I've been calling the oniric self uh constructed as a profinite set then we're going to review grth and de's one categorical the theory of descent and then uh going to try to present a new model of categorical mental imagery from the hyper Fantasia perspective using I think these very beautiful categorical sheave techniques of specifically grth and descent so descent theory is a super powerful tool and I hope you all like kind of what I'm trying to do there um then we'll pause at what are called descent conditions on hyper Fantasia pretty much how the sheets work and like uh why I think sheaves are really good for something like hyper Fantasia which is you know characterized by pretty vivid mental imagery and profound um episodic memory recall and then last we're going to actually model surprisal itself or self-information functorially as a category of descent data so if you've been following my work I tend to like to categorified so I'm just going to start with the the big results because it's going to take me a while to get there so I always start with the main idea so pretty much I'm going to so for me episodic memories seem fibered they seem like their memories over memories um as I think selfhood is fibered so I've given many talks on that five wearing the famous case of having both a retrograde and intergrade Amnesia um five has about a six-second working memory with a purely semantic memory and um not a lot of personal memory besides remembering his wife but everything else not so much so the selfhood of Clive is in question um what's the perception the frequency of perception of Clive I think that's very very tricky stuff so self it itself is fibered we're gonna consider these categories of memories over a fixed memory so I have the idea to use descent to actually represent hyper Fantasia so the idea is that you have these memory sheaves that are satisfying descent and to satisfy descent means that they form a stack which is a very particular sheath that takes values in categories and not sets okay and so then this payer at the end this fstar fee which I'll describe in these maps and is going to be our what we call the categorical descent datum so I call it this is our sheify surprisal so if you could sheify surprisal or self-information I think you do it this way so surprisal would be constructed as the sheif ification of memory all right so and you would say there exists the following The Descent conditions on Fantasia and it would be those cool all right so let's start with the Fantasia Spectrum sorry so this is what I'm going to get to I think they're pretty good ideas and then so let's actually dive into Fantasia itself which I think is very very very beautiful topic so what is Fantasia what is this Fantasia Spectrum um so it has to do with the characteristic of um you know um creatures having visual imagery and so the the Fantasia Spectrum itself is a relatively novel neuropsychological Spectrum used to classify to what extent an individual constructs mental imagery and visual IM imagination which is quite strange we're so used to it but it's very strange that you have these pictures quote in your mind that where do they actually exist this Fantasia Spectrum so this ability to imagine um visually has two extremes and they are quite extreme so a Fantasia is characterized by the absence of mental imagery and uh hyper Fant characterized by profound extraordinary mental imagery heightened visual imagination so detailed and vibrant imitates reality I myself seem to fall on this scale and the reason again I know of this is that one of my dear colleagues um introduced me to the Spectrum and said Shana I think we uh we were writing a paper together and said actually um I think we lie on opposite sides of the spectrum right know as I tell this story and um together I think we've written like the most amazing things okay all right so let's start on the a Fantasia side so when you say when you know the tendency to say lack you know it seems to say something negative but it's not really so a Fantasia has just characterized by the absence of just particularly mental imagery does it mean all imagery uh so it is an imagefree mind but it has conceptual imagination okay so if you were to say so the imagination of anyone who falls on this classification it involves other senses and Abstract Concepts so besides visual image and sensory details so perhaps this the way of thinking is and visualizing is analytical so um everybody knows my little girl Artemis and I talk about her all lot so if I was a Fantasia with respect to like saying someone said you know do you how do you imagine emis and I would say well I cannot mentally picture EMS but I can think about the idea of emis and there's emis all right so um perhaps like her fluffy tail and her being white like wouldn't actually appear to my mind but I would think about the idea of emis and I think about it very deeply right so contrast this to um what is hyper Fantasia so hyper Fantasia is characterized by sort of extraordinarily Vivid mental imagery and heightened visual imagination and also episodic recall heightened sensory details in episodic memory so you know um Shana what was your first ice cream cone and there's you know just Lush sort of Dolly like Exquisite um imagery that comes out of that so detailed and vibrant imitates reality that you actually say well which one um which one is quote more real the the one quote in my head or the one like outside of my head tricky and so I can actually picture Artemis so vividly as if I'm really seeing her in real life okay um and so for episodic memory these are personal memories sort of based on sensory perceptual data of which mental imagery is a particular form so that is why hyper Fantasia and the and how you recall episodic memory are so incredibly linked so with this um I I've started thinking about sort of the distance to and from memories so this is a concept um that I think would be exciting to explore which why I'm doing it and so if you ask your yourself a strange question like what is the distance to a memory um normally people would call this a timestamp that happened you know to you three weeks ago something like that what happened to you then is not what's happening to you now we call this a Tim stamp sort of a way that you can mentally file away an event and to keep it a little distant from you so but I have a question so for hyper fantasic um what if there were no distance to a memory I find this to be the case for myself people say well that happened then um that happened then it's not happening now I sort of have a problem with that sort of thinking that sort of logic um I question the timestamp that people are using to actually say a statement like that right um but let's imagine what if there was no dist to AC memory would that mean that it's always happening would it mean that it was always happening well what if that distance were actually not measured um by sort of the counting numbers or the real numbers with a linear ordering before or after so this is sort of the profound leave I want um everyone to try to imagine so an event happened to me two weeks ago if you were thinking according to the sort of like standard number line you would say well two weeks ago was before like now if like now is the zero point and so you know normally events with a time stamp have a linear ordering that happened then this is happening now that was yesterday this is today so I just want to ask like what if that distance were actually changed the metric on that distance was actually changed so if there was no linear ordering about before or after this would be very um curious so for me the distance seems the distance to and from memories seems more like it's a ptic so I invite everyone to sort of try to think of the distance to Memories as more non- archimedian um in the sense that there is no before after so if you been following my work I'm pretty Enchanted by the ptic numbers and things like this and so uh the P ptic numbers um actually have no sense of linear ordering they are kind of more fractal like now this property that you're sort of just zooming in but there's no notion of like bigger than or greater than um and I think that would actually be very interesting to try to evaluate memory on a time stamp of something that's not bigger or less than right so my claim is that the distance of or to or from memory is uh is patic and that's going to lead me down this exploration so the second um concept I want to talk about that I've been playing around with is the idea of a memory Continuum someone asked a question like what is the space of memories if anybody saw like the latest Blade Runner you see like there's like the Memory Maker she's in charge of actually making movies memories and she's actually um sort of kept away uh because she says she can't really leave this prism that she's in but she's construct she's charged with making memories so I want to ask the question like what is the space of memories right so uh you have them and my great colleague Chris field and I are always asking the question like why do you experience memories it' be very interesting to find the first creature Who had who experienced a memory not why you have them but why do you experience them all right so what is this space of them if you were actually put them in a category you could um I was actually more interested in like you know if you dive a little bit into that well what is the actually topology on the memory space Could you actually cover it with something can you think about memories in terms of open sense are memories topological do they sort of enter SE which it sounds like they do what sort of properties do they have and so to think about memories mathematically I think it's kind of a new idea um so not only what is the space of memories but what is actually the shape of memories what do these actually feel like look like and so I want to actually think about the concept like can you actually mathematize the duration of memories um how long do they last some memories seem to have expiration dates and some memories seem to be um you know holographic I'm really interested in how memories last how long they would actually last if they were actually considered with a different metric okay so for myself I notice a certain density of memories and what do I mean by that so density in the mathematical sense not not heaviness of densities um although that would be interesting like some memories are so heavy um they' probably you know implode some memories are sort of big light they float in the bathtub right so the density of memory is I'm not talking about the physical property but um what is often called emotional veilance sort of has this positive this plus minus charge associated with experiential memory so if memories can act like sort of black holes or memories are sort of like black holes they can literally warp the thinking if you would think about it so black you know actually in tend to if there's very I mean they're dynamical but when they're rotational when they're rotating and things um and spewing out these plasma Jets they can actually warp the SpaceTime around them I've always been fa fascinated by that um I seem to have memories that do the same thing so if they warp the thinking what happens well any object can be turned into a black p just have to squish it Beyond its shile radius so memories as objects something to be like to think about so density in the sense of okay like a teaspoon of a a memory that was as heavy as a black hole would do something but not that kind of density I want to talk about the mathematical density of memories okay um what would that even mean so if you take density from the standard form where you normally prove that you know the rationals are dense in the real numbers this is the sort of density that I want to think about so the rationals are dense in the real numbers which means that between any two real numbers you can always find a rational number which means between any two real numbers you can always find a real number so that seems like a light statement but then it sort of gets terrifying to know that between any two real numbers you'll always find another one um unlike the property say for like counting numbers or the positive integers between the integer the positive integers two and three I will not find another integer but between the real numbers I will so I thought hmm for me it seems sort of the same between memory a and memory B there seems to exist a memory C and I can continue this forever it seems for me between any the two memories there is another one and another one I can always find another right so the memory density and the memory Continuum I think is something that um could be looked at as a hyper Fantasia until I run out of memories right so between any two memories there is another one and another one I can always find another one until I run out of memories but that's where the hyper Fantasia kicks in right now between any two memories there exists a hyper fantasic memory right so this hyper Fantasia for me is a sort of memory Continuum so um one of my brilliant colleagues at Art Center um constructed this beautiful image which I think for me actually shows me um this is sort of what I think about as a hyper fantasic right so the density of memory that um I could be brushing my hair and in you know inside that memory of me brushing my hair my favorite brush there's more memories inside my hair memories inside of memories inside of memories inside of memories right um and it just keeps going on and on and on and on so I want to play around with another concept called miso memory that I've been thinking about um so is there actually a duration issue with the of memories um whereas the real numbers always exist so if I'm talking about between any two memories there's another one um how long does that last the real numbers seem to always exist seems that um between any two real numbers I will always find something but what about memories not sure about that um what is memory at the Miso scales this is the scale that Daniel and I we've been having talks about so sort of between the micro and the macro you have this miso which is sort of this slippery sort of inter like interstitial happenings and so what is memory at that scale I'm curious um is memory the only just the thing that sort of comes through with the macro side what would miso mean in terms of that is this like a an almost memory so I have almost memories at the Miso scale but then um what does that mean so what would a miso memory actually be well thinking in terms of density would allow more variations of what memory looks like so um sometimes you have memories that you know aren't quite out of the oven yet you have some memories that have been mixed some memories are fluid some memories can phase change and so I really think it's really important think about the density of Memories by opening up this new scale so sort of from micro circuitry of uh of neuronal processes micro memory this is sort of what I'm interested in so for me memory non archimedian this is I'm going back to the paics ptic metric it's like it's a fantastic metric in my opinion right so the ptic number system again just a brief recall is a non-archimedian system that is the completion of the rational numbers Q under the ptic metric repeated notes a prime number um interestingly so this is sort of the oz-like property of these numbers so ptic numbers are close if their difference is a high power of P ptic Notions have no notion of linear ordering their shape resembles a self similar set the fractal language I was talking about okay so just a little more technical so again for me the memories feel non-archimedian and so and formally the Ring of pic integers zp are written in base p and they admit infinite expansions to the left of the decimal point so pic integers take this form of sort of this sum a and PN where a an is um between zero and P addition multiplication are performed in the ptic integers with the sort of caring method used add ordinary ones um so the pic metric introduces a non- edian norm on the rational numbers and the fun part of this fun fact is that for a non- archimedian norm and the rationals every point of an open ball is actually the center so I want to think about that instead of actually having like a canonical Center in um in the archimedian case um every point of an open ball is the center so this sort of what someone would call a disorientation maybe it's sort of a superpower this could be the space from which hyper memory is made the fact that every point in this open ball would be centered so really what metric are you using do hyper fantasic and a fantasic use different metrics on the topology of the memory space this is what I'm interested in so are they using different distances to and from memory so at best you probably just have isomorphism classes of these timestamps instead of um um I'll just say yeah so for me memories feel also profinite this is a concept you've been following me I talk about a lot I'm inspired by schula's work so profinite set also called a stone space is a compact Hall door of totally disconnected topological space I've been Enchanted by these for a little bit it's also defined as an inverse limit of finite sets that's our notation the inverse limit of our SI for SI finite so the smallest um what is called light so a light profinite set everybody is just accountable inverse limit finite sets and so the smallest infinite profinite set it's actually called the one point compactification of the integers so you go all the way through the integers you join Infinity and then also the caner set which I'll talk about in a little bit more all right so fractality is this concept that I'm I'm studying um for my thesis work um fractals are such strange Beautiful Creatures they actually sit between burkovich spaces which are famous rigid analytic spaces and the pics so they're between these um and the pics are totally disconnected uh so baric spaces are these topological spaces like I just said in the set setting of non archimedian analytic geometry so fractals are sort of in between and so fractals have this um property it's very difficult to Define what a fractal is so they there's been some contention about the definitions back and forth and so but everyone agrees that fractals are a one property that they're self-similar at all scales so I want to ask if memories are self-similar what would it mean to actually have that what is the shape of memories in something like the dream space so we're going to call that like the onic self memories um what does it actually mean to have um like the memory space in an oniric setting that's we're going to explore that in just a little bit so memories seem local confined to some sort of particular area um but they also seem met up right so in some great work with my brilliant colleagues Hector manrique Walker um we're trying to figure out sort of a you know how would you construct the oniric self had this idea like H I think you construct the oniric self as a as as itself as a profinite set and so what we're looking at here is the different topologies and like on the memory space in the waking State um and in the Sleep State and actually the construction of selfhood in the the dreams space and then selfhood in the waking state so just to quote one of our passages here so we say in start contrast the oniric state events occur disconnected from one another in a profinite space that's compact hosler topological so namely of a finite Stone space where events Dreamt by our onir self are not stored relative to linear time so hence a plausible model for dream time would be a ring of ptic integer zp considered as a topological space so we're trying to figure out like an event in a waking State and an event in a sleep State um these seem to be stored differently and to construct different different narratives of personhood right that was Hector's idea how do I construct the notion of self um in a waking State versus self in a dream state so self in a dream state you know that you can be here and then you're all of a sudden you're uh you're you know flying with like V and then all of a sudden you're at AR one these are my dreams and um the question is you can get from one place to another very very very quickly instantaneous um or you seem to be nowhere in particular at no time and so the question was like uh you know if I had all these events happening in my dream space and I couldn't store them as dream I may start having waking waking State fights with people who did stuff to me my dreams I'm like Daniel how did you do that to me and and he's like Jan that never happened in the waking state I'm like oh okay so you have this very separate storage space which keeps dream things there and you're not constructing the sense of waking self based on the dream state of bents beautiful idea um and so the uh why did I say like okay let's construct the dream state as a profinite set well the hos dorf Criterion ensures that there's no simultaneous experience occurring in the oniric dream state you need to keep um you need to keep these events separated so every point in the profinite space uniquely represents um an event that has occurred onally um simultaneous experience is something that I'm after as well in my my work with my brilliant colleague Robert preter also my work with Chris Fields um so what would it mean to have a simultaneous experience that is something um that we're still working on more like that that is that is a waking dream but so you want to have these events being haos door that they're kept separate in the dream state so you don't have a sort of an Inception going on thank you Christopher no all right so the total disconnect in this condition actually accounts for a crucial difference between awake time and dream time so Chris and I um had this idea that a key aspect of the construction of waking time however is that is the assumption that between any two distinct events experienced by some agent a other things happened non- agent specific events occurred that a did not experience that some other agent B may have experienced okay so in the waking State um between any two distinct events experienced say I'm agent a and I'm giving this you know lovely talk for everyone listening and then I'm going to go give another talk and so the fact that in between these two talks for me something else you know happened to to Danel um and Vince in the onir state though I think don't have this property actually totally disconnected so that you wouldn't have this notion that between any two things that happened for me something else happened to somebody else all right so canonical example of a profinite set is the actual caner tary set so I have an image of this okay yeah so the tary set is constructed iteratively from an in as an infinite process from an infinite process I take this closed interval 01 represented by this solid red line and what I do is I actually remove the uh the um the open middle third from from the actual closed interval 01 leaving me with two intervals remaining so I actually have two line segments closed 0 to 1/3 Union 2/3 to one and what do I do I do it again so I remove the open middle third leaving me with the you know um remaining line segments and I continue this process infinitely so the caner set actually consists of all the points that remain that are never deleted from the original interval you can see there's end points of the line segments that are never deleted and so beautifully that caner set is actually uncountably infinite so it's the scary Infinity having the same number of points as the the closing interval zero one right so shown here is like just the first s seven um steps of the infinite construction it's always nice to be defined as an infinite process so this sort of a this is the caner set and so the points that remain are sort of like this caner dust um but if you add them all up it's the scary Infinity okay so this caner Turner set is nowhere dense has no interior points and it also has measure zero so it's considered negligible small so back to our um back to our paper so like so the on self is composed of these events that occur in a stone space um then the onir itself would be like caner dust everyone and negligible so the self similarity of the caner set it's sort of like fractal and it could experience what what we're calling these time Loops by the itself um but of course the cerebral neuronal micro circuitry is yet to be demonstrated which maintains alternative storage systems right so do you have these alternative storage system and what causes the switching between them how do you actually know to file event a as a dream event and um keep it separate from event B so strange and what would happen if this actually sort of flipped all right okay so the self-similarity to caner set I'll just show you this image um is the the fact that there is a p like the caner that is similar to itself upon magnification and translation of its pieces so if you look at this highlighted piece pink that I have like the pink highlighted piece so you can have two copies of that um diminished by a factor of a third and translated produces the original C right so very strange that these this set is self-similar and that's why I think it's actually very curious but the self-similarity is what I'm interested in in the actual memory space cool all right so let's get just a little Technical and then I will wrap up all right so this new model is um going to be based on these categorical sheaves like I showed the final the main results in the first slide so the idea is to model for hyper fantasic um model episodic memories as light profinite sets which means these countable inverse limits of pro of pro of um of finite sets so then we're going to actually use a claim that a theorem that Chris and I have in our paper um that entropic categorizations are actually condensed sets we'll I'll recall all these terminologies and then reframe recall in terms of representable functors so I think recall could have a very beautiful form as a in terms of representability and I'll go over that funter then you want to construct hyper Fantasia by a junction and Hyper Fantasia would be constructed as surprisal so Fantasia itself is sort of like self-information and so self-information by the way of the actual visual Spectrum or if you're a Fantasia um so then you would reframe further these memory sets as categorical sheaves and put the Descent conditions on them um and then you'll model self-information as a category of descend okay so it's like firstly um everybody always ask me why sheaves you know like why are you so fascinated by these things um well I um really believe in support gra and's work so Alexander graand always said that it was sufficient to study the category of sheaves on a topological space so he is so brilliant he really revolutionized just um algebraic geometry um descent Theory um really beautiful things in order to comprehend the topological space so all you need is a category of sheaves on the topological space love that so shees enable what well they enable this local analysis of the space they assign data to open sets you know you don't really get this local Global phenomenon a lot and then the category of sheaves on a space actually encapsulates its geometric properties like chology groups these capture topological in variance and structure so the category of sheaves on a site is actually called a par and topos a site is going to be a category with a topology so unlike a like a topological space but a categor version of this so a grath and deque and particularly a grath and deque topology on a category is going to be a set of morphisms which act like the open sets or the covers of a topological space right so this grath Indique topology is going to be very rich and um I just think he was really on to something here all right so a site let's define that real quick so a site is a category equipped with a grath Indique topology and so given a grth and topology J and a small category C you can Define the category of sheaves on C relative to J which is this sh C comma J and this is a reflective subcategory of the category C off um of set that's a funter category of pre- sheaves on C so reflective subcategory refers to the images image Maps reflecting each other so this you're going to have this funter from C to the category of sheaves of C relative to J given by What's called the composite of the Unita embedding which I've talked about before with the reflection that's called the shap ification and so this composite fure is going to be fully faithful which means injective and subjective on them om sets if and only if all the representable prees which are these set valued bunters are actually shees so descent is going to be you know the sort of generalization of the the Sheep condition from pre- sheaves on these higher categories so the topology with this property is really importantly called subconical so let's define a profinite set a finite set is actually a compact hos door of totally disconnected set as I said so it's defined as an inverse limits of finite sets so let's just talk about what an inverse limit is so you have some sort of index set um I and so the pair this pairing is going to be a partially ordered set and so say we're working in in groups and group homs so we'll let this AI be a family of groups suppose we have a family of homomorphisms fi J note the map there's AJ to AI for all I less than or equal to i j with so fi is going to be the identity on AI and then fi is is um fi precomposed with um fjk and so this pair AI fi is going to be an inverse system of groups and morphisms right and so the inverse limit is actually a subgroup of the direct product of the AIS so you construct a as the inverse limit which is all the A's in this um in this direct product such that so AI is fi J AJ so fi J of AJ is pulling out the AI component okay all right so a little technical sorry about that right so light for finite set is actually just going to be I'll just make this um nice uh so light for a finite set is just a countable limit of finite sets right so you can Define it in terms of size and weight um but countable sounds good so again an example of a light Prof finite set the the finite sets are the smallest infinite profinite set is the one point compactification of the integers and also the caner set all right so let's go back to what a condensed set is and then I will be done in about 10 minutes okay so uh what is a set so uh remember schul and Clon they had this idea of like how do I Define a sheath over a point super revolutionary very cool idea so um there's condensed sets and there's light condensed sets and so a light condensed set it's a it's a sheath on the category of light profinite sets so this is not a heart shorn sheath this is a categorical sheath right so you write it as this so light condensed sets are actually sheaves on the category finite sets and so they have this proposition that there's a more General way of thinking about these um a sheif on a category is normally a set valued funter so you let condense C and denote the category of condensed objects of C where C is a small category and then so the condens C can be represented as a category of small shoes on C great so what we need is this the the equivalently CATE uh condition so cond C can be represented as a representable functor which takes this form see op to set all right so so here's the cool part right so to summarize you're trying to get trying to get like um episodic memory um as forgetful functors all right so if you model memories as these per finite sets and you reframe recall as left working as left Junction which is kind of like an inverse map so um we recall the famous proposition that any set valued funter with the left ad joint is actually represented that's powerful so what is a forgetful functor right right so it's a map that sort of forgets some sort of structure in the original space um I think memory is all about forgetful functors what do I need in this moment what do I not need short-term memory long-term memory personal semantic you're alighting something in order to create meaning right so the forgetful funter for instance um from compact hdor topological spaces to all topological spaces we'll call this funter U has a left ad joint you can informally call it an inverse map from the topological spaces to the compact Oar topological spaces and it sends a general topological space to a compact Hoster topological space this is very beautiful it's called um a stone check compactification and it comes up in the war ofite set right so um the stone check compactification is in general very large but this is sort of like an example of um what you're forgetting right um yeah I'll just say that right so your like forgetting the compact condition and to get to topological space of great okay so let's get just a little technical but I'll just like kind of go quickly through these slides um so why is grth and deque interested in descent Theory what is this right so um a kind ofis scary definition but it's okay so the theory of descent can most aely be characterized as so from the point of view of infinity category Theory which I've given some talks on so desent is a study of generalizing the sheave condition on pre- sheeve right so pre sheep is a tool for tracking this local data sheep ifying it pretty much means you're giving very strict gluing conditions on these things so um you want to study the generalization of that sheep conditional preep to pre with values in higher categories so that's why we're calling this the infinity category point of view so those higher prees that satisfy descent are called Infinity Stacks right and so again I think like gluing and recall or um like gluing sheath gluing would be a nice mathematization of vivid profound imagery that imitates reality that's what I'm saying um so I think it's in the gluing conditions where the hyper Fantasia works okay so more generally descent Theory studies the existence and not and and uniqueness of an object in a higher category provided some inverse image funter which produces an object um um in some other category so you can sort of reconstruct um the the the object from a higher category itself I'll just say this all right so the non-uniqueness is only parameterized by equipping this object fstar U with an additional gluing data um which was we call kazai so this pair is called a descent data so you have a reconstruction procedure A View From the actual um data and so that's actually called a descent okay um so the section of descending is formalized as what is called a descent condition so again getting this fstar map so you have um an object U and a higher category CX and an inverse image functor right so F star is just an inverse image functor from the higher category to um a not a lower category C Y okay so think of fstar as an inverse image functor and that's what we're looking for so inverse image pulling this thing back and then this gluing data okay that's what we're looking for so to get very beautiful with this you could say every Infinity one category of infinity one sheaves is a a sub Infinity one topos of the infinity one topos of pre sheaves on a small category so um once again that something an object that you have is reconstructed from another object right so it's this sort of like local to Global phenomenon right um I will say um each Infinity topos is a reflective Infinity one subcategory and a localization which is sort of like a way of saying adding inverses of an Infinity one category and a collection of morphisms which are sent to equivalences by the left ad joint of the inclusion so precisely those objects such that uh the prees or the precisely the local objects um this is an isomorphism in the homotopy category homotopy um investigates these continuous these deformations of objects so The Descent condition is how you descend from the higher category down so hyper Fantasia is going to be this particular reflection localization that's what I'm thinking another one of my colleagues did this self-portrait of herself um in a black hole this is sort of like for me what I think about as dissent as a hyper Fantasia you have a memory upon memory between any two memories you have another memory so memory of self memory of what I was memory of what I could be since there is no linear ordering the time for the memory space can go both ways I think and so um you you know having like spaghettification in the black hole and having red shifted uh um red shift around the black hole and time sort of stopping but then you have that layered on the sense of like self so I think for a for someone like me if you're like oh describe yourself it be something like this and so uh the image itself has so many layers sort of on it okay all right so descent ingredients would be something like um so a more gentle way of saying this is that um that descent has a somewhat formidable reputation as you can see um amongst algebraic geometers almost everybody so in fact it simply says that under certain conditions homomorphisms between quasi coherent sheets um which is just sort of like a sheath of modules um over the structure sheath of a ring you have the structure um uh that that conditions um sorry that homomorphisms on the Quasi coherent she is can be constructed locally and then actually glued together if they Ser satisfy certain compatibility condition and then while the the Quasi coant sheeps themselves can be constructed locally and glued together by o isomorphisms that satisfy some something so going from um local to Global is actually um the the same notion that I was saying earlier in the category Ser sense of reconstructing data from higher category to lower category all right so you have this notion category theory of a category in which descent Theory works and these categories are actually known as Stacks right so what you really want to say is that um in working in descent you naturally work in fiber categories so that's why I said you know you want to fix a certain memory M and consider the set of memories over M and so um you actually call a fiber category is a generalization of a funter it's called a LAX two funter or a pseudo funter and then a pre- sheep is going to be a funter a pre-stack as a fibroid category and a stack as a category of she on the site so this is how everything sort of works so you associate with a fibroid category overse the data for a pseudo functor and then so Stacks are going to be the um the correct jaliz of she you're going to get this nice notion of the sheath of categories so if you want to actually say um so in a nicer way like what does dissent mean when you're going to you're going to hear this everywhere like in that sort of literature in arithmetic geometry uh so to say you know descent for quasi Coen sheaves these certain these sheaves of modules um or to say something satisfies descent if you're going to say quasi coherent she satisfied descent with respect to a particular topology fpqc stands for Faithfully flat quity compact just is a specific topology in the gluing map um to say something sad fies descent is to say that they form a stack okay that's all we're going for So when you say that you know the the memory sets satisfy um descent I mean that you can actually stackify them so this word subconical is again very powerful I want to mention it one more time when I'm talking about the topology on the memory space I'm hoping that you need something that's like um the topology needs to be subconical okay so topology on a category is called subconical every representable funter on C is a sheath with respect to the topology so every representable funter is a sheath and then a subconical site is a category Ed with endowed with a subconical topology great so fun fact um because this is fun all right the name subconical comes from the fact that on a category Seether's topology known as the canonical topology this is the finest topology in which every representable funter is a sheath very beautiful right so these Concepts of subconical I think um can be infused very beautifully into this notion of surprisal right so what does it mean to sheify a funter so the construction of the sheep ification of a pre- sheep on it on sets on a topological space can be generalized I'll just say so to sheify um um a funter right we're just going to add the sort of categorical definition of what restriction math is in the interest of time I'm just going to go over this all right so um if C is a site and um f from cop to sets as a functor then we're going to say that FA Cop to set is a sheif foration of f any morphism from F to a she factors uniquely through fa these are diagrams so there exists a sheep ification at fa which is unique up to a canonical isomorphism this is the thing that I'm actually using in my work okay and this is technical we don't need this okay um so let's get to The Descent condition so why is this actually important uh again why am I looking at like descending or pulling back memories because I think that's what leads to the vividness and also the notion that there is no distance between these things so let CBS site a f category over C should be thought of as a funter from C to the category of categories so you fix a category we want to look at the categories the category of categories over that category so a stack is morally a sheath of categories of C all right so if C is a site um the funter um from cop to set um we also consider it as a category fed in sets big big um Pro big note that f is a pre-stack if and only if it is a it's called a separated functor but f is a stack if and only if is it is a sheep so this is the reason why you need St this is the reason why I'm working with like sheeps and what it means uh that if something satisfied descent she becomes a stack okay so um there's an an archetypal example of descent that you work with the category of continuous maps and that category is fibered over topological Spaces by this functor P sends each continuous map to its co- domain and then um so suppose that F goes from from X to and g y to U these are two objects in the category of continuous Maps mapping the same object in to and the topological spaces so you want to construct a a continuous map fee from X to Y over U um yes okay so um I don't want to talk about this okay so in this because I'm already like out of time all right so in the new model the idea is again if you so memories as stack for the hyper fantasic and the sort of uh the sort of stacking is what leads to the vividness so um the The Descent conditions would be the following used descent to represent hyper Fantasia memory she satisfi descent and then this pair fstar fee would actually be the categorical descent data so therefore if you actually use this stack of vacation method then you get um surprisal as a sheif ification of memory right and so um what you can do is uh I don't want this all right so the way that um so I have some results that I'm sort of thinking of and so um you construct this funter from the category topological spaces over a profinite set and then um let me just get the final um the ideas right so there's a specific topology called a proel topology which is sort of like theel topy ology for um specific morphisms and um that topology is a subconical So based on some of's um very powerful theorems and and as work on perfectoid Spaces remember the subconical topology is very important that every representable funter is a sheath so schul has this Lima that says the fibroid category sending any perfectoid space to the category of um locally finite 3 Ox modules the structure sheet is a stack on the Vite this is a v called the the V topology on the perfectoid space okay so using these um uh using that limma and his other work on the V topology I thought okay we're just going to sort of shift these to the the memory setting so I'm going to say let y be a condensed set of episodic memories and the funter which assigns to an extremely disconnected space this is a profinite set um the category uh y over s is a stack for the proel topology the so this right here would be the way of actually showing that showing the actual descent conditions on the episodic memories second claim recall a topological space is a condensed set remember which is a sheep of sets on the proel site um the F categor sending any X and the category of continuous Maps to the category of locally free um locally finite free Ox modules this would be a stack on the protal site on condensed C so on the actual um condensed the category condensed sets on C all right so these results would be how do you actually get um um how do you sheify the memory space and these themselves would be The Descent conditions on Fantasia so these two claims would be actually like my new model Of self-information You' be okay you construct surprisal in this sheath condition and surprisal is actually just how far stacked your memory spaces and rethinking surprisal in terms of dissent in terms of what I can recall back because if I could actually theoretically recall everything I think I would have no surprise all you know um and I'll just say thank you okay sorry D I went a little bit over but yeah it was amazing Shana thank you in the last minutes I'll quickly ask some of the great questions in the live chat okay I'll begin with the one and only Dean who wrote If ptic as nonlinear do we suspend the concept of next do we begin at a place of all with all as our moves starter do we need thresholds I.E all puzzle pieces are present and each is connected regardless of order of application all pieces simply fit wow go Dean yeah I don't know if Dean's still there um yeah Dean please respond that's a great idea do you get rid of the idea next yeah I think so I think so that would be pretty phenomenal does this happen next or does it happen at the same time so Gia do you mean like all in the sense of uh um it's it's not that everything happens at once I like your idea of all I think all can be zoomed into and so yeah abandoning the notion of next that does sound a little radical doesn't it yeah there would be nothing next it's more sort of like it's more now so I don't know if you're abandoning it you're just sort of uh it's phase changing okay I love such a good question to the next question by Bert who wrote what about memories of imagined objects or events yeah so uh Bert that's actually what I'm trying to like poke at here um memory of an imagined event um so if you're like me that between any two memories there's another one and then between any two memories there's like a hyper fantasic memory I don't know I'm starting to call into question like memory of an imagination imagination of memory and so how do you how do you work that sort of discursively um and also like does the event get like when does when when are you when like are you able to realize that the memory of that thing was imagined I don't know um but yeah I think the questions that you're asking is sort of the mess that I'm in on this miso scale on this miso scale when does the memory of the imagined object turn into the memory of like a real object and so I think but you're in the same space that I'm why is it that the thing you call memory is just the thing that comes out and it's in the macro world but Dan I don't know what does it mean to have a memory of something imagined would we even know right and that's so I like your language there BT I'll just say thanks I'll look into that because on the hyper fantasic scale I I'm not sure if I'm doing that like all the time um so you know M inter acting says you're living truthfully under imaginary circumstances I think this is pretty much people all the time but yeah yeah yeah so in what that's why I'm we're interested in these sheaves in what layer is does the imagination the imagined object flip into okay that didn't happen store it over there okay and a question string of questions from Sonia who wrote could this lead to a typology of how hyper Fantastics quote picture conceptuality a space of abstractions and how a Fantastics a Fantastics conceptualize or perceive EG visually complex visual situations do a Fantastics abstract infinitely into the Future No grew SL ban spatio temporal ambiguities possible and Hyper Fantastics hold photically gr/ ble for experience all the time implying a different type of spatio temporality or conception of infinity yes Sonia Sonia hey all right yeah gr bling like okay that's yeah Fantastical I love that yeah yeah yeah so um yeah sure what whatever you're calling your spatial temperal construct and whatever that space is yeah I think um absolutely has to do with um the topology the typology of how your memory space is constructed yes on the a fantasic side you will have uh no problem uh working with just the idea of the space you aren't in a sense limited by trying to perceive the visualization of a tesseract this is a 4D spatial object can't perceive that uh but as an a fantasic I can go in the infinite process of working on the idea of the Tesseract which Sonia does so well right Sonia is the like master of discri ivity on taking a concept and just um looking at every possible part of it and how it reflects on itself um and so I think both of these both of these extremes are going to work with the uh absolute reflection of these two spaces so yeah the topology of something like an A fantasic in the memory space and the topology of a hyper fantasic memory space absolutely is going to reflect just like based on some talk reflect your Spa show temporal um like construct but also I think I'm interested in like the the frequency of the spatial temporal uh concept construct that you're working with and then even that itself would have like an expiration date so I love that space time for an a Fantasia you know is totally different than space time for the hyper Fantasia okay La last question with a short answer Dave Douglas wrote are a Fantastics actually devoid of imagery blind from birth would be considered separately given that their fantasies are kinesthetic or do they quote fail to access their fantasies oh I love that yeah that's great so again I'm I'm more versed on the hyper fantasic side which is why I went that way um I like that failing to access their fantasies and again and so I don't think any of this is sort of like negative it by any means they sort of you can say they go beyond visualization why do I only need to see in the 4 700 nomer range so I think through um I think they they they could be actualizing their own fantasies in in the analytic um in the fact that they don't need to actually be hindered by Like A visual representation that they can access it through the idea of it so say Artemis was my fantasy or something okay that I don't have to like visualize her that's weird but that I can um the idea of Artemis I have immediate access to so I really like that claim but I also think it's not a failure in any way I think they're actually getting the fantasy on the idea front such good questions thank you Shana it's been a great year with math art and all of our colleagues so looking forward to how this all continues thanks a lot thank you so much by thank you so much Matthew Brown discussing real time active inference for adaptive robotic control so put away the memories that you haven't had yet and let's talk about robotics thank you Matt yeah thank you Daniel yeah this is going to be a bit of a change in Direction although that was that was a really great talk I'm looking forward to reviewing a lot of the talks from the last um last day um but that was pretty pretty amazing from China um so yeah I'm Matt uh great great to be here um super excited to share updates about what we're doing um just to give a little bit of background on myself uh I I've been working in uh artificial intelligence for a little over 20 years um including time in the defense industry writing AI for video games things like Halo Bioshock Splinter Cell those franchises and over the last 10 years been working in more serious AI uh particularly uh in deep re enforcement learning and now in active inference uh notably uh built the first commercially available deep reinforcement learning platform which was bought by Microsoft in 2018 is now the autonomous systems platform there um currently I'm at a a startup at thought Forge I began about five years ago uh thought forge's technology started as a personal research project back in 2007 that never really ended exploring um biologically plausible alternate approaches to machine learning uh and in the over the last couple years have been applying it to Adaptive real-time control for robotics so that's what I'm going to be going through today um just a Qui quick uh uh brief agenda on on what I'll be talking about and go through a quick uh well not so quick I'll go through an introduction a lot about the highle concepts uh then dig into uh briefly the history of the project uh and the implementation uh and then go into some of the more interesting topics in depth here um Network self models embodiment synchronization a lot of the tools that we're using um to get these um robots to do interesting things and then um spend some time talking about uh what we've been doing with robotics so show some videos uh perhaps um and just uh explain a little bit about the direction that we're going over the next uh couple years so I'm not going to spend very long on this slide as is probably redundant uh you know as the topic toour I will just briefly emphasize the last point on the slide here um which is about um how the free energy principle provides a unifying framework that explains how adaptive Behavior Uh perception and action can emerge from the drive to maintain a stable internal state in a constantly changing environment this is going to be a a theme throughout the talk so I figure point that out uh and this one as well I probably not going to go into too much but it's worth um noting the last two points here uh about organisms don't passively respond to sensory input but also actively shape their environment through actions that fulfill their predictions about their sensory experience and then also about how active reference explains how uh organisms achieve complex self-directed Behavior by combining sensory process processing with active exploration um this is also uh going to be a common theme through throughout the uh the talk um now we're going to go a little bit deeper uh into what we're specifically using for our robotic uh models uh and the roots in cybernetics uh um in particular the relationship between the free energy principle active inference and cybernetics which might be obvious to many people um uh at the Symposium um cybernetics here being pioneered in the 1940s um also describes self-regulating systems that adapt to their environments uh and also kind of uh View agents as go directed systems seeking to main maintain equilibrium through feedback loops uh the goal here being to minimize free energy or surprisal in in uh the FP and active inference uh and you know active inference kind of operationalizes this through continuous adjustment of beliefs and actions um and these cybernetic ideas are foundational I believe to how the free energy principle and active view organisms as systems that maintain their states within a predicable range uh and also resists entropy and disorder by adjusting internal and external variables um over time now digging One Step deeper into the approaches that we're taking here um I'm going to talk a little bit about um the cybernetics of Ross Ashby in particular um uh cybernet Ross asy he formalized many key ideas um homeostasis in particular as the process by which an organism regulates internal conditions uh within a narrow optimal range uh in the face of Environmental changes and disturbances um he's known for um these are probably the two biggest things he's known for the good regular theorem and the log of requisite variety uh the good regulator theorem commonly is quoted as every good regulator of a system must be a model of that system um or more accurately every good regular must contain a model of that system um and with the law of requisite variety the the common quote is the only variety can destroy variety and you know another way of thinking about this is a system must be as complex or more complex V its environment in order to manage it effectively or if if one system wants to effectively uh manage or control another system it must be at least as expressive and at least as complex as the system it's controlling um and the active inference framework an agent maintains homeostasis by minimizing prediction error effectively predicting and compensating for uh environmental fluctuations U so this is how Ash's ideas uh really kind of come into play here ashp's emphasis on adaptive regulation through internal models aligns very closely with active inferences process of continuously updating beliefs and taking actions to minimize prediction error and maintain uh desired States um Ash's view of homeostasis as a self-regulating feedback system is effectively a form of active inference both involve an agent using internal models to anticipate and counteract environmental changes uh preserving stability and adaptability through continuous self-correction uh and Ashby's law of requisite vary here highlights the you know a lot of the successes of active uh when active is successful uh and of maintaining homeostasis here um relying on the system's capacity to generate a diverse and precise range of predictions and actions uh without sufficient variety of responses the system would fail to adapt leading to a breakdown in homeostasis and an increase of prediction eror predictive error um to exemplify these ideas uh Ross as Ross Ashby constructed a device known as the homeostat this was uh back in the 1940s um this was a demonstration of learning as adaptation as stability uh and he believed that behavior is adaptive it contributes to the maintenance of these essential variables uh and there's some various ways of looking at what he was doing here um some called it an isomorphism generating machine uh using feedback to generate distributed homeostasis via Ultra stability and and I I'll get more into that in a little bit um but that's kind of the approach that we've taken here at TH Forge uh and then there's kind of a long description here from one of his books about um what the homeostat was and how to construct it um and I'll also talk a bit more about how do you construct a homeostat in a in in a moment here um here's kind of a quick highle abstract view of home uh sorry Ashby's original homeostat uh and this is abstracted away in in a way that uh reflects kind of modern views of um mathematical networks here where you you can define a network as a set of nodes with connections and weights um and then he he defined uh Ultra stability as kind of the the the mechanism that powers the race of State space exploration uh which is used as part of this law of requisite variety uh and I'll talk uh in the next slide a bit about that um the network as a whole creates a dynamic attractor Set uh as a kind of self-consistent self-reinforcing uh model uh and the network can be seen as kind of a network of reciprocal constraints um the original the original homeostat only had four nodes here um constructed from recycled World War II bomb parts so this is pre-digital computer era uh and this uh there's a diagram here kind of conceptually explaining you know how how it worked um the original host stat itself I think is is I think at the University right now but the the the um description of of how it actually works is uh somewhat abstract uh I we used uh at thought for we used a lot of his um technical journals which were released by his family I Believe In online in um in the last 15 years or so um to kind of get more detailed um information about the properties and how they're constructed uh to get a more accurate uh reconstruction of the homeostatic um briefly I'm just going to talk a little bit about Ultra stability uh Ultra stability is the property of the individual node in this network that enables um the group property of homeostasis um the ultra stability here uh is talked about in in Ashby's books and papers um so I I I don't think it's necessary to go through a deep deep analysis of it um but it the ultra stability mechanism is kind of the the the local mechanism within a node that drives the active process of self-organization and criticality across the network uh and it's it's core to the function of valow that homeostat works uh and it is uh constructed by a double feedback loop with a low-level feedback loop that performs like a linear transformation and then a highle feedback loop um that updates or changes that linear transformation uh over time um this is just uh I love I I like including this this this letter uh this was a letter from um Alan Turing to Ross Ashby uh way back in when they were kind of debating on whether or not to go with analog Computing or or digital Computing here being uh ashb being supportive of the analog Computing Paradigm uh and uh this letter from tiing is basically saying that his uh his computer the AC was more General and more Universal than Ash's computer because uh you could Implement Ashby's computer on tring's machine um and it's just kind of a fun little piece of History here I I like including it um just to provide some context about the time when Ashby was kind of considering these ideas uh and why they were kind of lost to time in a lot of Senses at least within the you know Computer Science World um and I also like to bring this up because uh I think it's the first at at least in my mind it's the first time um that we really started conflating uh computation with cognition here uh and where where tiing had a universal comp utation machine uh while Ashby was kind of envisioning like a universal adaptation machine you know something closer to cognition uh and it feels very similar to a lot of the debate we're seeing today uh with whether or not deep learning machines are intelligent or you know if they can get to AGI uh I think that a lot of the conflation of computation and cognition goes all the way back to the origins of the analog versus digital debate uh within computer science um so the next question is how do you actually build a homo St and I took turing's advice uh and simulated we built a simulation of the homeostat and the reason why simulation is useful as opposed to like an analytical solution is because of the chaotic nature of U of how these systems reach equilibrium um uh and I used to have a quote here from ILO but um essentially you can't account for a lot of the chaotic interactions in the systems uh with loops algebraically uh the circularities and the the the the cha the chaos that's necessary for for these to stabilized require that you do this iter and that means um building a a a a simulation that you can kind of step forward uh and on the Fly generate the the the uh chaotic um interactions essentially uh as opposed to analytically um there's there's lots of references here with um you know the three body problem um uh a mapping of a line onto itself this this kind of concept of uh uh deterministic systems when you once you have three determinist determin systems interacting with each other uh you inevitably get chaotic behavior um that is kind of leveraged as the creative uh creative energy that drives the the um the learning process in this kind of a system um the techniques that we use to actually simulate this uh this is oops getting ahead of myself here um it come from a field known as software radio uh we started out by uh going through Ashby's technical journals uh his writings about how the homeostat was constructed uh and we started out by building out uh accurate analog level digital simulations um and then from there we kind of abstracted out all the mechanism specific Machinery in the analog systems that are not necessary for the actual function of ultra stability and homeostasis so kind of going from this kind of very very precise analog system to uh kind of abstracted implementation um of these these four nodes um once functional uh and this was simple enough to kind of uh further abstract into more kind of modern-day Network Theory implementations where we can kind of build out very very large networks similar to how it's done with deep learning uh that could be connected into arbitrary Network topologies as opposed to the original four node Network here uh and again you just to kind of reiterate here the network as a whole can be seen as a self model um each node in this network is effectively trying to predict its nebor as it seeks equilibrium uh and the the local Ultra stability process Tunes anodes local connections and the resulting positive negative feedback loops uh that uh that Define it uh and these overlapping Loops become self-reinforcing or self-repairing over time once once you've reached kind of um convergence with these feedback loops uh the final self-producing set of Loops is a kind of self- model where each node is trying to predict all the others and so once all the nodes kind of reach a global consensus uh this model this can be considered self-consistent and uh reaches homeostatic equilibrium um this gener this can also be seen as kind of like a generative model uh composed of emergent self-stabilizing homeostatic feedback loops uh and these Loops uh are maintaining this homeostatic equilibrium in a critical State and this is kind of an important note for why it is incredibly sample efficient which is going to be very important for robotic applications here U the idea here is that maintaining this homeostatic equilibrium at a critical state uh makes it very very sensitive to prediction error you know the smallest predictive error uh can trigger a a global structural change you know um uh so you don't need to back propagate with a large data set in order to change the the the model uh the behavior of the model um a single misprediction can effectively trigger a cascading uh set of of redefinitions where the whole you know the whole network destabilizes and then Recon converges on a new attracting set Um this can also seen as a form of synchronization and we'll get into that a little bit later um and the prediction error reaching zero kind of marks the establishment of an of a new adaptive synchronization manifold uh and the synchronization first starts with immediate neighbors uh at each node and then spreads through the entire network uh as a set of like compound hierarchies of feedback loops um looking at the time here I sometimes I can I could show actually uh I could switch screens and show demonstrations of the maybe I I'll share some of the I'll wait a bit later to do uh demonstrations of the original homeostat or kind of larger homeostat um just so that we're not switching screens too much um in case that's useful um all right now we're getting into some of the more interesting stuff Le uh less talking about history uh of cybernetics um so the the homeostatic Network itself seems like a lot of work uh in terms of you know um there's there's a lot going on there and it's not connected to any environment it's just learning itself in in coming reaching to convergence without kind of any um sensory or motor actions so um we have this model that can adapt to its internal structure to reach homeostatic equilibrium through a kind of self- synchronization process next step is how do we actually get this to drive uh Behavior Uh and this took you know a couple years to kind of work out the engineering uh conceptually it's pretty simple um but it will get very complicated very quickly um the short version is that the sensors and the motors uh um basically we Define sensors of Motors uh and they Define how an agent is embedded into an environment uh and the network topology separately kind of defines the Adaptive capacity of the homeostatic agents model uh the sensor definitions themselves here um are are uh kind of defined externally from the model itself uh they provide a mechanism to influence the behavior of the system by providing priers in the form of preferred sensory States or preferred environmental States uh and these act as te teody dnamic priors kind of externally defined uh Dynamic set points for the homeostatic adaptation and these sensors can be seen as input attracting sets which can even be modified in real time or animated in real time to generate you know different kinds of behavior from the same model um at runtime without having to change the model structure or anything like that uh the motor definitions then provide you know kind of critical affordances for how the agent can modify the environment to help provide a basis for future sensory predictions um and the third state of the sensors become the preferred outcomes of the motors through the this reaching equilibrium uh State and the model's Behavior ends up being biased as desired so this is how we get it to do useful behavior in a robot for example um from something that is seemingly just reaching equilibrium and I guess one one important or Salient note here is that in a real world environment the you know there it's reaching pure homoy equilibrium is is actually quite difficult unless you're unless you're in very special circumstances um so to expand a bit more on on that um the network diagram here is a is kind of an overly simplified version of our homeostatic network but meant to kind of visually explain how to think about these uh and in particular here um embodiment of of of these models in kind of an environment here um coupling it coupling a model like this to an environment through sensors and Motors is kind of a form of embod here meaning there's a kind of a specific time and place that this model is operating uh and there is um the there is kind of a temporal component here where um the the uh the model's sequence of sensory Behavior actually is relevant you the the um the the inputs to the system are not independent which is kind of different than traditional what you would imagine back propagation based perceptrons how they function uh and one thing that's interesting about this uh and and and it's worth pointing out here is that there there's no forward or backward no feed forward or feed backward um uh directions in this kind of network there's no front or back or top to bottom um the only thing you can really kind of uh Direction you can kind of take from a model like this is distance from sensors or Motors or or how close to the environment a particular node is um otherwise relative location is pretty much everything you know where as as an individual node where am I relative to the rest of the network uh and it's it's common for people to think of of machine learning models these days especially if you're used to think about deep learning or traditional perceptron based back propagation models uh to think of it as a sequential flow of processing like uh I take in an input I perceive or I do planning on that input and then I generate an action um that's not how these Network networks uh function uh instead um everything kind of happens all the time and everything's overlapping uh what I mean by that is um this is not a sandwich model I guess is is the term time simply advances and information moves from one node to another All In Parallel you know so in in traditional deep learning it's kind of an atemporal process where information goes from the inputs to the outputs instantaneously uh and uh there's no internal time unless you're working with recurrent neural networks and there are ways of adding a temporal aspect of it but the temporal aspect is really a fundamental feature of of these types of notes um what that means is um the the network itself the the homeostatic feedback loops come to be dependent on expected sensory signals uh meaning meaning the the the deeper parts of the network are generating predictive signals forward to the sensors uh and so um in order for this to function you the the deeper parts of the model have to be actually actively predicting the sensory inputs that that are coming in um this I believe also means that it ends up becoming grounded in the embodied experience itself meaning the the homeostatic self-maintenance of this network is composed of environmental signals from the Environ from from The Real World or whatever whatever embodiment you you you you've put this model in um and as part of its kind of predictions and expectations meaning the the actual sensory experience is what what is being predicted and so what the models actually generated is um generating is is derived originally from this uh essentially embodied experience um the expectations and predictions for the consequences of motor actions are kind of embedded in the behaviors uh that are being generated uh embeded in the signals sent from this uh to the sensors from deeper in the network at the same time as these kind of predictive signals are being sent to the to to the sensory regions actions are being sent to the motors um and these actions at any moment are not based on the latest input but based on predictions about what the model believes is going on in the world world and how best to to kind of satisfy the sensory inputs or get the the model into a preferred state or a lower energy state or or lower surprisal State um these kind of these motor actions are accompanied in parallel by sensory predictions that are all moving kind of to the to the to the this the uh the I don't want to say the front of the network but the the abstraction that defines the embodiment you know to the sensors and Motors um in kind of in parallel um this is this may be familiar to people who are who are um have read about perceptual control theory and predictive processing and very much uh along those same exact lines in terms of what's going on here terms of um actively generating predictions to counteract sensory uh sensory input um the mo the the motors here uh end up being kind of um a critical part of making those sensory future sensory experiences um easier to predict or um as part of this whole process another way to look at this this homeostatic process is through uh synchronization um the the homeostatic process can be seen as generating an Adaptive synchronization manifold between the agent and the environment um based around minimization of prediction error basically minimizing the difference between what I'm predicting the environment is doing and what the environment is actually doing this is kind of a synchronization as prediction uh process uh where when the two systems are synchronized they can be thought of as predicting each other you know um you know anticipating the other individual nodes are using uh Ultra stability to uh predict their neighbors and then when at equilibrium all the nodes are synchronized through these feedback back Loops uh the synchronization between the noes spreads to whole network synchronization at this point uh ultimately uh this synchronization spreads across this the sensory and motor abstraction uh between the environment and and and uh an agent so you kind of get a um agent environment synchronization process uh through this um looking at another way uh these overacting feedback loops interact and synchronize with each other as part of the homeostatic self-maintenance as these feedback loops extend from Motors through an environment back to the sensors the model gains the ability to anticipate the consequences of its own actions uh and the the the capacity to synchronize internal states with something in the external World um you know without direct communication is is kind of critical to this the synchronization means that the internal States become uh uh they they they come to resemble internal states of the other system um when and when at this kind of synchronized homeostatic State it's also in this kind of critical State sensitive to local prediction errors um and this like I mentioned before this is kind of critical because that allows internally the model to to completely restructure itself or redefine how it predict Century experience um at the very slightest misprediction um and uh this is this is very very critical for sample efficiency um uh with to contrast this with you know deep learning models or deep reinforcement learning models you need um a a very large amount of data to kind of change um The Domain the or the way in which the the models are are behaving in the world while in these kind of models it can kind of change at the uh with the slightest slightest uh one piece of data can kind of trigger Global uh Global restructuring um so one interesting aspect here is that in order to maintain a synchronization manifold the network must be able to converge to equilibrium faster than the environment can destabilize it and so this kind of goes back to um ASP theory of uh the good regulator theorem or the law of requisite variety where the model has to be ex more more complex than the environment is trying to to control um and has to be at least as expressive as the as the as the uh the environment is in terms of coming up with Solutions um and you know uh yeah actually I was going to go a bit more into synchronization uh and how once once the network is synchronized with an environment um once two elements internal to those systems are synchronized they can kind of coordinate without direct communication and this is kind of important to the whole process as so that the and Motors abstractions don't have to be the direct link between uh an aspect of the model and an aspect of the environment one synchronized those two pieces can coordinate in terms of their behaviors will be coordinated without direct communication between the two um so now on to some more fun stuff um the killer application of all this is robotics uh other early Avenues we looked into is uh for example applying this for an no detection um where you can kind of use the network as a thermometer for surprisal um the other way you can kind of use this applied to like modern deep learning is for like a model dri drift detection where you can kind of uh use the model to um detect when a deep Learning System is being used on a uh if if the input stream to a deep Learning System is being is diverging from the model uh domain so like if the sorry the the training domain so if the data set you used to train it on is starting to differ from how it's actually being used uh this could kind of uh be useful for that um but really I feel like robotics is really the killer application for this especially when you consider the embodied nature of it um you know this the embodiment is kind of a requirement for how this model Works in terms of having to be embedded in a an environment with uh this could be a virtual environment as well but it has to have some sort of uh sensory motor coupling um to something external to itself um also the the ability to kind understand context over time and understand that um even for a sensory input that is the same as some other uh uh a sensory input at another time the the the the things that have happened previous to it are relevant you know um and so building up this context to awareness uh through interaction with environment is kind of critical to how our models work um but also very important for for robotics uh and then real-time adaptation um of the model being able to update the model in real time as actions are being performed um is kind of a really useful technique in robotics in order to enable many uh dexterity uh and and adaptation in the moment um that is missing from a lot of robotic applications here uh and specific to deep reinforcement learning and deep learning ways of doing robotic control there's this uh field known as Sim toore and so uh Sim toore problems come from uh deep re enforcement learning is is where they're commonly referred to but it's where the models are sensitive to simulation Fidelity uh the idea being you've trained a model in some in simulation um and how the model behaves is very sensitive to the specifics of the the physics in that simulation or you know the actual Fidelity of that s uh that simulation and so moving to the real world often shifts the input domain uh and deep reinforcement learning and deep learning models are not usually robust for these kinds of changes uh unless you you drastically increase the amount of data through domain randomization uh which causes other problems such as the cursive dimensionality there's all sorts of solutions to the sreal problem for us we sidestep all this by leveraging realtime learning and adaptation so um when the model is in presented with something new or unusual in the moment we we don't we don't separate the the training phase from the deplo you know the the inference phase it it as it's inferring it's doing the updating and the learning um and this this kind of solves a lot of the sentto real problems um as the model is able to adapt to whatever differences uh happen in the real world that we're not represented in the simulation um so this is a lot of re uh a longwinded explanation of why we went after robotics for applying these these kinds of models um to get into more kind of robotic side of things in terms of technical detail in terms of how we're actually implementing this um for a lot of our interoperability between a lot of different robots um the model itself doesn't really care what kind of robot you apply this to uh so you get a lot of Hardware agnosticism where you can take a model built for one arm uh the a robot arm and apply it to another robot arm for example uh and the model will adapt to any kind of kinematic differences between the two that kind of thing um this is a diagram of how we fit into Ross Ross is the Robot Operating System and it's it's the it's the open- source uh academic uh software framework that is used uh across that supports a lot of robots it's not used in a lot of industrial settings um but it's incredibly flexible in its ability to experiment and try new things and so we've kind of used this as our kind of be head uh robotic stack implementation um these uh homeostatic models are particularly small uh they only operate like the uh typically about 30k of memory so um they they can operate on edge devices you know no GPU needed uh no Cloud needed uh in fact you you want to run it as close to the robot as possible to minimize latency in the control um phase uh and just runs in in a single CPU um because of our our implementation of ultra stability and this homeostatic networks are are quite efficient uh model update times are sub millisecond so this enables us to um update the model and update the the uh that synchronization manifold um as a task is being performed you know uh We've clocked the networks being able to run up to 80 khz um but for most robotic control algorithms 500 HZ or th000 or 1 khz is typically fantastic and most robots don't support anything above that um and there's you know there's Al also some other interesting aspects to the mo the the the technical side of things where the models are small enough we're talking 30k here where um analytical explainability is possible where we can take a look at robotics action and actually analyze the homeostatic Network to figure out um which sensory inputs or which you know which motor out you uh which sensory inputs led to a particular decision um and because the models are small enough we can kind of use different kinds of network analysis approaches uh to figure that out uh on the more technical detail side in case it's useful the our models are built in C++ and we have a python wrapper so that gives us a lot of integration to capabilities to lots of different robots um which is is is useful and fun um so just going to go through some of the applications that we've done in the past uh and I could I could bring up a demonstration and and do it live U this this was a uh a use case we did uh for a company for uh adaptive inspection so they had a use case where uh they have autonomous vehicles going through um going through a field and they needed a robotic arm to kind of uh compensate for um movement or unexpected you know orientations and and configurations of of the of the uh autonomous vehicle base um without knowing ahead of time what it was going to encounter uh and so yeah this was kind of a a fun interesting uh pretty simplistic model we built for them uh and you can see kind of a fun video here of it in our lab kind of tracking this this QR code um and uh going one step further in terms of uh of difficult this was another use case um from a different customer um they had a challenge of um they they were seeing an injury with um turning High Press gas valves and uh they wanted to automate this with robots the problem was that they never have a good accounting for what kind of valve or what kind of uh you know thing that needs to be turned what it was going to look like what the shape of it was going to be if it was going to be rusty and difficult to turn or if it was going to be covered in oil and slippery or if the edges were going to be worn off um so they needed something that can adapt to this high level of variability um in real time um without access to the internet because this was on you know remote locations uh and so we built this model for them um and you this is just kind of a fun little video of it um we've turned this into an interactive demo that we've brought to conferences which is um kind of fun here this is uh this is just recorded on a cell phone you can tell it was it it's it's real based on the production qualities um but this was that that that valve turning model that we we created before um except we've just kind of adapted it here so that we can you know show off people torturing it um in real time live as it tries to kind of turn these bolts of various shapes uh and sizes um you know some interesting things to note here the original model that we built for this um only took about uh 15 seconds to train three examples essentially uh and it was only it was only trained on a single um bolt shape here uh the square in fact uh and then what we do is we kind of show how we can kind of generalize to all these other different cases here in real time without having seen any of these examples before uh and at the beginning of the video there you can kind of see on the screen um uh our our internal it's our internal debugger but it's a fun visualization because you can see the the homeostatic Network um learning in real time as you kind of move this um this uh panel of bolts around um I'm just stalling a little bit there we go you can see those little those little squiggly lines there on on the on the screen um that's the that's kind of a Viewpoint into the 30 of the nodes of the network kind of um updating in real time as we're as we're kind of having it turn these knobs here um getting into the less interesting stuff but just to kind of show uh where we are as as a technology um this is a platform we're building uh currently this is used internally um to develop new models uh and we have um some some interesting tools here that allow you to um effectively search ranges of topologies to to experiment with different kinds of uh Network topologies for particular task um and then see what the performance looks like in in simulation and kind of iterate on this uh eventually you'll also be able to uh connect your robot to the platform and download the model directly um so that you can go straight from um designing uh uh a goal specification we call a goal specification but really what is what you're doing is you're defining those T Dynamic priors in the sensor uh sensor definitions as part of that embodiment uh definition uh and then you go and you evaluate what topology is optimal for the for that for that definition and then you can deploy this directly onto your robot uh in the real world and try it out um next steps or the direction that we're heading um is is kind of interesting um right now a lot of the models that we're building are very kind of small and uh task specific um but we're building them in a way that is recomposed so the the idea here is instead of building one model that does a task like turning the valves this that that task is actually composed of three smaller models and the idea here is we're building up a library of recomposed submodels uh so that we can kind of recompose them in uh in in simple ways to go after new tasks uh so the idea here is we can take um right now we only have a handful of submodels but over the next year or two building out this larger library of submodels that can be recomposed to go after a much wider variety of tasks uh and um so this this diagram here is is meant to show how we're kind of looking at um building up this framework of recomposing models uh eventually that platform you saw this one here um eventually we'll probably open this up for the public uh to go and compose their own models from these recomposing submodels uh and um then we can even string together you know many models together and and um have a framework for deciding when to use which model um because they're only about 30k of memory uh you can really kind of load up uh a robot with you know let's say a thousand models for example uh and you can kind of get a lot of generality just from um a collect of sub problems being solved individually in in a robust manner um so I went through this talk pretty quickly I fig I'll just end with some current work that we're looking at uh right now we're working on a um this is just kind of an early look in simulation of of an insertion task um we're working on uh what's interesting about this is that we can't use vision for a lot of this because once the insertion process happens um the the arm kind of includes the view of it so we're incorporating uh four sensors uh so that we can kind of Wile you know tight fitting pieces into a um a housing for example uh and you really kind of need to build in um uh this kind of nuanced uh feeling of you know wiggling something into place this comes very naturally to humans you know um in terms of like filling your way through a problem uh and so we're we're kind of um working on building out the subm models needed here for high frequency Force torque sensors for wiggling uh things into place um to to kind of broaden the amount kind applications that we can we can start attacking with this approach um so that's that's kind of uh some current work that we're working on um I kind of blazed through that really quickly I could show some live simulation demos but um maybe questions maybe it makes sense to just move on to questions thanks Matt that was awesome I'll ask some of the questions in the chat maybe we'll we'll still have time to to see the demo but that was that was really exciting um okay I'll start with black cat who wrote very cool how would you say interoperability challenges change from software to Hardware that's inter question there's different ways to think about it um moving to software things I think get a lot easier um as Hardware is always hard uh and always timec consuming um for us we're using Ross to get a lot of interoperability between different kinds of robots but Ross is nonoptimal for for a lot of lot of situations uh and so there is going to be some sort of low-level drivers that we we'll have to to write that are kind of going to be Hardware specific but for the most part most robot uh uh manufacturers or oems they have an interface that is that provides some sort of joint control so the models right now are uh generating effort based controls uh that we then convert into you know positional based controls because that's the most common most robots do not support effort based controls um but pretty much all robotic arms in particular have some sort of joint space control contr uh interface so we can basically all we need is kind of a driver to translate the output of the model into uh joint Space controls for that particular robot um the other thing that's kind of interesting and you don't get this with deep enforcement learning is that um we don't actually specifically tell the model um what each motor output does we we we do have we do specify how many motor outputs there are from the model um uh although I guess we could overestimate we can give it more more actions larger action space than it actually has and will'll just kind of learn to ignore or it will just there will become no correlations between the unus actions and the sensory space so it ends up being unnecessary um but because we don't actually specify what each motor action is doing um it learns that uh meaning like it it will in the first couple uh of milliseconds of interacting uh with the it considers the robot part of the environment let me put it that way so in the first couple milliseconds of of controlling this robot it will build up those correlations of oh if I apply a positive effort on this joint it moves me away from my target or if I move apply negative it gets me closer to it kind of figures out those correlations and as you know as a process of it learning how to get to equilibrium if that makes sense um and as a result of that we get a lot of um Hardware agnosticism so with that that valve turning model for example it was originally built for a different robot it was originally built for a he heavy six degree of

Freedom arm um the customer ended up deploying it on actually a different heavy robot with with an additional actuator um and the model can kind of you know we basically added an output to it um but the model can kind of adapt to that situation there um for those videos you just saw and and at that conference we then took that same model and deployed it on a different manufacturer uh with a six degree Freedom arm uh and it kind of just figured out you know what it needs to do differently uh that robot is really nice because it fits in a suitcase so it's really great to travel with so from from the uh moving to software and especially moving to this particular way of looking at the interface um it buys you a lot of interoperability bonuses that you don't get with other approaches um but there is still you're still dependent on the hardware oem's driver level interfaces um but generally those are all the same uh with the the interfaces are all the same even if spinning up the robot might have different mechanisms awesome Six Degrees of actm I'm I'm going to Cluster two questions together so Arun asked how much of this is or will be open source and Alexander Robbia asked wonderful talk Matt I'm a huge fan of the homeostat as you know have you given thought to perhaps translating the homeostat to its core set of equations dynamics and contributing them to open-source projects like NGC learn so others can appreciate its value oh great question um and great great I'm glad to hear Alex is on the call um these are both great um I I'm always torn in terms of like what to open source and what not to open source um I think eventually we are going to open source a lot of this um right now the intellectual property of of this approach is one of the things that is most valuable about our current business so from a business perspective in the short term we're probably not going to open source a lot of this we are expecting to open up our platform to to a wider audience in um over time um I and actually I should follow up with Alex I we have actually kind of distilled down um our implementation of the homeostat at least you know there's a lot of inter interpretations I want to say there have been five or six implementations of the homeostat over the last 80 years or so uh and they're all different which is kind of interesting uh there's a lot of room for interpretation but at least our implementation of it we have kind of distilled down to a relatively simple set of equations and they they they are very very similar to kind of the the um equations using the perceptron with a couple of very subtle changes um the the the subtle changes here having drastic effects uh one one is that we've added state to the to the to the node um which really changes how how you think about these things as perrons are generally stateless and so the adding of the state there kind of adds that temporal notion so even though it's a subtle change it has drastic implications for how how the action model Works um but uh and actually I might I was trying to see if I have it as an extra slide here in in the talk um I might follow up with Alex and and share uh share those uh the equations I'm I'm less concerned about um sharing kind of you know the approach and and the and the uh the uh equations for for our definition of the home stat just because these are very very difficult to implement uh and I would kind of Welcome more ideas and activity in the space and I need to take a closer look at NTC learn uh I've I've been I keep seeing uh uh things being added to NGC learn uh and it's it's relatively new package but there's a ton in there uh and I really kind of need to take a look at what's in there um and if we do open source it does make sense to maybe integrate that as potentially an option in NGC learn um a lot of it depends on whether or not the interface is compatible with what we're doing and I I say that not because as as a limitation of NGC learn but more of as a limitation of how people think about machine learning models as kind of um most most Frameworks that are built around deep learning um their interfaces kind of assume uh a temp you know uh Behavior where each input is kind of independent uh each input output relationship is independent of the others um but I think it's just it's a matter of me just taking a look at NGC learn and how it how it's structured and how to actually interface with it but I think if we do eventually open source um the code um that might be the the best Avenue to do it just because that way it would be alongside a lot of this other great ideas for nonb back propagation based machine learning thank you Matt and and also really appreciate like sharing that Insight it isn't a a open and shut and just because we can open source something doesn't mean we could or could in that way or should tomorrow or whatever happens to be so so hearing your your well-worn experience about all of this is is really insightful um okay another question from Alex he wrote how do you go about designing the structure of your homeostat net is it a random graph that you instantiate or some other graph type does it depend on the robotic control problem that is a great question and just right off the stat off the start I I'll mention that it's unsolved like it's something that we're still working on um in terms of how we do it now um so longer term I would love to start with like a random graph and then have some sort of um metric for evaluating what graph is better than another graph for particular use case and and we're probably pretty close to there and then we could have some sort of mutation function and that that way you could kind of evolve what the right topology is for a particular graph I think that actually has a lot of promise and I'd really love to get into that kind of stuff right now um what we're doing is a half step in that direction um we have found that the topology has a big impact on how good the model is at performing particular task meaning you can take a random model and then try to have it turn vals for example um but often it'll do a pretty poor job or um it'll be slow or it'll be a little wobbly um and and so really kind of fine-tuning what the topology is for a particular task is is critical if you actually want a really effective robotic control model uh and so the way that we've been solving this is um for for specifically robotic control we've come up with a a way of parameterizing the topology um and this kind of constrains the search process for what is finding the right topology and then what we do is in that platform you saw in this platform the primary thing that this platform is doing is providing an interface to do that topology search so in the back end here uh when you actually are trying to find what is the optimal topology for a particular use case what's actually going on is that it's using that um that kind of parameterized search space and doing doing a big search U and so we're actually leveraging some Cloud compute here for this not a ton uh it's not particularly expensive the training process only takes about 15 seconds for per model so um so what we're we're using any scale which is built on an open source technology called Ray uh which is uh really good at spinning up let's say thousands of CPUs at once uh and so the idea here is we can go and spin up uh 20,000 different topologies uh and then evaluate of those 20,000 which one is optimal for a particular use case and this process takes about five minutes uh and so uh we're trying to get that as as short as possible so that this is kind of a compilation time interaction as opposed to you know training the model for a day and then coming back and seeing how it performs you can go and like explore 20,000 different topologies and figure out which one is optimal for your particular use case so that's our current approach which is kind of like a brute Pro Force approach um but longer term I would love to get to that evolutionary uh approach where we kind of start with a random model maybe we can estimate what a good starting size is or maybe we have some sort of um uh some you know uh heris for deciding whether or not to shrink the model or or grow the model um but you know the to contrast it with deep learning um this is one of the downsides of the approach is that the topology itself is uh is very critical for good be good performance as opposed with deep learning you can often just go with a larger model and just give it more data and you can kind of get the same performance as a smaller model um I think this makes sense uh from a biologic biological standpoint in the sense that our our brains are not just randomly connected they are evolved over time you know uh and you can imagine how some brains might perform better at certain tasks than other brains um and so I think this this topology dependent behavior is not necessarily um problematic or or wrong this just something we have to deal with awesome I'll read a comment from Scott David this is as most emissions from Scott's language model it's not necessarily a question so if you have a response go for it then I do have a question that that also connects with Ian and little interception so Scott wrote it is interesting to consider the hardware software question of application of active inference in light of the the historical Encounter of Claude Shannon with Von noyman when Shannon was encouraged to apply the second law of Thermodynamics to his quantitative theory of information thermodynamics at the edge of tangible and intangible information domains the ubiquity of the expression of the second law results in it being a foundational Vector for interoperability including across tangible hard and intangible soft domains there's a lot there I might I might need to take a look at that in text form and and come up with a coherent response um textually um but there's a lot there um there's there are a lot of information you know Shannon level information constraints in the you're right in at the hardware level especially when you're talking about um update rates you know for example um the model the model as a part of its embodiment has a temporal aspect to it too in terms of um the frequency of updates of the model has to be consistent like you can't update it you know 100 times per second you know for one second and then a million times per second for the next second so there is um there are kind of interesting information Theory do you know problems here with you know nyis theorem like uh as the sensory inputs are coming in um are are you getting sensory inputs at a high enough frequency to actually makes make interesting inferences about them or is on the robotic side there are absolutely uh control constraints you know if you're only able to take an action once per second there's you're very very limited in terms of what you can do with a robot as opposed to if you can take an action every 500 every you know 2 milliseconds uh there's a lot more Nuance uh in in in the feedback process that of in the control so there's absolutely some information theoretic stuff um on on that interface in that embodiment you know definition um for the most part I haven't I I don't want to say we've done anything particularly Innovative on that side it's just these are all things that you kind of have to understand as part of like the constraints of of the implementation um but uh definitely send me the text for that uh that question and I'll try to come up with a a more coherent and thorough response um there was a lot in there cool and the historical part was was was epic um oh here's a a question so Ian not sure which parts you listen to from prior maybe Matt give a first pass but I was really interested when you mentioned the difference between the position sensors and the effort sensors so how does that play out in robotics and then like Ian to the interos perception and introspection angle in what ways do does our postural adjustment relate to positional inference and effort inference that's a fun question um unfortunately I have a feeling my answer is going to be not part not too interesting um in terms of we're not doing anything particularly Innovative here either um the the outputs of the model are almost unitless uh in the sense that they'll they'll kind of adapt to their embodiment in terms of like so we've kind of been interpreting them as efforts or like you can think of this as like torque controls um the for robotics this is problematic though because first of all like I want to say 90% of robots or 95% of robots don't support torque based controls that requires there there are just a few robots out there that require that and for the most part they're they're confined to research applications um the reason but you know the reason why they exist and the reason why there's a lot of interest in it especially with deep reinforcement learning is is there's if you can go to effort base controls or Torque based controls directly to the robot there's a lot of nuanced Behavior you can get you can get really kind of uh compliant Behavior where the robot's applying just enough Force to perform a task that way if you bump into it it's it's not going to hurt anything and it'll it'll be compliant and you can kind of get a lot of nuan behavior with torque based controls but the reality is that 95% of robots don't support that they just support either velocity controls or or you know positional controls of some kind uh and so to to solve that you know we're we basically convert the effort output of the model into a relative positional change and then we basically integrate that relative positional change over time to get an absolute positional control um so not particularly interesting on that side but uh it is kind of a technical note that's that's a bit challenging on the robotic side of things um it does I you know I hadn't really thought about it enough from an active inference perspective in terms of like what does that mean for um you know the the attracting set that you're you're you're bringing in a lot of it is because it's it's somewhat disconnected from the sensory inputs like um you know you you know what I mean like the positional controls end up impacting the environment in some way and then you know connecting the dots kind of indirectly that affects your sensory inputs um but they the it's not like a direct it it's not like the the positional versus effort outputs of the robot are directly changing how our sensory experience happens it's more that like it might change the the effectiveness of or the capabilities of your robot uh to it to to alter the environment if that makes sense um but maybe I need to think a bit more about that because that is kind of an interesting aspect that I I hadn't really considered enough how does that play out in the body Ian oh yeah that's really interesting I've joined this conversation late but thanks for including me Daniel um so yeah I was kind of listening to that with interest and thinking of you know how we as a human something simple like the the concept of being hry hungry and angry at the same time and how that affects our behavior when you've got compete competing kind of um uh interests or physiological variables that you're trying to deal with and then thinking about okay a robot something simple to start off with like maybe a um one of these robotic lawnmowers that needs to go back to its docking station to charge up for fuel um so it you know it needs to end the task it's doing and move back to its docking station and charge up but as robots get more sophisticated and they're trying to please humans more but there will be some kind of you know I don't know some limit to its um capabilities its internal processing or power that's and maybe at some point it will say it's you know absolutely have to weigh up all of these and come to some computation about what it needs to do next to satisfy its own internal this honestly I think ties into some of the stuff that Alex has been talking to about with like moral competition as well um if I'm gonna be honest the robot control Bots we're making are far less complex than an actual biological entity like a human um but you're absolutely right because you're going to have all sorts of here right now we're kind of having these kind of first layer sensory inputs kind of defining the tele damic priors but you can imagine like a hierarchy where you have like other models that are feeding outputs that feed into your moto controls because these motor controls are are kind of the the end point for for modifying the environment but you can imagine why you would be reaching for a a cup of water is is driven by some other network that's saying oh you're thirsty um and uh and so we just you know I'm just not nearly there in terms of complexity yet but I think you're right eventually we will be especially if you want to have robots that are fully self self-sufficient fully autonomous know when they need to repair themselves and then going out and Performing those repairs uh whatever that might be maybe it's charging um I think you're absolutely right I think that's the future uh longer term I love it thank you Matt great times till next time yeah likewise hello octopus and welcome to the fourth active inference Symposium thank you for joining us um we're looking forward to your session titled cognitive bations and mathematical Explorations so in case you need a reminder of what the abstract for this talk was and I had to look this up this morning uh thinkers and thoughts have a co-evolutionary relationship this relationship is especially clear in the arena of mathematics where mathematicians forage for mathematical statements and structures in a process governed by complex and potentially unstable rules we will give a partial mapping of the structure of this space which necessarily relies not only on rigorous mathematical logic but also on subjective beliefs metacognitive juristic and creative play so I've done a magic trick here all of these things are claims that are in need of evidence from my perspective and the goal here will be to provide some evidence in that direction but also to engage in conversation so in the construction of expertise I have no ambition here to play the role of expert there are certainly qualifications and evidence that I could point to my experience as a math professor for seven and a half years my lived experience as a neurodivergent individual for 42 years my my time spent in an octopus identity for the last 3 years openly and none of those things makes what I say here more valuable than what you think about them the only request I will make is that I've given a lot of effort into thinking through these thoughts some of which were quite difficult that showed up for me and I ask that you will give some level of effort towards finding the value in those thoughts and how they can be mapped to your reality so let's move from the abstract to abstraction when people think of mathematics in the modern sense they often begin here as both its value and its difficulty we might imagine that Bob and Alice both have some cows Bob has two and Alice has three and they're curious how many cows they would have if they combine their herd Alice can bring her cows to Bob and they can count them then if they have the physical process of counting at their disposal a cognitive tool of onetoone mapping between objects and objects but they can also do it conceptually one of the things that abstraction enables is the ability to move information about physical perceptions more quickly than the physical things themselves can move in other words Alice doesn't have to bring her cows with her if she can bring the concept of three or the idea that she has three so this is another aspect of mathematics that is often highly valued and highly contentious in relation to the ways in which we realize and reify scientific practice education and other ways in which education other ways in which mathematics shows up in our lives is how are we actually giving a physical body to the concept that we are hoping represent the fact that two apples are conceptually in some sense the same as two cows or two fingers is one level a level above that is the that we can attribute symbols to representations anything from two interlaced eyes in the Roman numeral setting to two in the Hindu Arabic numeral setting to words like two and do to two dots to two fingers the two fingers are physical yet somehow they're also a basis of abstraction because the idea of abstraction includes the fact that there must be something concrete to abstract and and this two fingers I've included on the page of abstract representations partially because I like the idea of things having bodies I like the idea of a number two having lots of different bodies how many bodies can one idea have but also I like the ambiguity here between the two and the three because in some systems the way that I'm showing this here would be interpreted as three and in others as two so with that ambigu in mind there is no slide three please do not discuss slide three later we will not bring it up again and in fact it is not advantageous to you to do so because the claim that there is no slide three is an idea wholly owned by eight arms nine brains and if you want to discuss the fact that there is no slide three there will be a licensing fee involved thankfully some ideas are open to discussion with no um TR my financial transaction involved and one of them as we finally come to slide four is a very important Concept in mathematical pedagogy called the rule of four I think this encapsulates thinking that might have been going on for a while before it was formally published but the first time I encountered it was in the excellent calculus textbook by Hughes Howen gleon and McCullum at all this was the first calculus textbook I looked at When I Was An undergraduate student student and it was also the first calculus textbook I looked at when I was a beginning Professor beginning to teach my first Calculus class I'm afraid that far too many students probably completely ski this section uh which is in the pre Preamble uh but one of the key Concepts that it elucidates is how to think about mathematics and it's not saying this is the right way it's saying this is a way and this is a way that the textbook uses quite profitably it's called the rule of four and it says essentially that when you encounter a new mathematical concept there are four distinct but interrelated ways that one might think about it you could give a verbal description or a definition um you could draw a graph or some type of diagram you could use numbers in the sense of data or you could use a symbolic representation like a formula or an equation and the extent to which these are distinct and interrelated highlights the value that each of these may have to individuals as a latching on point for thinking about difficult mathematical Concepts and calculus it also eliminates some of the privilege that might exist around specific representations around the idea that symbols might be real math whereas when we explain them at a high level somehow that isn't real math for instance if we look at the types of mathematical works that mathematicians actually produce um it might be surprising ring to undergraduate students what percentage of this is not what they would traditionally think of as symbolic and what I would mean here is equations and formulas like we get to page three and finally we see something that looks like a symbolic mathematical expression but the majority of the work here is spent on Words and diagrams visuals a large portion of this is in the theorems of course which are something that is uh a very individual feature of pure mathematics but let's not get bogged down there let's simply say there are many different representations and one of the challenges of an effective instructor is to map between these different representations so that people who might feel most at home in one of them can use that as a leverage point for accessing the other ones uh based on my background I'm I'm a defro math professor voluntarily also a chaos magician and so I'm very very tempted to try to build a funter between this rule of four originating in mathematics and this four quadrants model of Human Experience Advanced by Ramsey Dukes in his book known as sop copy um which says that there are these four lenses through which human beings can process and structure experience there's art religion magic and science it's so tempting to try to do something like say the symbols are the magic and the words are the religion and the numbers are the science and the graph is the art and I'm going to resist that Temptation because I think it's it might be a useful initial temptation but the resistance of the Temptation provides its value instead I'm going to hold this as an idea we'll return to it in a moment where it will play against some other Concepts that we're likely to encounter I will stop here for a moment and say something in promptu though which is um there's a practice in this world of chaos magic called belief shifting where the magician consciously and intentionally adopts a specific belief as if it were something they believe to be true and there's a very related practice in mathematics which is the construction of the axioms or the adoption of the axioms so in mathematics proofs once you adopt specific axioms or specific assumptions the conclusions should follow not without effort but in a clear logical flow mathematicians are capable of adopting different axioms and this produces what I'm going to call a cognitive bifurcation so a cognitive bifurcation since we've already split things into four they may be too much let's try to get things back together and just split it into two one place that a cognitive bifurcation might show up in the history of mathematics is in the different types of geometry the geometers think about so one of the features of modern geometry is the realization that the parallel postulate in ukan Geometry is not possible to prove from the other axioms of ukian geometry and so the parallel postulate says that given a line and a point not on that line there is exactly one parallel line or there's actually one line that passes through that point parallel to the original line and if you choose not to adopt this belief as ran played with in the idea of one of his lectures um it opens up a whole new world of non- idian geometry both spherical and hyperbolic so the axioms that we adopt and the beliefs that we intentionally choose to consider are powerful tools that can open up new conceptual playgrounds but also quite dangerous Alice and Bob probably feel very safe in this world even if we don't admit a slide three we admit a number three to describe how many cows one of them may have and then the question is are these all the numbers and and if they are are not in which directions would we be willing to consider perhaps we're asking what comes after five or after the number after five and like a child we realize that we can continually add one and there is no a priority upper bound and we come across the idea of infinity perhaps we start to look for what could be in between the numbers and so we discover numbers like one and a half or 1.1 or perhaps we make a dangerous leap in the other direction and we admit the possibility that there is a number which says that you have no cows and why you might need such a number is an intriguing question but let's say that we do have one when we introduce a number in this way we need not only to consider the fact that the number exists we need to consider the ways in which it will interact with the other numbers that exist in other words the rules so as we begin to if we want to we know how to add numbers that were 1 through five but how do we add the zero and this is a cognitive bifurcation because now we have a symbol that plays by its own rules and we've opened up a whole new world of numbers to where when we added positive numbers together they always increased and we were always heading to the right the fact that we can now introduce a number where we add it and identity for the binary operation addition is uh an interesting an interesting split that opens up more possibilities on the leftand side and I would say this is a cognitive bifurcation it says in the definition that I'm adapting from Wikipedia a bifurcation occurs when a small smooth change made to the parameter values of a system causes a sudden qualitative or topological change in its Behavior it's hard to imagine a change smaller or smoother than putting the circle here and declaring that it doesn't do anything on addition and yet this an immense cultural controversy when these numbers are being introduced so I'm going to make the claim that mathematical thinking systems are complex systems I'm going to give the highly unsatisfactory proof by Authority that this claim is obvious one piece of evidence we could point to is that the existence of a parameter bifurcation almost immediately suggest that we're dealing with a complex system and yet that is not quite the right way to think about it especially if we're going to be looking at it from the active inference context so let's ask for more evidence and let's not accept proof by Authority what evidence do we see for the belief that mathematical thinking systems are are complex systems and one way that we might conceptualize this through a model is at the agent level where we think about the types of agents who are involved in systems I've heard that there would be people here who like ants so I've made sure to include some and uh here's my ant hill and here are my ants and I have these ants foraging for formulas that they bring back to the colony which then feed other ants which are a part of the ecosystem the formulas and the symbols that they bring back in this mathematical world of semantics is is more than just symbols right it's something that actually contributes to the survival of the colony but in order to do that it must be converted into other things so it' be easy to just try to be efficient through this and reduce it as much as we could to something useful I'm sympathetic to that Viewpoint because mathematics has to be useful at some level to human existence to justify its existence I believe so if we were looking at this on a belief level on a on a probability level we might say something like with certain axioms as priors mathematical theorems are simply beliefs such that an agent in order to survive who quickly will quickly adopt correct beliefs with probability ility one if you want to make 2 + 3 equal 6 you will quickly run into issues in which everything collapses and it ceases to be a useful tool and and I'm not opposed to this model I just want to say it's oversimplified because many of the beliefs that mathematicians choose to adopt and choose to spend their valuable resources is thinking about don't have any immediate mapping to Applications for instance the question of whether or not zero is a natural number is something mathematicians will argue about what is the utility in foraging for this belief perhaps you can think of one um but I think it speaks to something else that could be happening that mathematicians throughout the centuries have seen which is that the claims of truth that you make are not entirely limited by the axioms that you assume and by the process of logic there's there are larger questions here cultural questions uh of identity ontological questions of what is real what does it mean for zero being a natural number to be a true statement mathematicians find it useful to disagree about this the conversations that we have about these things make us better mathematicians I feel I'll never forget when my PhD adviser explained to me why he believed that zero was a natural number but there's no mathematical proof that can provide this it's an axium that you accept or do not when you define these numbers and when we talk about types of numbers there's a famous cognitive varification that occurred at some point in history which is the division uh the discovery of the fact that not all numbers are rationals not all numbers can be expressed as a ratio between two integers if you're not willing to accept this then you cannot come up with a solution to x^2 - 2al 0 and of course you could by authoritarian F Fiat simply declare that there is no solution to x^2 - 2al 0 there's a legend that the mathematician hippasus who discovered that the square root of two was irrational um was thrown overboard by the pythagoreans so that he would not share this information and I imagine this as a kind of cyberpunk reimagining of ancient Greek geometry not as an insult in any way to Pythagoras who I think would appreciate this um but to imagine that there is a geometer who is not opposed to the idea but opposed to the pathway by which it came through and willing to use Force to ensure that that dangerous doorway stays closed so another model of math person perhaps that we could view is as a boundary between wherever math comes from and the quote unquote real world and the real world is in full of Agents already that we can easily map to there are institutions educational institutions governmental institutions um professional societies there's politics where the government may decide certain levels of funding for those institutions business which may choose to fund the institutions to uh insert itself into the process of politics there are funding sources there's media one thing that occurred to me as I Was preparing this talk is we don't often think of mathematical journals or scientific journals as media but they are they're media that's made and constructed like any other and then is the question of where math comes from where does math come uh this is what I've wrestled with a while one reason is it opens up some very some doors that from the material materialist utilitarian perspective of like a School of Engineering feel really strange where did Alice and Bob first get that idea of numbers and does doing math involve some type of contact with these strange things that could happen we don't see very much of this in our schooling which is often about producing symbols uh producing people who can follow precise rules with respect to symbolic manipulation and at that point it can become a type of overcorrection and I'm sympathetic to this door that uh I'm sympathetic to this perspective that opens the door a bit more to the strange not only by my own temperament but also through the lived experience of people who accomplished very effective physical and Engineering uses of mathematics but were willing to express that they were sympathetic to this Viewpoint so Hinrich Herz For Whom the the unit of a cyclic repetition and a computer processing unit is AED for says one cannot escape the feeling that these mathematical formulas have an independent existence and an intelligence of their own that they are wiser than we are wiser even than their discoverers in other words if the intelligence is in the symbols and not in the humans who are trying to work with them as tools this provides a very different perspective on the creation of mathematical thought and on the ways in which mathematical thought not only originates but also can be applied it also points out to some extent the horror that we can be doing when we do this non-consensual cognitive bifurcation of a person of people of persons into math persons and not math persons Often by their ability to match some preconceived rule to arrive at a correct answer often with an overemphasis on speed so as we think about the fact that the people the persons the mathematicians the math persons may be foraging for these symbols and formulas to bring back to the real world it's what happens if we begin to give if we as Herz did begin to give these symbols their own ontology and their own agency to realize that they are also likely out in the wild searching for the people to whom with whom they will engage productively so this takes us back to our stranger ideas where we intentionally left the door open and here I'm indebted to geard mayor who in his article magicians of the 21st century um pointed to Ramsey dukes's four quadrants model as splitting reality into the irrational and the rational intentionally left the rational box off here because I wasn't sure where to put it I would put religion in the rational category because there are very clear apologetic logical arguments it's clear that religion at least attempts to play in the same sphere of the rational science does whereas art and Magic would be uh in the irrational category and I am intentionally echoing the splitting here of the real numbers into rational or irrational with whatever consequences this might have the quote hinr Hertz again outside our Consciousness there lies the cold and alien world of actual things between the two stretches the narrow Borderland of the senses no communication between the two worlds is possible excepting across the narrow strip for a proper understanding of ourselves and of the world it is of the highest importance that this Borderland should be thoroughly explored and the problem as mapmakers in an Ever evolving and some would say chaotic territory certainly highly Dynamic of this moment and SpaceTime where we find ourselves is where to draw where is the border if this is the border that we're supposed to be exploring for a proper understanding of ourselves and of the world how do we know when we are on the border and when we may be inadvertently Crossing it and how can we enrich our tools for finding each side and productively bringing ideas from one side to the other or whatever Travelers are wishing to cross whatever we call them I've late recently been reading this book symbolism by Alfred North Whitehead and he speaks to this but he speaks to it in a different way and in one that I would challenge which is very intimidating to be honest um but in this in this essay on symbolism he talks about essentially two types of symbolism the symbolism of church and the symbolism of language and algebra so he initially says that there is a symbolism of the medieval cathedral but because that symbolism can be dispensed with it is on The Fringe Of Life and has an unessential element in its Constitution and then he says and on page two here there is another sort of language a written language which is constituted by the mathematical symbols of the science algebra in some ways these symbols are different to those of ordinary language because the manipulation of the algebraic symbols does your reasoning for you provided that you keep to the algebraic rules how can something do reasoning for you if it is purely mechanistic and has no agency and then he says this is not the case with ordinary language which puts up that boundary again between words and symbols that the symbols can do something the words cannot so I would disagree here with how he draws the bifurcation between Cathedrals and mathematical symbols uh Pythagoras would also have quibbled with that pythagore for the pythagoreans mathematics was very much a religious practice practice and they believed that numbers had a spiritual Dimension but that takes us to a very weird that takes us to very weird places very quickly uh I tweeted this I don't necessarily believe it but it's you don't have to tweet everything you believe or vice versa uh so let's go to slightly a bit more out there but this is coming back to relevance I promise it has to do with the notion of sovereignty and so here I'm looking at the article sovereignty in the UFO by win and Duval where they talk about how sovereignty is constructed and so this might go back to our definition here of a quote unquote real world an institution a government a nation state even an individual as being Sovereign has a particular Construction in the world of our experience but this has not always been the case I loved this quote that they said anthropos anthropocentric sovereignty might seem necessary after all who else besides humans might rule nevertheless sovereignty was LE historically sovereignty was less anthropocentric for Millennia nature and the gods which nature would be in science and the gods would be in religion were thought to have causal powers and subjectivities that Ena them to share sovereignty with humans if not exercise Dominion outright authoritative belief in non-human sovereignties was given up only after long and bitter struggle about the borders of the social world so again we're trying to place a border in which who what could be Sovereign depends on who what should be included in society and there's a great footnote anthropocentrism need not mean all human beings since historically many physical humans were not considered human socially and let me go back here and say that perhaps there's a similar construction occurring perhaps this division some humans are human and some are not some we would not recognize that today but we do recognize this bifurcation into the math persons and non-math persons and I would say it's equally as social and equally as artificial and equally as dangerous so where are we now U so I've opened some doors here perhaps some of them should have stayed shut but regardless they're open for the moment and we can shut them when we wish hopefully I love this quote at the end about sovereignty in the UFO the UFO can be known only by not asking what it is uh because this has been totally obverse to my process here I have tried to know mathematics by asking what it is by struggling very hard with its identity as an egregor as a concept um but maybe some things can only be known by not asking what they are so I'm sure that this has perhaps been an interesting metaphysical exploration of some different historical ideas and ideas of History how is this applicable to ourselves as Scholars in an active inference environment well what I would like to point at is the ways in which some of these ideas that are already Dynamic may be changing very rapidly at this moment so if we look at the identity of mathematician if we are solely confined to a meatspace identity that has connotations but what do we do with someone like bluecow and I did request blue Cal's consent to their likeness and their thoughts uh the creator of the first hyperdimensional Computing OS and the super prompt a prompt God uh someone who is a self-proclaimed mathematician who does impressive mathematical work in prompting in prompt engineering of mathematical prompts via GitHub and whose appearance is uh an anime person dressed as a cow um can they be a real mathematician in the way that we currently conceptualize math and are the mathematical ideas that they're bringing ones that we are prepared to reckon with as math and then when they post things like this what do we do self-identified skitso and it's helpful here perhaps to remember that skitso means split another scholar in this emergent might one might call parallel or underground genre whose work I respect is Sir Loop and Watson his conjecture soon you'll be able to buy an AI hacker for less than $100,000 he'll or she do he he he'll or she do anything for free you'll need to pay only once and it will last forever this is exactly the bargain of the theorem once once you have proven a theorem once with a finite amount of human effort it remains irrevocably true and will do whatever you ask it to do for free forever so if we believe there is a world where theorems come from what does that say about other objects or other constructions that might share their properties I don't have a conclusion if I were going to have a conclusion it would be going back to the thoughts of slide three but one reason I don't have a conclusion is that this is not the end of a completed research Journey but the beginning of something we are all experiencing together so I'm gesturing at what I know and then the next question is what do I hope to know well one challenge that I think this immediately offers but also an opportunity is what does it look like to be an institution in a world where these agents these human agents are seriously considering the same ideas as you are and perhaps offering insights that are not being produced at the level of your institution I don't this may be Fringe but I think it's to be taken seriously and I offer it because many people will see the work that for instance Terry TOA is doing on leading larger mathematical teams that use AI to modularize math and prove more efficiently in an engineering type framework different from the idea of the individual math um it's less likely that the average academic I believe would encounter these types of people who are also doing interesting mathematical thinking not only about the mathematics itself but about what it means to present oneself as a mathematician or to present mathematical work so I hope these thoughts have been interesting to you and I'm excited to see what they may have brought up in the form of questions and comments thank you so much octopus um I really appreciate the number of different ways that you've kind of viewed this topic um if I may start with with a question sure so what as you were talking about this idea that the formulas maybe exist and that they're kind of waiting to fit on to something it was reminding me of a um a song by the Icelandic uh artist Bor I'm not sure if you're familiar with her she has a a song called modern things where it says all those modern things they've always existed they've just been waiting in a mounting for the right moment um and it's their time now and um so yeah I was wondering you know what what do you think about um something like the global economy which it's um it kind of acts like an entity in itself so it's you know it's the Invisible Hand of the market for example but it's the humans that kind of assemble around it and imbue the meaning on the all the uh interactions and all Dynamics yes uh I think again that there again there might be a border uh for instance if we view economics as purely human transactions of material Capital it misses a lot I think a lot of economists would say that but symbols are important even there if you look at a unit of currency it is imbued with symbols that are highly nonfunctional almost overridden with non-functional and one of the ways that one can assert Authority and or immortality is to issue a coin and that's what we see now with cryptoeconomic systems is a debate around the individual sovereignty and the right of who is allowed to issue something that can be used as a unit of exchange U but that's not a new debate there were certain Roman citizens who had the authority to issue a token and then the question is uh where does the value of that token come from ultimately right so again it kicks back very quickly to questions around uh beliefs and identity and sovereignty I guess the thing there is when something is real to me it must have more than just a material component thank you um Daniel other any questions from the chat there's a lot of great commentary octopus it was a great presentation do you have any final comments oh um well I really appreciate the invitation to be heard and the ability to show up as octopus that's so meaningful to me and U grateful to the organizers and the audience and looking forward to seeing what conversations can emerge from this beginning awesome thank you till next time see you oo thank you welcome to the fourth active inference Symposium Dr gomo rostini we're very pleased to have you here and um looking forward to your session titled PL ebo NOS ebo and contextual effects the missing the missed Link in clinical practice over to you thank you for the presentation and for the invitation to use this talk it's a pleasure for me to be here for the hativ imposium and today I talk about Placebo noo and contextual effects showing the Mis Link in clinical practice as I mentioned I'm an Italian physical therapist I'm involved in clinician and researching teaching session I come from Verona that is a city in the north of Italy and it represent the city of love on Romeo and Juliet and if you have never visited my suggestion is to do it before to start I want to declare that I have no conflict of interest for my speech I Have organized my presentation to discuss consensual effects personally we will analyze the definition and the constituent dimensions of contextual effects the mechanisms of action the impact on outcomes and treatments next we will discuss the possible link of contextual effects with the active inference and the implication for clinical practice and research finally we will conclude with a take on message contextual effects are defined as the effects of the context around the therapy and are generated by contextual factors contextual factors were recently defined in international consensus in which I participated in the slide you will find an accept of the final share definition where contextual effector are the perceived views that affect both the patient and the practitioner and can arise from previous experiences and the immediate Dynamics within the therapeutic encounter or a combination both and from a clinical perspective contextual factors fall into five broad categories or Dimensions that you can see represented in this graph on one hand we have the characteristic of the clinicians with their professional reputations the appearance the beliefs and behaviors on the other side there are the patients characteristic with their expectations preferences and previous experiences between the clinicians and the patients there is the relational Dynamics in terms of verbal and nonverbal communication what we say and how we communicate in in addition to the relational component the treatment characteristics also play an important role they represent the rituality of the treatment finally The Fifth Dimension is represented by the characteristics of the setting within which the encounter is being delivered such as the environment the architecture and also the interior design well all these five Dimensions inter acting with each other influence the overall therapeutic outcomes the question we have to ask ourself is through which mechanisms does this happen I will tell you in the next slides contextual factors have been studied for their ability to induce Placebo and noo effects which represent the effects of the psychosocial context around the th and to explain how this occurs I want to start with a clinical example this illustration show a common clinical encounter in which patients with the muscular skial pain receive a therapy such as the joint immobilization therapeutic exercises or an ultrasound injections and during the administration of the therapy patients are influenced by the specificity of the treatment but also by various different psychosocial cues such as the clinicians words The Sight and Sound of the health care devices the smells of the hygienic and also the taste of the drugs when receiving for example a painkillers well all these cues can influence the patient's perception shaping the experience in a positive or in a negative way triggering Placebo or noo threats thus the positive use of contextual factors can produce Placebo effects when these elements are used negatively it produce No Effects and the clinical effect of contextual factors can be explained through conscious and unconscious mechanisms of action such as expectation learning processes personality traits mindset and even if genetics however among the most important psychological theories expectation classical conditioning and social learning are the most widely accepted the expectation is a conscious phenomena through which the patient expect that something will happen in the Futures when need to consider patients for example who are waiting to receive a therapeutic exercises or manual therapy for a muscular schal pain well they will predict a positive therapeutic outcome by evaluating all the contextual factors around the therapy we offer as a clinician such as our Worth or our behaviors instead the classical PL of condition is is an unconscious phenomenon concerning the past events in fact patient can learn over time to associate a certain therapy such as for example an injections with a specific contexual effects that represent the rituality of the technique created by the needles that produce a certain outcome which can be positive or negative but patients can also learn from others they can learn from positive or negative experiences of other patients family members friends or from information they get from social media or blogs regarding for example muscular skial pain but how the contextual Factor Works in practice I will use this image to explain this mechanisms of action briefly whenever patient interact with the healthc care context there are always influenced by the five contential factors and these elements are encoded by the patient's brain who interprets them and provides an evaluation and a prediction of their meaning and if this inter ration is positive Place effects are triggered if the interpretation is negative No Effects are stimulated and these effects are sustained by a Cascade of psychon immuno endocrinological eventss at multiple level of the central nervous systems in terms of the release neurotransmitters for example negative Contex of factors can activate the release of kyin and the deactivation of op amine and the opioid systems instead the use of the positive Contex factors can facilitate the release of a Pates andoc cannabinoids and dopamine and also different brain areas can be activated for example the dorsal lateral prontal cortex the anterior singulate cortex and the g peral grade and the dorsal Horn of the spine are modulated and all these specific neurological p way translated into changes in clinical outcomes and using for example muscul scal pain as an example if the context oper is positive the patients will perceive a relief of symptoms instead if the context adopted is negative the patient will have an aggravation of symptoms and contextual factors interacting with the treatment specificity are able to influence more the patient subjective outcomes or the illness Dimensions than the objective outcomes that represent the disease component for example pain disability experience satisfaction represent typical example of outcomes influenced by the contextual factors however not all the changes of the therapeutic outs are due by contextual factors or by the specificity of our treatment in fact several confounders can explain the improvements or the worsening of the patient clinical condition an example are reported in this graph for example the natural history of the disease can explain the natural flation of the of symtom or the regression towards the means which represent a statistical phenomena or for example the presence of unidentified intervention as in the case of patient that present a muscular spal pain or receive a manual therapy or therapeutic exercise and concomitantly receive also a painkiller another question we should have ask ourself is what is the weight of contextual factors influences the outcome and for a long time it was a common opinion in the scientific community that about 33 to 30 35% of the total therapeutic effects depending on the contextual component and even if it me seem limited it is actually not because it is a component influencing clinical outcome that we should know and we should use in our clinical practice and the international debate on the weight of contextual effects has continued to the present day without find the conclusion several recent systematic reviews with the with the metanalysis has assessed the weight of contextual effects what the different authors have observed is that the contextual effects can vary from small to large and report here different example you can see what happen for non specific loic pain neck pain fibromyalgia and also netis and the heterogenity in these results can be explained for example by the characteristic of the patients the clinicians by the characteristic of the condition analyze the interventions considered and also by the characteristic of the statistical metric adopted so some student has also provided data on the contextual effects for treatments for example in this meta research the contextual effects of of different treatments was evaluated and what we can observe is that the contextual effect is progressively higher in treatment with the high realistic Futures to go into details in psychological treatments the the contextual effects account for the 38% in pharmacological treatments it accounts for the 61% and in the physical treatment such as manual therapy oron techniques it accounts for the 65% 64% and having reached this point I think that we need to understand how contextual effects can be related to the theme of this coner this conference the active inference and to understand this we need to start with a clinical example we need to think about the management of patients with persistent muscular skial pain who have negative expectations based on their previous experiences psychological characteristics or personality traits and for example if you consider a patient with the neck pain that in the past has been exposed to exercises and learn that these exercises produced pain consolidating a negative internal mage toward this in our patients Clinic we offer exercises to our neck pain patients in accordance with the best evidence and as a result of this direct experience the patients can have two main outcomes at first for example it can confirm that exercises can produce pain or the patients may not perceive the pain violated their expectations and this violation of expectation is interesting because it can result in two main scenarios one adapted India where patients Mak it makes an update of their expectation and on their internal model and then this functional one where the patients consider this violation as an exception to the rule or as a results to the chance and therefore maintain it and this is referred to as a cognitively immunized patient and we suggest that such difference in updating responses can be understood at least to some extent from a basian perspectives and in fact the ability to make an update of the patient expectation can be interpreted in the light of the basian model and will depend on the strength and accuracy of the patient prior beliefs and behaviors and from a bayian perspectives our brain is concept izes a predictor machine that generates predictions know as PRS about the expected sensory inputs and the integration between the prior and the sensor inputs results in a posterior the percept which can be more or less influenced by the prior and by the sensory data depending on their level of precision and let let us think about the patients with the pain who has negative prior or expectations toward exercise they represented the sensory and if this priors are very precise and reliable it will be much more difficult for the patient to rely on the sensory input given by the exercises and the prediction error is really minimal and the patients consider the experience as a result of a set of the chance or an exception of the rule existing in maintainance or the initial expectations instead if the interpretation are less precise and reliable it will be much easier for the patients to to reely on the sensor inputs given by the exercise because the prediction error between what it is expected and what it is experience will be more significant will contribute to improving the patient expectation by generating an update and I think is very fascinating and complex but it does alls us to think about how we can use the context to challenge or making updates of the patients prior and I will try to help you to understand this point with another clinical example with this bium model in mind it is possible to better understand the clinical scenario presented in this image where we have a patients with the muscul kidal pain of the shoulder who has had a negative PRI in that in that situation the patients prefer not to rise their hars because they have been told that the tendons can stamp on the other hand we have the clinicians who tries to challenge the patients prior to the tools that he has at disposal for example the exercises the man therapy the virtual reality and the education but the clinician also has another secret weapon to use that this call has the contextual factors made by words cues behaviors rituality that can influence the patient's brain and the patient prior and as an outcome of this challenge we can have for example an expectation violation following by update which is realized when the prior with low Precision of our patient is updated accordingly with the newly acquired evidence the patients leave the bre the arms without any pain and without any types of lesions another possible scenario is represented by the expectation violation followed by the maintainance of the dysfunctional beliefs as the consequence of one having a highly persued prior which is considered really reliable and therefore the newly acquired information and this confirmatory evidence is not sufficient to disprove and update the such strong prior and for example in this case we have a patient with a miscal pain that during a clinical section manag to leave their har without any negative consequences to their shoulder but still believe that leing the arm will possible create and lead to the shoulder to break are likely to have for example a very high precise prior and might discard the positive evidence for example the success in lifting the har which might might be classified as an exception instead of the rule so try to pull the string of the speech what might be the implication for this from this talk to our clinical practice the use of contextual factors represent an opportunity to Aras the evidence based therapy we know it as a clinician but as a clinician we also need to consider that using contextual factors such as words touch the rituality of the technique and environment could help to challenge the patient prior and improve the patient muscular Cal pain however this observation that arise from the clinic need to be supported by data and more study to substantiate their effectiveness thus researchers are called up to integrate contextual effects into the study on muscal pain using for example the basian brain and the active inference L and to do is this it is important to produce experimental work in the lab in the clinical work in patient to validate these theoretical models by using for example subjective outcomes but also some objective physiological metrics we are in the end so I concluded I want to bre in you wishing you to remain curious clinicians capable to see the things from different perspective and asset the complexity presented in the clinic and also within this image let's not just focus only on the details the the technique it is important but it is not all let us always remember that the scenario in which we operate it is much more complex and both the context the moon and the confounders the sege in the sky comes into play only if we Embrace this opportunity and this complexity we could find new strategy to best manage our patients with muscular skial pain thanks for your attentions thank you thank you till next welcome John Bo PhD to the fourth active inference Symposium we're really pleased to have you here and looking forward to your session cogar ecosystem facilitating group cognition at scale and over to you all right thanks Ian um again my name is John Bo I'm a a fellow at the active in Institute active inference Institute um and I will be talking about a cogar ecosystem to to facilitate cognition at scale as Ian said uh I do want to say that uh the links for this there's a bunch of links on The Institute project page uh I think I think these slides will eventually be made public but in the meantime if you're interested you can just simply search on cogar and Koda or cogar and my last name bo and you'll find a The Institute project page there's lots of links there all right so what is the cogar ecosystem uh cogar means cognitive narrative and in concept it's uh it's an open- Source online extensible ecosystem uh Technologies tools libraries to facilitate group cognition uh especially for large groups and although there are many many use cases uh uh the the use case that's that one of the Prime use cases is uh deliberation strategizing Collective problem solving and decision- making in the large group setting kogar essentially helps what I like to say is it helps users to tell their story that is convey their beliefs uh to a group about how the world Works relative to some situation and even more specifically convey their belief model to a group make make the belief model explicit and transparent to the group um so they might uh they might answer or address questions like what's the problem or situation what happened how did it happen What might happen next why did it happen who does it affect why who do is involv what does it mean what should be done what can be learned of these kinds of questions um cogar allows PE allows users to tell a rich story and by Rich I mean potentially long complex nuanced Dynamic conditional uh uh um quite different quite in comparison quite different from some kind of online collaborative architecture that for example asks you to uh a yes answer no question or ask you to give a thumbs up or thumbs down on some issue uh no this is uh this is this cogar will allow a user to convey very rich information and there's a backend computational system that assists the digestion of that information for the for so that the group can better understand the the many different stories so the computational system assists and guides users in construction stories uh helps the group digests the information make sense of it um uh assesses stories for their quality you know is is the story on topic is it comprehensible that kind of thing and then the backend system also performs inference for queries that uh that various group members might have about a sto story or for a Storyteller it is important to say that the system the computational system does not predict the future that is it's not going if a bunch of people if the issue at hand say is garbage collection in a local community this the uh computational system is not going to assess stories and then predict which of them is most realistic that's not really its purpose what it what its purpose is is to convey to make transparent uh and explicit the beliefs of members and uh the members when they create their story when they tell their story they include what they think will happen in in the future the pred they make pred the users make predictions um and also the group assesses the realism then of Any of of the stories that are submitted so what is a cognard group uh a group is is a set of people that sense their world process information anticipate outcomes evaluate options make decisions and in other words we can think of a group as a as a as an organism that performs cogn um and then the question is how what what design of cogar might actually facilitate functional cognition as opposed to dysfunctional cognition so examples of groups could be you know every group a informal group who like polka dot shirts uh a a social club volunteer group Civic organizations professional groups corporations nonprofits political entities such as cities and more uh so groups can range from formal to informal flexible to structured temporary to permanent small to large autocratic to democratic the the the platform is really a tool that can be used by for many use cases by many kinds of groups for many purposes there are of course some rules to keep it all functional and transparent and when I say large group what does that mean uh uh often in the literature when they talk about a l large group u u deliber ation and you know cognition they might be thinking of dozens of people but uh here we're thinking about perhaps dozens hundreds thousands millions there this should scale up to you know cities states and perhaps even a nation who knows you know if if I should say if cogar were ever built if it works as advertised uh you know this is all possible and and maybe I should also say at the moment this is just there's I've published uh two or i' I've posted two story two articles on cogar so far two research articles and um that's all that's been done so this is this is at the beginning this is the a vision really of what this could be I'm I'm presenting a vision um so the question is what kind of design for cogar would allow some large group say a 100,000 people to to to um gather online synchronously or asynchronously and share the potentially complex Dynamic beliefs belief models about some situation how could that be done that's the question here and I obviously I have have some ideas on how that could be done so uh but first maybe I should say that that the cogar project is inspired by uh inference and in particular active inference Pro from you know from my U from the vantage point of this project active inference provides a description of functional cognition that is as I think everyone probably knows by now watching watching the seminar that uh cognition we can idealize it as a maybe a four-part cyclic process of predict act sense and learn and and it involves um many components for example memory recall attention um and this the entire picture of that process can be um you know that as it plays out in a human or an agent it could be either functional or dysfunctional for example uh an an agent that that that cannot predict the future with accuracy is going to make decisions that are suboptimal in gen you know typically so that would be a a an example of dysfunctional cognition if you can't remember what happened or you can't uh predict accurate accurately what's going to happen in the future those would all be IND indicative of dysfunctional cognition so an active inference an agent favors choices that ensure the greatest resolution of uncertainty under the constraints as preferred outcomes are realized and all of that provides you know a framework such that um that the components of cognition could be recognized can each component can be um focused on as as a design element and the and the and the and the process of cognition in any group could be assessed whether whether that is a functional or dysfunctional or somewhere is in between and of course that would be a you know multivalued scale so what is cogn what is cogar really um cogar essentially is designed to be part of the cognitive architecture of the group that is using it that any organism that performs cognition has some kind of cognitive architecture for for a for a animal that is you know involves a nerv nervous system it involves uh you know can for a human it can involve external objects such as Computers and Tools and um for a group it includes a set of norms rules mechanisms in institutions sensors analytic processes systems and any other procedures that facilitate or guide cognition and then Co cogar is obviously designed to be a component of the cognitive architecture of the group that uses it and viewed differently and I'll talk about this more in a moment cogar technology provides a means to digitize and make explicit a person or an agent's belief model and in that sense the the some of the technology behind cogar could be used far beyond cogar could be used in medicine psychology and you know unlimited number of other applications of of how you would take a you know an agent or a person's belief model Make That explicit in a way that computation can occur and uh and you know include that into some kind of inference models or generative models so cogar stems from some past work that I've done and it's this past work that cogar and this past work are related in that all of it is about cognitive architectures in the past work I was looking at Social systems as cognitive architectures and by social systems I mean economic systems Financial systems governance syst systems I view those core societal systems as the as a means as a as a part of the cognitive architect architecture of the society that is using them that is we we think through our um governance systems we think through our economic systems we make decisions that that affect our world through our economic system for example and um the previous projects asked this question out of all conceivable societal systems designs and I don't mean conceivable in you know capitalism and socialism and the way it's been implemented in the past what I mean is like Blank Slate out of all conceivable designs for uh economic Financial governance systems which ones but might be most fit for purpose and that purpose is cognition in the active inference set so reduction of uncertainty accurate prediction of the future uh attaining those conditions that that allow for thriving of the population uh those things and I propose in this past work uh a path really of how such a project could be implemented how you know what it would do how would go how how would you test new systems um um and so on so I referred to this earlier work and again the links are all on the cogar project page if you can find that all right so what is uh the maybe the core element of cogar is um what I call a story graph and essentially it's a meaning representation it's the it's the explicit representation of a user's belief model and this of course is a very difficult problem to solve how do you how do you take some belief model that is in a is in say the mind of of a person and put that down on paper so to speak such that computation can occur with it inference can occur with it this a very very tough problem but I think there's there's possible ways to to move forward with it um I think I mentioned or I should mention that there was two papers published on cogar first paper is more of an introduction overview the second paper is really all about it focuses on the potential meaning representations um so story graph is the sort of the core Innovation on which cogar is based and it's story graphs are used to curate understand reason with you know conduct inference on high quality information about human experiences beliefs and expectations and essentially the way it works is um a user might type in or say um some passage a few sentences perhaps a thought and uh that Passage would be translated into a what I call a story graph fragment just a small little graph of meaning graph and that little fragment would then the system would merge that fragment into the growing story graph the overall story graph and then eventually by by uh by um inputting passages after passages after passages the story graph gets built up the meaning representation gets built up and then once it's complete or once it's almost complete then it can be translated into alternative meaning representations that might be appropriate for various you know logical models or other kinds of models probabbly istic models um can maybe be translated more directly into some types of probabilistic models and could also be translated into a natural language so in in Time Event you know if kogar is successful over time it should be possible to have the system read back your story like tell me tell me you know read this back to me and let me make sure that it's what I think what I'm intending it to be uh even you know in the far future it might be possible to make animations as an output of this you know this a little movie of this is what I think happen and this is what I think will happen in the future um so what is a meaning representation I have a picture here of a um of a abstract meaning representation that's a very common one used in the in the in the research world the in in natural language inference there's uh in the research world there's lots of different types of of meaning representations this is just one of them I'm using as an example and this is for a story excuse me a sentence a similar a similar technique is almost always impossible to apply to other crops such as cotton soybeans and rice and you can just you I won't go into detail about this but you can kind of see that there's a graph and various words are connected uh to uh to each other ideas are connected to each other concepts are connected to each other and this kind of meaning representation is used throughout research and the story graph you can think of the story graph as a much more complex version of this it consists of layers uh you know one layer might be um predictions that the story is making about the future you know that that could be one uh layer um but story graph is actually more instead of sort of static picture like this static graph story graph is really more like a movie it's a evolving graph pth really and and it involves High dimensional vectors of word meaning so there's it's far more complicated than this but this gives you a really rough idea of what I'm after uh cogar has many use cases it if if cogar were successful it would it could be ubiquitous in in in social life um so a group and there again could be all kinds of different groups many different groups Divi defines a group event uh it's its topic its purpose how how decisions were made for this event what kind of methods and rules will be used uh the group then invites people to participate and Communications in this uh for this for this group event could be many to many many to one one to many you know I have examples of all of these in the paper but um and I'll go over a couple shortly um use cases could be decision-making customer service emergency reports public polling instructions on how to build a you know a shelf or something how to run a machine system descriptions of how you know how this system works educational uh U use cases and more and the later versions of cogar could include role playing strategizing um you know here's what I believe here's what I think this my my competitor believes and I'm going to do this you know how what would my competitor do if I this or you know all kinds of various versions of St how do I get the girl should I should I wear a suit and tie should I wear jeans how should I act you know like whatever the whatever the the the issue is at hand so I this is a example use case for decisionmaking that i' I've written out a little text here but obviously this again this would be passages that a user is inputting into the system and um and a graph is being created you know as proes are implemented so this person says you know whatever this issue X is we better first address issue y that caused X and issue y started when blank and it has the effect of blank and if we don't solve y then we can't solve a and this is likely to occur that happens and you know just on and on and on about some kind of conditional story complex conditional story and um in that in this kind of setting where there where it's uh group decision- making what could occur is that um uh at the top of this craft here uh story input and editing use there could be you know 10,000 users or something 10,000 members of this group they each uh create their story and and uh you know the the system would be helping them create the story too to make it easy um and then after say the first round of input the system would would perform a bunch of computations assess the what the variety of different stories are report back on the on how stories cluster with each other and all kinds of other things story quality and you know story components and all kinds of things so that the group can better understand what the picture of all these stories is and then there would be opportunity for group dialogue feedback story digestion asking questions users asking questions about different other people's uh stories and then after uh that is finished then we can go back to the beginning and uh users see everyone's story with and all the results of of Assessments and then maybe they want to change their story uh to to reflect what's what has been said maybe I I'm involved in this in this group and I I put down my story and then I eventually get to see what other people are saying and I go and I think oh that's a that's a good idea I want to include I want to include that piece in mind I like that or I don't like this or I'm going to comment on this one of the this part of this guy's story so over a series of rounds the idea is hopefully that the belief the belief of the group would start to CES around you know core models you know the maybe maybe one person story really captures the the group sentiment or maybe maybe the group as a whole the synthesis of their stories captures the group's sentiment and um and over a series of rounds the the thinking of the group the belief model of the group starts to coales around central themes and then there could be voting at the end various kinds of voting that would be possible another use case is emergency reports again this is a text passage but it would be formed into a story graph but somebody might say hey I might call a utility and say I just I just Witness an accident uh near this intersection a large truck began swerving uh it was wet shouldn't have been going so fast you know uh there have been many accidents at the corner as a kind of an aside A truck hit electric pole the wires started Sparks in the grass there's a school nearby uh you know all of this there's all kinds of implications about this story right the implication is is that that there's some kind of danger maybe the maybe the the operator should send a uh fir truck a ambulance you know send the police you know something like that there's dangers to the there's implied dangers to the utility workers um the this is you know just a short little paragraph but it's actually quite rich and there's opportunity for many kinds of there's quite a lot of of of of information that can be inferred from this little story that isn't said explicitly um in the second paper I list a deera of about 20 uh aspects of what a what a meaning representation should be like or should be capable of doing uh I just list a few here but it should be graph based to you know to be amenable to computation uh should be capable of handl handling document like stories uh you know the equivalent of say you know a few pages of text or maybe even a chapter of text in a book or something like that and hand and should be able to handle a wide variety of linguistic phenomena uh should be reproducible in the sense that similar stories should make similar graphs and there's more and more and more should components should be compositional um uh and reusable and more I won't go into any more detail on that but there is a whole list of of requirements so the last couple of slides are just sort of to you know fun to think about how how some of this might be done and what might happen in the future uh are there applications of category Theory here um this last paper on group interactions belief sharing via sheath and topos Theory I think is actually uh uh might be quite appropriate because that's exactly what is happening in this group setting the the group is sharing their belief models those belief models are are are are the sh you know the the the being being they evolve according to how the group thinking is going they're they're unifying generally the group would be unifying around one or a few stories and I think all of that could be studied mathematically uh using some ideas from category Theory but also in category Theory you know one of the ideas of category theories to reduce the complexity of a of a project or you know system so that's important because this is obviously a complicated system uh CET property graphs might be a basis for story graphs uh funs might might be allow for graph to graph graph to text and other kinds of Transformations um some ideas from category Theory might allow us to zoom in and zoom out this if you can imagine some very complicated story graph but the user just wants to focus in on just one little piece to verify that the system understood emmer her correctly and for that you might want to do a kind of a zoom in or zoom out summarization of some com some more complicated components of a story graph so there's applications for category Theory um and there's uh I think applications for friston's U from pixels to planning paper on renormalizing generative models um that I think there's potential there too because uh graph really represents a path you know an evolving path of the graph itself is evolving right there's it's like a movie as I said earlier and uh so the renormalizing approach might be potentially useful there um and there's many many opportunities for agent-based agent agentic models of for example group members understanding each other's beliefs so that's uh that's about it um yeah and again the project page has all kinds of information about this thank you John Ian if you have a question otherwise I can read one from the live chat oh yes maybe a quick one before the live chat so it's a you know lovely optimistic um sort of view that every member's story can be heard and understood and contribute to the the collective um and it's it's sort of making me think about you know C citizen kind of involvement in cities and knowing what issues are important to a community um then is there a way that it can help with sort of this perennial issue of top down opinion and bottom up needs and wishes um so how would you wait how would you wait let's say the the the stories from um you know opposing members right of a group if there's yeah right um so so there's the bigger issue that you know around what you've just said and that's how do you keep this whole thing to actually represent functional cognition like and uh obviously if one party has enormous power and another party has little power that's the you know the the cognitive process is not going to be functional so um it quite clearly a lot of work would have to go into the design of this to to prevent uh uh you know one party from dominating the entire discussion for example and there's lot and and work would have to go into preventing you know hijacking of the process for some you know dysfunctional means or purpose but uh it you some of that is quite addressable so it should be I I it seems to me that it would be quite possible to create a system that that is fair uh if everyone has only one voice that's one way to to prevent um one you know one person from dominating the discussion that is if this on a very simple level if there's not Bots you know if a person can't create uh imaginary Bots to be broadcasting their version of what should happen then the person would only have one voice and and over time maybe a group would get to know its members a bit and some members would be more respected than others just naturally because they've contributed in the past they've had intelligent things to say they've you know they they are experts in this field or whatever so um we only have a minute here so I can't I can't go into much further of how that would you know the ways that kind of fairness and transparency and all that kind of stuff would be rest but I I would think that there's many opportunities to to do so thank you John awesome presentation and looking forward to continuing the collaboration so see welcome William Gart to the fourth active inference Symposium we're looking forward to your session titled introduction to NGC learn a computational neuroscience libr Library over to you yeah so today I'm here to talk about um The Knack the N adaptive Computing Labs uh python n tool called NGC learn uh that is really focused on getting your ability to prototype your ideas and work with your ideas in the field of computational Neuroscience quickly and efficiently in Python um so before we're we're going to start off with all this first we kind of want to briefly cover what we mean when we talk about computational Neuroscience so this is just the study and investigation of brain function using mathematical computation and tools and theories um we have a focus on uh biologically plausible and biologically grounded systems and neur uh neurons and neuronal systems and we pay very high attention to the uh constraints and Dynamics uh that are found in the real world when we talk about these uh for instance we model things like you we have the resistance for membrane potentials we have currents we have a bunch of other various uh internal components to everything that make sure that we are modeling Faithfully uh all of the neurons and components that we can find inside NGC learn um and it we don't just take from humans uh in fact there's a couple of various things that we have that are based on uh findings uh for things like rat hippocampal neurons and whatnot for the stuff like synaptic plasticity and transfers um and so yeah so kind of look at this biological interpretation of the world and so one of the main things that we also like to look at is like what is realistic when we talk about an interpretation of our world and how biology looks at it and so like for instance our human eyes it's been studied that they process think information at about 60 frames a second we can certainly detect things that move through our visual field faster than that but as far as like visual updates and whatnot once you pass 60 frames a second or 60 updates per second we have a hard time distinguishing and then also processing for a lot of things in our brain is not just we see an image we process the image once and then we just sit around idle until a new image comes into our brain um a lot of the time what we'll end up with is we'll end up with a continuous stream of data coming into our brain and then we have to do continuous updates and processing on that and it's that continuous update and processing that we're really are trying to uh hone in on and identify here uh we also uh so we do all of this inside of a biological framing uh so we use neurons now these are not necessarily neurons like deep neural networks like to talk about neurons where it's just an abstract point in space uh we actually model them as uh physical objects uh that have things like internal States um and that also uh embody what they look like in brains and to just other neural structures in the body um we have our uh we have our like this cell the center of the cell we store a lot of our values we have synopses that connect these to other neurons later down and one of the key things that our library very much highlights is the use of spikes in spiking neural networks in order to actually achieve all of this happening um what this means is that instead of working in continuous values where you have a real valued image coming into your network um and then it passes through your entire network as a bunch of other real valued numbers uh images as they enter into our networks tend to be converted into what are known as Spike trains uh which are just a bunch of discrete either ones or zeros on whether or not there is signal and then from uh from neurons to neurons in our systems we tend to transmit these things only as uh spikes so everything is dealt with with discreet uh signals rather than continuous signals um and then the other thing that we really heavily rely on in NGC learn is stateful modeling uh so as we're dealing with these models that are constantly taking in signals over time um it's to be expected that we require many signals and many iterations over the signals uh before we will actually produce any given output what that means is that all of our neurons have internal States and are stateful and so we can for instance stop getting any input and we can actually still watch what happens to the neurons and our models as they basically return to their rest positions um and so we have our biological components that have our internal States and then these states do change over time and so we tend to think about all of the models that we build with NGC learn and just computational Neuroscience in general uh somewhat akin to State machines uh with the idea that we progress Through Time by changing the state over our own model and then there's certain actions that happen that can affect our model and transform it from one state to another um and so that's kind of the basic framing behind what we're going for with this right we're not trying to be the next tensor flow we're not trying to be the next a deep neural network library uh what we're trying to be is we're trying to be a library that is specifically catered to computational neuroscience and the biologically inspired and biomic systems um so why do we need this uh so building these complex systems has a lot of overhead if we need to make a system that tracks all of the different interal states of all models uh and can communicate between different components easily uh that can run efficiently um so at the bottom here you can see here so in one of these models that we built early on uh we had a pretty standard d uh spiking neural network uh that was not big by neural network standards uh that took about eight hours to do an Epoch of mnist um just and that is due to the fact that when we are looking at an individual image in amnest we have to go over it I think this model was going it was doing 200 forward passes effectively over it basically we take the image and we convert it into a spike Train That's 200 time steps long and feed it through a model um in addition to this um because in our brain we don't have really the concept of mini batching you don't perceive 18 Different World Views at a time um and then update off of them uh or you know 8,000 or 500 or however many big batch sizes have gotten to the these days uh we do everything in a continuous online updating um and so because of that we can only look at one image at a time so if we're doing roughly 200 forward passes per image one image at a time this can get really really really slow um and up to eight hours for us to do an Epoch over mnist um and so when we looked at some of the other existing libraries out there like Brian 2 bindet snn torch and whatnot they all worked and they all have their strengths uh to be able to mement these systems but some of them very much operate uh like blackboxes with the idea that you can modify parameters on the outside um but then you're entrusting that your various existing Library uh is doing everything you expect it to mathematically inside um and what this also does is this makes it hard sometimes to actually either dig in and see what those Dynamics specifically are or even just want to modify some of those Dynamics and so what we wanted to focus on was building something that was incredibly modular uh but still fast right so you can usually you have this trade-off between you can have a system that's incredibly fast but incredibly Limited in what you can do with it or you can have a system that is incredibly modular and can do a bunch of things but gend to trade off speed and so we were kind of aiming for the middle of that with errored more to while trying to expand it to make make sure we didn't sacrifice too much speed to be able to do all of the modular stuff that we wanted to be able to do um and so with that we have some highlights of NGC learn here so first off this is an open source project it's available on pip it's available on GitHub uh we are an active development base which means that if you write us messages on like if you put errors or message notifications in GitHub we'll see them if you email us we will see them we will respond to them we can work with you on them um it is uh so NGC learn specifically uh is built using Jax uh which is basically built on the underlying Library called The Accelerated linear algebra library or xlaa which is a bunch of C of cpython bindings uh wrapped around um basically C functions for doing linear algebra and what that allows and so those are really fast because we get to use c um and then on so Jax is built on one step on top of that which basically creates a bunch of numpy bindings for said cpython functions um and so Jax has this really really uh cool and uh component to it uh called justtin time compilation and what this allows you to do is you can put a pure method uh into Jax's just in time compilation and it will give you a pure method out but when you run that method it runs entirely in C it does not actually do anything in Python and so the advantage of that is if we can take our giant models State transition functions and make them be pure functions where you give it an initial State and it gives back to you the final State we can pass that into Jax's just in time compilation and basically take the entire model and its entire State transformation and we can do it really efficiently and see um and actually as a matter of fact we can take that and we can do state transitions uh hundreds of times before we ever return to Python and so this is kind of how we get that speed up and in our testing that same model that took eight hours to do an Epoch over mnist uh we got down to roughly 15 minutes uh 12 to 15 running on the same Hardware um for to do an Epoch over mnist which is you know a 40 times speed up there uh there have been other examples that we've had that have gotten up to a 50 times speed up uh this is not me promising that you will achieve a 40 or 50 times speed up in your work this is just me saying that like sometimes you don't realize how fast it can be running um we also wrote this so that way if you want to be writing your own new components because we all know that you know as models are great we can work with those those are great but the fun part about science is actually getting to build new things and try out new ideas and new modules so we made it really easy to write new stuff uh write new components write new basic like write new learning rules write new you know neurons all of this stuff is really easy to write and they wire directly into the compilers without any problems um and then the one of the other things too that we have is that it interfaces with uh Intel's Lui 2's uh simulator uh now the Lui 2 is a neuromorphic chip uh developed by Intel for doing neuromorphic Computing uh this is basically taking spiking neural networks and putting them at purely a hardware level um which is where all of the Energy Efficiency that you've heard people talk about with spiking Dural networks comes from uh is the fact that when you put it onto a neuromorphic chip it's really efficient um and having worked with physical Luis and whatnot it can be somewhat of a struggle uh to actually get those chips running and get them working well and so what that kind of comes down to is when you want and you actually get access to one of physical pieces of Hardware you want to make sure that your model probably at least Works in their simulator that they provide um I will tell you that the simulator does not onetoone match with reality but that is uh something that Intel knows and is aware of and I presume working on um but what we allow you to do is that we have a we have a sub project of NGC learn called NGC lava um and what that gives you is a couple changed object classes and a couple more restrictions that you have to follow while building things um and then after that you can run NGC models directly in the Lui 2 simulator um this means that you can take your model and you can run it on a GPU you can run it on a CPU and you can run it on the Lui simulator all without actually having to change any of your code base you're basically just like changing a flag somewhere on where it's running and that's all you have to do for that so how do we actually build models in NGC learn because now that I've it to you is this great tool let's see if it holds up with how easy it is so overall it's a modular system right so the goal of NGC loarn was to be incredibly plug and play and also it has to be effectively one giant State machine so we can see on the right here we have this context block now for us a context is generally our model uh but sometimes you'll have multiple models that you define or some models that are defined inside of one giant overarching context it's just the container that holds everything now the parts that we're the most interested in the things that actually do all of the updates and hold all of the fun bits of research are the components um and so components tend to be consist up of a couple of things so we have parameters uh and we have compartments and then there are the cables that connect them and we'll talk more in detail about what each one of those are in a little bit but that so but the components are the new things right if you build a new neuron if you build a new weight Matrix with a new learning rule or whatever those are what you're building you're just building those little tiny parts and because of the way that they interface with one anothers you don't have to worry about oh does this have the right number of outputs to line up with the right number of inputs and whatnot if you build the components in such a way that uh you follow just the standard practice that we've set out all of the data will flow from component to component without any problems so if we want to actually talk about the components these are the backbone of what you're doing so on the so on the left here I have the Leaky integrate and fire neuron now this is one of the most common neurons to see when you're starting out in spiking neural networks uh it is the idea that you have uh an internal voltage that slowly builds up passes over a threshold and then depolarizes into mitzah Spike um and does that by get taking in current and you know V equals ir and you do current times resistance and you get a voltage and it slowly builds up your voltage so inside of your LF neuron or our leaky Integra and fire neuron we have a couple of fixed parameters now for us fixed parameters are values that don't change during runtime these are the static parts of your model but they're not but they're not fixed across all of your comp opponents and what I mean by that is that I can have a leaky integrate in fire neuron like with a voltage threshold of one and I can have another leaky integrate in fire n with a thrt threshold of two um and you can build both of those it works perfectly fine but the one and the two can't change um well it's running uh because just of the way that it all works out internally with the compiling but for instance here we have some fixed parameters we have neuronal units the when we talk about having a neuron or a component that is a neuron it is more in reality a layer of neurons so for us and uh neuronal units is just the number of leaky integr and fire neurons that exist inside this little component or cluster of neurons uh they all have a voltage threshold um then we have towm RM a V reset and a v rest uh these are just all of the various parameters that you need to get a leaky integrate by neuron running and then below all of these we have the compartment values now this is where you actually store the stateful part of your component this these are the values that change over time these are the values that we are transmitting to from component to component and so for instance here in our leaky Integra and fire neuron on the left we can see that we have current coming in on the left we have this voltage value that kind of floats around in the middle because we're not expecting to put to have it as an output uh we have our output Spike which is just whether or not our voltage did Spike and then we have time of last Spike uh which is just well the last time step that it spikes because there's multiple learning rules that require that and so what this so what we can see here is that if you can kind of think about the various parts of your model that you like to build the neurons the weight matrices any other auxiliary system that you have in your model if you can kind of fit it to a component it's incredibly straightforward to get these to be built and then once you have them in component form you can then plug them into any part of any model so then uh what we're going to talk about here is we're going to talk about cables now cables can be a little bit confusing at first uh so cables are what actually connect our components together so if we look at the top we have component a and we have component B and basically what this is is that and then they're connected with the cable and so we have our output compartment which is the brown box that's on the right side of component a and then we have our input compartment that's on the left side of the compartments box for component B and then they're connected with this cable and so the idea is that data will flow from the output compartment of component a and get copied into the input compartment of component B and so this is basically how we get data to move from one to the other without necessarily having it know where it comes from so this was one of the main things that we wanted to make sure you uh was a founding principle of kind of what we were going for when we looked at the component and cables is that uh a component Downstream so in this case component B it doesn't necessarily need to know and it shouldn't know where the incoming values are coming from so these cables are all oneway cables you cannot go backwards up the cable uh basically the components are just told hey this is the value that has been transmitted along this line and thus this is the value that you have and you don't get to know where it comes from now what this does mean though is that sometimes sometimes what we'll have is inside we'll have a component that knows that it's getting values from two places for instance in the bottom explanation we have component X and component y each one of them have an output but we're trying to put both of their outputs into the same compartment in component Z now if we were to do this without the summation that you can see what would happen is the value from component X would come down and it would get put into the compartment and component Z and then the output of component y would come down and it would also get put into the compartment of component Z effectively overwriting component X and so but what we do have is we have this concept of operators which are just simple uh Transformations that can be done to the data that are being transmitted along cables so in this case summation uh this is you just have two you just have two pieces of data that all flow into this little summation operation that just flows into your output compartment um it's important to note that the value like the Intermediate values inside of things like summation are never stored anywhere they are purely just an inline operation happens yeah so that's all you actually need to build a model right is you have components and you wire them together that pretty much talks about everything you need to know for the physical parts of your model and how they talk to one another but that doesn't get us anywhere if I just hand view a model that has a bunch of wires around it and a bunch of values in it but I don't tell you how to actually change any of the values that's fairly useless um in addition to that it's not a particularly great you know just research project we all care about how do these models adapt over time we all care about that whole temporal and evolving nature of everything so with that in mind we want to talk now about how do we actually transition a model and uh what goes into that so for transitioning a models ACR so we have state transitions across just a component so if we just zoom in on what a component is we have component a here uh where we have our fixed parameters across the top here you can see we have four fixed parameters and then we have two different state transitions that we're going to allow we have the reset transition and then we have the advanced transition and so in this example here we can see that the reset transition takes two fixed parameters in and sets values via like the dashed lines that we see sets values into all four compartments or we have the advanced transition that takes in again two fixed parameters but different fixed parameters than reset and then it also uses three of the compartment values we can see the input compartment value on the far left and the two floating compartment values with their solid lines that go into advance and basically this is we have written an advance function so basically this is just we're doing a state transition called advance that requires two fixed parameters and three dynamically changing compartment values Advance internally does some transformation of this data and it outputs two values one of those values gets looped back into the floating compartment that we see on the right and the other one gets sent to the output compartment that's also on the far right and that's really all these State transitions across components are when we write them internally and we'll look at some of what this looks like in a little bit um what we'll notice is that each one of these uh transitions is relatively simple right it's just you just take values in do TR do something to them and send values out and that is really again you'll see this throughout all of NGC learn that is basically how it all works under the hood is that we just we take values in we transform them we send values out and we don't necessarily need to know where those values are going or where the input values came from that we just know how to move them around and it's important to note here again that the parameter parts are fixed uh it's only the compartment values that we can change so you couldn't see Advanced draw a dotted Arrow up to one of the fixed parameters because they are once again fixed so now that we know how to do a transition across one component what if we wanted to do a transition across two components and I'm aware that the parameter little circles are missing their arrows the images got significantly too cluttered uh when I tried to draw those arrows in as the diagrams get smaller um but what we'll notice here is that we now can take a group of components and we can talk about taking the advance action across a group of components and so what this basically does is we will do the advanced operation or transition on along component a we will then transmit all of the data along all of the cables going between component a and component B so whatever that outputed value of advanced state was would get sent into its output compartment and then sent to component B and then component B would once again also do its Advanced transition in which it would take on all of its values and compute its own output value and you could chain these together at infinum there's no limit to how many of these you can chain together um I will say that eventually it's probably you know might want to look at making your model a little bit smaller but there's no actual limit to it uh there's also no limit to how complex you make these uh there's nothing that says that you can't have recurrent Connections in your uh cables uh there's also nothing that says that outputs or inputs need to actually have values you can clamp values to them um which is basically the process of we just set the value and then just let it go but there is one thing to note especially when talking about recurrent things and recurrent connections between neurons and groups of neurons and uh what that is is that the order of operations in a state transition is deterministic and thus uh it's also something that each one only goes once so if you have component a why into component B and then component B why into component a it will still only execute a transition across a and then across B it or across B and then across a it will never do it will never actually Loop itself um it will always do one and then the other and so now all of this really really works itself down to compiling so the compiler of NGC learn um is not actually found in the main repo of NGC learn so NGC learn is built on top of what is known as NGC stimulation library or NGC Sim lib for short um and that is where all of the I don't want to say messy or ugly but that is where all of this code behind the scenes that are making everything work exists um and that is all of that Library can be found in that little tiny compiling arrow that we see down here now what we for the way compiling works though is we have our state transition right we have the state transition that we talked about in the previous slide and we want to turn it into a pure operation and what we mean by that is that it uh the same inputs will always produce the same outputs and no value outside of of the outputs of the method are ever modified and so what how we can do that is with the way that the transitions all work is we can just pass in the entire state of our model and then based on the transitions inside each component they know what compartments they have to be looking at and so they will look at those compartments and they will then update and modify the values in the model and then it'll move from that one into the next one and from the next one into the next one and so on and so forth until it's actually done and so these are these produce pure methods and if we think back to what I talked about with Jax all along at the beginning uh that means that we can pass these compiled methods into Jax's just in time compilation and so what this gives us is this allows us to basically call Advance or reset or any transition function that's defined across all of our components at one like effectively at once in one operation here we have it as Advance the model and so this is truly where the speed up of NGC learn comes from um even if you do not use uh Jax Jax or if you don't use Jax's just in time compilation because you're debugging um it is still a fairly large speed up just to go from python being written like the way it is into compiling into just running the compiled operation um so if you really want to look into and care about how are we getting the speed that we're getting look into the compilers um and once again the compilers are actually located in NGC learn NGC Sim lib which still can be found through all of The Knack laabs GitHub Pages it is also public so now what does this actually look like in code uh and so I wanted to take a brief look um because pretty pictures are really are you know a tool that helps us understand the thought process behind the models and they understand how the models actually work but sometimes people will show us really pretty pictures and then when you go to actually implement it in code it looks a whole lot different or things just don't quite or not quite as pretty of a picture uh so we do have a short section here where we're going to talk about all of the different code that we do have so here's an example of a component in NGC learn uh specifically we are going to be looking at the D synapse now all a d synapse is is it is a weight Matrix uh that has both a resistance scale value on it so just a constant multiplication scale factor being applied to everything um as well well as a bias term that is added to everything now so here we have some compartment values on the left so as you can see uh defining compartments is incredibly easy so now uh everything right here where you see self dot is all inside of the Constructor of our object um so here we have self. inputs is just compartment and then some values for us that's the prev Val so that's whatever we want that's a it's a matrix of all zeros of the correct shape uh and we have self. outputs which is just again another compartment defined with post vals and then we have our weights now weights are just a weight Matrix uh which is defined previously in the function um but it is just again simply compartment value and what this does a bunch of stuff under the hood but this basically actually Associates so when you build one of these objects when you build a dense synapse and and construct it this will actually build a bunch of compart this will build three compartments under the hood and basically register them inside the D synapse and Link everything together correctly so that way we know that they exist and we can wire into them and it does a little bit more than just python knowing that the object has these fields in it they exist and they have their own set of properties um and then we have our fixed parameters which if you look look exactly like just values that exist on a class right we have self. shape which and self. r scale which is just a resistance scaling and I'm sure everybody here who's worked with python is very familiar with what that looks like and so finally we have defining our state transitions now there's a couple things to note about this we'll notice that our state transition definitions are static methods and what they are really is they're just method these have to be pure methods to Define our state transitions um so here we have advanced state and we want to still be able to use our scale inputs weights and biases inside of advanced State and have them all correlate to the values that exist inside of our component but due to the way that the compiler works is we still need this to all be a pure method so how do we combat that well we combine that by just doing effectively pattern matching where as long as your names for the various inputs so our scale inputs weights biases whatever as long as those values are found inside of the component that this transition is acting on uh it will automatically grab those values out of those components when it's doing the transition and so here we can see this is a pretty straightforward matrix multiplication with a scaling and a bias term uh to compute outputs and we'll notice that once again outputs is something that is keyworded on the component and because when it's done resolving its transition it needs to know where to put the output value or in this case outputs of the state transition like what values are changed and so it knows how to put outputs back into self. outputs and so this is kind of the key to how all of NGC learns compilation and just plug-and play models work with these State transitions is that you can Define these methods that can pull in values from the components that they are appi being applied to without actually having to have an instance of the component itself now there's one other F uh aspect of these that's not actually shown in this example and that is parameters at runtime so for instance let's just say that we wanted to add the current time to our output I don't know why we would want to be doing that but let's say we wanted to so if inside Advan State inside the parameter list we had another value say time or T and T doesn't exist on the component that this transition is being applied to what we'll see with this then is that late it will and we'll see this a little bit later in an example what it will want to do is it will say hey I can found all of these values but this specific value T is not found in this component you need to give me me T for whatever I need to do for this state transition and so this is how you can get runtime parameters to be passed into your state Transitions and modify them that way so now that we have a component and we kind of figured out how to define a component and I should note that we can Define as many of these State transition as we want because they're not actually bound to any specific component we need to do something so how how do we actually Define these components in a model because generally we want multiple of them in a model a component is one piece of a model is not very useful so in order to do that we use wi blocks in Python uh for us we call them contexts uh so we do a with a context and then we give our context a name in this case we've named it model and so what we have here is this is a WID block and so what it what this does is this is just saying everywhere every time we Define a new NGC learn component or initialize a new NGC learn component inside of this width block it will automatically be added to the model so there's no model. register various built things no model. components list that you need to make sure is up to date nothing like that if we call heavy and synapse here inside of this width block the heavy and synapse exists on your model now what this does mean though is that the names of all of our synapses are required to be unique uh but I feel like that shouldn't be super challenging because in our models just for clarity and human readability we try not to name things the same for instance it wouldn't really make sense if I have three different parts of my model that I've named Z1 so with that we can see that we can define components like this but what if we have complex models in which we have one overarching model with submodels that run maybe at different times or at a different frequency and we want to be able to have those still be a part of this bigger model the bigger scope of the model but run on their own well luckily for you we can do that so components are sorry contexts are fully nestable you can Nest them as deep as you want and when you actually look at the structure internally you'll notice that it follows very similar to a file structure where everything just ends up with pads based on context name and so you can Nest these as deep as you want generally in my own personal use I've never found really a need to go more than an extra layer deep but there's nothing that stops you from going three four five layers deep um and so as we can see here though it's incredibly straightforward to actually build the model uh this here is built a model with two heav and synapses in it now I'll give you a brief note there's a bunch of other components that are defined below this to get the model to do something um but those all look pretty much exactly the same as this just with different objects and different components being built at every line so now we have our components we have our components defined inside of our context but they don't do anything they don't they don't interact with one another if we think back to the way that we get components to interact with one another we have to wire them between each other right we have to get data to flow from a heavan synap into ANF cell we have to get it to flow from an input placon neuron into a weight Matrix into an LF cell we have to get all of this data to flow so we do that with cables and luckily defining cables is incredibly straight forward so we have overridden the leftand bitshift operator basically uh to say hey if we're going to read the top here so unfortunately we end up reading right to left with this is we're going to take the z0 outputs whatever's in the outputs of z0 and we want to wire it directly into W1 which is one of our weight matrices inputs or inp and we just write that line and that cable has been connected everything under the hood is done and then and I should not I should mention all of this is still inside the width blocks found that builds the context of our model anytime we're dealing with building parts of our model it is all done inside of the width block uh and then on the next line we can take z1i spikes and we can output we can send that into W1 I's inputs and we'll notice that this pattern of just taking an output and wiring it into an input um is pretty easy to follow and so if you build your component with the inputs that you're expecting in the outputs You're Expecting and you make it and you make it really clear to yourself what you need to with that when it comes to wiring up your model it should be incredibly straightforward but if we think also back to the wiring example there was a point where I took two outputs and wired them into one input so we can see that actually happening inside the second part of this where we connect two compartments into a single destination so we can do this with for us the operation summation and so what this will do is this will add together any number of compartments that put here so in here we just have it as two but if we had four compartments that we wanted to wire into there uh it would be easy now one of the advantages of doing this instead of just having two input compartments inside of z1e is this allows for a variable number of connections so if inside of one model that you define you need z1e to have three inputs or three values being added together or inside of a different model that you're defining you have Z1 e needing to have seven inputs going into it you don't have to build new components there's no additional logic that you have to do you just add them to this list of parameters that are being passed in and you're done and another thing to note is that these parameter chains are entirely nestable as well so for instance if we wanted to invert or negate uh for W1 ie's output value so it would effectively be W1 out minus W1 IE outputs you just put negate around it and you can just Nest them down so if you wanted to sum a bunch of things together and then negate it you can if you want to negate two of them and add them it all works um and I should also again mention it does not actually store these values anywhere so they are never actually inspectable outside of just looking at the value of the compartment so compiling so compiling an you see learn we didn't want it to be a painful process we didn't want it to be something that you had to make sure that everything was like touched the specific right way where you're holding it in with all your fingers and then it happens to work so for compiling it's just a list of components and a key so if we think back to the example at the beginning where we had Advanced State as the name of the transition method that we have that's all we need it's that name Advanced State and so what this one I'll do in this line of code here is we'll do model. compile by key and so this is on your context model the whole model uh we give it a bunch of components W1 W1 IE and so on and so forth and then we tell it compile Advanced State and so what this will do is it we'll go through and this just if we think back to the com the thing where we like composition of functions of everything together it will go W1 goes through its Advanced State and then W1 IE goes through its Advanced State and then W1 eii goes through its Advanced State and it goes down the list now obviously if we reverse this whole list that will change the output it won't change what it's really doing but the order that the advances happen in does matter and it is dependent on the order that we find here and so and the other Advantage is is that we can compile many of these so in this case we can see that we go through all two layers of our model here uh and all of our neurons in our model everything that's a z is a neurons but let's just say we only wanted to go through the first chunk right we only wanted to do z0 and then we just wanted W1 and then we just wanted like z1e that's all we wanted to compute well all we would have to do is we would just have to put those three values in a compile argument and it would just go we can we can comp we don't have to compile every component in our model and what this means is that we don't actually have to have every transition defined for every component for example when we're updating weight matrices we tend to call it evolving them right they're changing in time they're evolving we don't evolve neurons it's not something that we do it's not something that makes even doesn't make sense in the context of most models so from that standpoint we would only call evolve on the weights or we might only call reset on the neurons if our weights don't actually if our weights never have anything static in them I mean variable in them there's just like there's a bunch of different things where we can pick and choose which components and the Order of the components that these operations happen in and so in this again just to fully explain we do have two different output values coming out of this compiled key this model. compile by key as well as one hidden aspect that comes out of this model. compile by key so but first the AR the outputs we have this Advance now Advance here is the pure function returned by the compiler uh so this is the function that takes in the model State and all of the arguments it's expecting and will return to you the final model State advv ARS these are all the arguments is expecting so as it goes through and it compiles down W1 W1 iie w1i and so on and so forth it's slowly collecting a list of all of the keyword arguments that it needs to actually compute this transition so W1 requires T and z0 requires T and DT and z2e requires a j what that means is that it would end up you would have a list that is we T DT and J it if if two things require the same parameter it does assume that they're the same parameter sorry the same yeah like the same keyword argument um so if you have t used across all of your components you only have to pass in t once so that's everything that's visibly returned by this function in addition what this function also does is we can now call model. Advance State whatever this compile key here or an optional name flag that's not currently being used we can take that advanced State and we can do model. advaned state and it's the same as calling model do advance and this just has its benefits so you don't have to necessarily keep track of the actual Advance function if you have a bunch of them you can just name them so that's everything on actually how we get the compilation to run so what if we want to perform many of these State transitions in a row and what I mean by that is if we think about a spiking neural network we have a forward pass where we take one look at an image but also in spiking neural networks usually we have to do many forward passes up our image before we actually can before we say we have an output and so when we look at that we're doing like 200 forward passes before we ever actually care what the output value so because of that we don't want to just do a state transition from t0 to T1 get T1 plug it back in over here to get T2 what I want to do is I want to plug in t0 here and then do T1 T2 until I get to T200 and so for that we have scanners now the

NGC learn scanner is built on top of Jax's scan function uh which is effectively a method of doing a for Loop inside of their compiled so inside of their just in time operations and the way this works is we have our scanner we have an observe function for us but this is just whatever you want to name it there's no actual hard fast rules on what this needs to be called we have compartment values or the model State and then we have arcs and so what the arguments are is these are the keyword arguments for that specific Loop in time so t0 so at t0 we would have ar0 t is equal to Z and DT is equal to say one at T2 or the second Loop through so yeah at T1 or the second Loop through this R zero would be one and so on and so forth and we Define that down here in Loop ARs and we'll talk about that in a moment but then we'll notice that inside of this we're doing two actual transitions for every Loop we go through and we advance all of our stat forward and then we do an evolve step over our model and this was done on an stdp or timing dependent plasticity trained model um which means that we do an evolve Step at every single forward pass we do one for for every forward pass over an image we do an evolve step over the image and we'll notice that this gets a compartment values because this is a pure function that takes in the component the the model State and Returns the new model State evolve as an another compiled method that takes in the current state and Returns the final State and then we just return the final State at the end of that Loop and it just loops and this will Loop over for however long you give it Arguments for so if we give it 200 Loops worth of arguments it will Loop 200 times and this part right here at the end is just how you can actually get an output out so this will return uh the compartment value located it s ra the raw Spike output of z1e at every time step will be what underscore s is equal to and finally uh to call it we just call observe it's whatever the name of this function is so if you named this Fu it would just be model. Fu we T to use slightly more of that useful name so here we have observe and here we have observe and then the loop arguments is a 2d Matrix where every row is uh for every row is just the list of arguments that are found here so this is a 2 by 200 sorry a 200 by two Matrix basically and that gets us our state Transitions and so this is how we can look at a model many many this is how we can transition over many many many steps with without ever having to come back to Python and since this is fully so scanners automatically put things inside of just in time compilation just for the way that they work internally for JA but what this means is that we can go from an incredibly slow forward Passover one image to a compiled forward Passover one image to a just in time compiled pass over just like one pass over an image to a just in time compiled 200 passes over an image and just do that in a single function that we have defined this is where all the speed comes from finally uh I just want to mention monitors uh so what a monitor is is sometimes we have a desire to actually look at well the internal state of our model as it's running and when we're doing things with just in time compilation we never get access to those internal States because they all exist in C and they never exist in Python and we can't look at them very easily uh again due to the way that just in time compilation works you can't put print statements in compiled code it does not work so what this means is that we have monitors and so the way this monitor works is it's it's just a component like any other component in NGC learn admittedly it does work differently under the hood but that's fine what we have is we have a monitor by name and then we have this default window length what this is doing is this will keep a rolling window of the last n values that you define so in this case 100 so it will keep the last 100 values of whatever you uh tell it to now this can now at some point you might say well why don't we just make this really long and do it for every value well that gets really big really fast if you're storing floating Point numbers and you're storing millions of them you can run out of memory in your computer um so you do have to be slightly careful with that but generally if you just keep the things that you're looking at to relatively small amounts you'd be fine um and the use case for this we'll notice too is slightly different with the way we wire things into it so we'll notice that there isn't a compartment value right here this there's no M do compartment name we just wire directly into the Monitor and this is why I saying they work slightly differently um but as far as the outside user is concerned you just wire into them like any other compartment you can think of it as one giant compartment that stores everything so here we take the spike output of z1e and the spike output of z1i and we wire them both into the monitor now somewhere in between this we have run like the observe function that has gone through and done a forward Passover and image now we want to actually look at the states as we're running and so what we'll notice here is in view the contents of it we can just do m. View and we just give it the compartment that we want to look at and this will give us the concatenated Matrix of all the spikes so if z1e has 10 neurons in it it will produce one by10 um like sets of neurons well basically just 10 by comma um spikes either zeros or ones and so as it goes through the output of the monitor will be 100 by 10 because that'll be all of the compo all of the spikes for all 100 time steps and so that gets us the ability to look at models as after they're after they have run but the internal states of the models as they're running um I will say there are a couple future things coming with monitors about the ability to actually get them to print out and integrate with Matt plot lib and getting it so you don't even actually ever have to view you can just make plots of compartments and they will just plot nicely for you um it is currently working mostly in our development Branch finally how has this been used right I've been marketing this to you is like hey this is a thing that we have made and it's really cool and it's really fast and it works really well but let's see it actually work in action so the paper that started this all was this paper called neuro coding Frameworks for learning generative models by Dr aoria and kefir and so this was kind of the founding idea for NBC learn this was built in the original version of NGC learn about two years ago um and it built this incredibly complicated looking model here uh that has both excitatory and inhibitory connections as well as a bunch of lateral connections every line that we see here is part of the weight matrices and sometimes depending on the way that the connections go it's multiple sets of weight matricies um this is a model that you can build in NGC learn and it will look no more complicated than a bunch of little brackets telling comp compartments where to wire to in addition to that uh We've reproduced uh deal and Cooks paper um from 2015 called unsupervised learning of digit reconstruction using Spike timing dependent plasticity uh this is another model that we had this is more of a historic model that we have built uh inside EDC learn and gotten up and running without any problems at all we have uh one of my papers actually with my advisor Dr arovia called time integrated Spike timing dependent plasticity uh in which we built a patching model uh that does what it that basically is this whole structure where we have a sensory input layer an image um and it breaks up into a bunch of different sub components or sub models basically that all learn small patches of our image and then we have one final part that Aggregates them all together uh this sort of model is also very easy to build an NGC learn and doesn't there's no actual time difference between running some of these um and so again this is just something that we can build and this is all able to be built with components that already exist in NGC learn if you wanted to build uh some of these models this one the ti stdp synapse isn't quite in there yet uh but this model is fully in there you can build this with every component that we have inside of NGC learn um and it doesn't uh without any without you having to write anything yourself uh but even if you wanted to rebuild this whole model inside NGC lar and writing everything yourself it still would not be that challenging um in addition to that we also have this contrastive signal uh dependent plasticity uh which is a self-supervised learning and Spike neural circuits Again by my adviser um uh Dr Aria and this is again another model that like this whole thing gets expanded to this which does this and this model is actually inside NGC learn as well all of the components that are needed for it are inside NGC learn and you can build it and you can look at it and it's all linked to in his paper and what that kind of wraps me up to here is the NGC learn Museum or just NGC Museum um and so NGC museum is a public repository for NGC learn that houses the buyer medic the brain inspired Computing and the computational Neuroscience models basically everything that we've built uh for all of our models can be found here um it makes a very convenient place for if you have code that you want to publish somewhere um you can publish here as just a hey here's your paper you can it's all it is organized and it's working on being more organized um into uh different groupings of stuff there's an whole exhibitor section for it so if you have multiple pieces of work that are all done using NGC learn you can you you will have your own you'll have your ability to have your own folder that you can all that you can organize however you want that will have all of your papers that you've built together uh which would mean that if somebody is looking for similar papers that you have done they can come here and they can see ah here is a spike timing dependent plasticity paper that you've done and here's four other Spike timing dependent plasticity papers that you have also worked on as well as here's just other Spike timing depend plus disy works by other authors uh that would you know allow you to cross reference things and just have uh a good uh community-built platform for uh models because as I'm sure we're all aware it's really frustrating when you read a paper it has a really interesting method you go look to reproduce it yourself and the code is not available or it's in a get repo that's been deleted um I think we've all been there and so from that standpoint we're our goal is to it so that that doesn't happen again and that everything built with NGC learn can be found all in one place so we can all see how much it do uh and finally some upcoming features uh reinforcement learning uh so once again this is actually uh currently in our development Branch uh so we have grid-based environments uh technically you don't need to make them grid based it's just the easiest one to visualize uh so it's using components so with the way that components work uh if you can basically chunk your environment into its own individually Run State machine um you can display it and run it in NGC learn uh currently we have rat water maze rat team maze another model that I'm currently using for paper lits not published yet uh that are all navigation based tasks that have to perceive the world around them and uh get updates I will say right now it is only limited to 2D uh you would not stop you from using it in 3D you just might have to see how to visualize it differently um and since it is built up out of components it is fully compilable uh and what that means is that we can compile the rat tze into a c function and integrate it directly with the compiled version of our um agent meaning that at the end of the day you can do your entire training over your RL environment in in C never having to touch python uh and I will say entric and World Views are both available so obviously these are rendering the world view and then the little highlighted section is your egocentric view both of them are very easily attainable um yeah and so with that uh there's some links to all everything and if anybody has any questions please don't hesitate to ask um slightly early on time but that was kind of the goal thank you William okay just while I'm getting back into the game here maybe just share a little bit of your journey to working on this like where are you at in grad school how how did you come to be working on it yeah so I am a fourth year graduate student working under Dr Alexander aoria at the Rochester Institute of Technology um here in New York and so this kind of all started so my journey on spiking neurl net started about two and a half years ago or so and very rapidly I found just like the speed problem to be incredibly annoying on like how do I actually get this to run in a way that's efficient um and as I started moving to bigger and bigger models specifically RL and some other models uh it started getting really really slow and so around that time so around like two years ago or three years ago or so uh Dr abbia had started uh working on the first iteration of NGC learn which is what his original paper this one was written using um and then I got a hold of it and it it works it works beautifully um but there since that P came out there were some pretty large advancements in the just computational things Jax had really taken off a little bit since that had come out um it was all originally built in tensor flow uh which works but it really felt like we were hammering a square peg through a circular Hole uh to get it to do what we wanted and so starting about a year and a half ago now uh we kind of set out to make NGC learn what it is today and then probably eight months ago it really kind of took off on where it is with its compiling and the speed and we've been using it and developing it ever since cool epic okay I'll start asking some questions from the live chat and if anybody wants to add more questions they can go for it okay okay I'm going to read a possibly unrelated comment but if you have a thought otherwise it's not necessarily A a direct question okay Scott wrote lawyers say that contracts are a meeting of the minds singers and chuses experience aesthetic amplification of group identity hear the meeting the meeting of voices in storytelling equals a smart mirror for group identity that's that's okay it's interesting I do I do kind of uh thinking about it is like just my initial gut reaction to that is that uh I think especially when we're doing science sometimes we can get a little bit too bogged down in being scientists and not enough time in being like freethinking and actually just coming up with like new ideas right we get too ingrained in like ruts of how things have been done in the past and so I think I think that's something that we all do need to work towards more it's just being a little bit more embracing of just new ideas that might not work awesome okay first question mobina wrote does it only work with discret value spiking no so I'm going to flip through a couple of slides here so when you go to these wires here there's nothing that says that the wires actually move discrete values but the advantage that we run into is that the fact that they can move discret values so when we look at here so the weight outputs so like w like the weight weight outputs are actual Matrix multiplications of weight outputs we can see that here this is this is not a discret value but for us when we're dealing with spiking mural networks the input coming so the short answer to the question is no it works fine with floating point values as well um it just also works with discreet values which many other systems do not okay to give you a a breather I'm going to read an interesting quote that your last comment reminded me of it's from Henry David thow in Walden he wrote I had not lived there a week before my feet wore a path from my door to the pawns side and though it is five or six years since I Trot it it is still quite distinct it is true I fear that others may have fallen into it and so helped to keep it open the surface of the Earth is soft and impressible by the feet of men and so with the paths which the Mind travels how worn and Dusty then must be the highways of the world how deep the ruts of tradition and Conformity it's a good quote it's a good quote I've seen his cabin I've been to I've been to that Lake cool like I had a sense that you would yeah yeah it's it's a really cool place okay reading down viit your colleague and previous presenter wrote two questions so first question I saw that Jax also has a method. lower to lower the pure function down to machine code but not compiling do you see any fruitful exploitation of this function in the context of NGC simulation Library um possibly so that could get us definitely some of the beneficial parts so right now when you compile a function and then it throws an error debugging that error is a nightmare uh because it's all compiled under the hood I would have to look more into it but if we can just get to machine code that would be really cool uh like if it didn't end up like if it ended up skipping over like it's going into C but I'm not sure if it does um on top of that something else that would be kind of interesting is and I've actually spoken to my father who also does computer programming about this is if we can get it all written into C or machine code theoretically you should be able to precompile this which means that you could get your entire model and all of the state transitions as a program basically that sits on your computer without actually ever having to touch python which I also think would be a really cool place to n direction to go yeah I think I think the presentation and the package architecture reflects it like many octaves of computer science you're really getting down into the the compilation details but understanding those is critical because the point you mentioned sort of by The Bu that there's the possibility with only some constraints to run a model on CPU GPU neuromorphic that that that's an incredible cross um deployment of of potentially identical models so you could have yeah certain models deployed across Hardware to better understand the hardware forget the model or of course for sure inside yeah inside of sim so inside of the tutorial section of like the documentation for NGC learn there's a part with NGC lava and it actually will lock you through how to run it run the same model both on the the low he simulator and your computer like in the same time um and you can look and it will get the same output like it's got some Randomness to it so like your filter might be in a different spot but it will produce the same model yeah I mean what is it even like to or how is it similar and different to be programming on a neuromorphic simulator so the simulator is relatively straightforward actually going on to the physical Lei itself and this is why we have to stress simulator everywhere is a challenge so first off Intel's neurocore which is like their propri rary code for running on their Lei chips uh well it's proprietary code which means that I don't have easy access to it um like I have a couple working groups that I can have access to it through but it's not something that I can just put on my own machine and test um in addition to that I micro code is needed which is the basically what is usually used to like Pat CPUs and gpus and whatnot but the micro code it's the level between like machine code and it looks somewhat like assembly um you need to write a lot of your models in that which gets complicated so until we get a good translation layer between Python and either C and then C to that layer or between just python to that layer uh the physically putting it onto Hardware still is a hurdle yeah again super interesting how this is really spanning the gamut with different layers of computer science and and bringing in all these unconventional um compute okay another question from fit have you also considered automatic operator such as when we add two compartments A and B together instead of summation of a comma B it is a plus b and the library would wire them to the summation operation uh yes uh so yeah we definitely have considered that uh you run into so it works so so the long story short is like that can work the problem that you can run into is it can become a little bit syntactically uh weird to look at when you have compartment plus compartment meaning one thing and compartment. Value Plus compartment. being another thing like basically the problem is that if we start overloading those operators it's you now have to be careful like right now if you try and add two compartment together uh and you forget to do like compartment. value um it'll just error because compartments don't have an addition with one another but if we produce an addition with one another um it starts to make it's you have a lot more silent errors so until we find a good way to overcome that problem it's currently on pause but this brings in a lot of interesting category Theory type question s about concatenating nested systems and putting them in parallel in series and what are these operators what does it really mean when just to have a certain symbol and then what do you really want to do what are the operations supposed to and that's kind of been why like we overrode the shifting Operator just because I wanted something there that was not just like dot connect um but beyond that it's I've been a little bit hesitant to override okay I'll read uh two comments from Professor Aria okay first it's just my lab so far yeah but first first uh a historical not I guess the nod from will to the old many years ago NGC learn was nice wink the original NGC learn was written in tensorflow and started by offering generic customizable support for predictive coding course no and then second note um I posted a link to where we've curated uh many implementations of active inference and Alex wrote we would love to have Community contributions translating these active inference models to NGC learn we would love to feature and maintain these in the NGC Museum I mean that was an incredible insight to have a package Centric Repository yeah yeah the the whole goal of this so all four of these models that are shown here are all in the museum already or on their way to being in the museum and yeah the goal is just so that way like if you guys have this repository if here's all the papers well what's better than here's all the papers then here's all the papers and the code to reproduce them all in one place yeah um okay you you you mentioned a few pieces of this but uh mobina wrote what improvements are currently being explored to enhance NGC learning yeah so uh again one of the main ones is just getting reinforcement learning up there and available for people to use easily um in addition to that there's an overhaul of our visualization tool the tools that I've want to have so we already support um things like the raster plot uh all of like the for like plotting Spike patterns uh we support obviously Matt plot Li does all of its stuff and since a all of our NGC learn stuff is built inside of numpy it all plugs and plays into that really easily um but we do have um a couple other just additions to all of this that we're trying to get up and running basically there's an overhaul to the visualization um in addition to that I'm sure there's other projects that I've been thinking about but I don't have my I don't have my notebook with my list of plans in front of me but those two are probably the first two on my list is the reinforcement learning and then visualizations cool and this is a question and the help section and the library built help section yes the lobby of the museum um yeah and then the art outside um here's a question that we've discussed with each of the previous package developers today uh active inference .jl and and the RX andur like what was the most challenging part of developing this out um definitely the compiler and more and it's not the funny thing is it's not the compiler specifically that was the most challenging part of developing this out it was how did I want the user to actually interface with it because of the most complicated things and I can go back to here right was getting this to all work in a way that didn't feel clunky so and it still isn't perfect it's just the best we've figured out so far but the the way to define your compartments and your fixed parameters and then write transition functions that can pull them into into into itself and do the operation getting that workflow down in a way that made it so that way if you want to build a new component it doesn't take very long was a challenge um that was probably the most challenging part um the next one's definitely just actually getting the simulator to Plug and Play everything together um I could give another like thre hour talk on how the simulator act like the NGC Sim live Works under the hood um because it's it does a lot of metaprogramming and pattern matching and function generation like your actual compiled method that you're running is like four step three steps of com of like dynamically created functions deep basically um and so getting that to all work without bugs was hard um and it's been doing pretty well uh we've only had two very very minor bugs in the past like five months so I'll call that a success in anology we love minor bugs oh yeah of course um okay I'll read another question roric wrote is there an active outreach program to the museum as such treasure ought not to be buried so we're not actively going out and trying to seek things but it is open source and anybody can make poll requests into it um so and we are and whatever we for like our lab or like for our own personal research reproduce a model um or either a baseline or just as like as we're learning something new that also ends up in there but um I think as the going out to people specifically and being like Hey we're going to it's a lot of it is is we're we're trying to get the people that use our tool to just when they've published the paper using the tool put it also up there that's a very humble answer also you're doing the active part now so that can't be discounted okay I'm going to read another this is not a question from Scott but just if you have any thoughts okay he just wrote things get interesting as our new computational and information processing systems computers internet AI etc etc etc more Faithfully reflect human cognition active inference and biophysical models than are existing institutions and organizations the latter are historical artifacts not yet improved and socialized to new learnings about physics supported flows I mean I think I think this is just a my first thought of that is uh many yeah many many modern AI systems and giant structures that are made uh do not reflect well on how on like are not faithful to biology um and we definitely have a we should definitely have a goal on making them more faithful to biology and uh getting a basically there's a bunch of things that like biological models have uh can achieve for us uh with one of the biggest ones just being the Energy Efficiency of neuromorphic computing and the fact that we have things like chat GPT and whatnot that have uh you know use power draws of small cities to train uh kind of goes against all of that and so I do think that like this is definitely a way of the future I don't think it will ever beat out everything but I do think it's definitely something that we do need to continue pursuit of okay I I have a question what is the putative or possible title or topic of your dissertation overall like is this one puzzle pie or is this the so so yeah so I've done my dissertation proposal I off the top of my head I do not remember what my proposal name was uh but so I do uh spiking representation learning uh so I up until now most of my focus has been specifically on the uh learning rules in credit assignment plasticity rules of spiking neurl that specifically Spike timing dependent plasticity uh this is my paper um time time integrated by time independent plasticity um and then uh I wanted to do all of this with the direct goal of putting it into reinforcement learning uh which is why one of the reinforcement learning parts of NGC learn is on the way is because I need it for my research and since other people have also expressed an interest in it for their research I'm like let's all put it in the library together um and so yeah so I've got that and then the actual uh reinforce and then all of this basically played together to go with the representation learning of how do humans actually represent stuff because I'm not a fan of directly going from like pixels to thought process I want to go pixels to representation to thought process and so that's kind of where I've been at right now in my career path cool okay another comment from Alex he wrote we should also mention that njc learn is going to offer support for dynamical systems and even more detailed support for differential equations beyond the basics we have now smiley face yes yes that is that is the thing too that we are on on track for we've got um currently a bunch of different integration methods that are in there um but they don't allow they allow for some complex dynamical systems but not every complex dynamical system that we want and so the goal is to be able to put all of those in there too um pretty much anything you think of that can be represented as a state machine is something that would love to be able to support um just because at the core of it it is just a giant State machine moving pieces around and so while it works really well for spiking neural Nets it works really well for so many other things that use uh that are you know a dynamical system something that changes with time yeah from the active inference side when I see that well first the reinforcement learning and the ability to have the egocentric and the allocentric I think it is very fascinating and it makes me wonder how how do epistemic value inferences come into the picture when we zoom out Beyond utility or reward maximization and include these networks with structural fixity or with with a dynamic structure doing learning that's kind of just an interesting yeah for sure for sure I don't I don't have any great Insight on that right now other than it sounds really cool and I would love to be there working on it cool well I I I don't know if I should say that I do or don't hope your PhD ends before then do you have any uh I hope to get work on I'm pretty sure even after my phdm I'm still going to be working on this so I it's all good cool do you have any uh last comments or thoughts to add uh not really just that if you are ever trying to set up NGC learn and it's you've you're struggling with anything don't do not hesitate to reach out um to either uh myself or my adviser Dr Oria both of us are the on the main Dev team of NGC learn and we would be more than happy to help okay I'll just read a closing comment from Dr Robia we really hope the active inference community and audience considers Community contributions to NGC learn especially if they want to go beyond using backprop slst standard deep learning also consider using it in the classroom to teach computational Neuroscience cool all right excellent thank you William farewell hello welcome back we're here with real Bradley Lisa and recorded Stefan Bulman there will be a pre-recorded talk with some discussion too so thanks all and looking forward to the talk go for okay okay this coming through yes he wonderful well then um yeah welcome everyone uh to my presentation about neurodes um my name is Jeff bman I'm a senior research fellow uh in at the University of Queensland in Australia and sorry I can't be there live but the time zones are just too difficult um I want to start with why we build neurod desk and then introduce what it is but we saw a couple of challenges in open signs and one of them is really non-reproducible workflows so we have um different great tools uh they run on lots of different data sets but the challenge is that if underlying dependencies change they produce slightly different results and that's really not great if we want to infer anything on this data the other challenge that we saw is infrastructure disparities so lots of people have access to different Computer Resources so be their own notebook the HBC cluster or their cloud provider to process their data and wherever they want to run data analysis they need to come and install their software again and the software will be different across these installations and often costs a lot of time again produces different results but also costs a lot of time in maintaining these different installations the other challenge that we see is data management so when you acquire experimental data then from an MRI scanner for example or behavioral data then we have to organize in some sense and there lots of different tools but we wanted to see if we can make that a lot easier and then also data storage and privacy so where we store this data how we collaborate with others how can we make this data openly available to others these are really questions that we had and thought well can we somehow make this nice and U nicer for people and easier to use so why is it so difficult um in the end all the software that they use for neuroimaging analysis are are software pieces that depend on other software pieces and these we often call libraries and what they do is these libraries provide functionality that you don't want to reimplement for example it's you have your software here that does some brain Imaging analysis and then uh you don't want to reinvent how windows are drawn on a computer or how data how data is plotted or how images are processed so you want to use what other people have already built to do these functionalities and then what happens is your software starts depending on these because you're using functionality from these and this usually works quite well and the reason is that we have in Linux for example we have package managers like EP yam or if you use Python for your work we have cona and pip but pretty much all languages systems have some kind of package management system that keep dep pendencies compatible with each other and that's really how it looks like it's like a puzzle piece where if one thing upgrades another thing has to upgrade as well they have to have matching versions and then uh we're hoping that this all works the challenge is with a lot of the scientific software that we work with it's often build by scientists who don't have a lot of time they're not really software Engineers a lot of the time so what happens is they have unpackaged scientific software so these scientific software packages are not available in standard package systems and that means that if the operating system upgrades the package manager doesn't know about this software because it's not registered there and then what happens is the package manager changes something and because it didn't know about the software then it breaks and that's happens unfortunately quite often with scientific software and it leads to two problems uh again here's our revability problem because if we just depended on a library and even if it doesn't break let's say it just changes functionality our results will be different and that's really bad for reproducibility and also this means portability because if our software is tightly integrated to the system it means we can't install software on our notebook and then run exactly the same software on an high performance Computing system because that will have slightly different dependencies so we need to install it again often compile it from source code which can be a really a lot of work so just to show this reuil problem in a little bit more detail um this is a a great article that looked at software that depends on gipc gipc is a very fundamental Linux library and what they did is they varied only de Library version um and they left the main piece of the software intact so they used free serfer and lots of other pieces but I'll Focus here on free Surfer um so it's the same software installation just in different operating system just in different Linux versions in the end and what happened is in gilp C the exponential function for example was altered and was made more efficient but unfortunately this meant that the Precision of this function changed so now you you would think this doesn't matter right it's like very uh a couple digits after um uh so yeah so it should be quite insignificant unfortunately it's not because free serer in particular has a it's a long running pipeline where we sment the human brain and this means that if you have small changes this accumulates over thousands of processing steps so this means in the end here they found differences in millimeter up to 02 millimeter in the cortex where um this is a change that's only based on the software version of this underlying gipc so free was identical the only thing that changed was the operating system and the thing is is this was even statistically significant in some areas and that's really not what we want because if you acquire patient data on one software version and then you have matched controls on another software version then that would really lead to significant results that are actually meaningless because they're just based on computational noise so the organic Sol solution that I've seen in my career is that we just have people set up different computers and don't touch them don't try to update them because people are aware of these problems and this is not great because it's it's not a great use of resources and also it means we can't easily take our software and run it in different computers and this is this portability problem where let's say you want to run the same software on your notebook on your lab workstation in a secure environment on a cloud provider on a high performance Computing cluster or even on the instrument where you're acquiring your data like your MRIs can't just easily uh build an algorithm on your notebook and then run it on the MRI scanner it's currently not possible and but that would be cool right for for translational studies and patients it would be fantastic if you had a model um that you build on your notebook you validated on lots of data and then uh you can actually deploy it in in real clinical environments that fantastic so this is really the problems that we're trying to solve with this Pro with this project and we adopted one solution for this which is called software containers so I quickly want to introduce what software containers are and how they work because our software is based on all of this so software containers if you've never heard of them a very simple thought pattern is um they're just boxes they're boxes to put software in and they're really like the stuff where you put food in and put it in your fridge it's fantastic because it keeps things separated and you can move them easily they're very portable and they don't depend on each other and that's really what software containers do they use specific Linux kernel features called namespaces and cgroups they isolate processes and dependencies from each other so this means we can put everything our software needs into a standardized box and we can move that across operating systems and what it basically changes from our mental model when I said earlier we have this Lego piece where this this piece depends tightly on the other things in the system this now changes to where we take these dependencies we put them into the same box so now we actually don't have a dependency on the system anymore um the Box got a bit bigger because we have these dependencies not included but in the end storage is cheap but experiments scientific experiments expensive so this is a very low price to pay for a lot of great features and the whole Cloud world everything the internet depends on today is based on software containers so this is really a very mature standard that has been adopted everywhere in the Computing industry so in science it we're just we should just do it as well um so what we can do we can package all of our software in these boxes and then because we have these standardized interfaces now we can run this on Windows uh Linux and Mac on Linux this is a standard kernel feature so no performance penalties very easy to run on Mac and Linux we have some virtualization technology that we need but that got really good over the last years so you will not feel that it's slow it still feels very native but there is some virtualization there is some overhead but it's tiny compared to what we had a couple years ago so then let's come to what we built um before I show what we've built in neurodes is I want to acknowledge all the people working on this project because yes I'm giving the talk here today but I'm done just one of many contributors um we started this project really during Co because we were all bored and sitting at home and we said well let's let's solve that problem why not right we have time um so we got together couple people all highlighted in in in bold here and said okay let's build this we built a prototype during hackathon at the obbm and we got funding from the aidc to build a bigger version of this and then yeah we got lots of people involved um slide change sorry um we got lots of people involved from involved from across the world and lots more supporters like the big cloud providers AWS uh Google had supported us in the past as well um and then also the uh the university Cloud providers uh like egi Jetstream uh ADC nectar Cloud so that helps a lot and and yeah we're currently funded by the mar andan Zuckerberg initiative and welcome trust to build that uh system out further over the next years so this is what we built so I'll start here at the container level so this is really the software containers that I introduced earlier where we build a software container for every piece of software that people want in this so here we have a couple of Neuro Imaging pieces so FM prep free Surfer field trip ends and then we can control everything in there from the binaries the libraries that we have in there up to the actual software that people want in the end like m& which then depends on python which then depends on cona and so on so we can control all of these dependencies and then if you've used the software container before they are fantastic but they're not always easy to use like you have to run specific commands like Docker run or if you're on an HBC you need to run Singularity or obtainer run and we didn't want that people have to think about this and we build an accessibility layer we call it but in the end it's just an abstraction across multiple container technologies that we use and it's also a way how we can stream containers to end users so people don't have to download these containers so this this little block here was actually the most of the work that we spend on how do we get these containers to end users and how do we make it easy for people to use that so but ultimately what this allows us now to do is we can run these containers on any system so we can run this on mechos on Linux on the cloud on windows on high performance Computing clusters in jupyter notebooks like Google collab on the terminal pretty much anywhere and we try to indicate this here with this little script that goes between systems and that's now the thing that wasn't possible before we can code up an analysis on our notebook and we can move it across all these systems and will perform identically and behave identically across all of these systems so this really is fantastic for collaboration for translation of of results and um for reproducibility ility general and then the community uses that interacts with the system but also finds problems bugs and then reports back and contributes new containers new things that people want to use in the system and that's really the ecosystem that we've built so the best way to get started with neurodes is go to our website neurodes your computer and it depends on your use case if you have data patient data you definitely want to run this locally if you have open data you can use a host version and you're fine and then we have a more fine grain table down here where you can select what your compute platform is what your operating system is what the interface is that you use and then you get guided to the documentation where we guide you through how to run this on your system so in the end what you're getting is you're getting a virtual desktop which is just Linux desktop with all the tools installed and the cool thing is all the soft is ready to go so you don't need to install anything everything is there um we have no dependency issues so we can even have the same software in different versions so for example free serer is almost impossible to install different versions of that software on your computer because it will replace dependencies that that other pieces depend on but because we we wrap all these dependencies in a software container we can actually provide all versions of neurodes of uh free server for example going back over the last years without any problem because of this containerization the analysis reproducible uh down to the software level we're not reproducing anything Hardware effects I can show this later uh but yeah if there's like a CPU difference between Intel and AMD or arm we we don't control that but down to the software level it's reproducible and then this whole uh system is only 1.5 gab in size and the reason is we can stream all the software we need on the in here so when you actually click on a software it is streamed live to this endpoint you can also choose to download it but then we realize we look at brain Imaging data and and check if things worked or didn't work then also we wanted to support high performance Computing systems so high performance Computing systems they don't often come with a desktop interface so you need to work from the command line so we build everything to integrate with these HPC systems so you can easily run neurodes on there often you can't install things on there so we took care of making that possible so you you don't need any elevated privileges if you have Singularity or obtainer already installed on your HPC and then we also support Jupiter uh and notebooks computational notebooks in general because we think this is a great way of documenting and anal analysis experimenting collaborating with other people and sharing results in the end so we really build it into Jupiter lab as well and provide with this a very nice interface to people that they already used to that I know how to work with then it's also very easy to contribute new software to it it's all automated on GitHub so we have GitHub actions where people can propose a recipe for a software this gets automatically built tested and integrated into this whole system and made available and then we have a system which is called cvmfs this the streaming of software that I mentioned it's developed at CERN so CERN used cvmfs to stream software and instrument data across the world and we use exactly the same system system and also collaborate with the people who run these servers so we have a a primary server where we ingest all the software and then we distribute it worldwide so wherever you are in the world you should get a quick access to this software and um there's a client in there that localizes where you are and gives you the closest server and even if a server is down it just goes to the next server so it should be a quite Reliant system and we're using this in a couple of years and had never had any issues with that and then for the end user it just appears magically that there they just see the software there and when they started it just magically works it's just on the first start it's a little bit slower because it has to download certain things but then on the second start it's CED locally and it's it behaves like it was installed locally so the update in the community is pretty impressive and we saw really big growth over the last years so we have more than a thousand monthly users on our systems uh that come from over 60 countries so really worldwide adoption we see that more than 16,000 people downloaded our main soft container that provides a lot of these things it's used in university courses for teaching it's used in conference workshops because it's fantastic for standardizing teaching material because you can't you don't have the problem that students have different software requirements on their computers they all log in into either a cloudbased system or they install our software container and then they can run all the same analysis which fantastic and we also have various collaboration models that we support with this because uh we realize that in some instance it's nice to have a central compute instance where people all log in and they all work on the same data this is fantastic for uh open data sets this is fantastic if people are actually all on the same institution and they all collaborate on an Institutional server and that's really useful and used a lot also for teaching obviously that's a really great setup because teachers can then log into the other uh students noes and quickly have a look what's happening there if they if they struggle if they have a problem but then we also have have this Federated model where every in let's say researchers are collaborating from different institutions they just install it on their infrastructure on their computer and they run analysis on their data and then they can for example aggregate high level results in a publication uh without sharing actually any confidential data so let's say brain extraction or brain uh let's say cortical thickness measurements right so you could easily share those without actually exposing the patient data that's underlying this and then then that can be shared across different institutions so that's a really cool model as well that we see often used to our our project and with this I claim that we made maintaining and installing scientific software fun again so it's now very easy uh basically you install neur desk on these different systems and we make sure that the software runs on these systems so even very recently we managed to get our containers running on an MRI scanner so this is uh uh Works particularly well for the seens ecosystem so seens has a software platform called open Recon so we can actually build a software container that can directly be integrated into a reconstruction running on the system and um that is really closing the loop so now you could yeah acquire data build a cool model and then deploy it on an analiz scanner where patients can directly uh be scanned and results end up in the PEX system for people uh to look at so that's really uh great progress with the last months that we've made there so I want to quickly show the open data workflow in neurodes so this is something that we always focus on a lot we want to make consuming and producing open data a lot easier and what we've built there over the last years is that we support a lot of tools and databases that make that possible so I'll start with importing data so usually if you produce open data you would start from for example an MRI scanner or an EG system or you would download data from for example open Euro or the open science framework and then we have a tool in there that's called data that can handle a lot of these uh cases where we can download data from these systems also xnet we support which is a a tool commonly used in the Neuro Imaging space to Archive data and then we of course support any cloud provider or any local connection so pretty much anything this is not an not a complete list this is just C of examples what we support and then once the data is in the system we can process the data using pretty much any software our there if the data if the software is not yet there it can easily be added we currently have more than 100 pieces of software in there that that the community wanted and then we can organize the data and we can document our analysis so for this we highly work with the bid standard with the brand Imaging data structure but also bits has been extended to the EG world so we support this as well we support Jupiter as I said as a way of documenting this data but you can use also your own script system whatever you're used to you can install that um and then we wanted to make it very easy to share data with the community so we support Google colab for example so you can write your analysis in Jupiter and then you can provide for example a Google collab notebook with your paper where all the analysis run and we stream the software into Google collab so with this we can even run soft that wouldn't be able to run in Google collab so for example because you couldn't install it because Google collab doesn't have enough disk space for example to install a certain piece of software and we can actually run the software in Google collabs through our streaming service but also we made sure that we can easily upload data back to open Euro again through data lab or we can share data out to OSF or that we can share our analysis quickly on GitHub and then people can start again from this and can use our data that we produced or our our derivatives that we produced and then start the cycle again and then really yeah enrich the open ecosystem out there so people often ask what is our sustainability model uh will it be around in the next couple of years and of course that's a very good question because if you start adopting a framework like this you invest time learning it and you want to be sure that it's around so we try our best there uh so we as I said we have been funded through an a platform Grant in the past we're currently funded by the mar and Chan Zuckerberg Foundation through welcome trust and we're applying for uh grants left and right to keep the system growing and developing but of course Grand funding is very competitive and of course Brands always want to fund things that are new they don't want to fund things in maintenance mode so that is definitely uh something that we anticipate so that's why we really want to build the community so we already have 10 more than 10 active contributors we have a lot of the infrastructure hosted through through the community Through the open science grid uh the egi foundation where we don't have basically Cloud bills for us which keeps our costs very low which is good and also if we have help us from the community we can also lower our costs because we don't have to do certain work and then we provide neurodes as a service as well so this goes through a non-for-profit um Foundation which is called the Queensland cyber infrastructure foundation and what they do is they provide new desk at a certain cost for for example a conference for a workshop or for a university so let's say you don't want to use the open source software and do everything yourself you can go to qf and say I want this professionally managed I want a service contract I want an SLA I'm happy to pay money for that and I don't I want to use cloud resources for example and then qf can can provision that and set it up in a way that it's really at at scale and working really well with with certain guarantees as well um and then this money flows back to us to develop things further and then we also have a lot of support from cloud providers so as I said we heavily run Cloud Technologies so we had been supported by Oracle cloud in the beginning then Google cloud and now currently our main sponsor there is AWS and the research clouds like egi Jetstream aadc which support us with Cloud resources and make that all possible so I want to quickly show you some results from this reproducibility aspect because I'm always claiming that neurodes is reproducible but is it and how reproducible is it so this is something that we looked at so PhD student in my group took that original paper that I showed earlier from Tristan clar where they found these reputability issues with software so we took that paper and thought Oh tested can we reproduce the nonreproducibility of software so the analysis is the following uh we picked this part where we have data was scanned from an MRI scanner in anatomical space what we have to do is we have to register it to the m& space so this means we align it to a template space which FSL with which a with a tool called FSL flirt and then we take this data and we segment it so we segment subcortical areas and we measure for example the size of that so it's a very simple pipeline where you think that should be pretty easy to reproduce so what we did is we used a very similar setup than what clar used in their paper where we have a local installation of FSL so FSL is the same but then once we run on gipc versions 231 so on a Ubuntu 2004 at the neurodes which chose a quite old version of yuntu uh with a quite old version of GPC but the cool thing is the Gip version is consistent whereas here gdpc version is 2.28 so something in the middle it's on an alma Linux 8.5 and then also what we varied is we V varied the processor so here it's an an i7 and here it's an AMD epic processor so let's see what happens so we of of course we compare these different systems now let's see what happens so the first thing is image intensity differences so the first thing is we were able to replicate the problems from the paper so if you have a local installation with two different GPC versions then you get different results and this is really intensity so we we only focus on the intensity in the area where we will later do an analysis so this actually the intensity change as whole brain but just to show where the problems are so we basically see that the the intensity of a boxel is like up to one one integer value different and this is small because when you think about the images are usually like 0 to 1,24 so a change of one doesn't matter much but still it's a change and it shouldn't be there so the cool thing is when we run this a neurodes where we control GPC the gipc version this effect is not there there's nothing there and that's exactly how it should look like there shouldn't be a difference between this it's a fully um deterministic analysis there shouldn't be any variance between that and that's cool so then when we look at the classification differences so this is now when we segment this data um we see again in the local installation with different GPC versions we see quite High disagreements between labels and this is now dice score so this is between zero and one and a016 dice score is quite a lot when you think about it um so this means we actually disagree in certain areas quite a lot between segmentations and again that shouldn't be the case now in desk you still see something there however note that the scale is an order of magnitude scal down so we had to scale it down to make anything visible there but we wanted to show it and why is that why is neurodes here showing some non-reproducibility because I just claimed that we are reproducible right and why are we suddenly not and we looked into this so we looked we traced the executions of this program um and we found that as expected in the local installation the software diverges so we're doing different things on different computers with different gipc versions and that's why we're getting different results so that explains it it's great on neurodes we're not diverging we're doing exactly the same thing on both computers so this means on a software level it's reproducible but why is there still a different result and the reason is that we use different processors so was an inut processor one was an epic AMD epic processor so different instruction sets different floating Point precisions can still have an effect that we cannot control for even if we do everything the same we get still small differences they're tiny but they're still there so it's something to to note that 100% reproducibility will only be possible if you also control the hardware but then in the end you have to yeah have a trade-off there right when when is when is producibility enough uh and when how much effort you want to put into having reproducible results there so yeah that's just uh our results there so a little bit of an Outlook what we're working on next so there's a lot of things we do work on but one thing that I really want to see live in the next years is interactive papers because that's what we can do now we can have an analysis in a Jupiter notebook where we describe what we've done we can show all the results we can show all the analysis we could run everything and everyone could rerun it as well and then base their analysis on it and Fork papers and really say well my anal is based on that paper and have open data for pretty much everything and really uh yeah make it easy to to contribute to this open Science World and we already see some adoptions from this from from bigger journals like eive they already have like some of these executable papers so we're seeing this coming through there's still quite a bit of work required to make this easy and and very doable but uh our neurodes paper for example we already did that so the neurodes paper you can go and reproduce it and you can reproduce our reproducibility analysis for example which we thought was quite nice so with this I'm at the end uh of my talk so I just want to highlight that um it's an open community so if you uh we we need we need Community we need Community engagement so if you like what we've done if it's useful for you get in touch help us build new things find bugs and um yeah we also have a little bit of time still left to show neurodes connection if this is something you want to see go to our website um neurod des. org click on hosted and then you see we have different services and I said we choose the place service with different regions Australia us and Europe so if you're roughly in one of these regions pick that one so I said I'll pick Australia I just want to uh quickly highlight here that all of these servers require some kind of a login Australia you need to be an Australian researcher because that's how the service is funded in the US and Europe you need to have a GitHub login at least um so I'll click Australia and what happens is we'll authenticate you with have an account in case of Australia we use an Australian access account in then you just say Okay I want the default environment there shouldn't be anything else available for you this will start up uh a jupyter notebook on cloud resources um that are in that country so you don't want to put patient data there of course but you can put open data there and in terms of storage uh you have a home directory which has about two gigabyte of storage you will get a warning if it's full and then nothing will work anymore so you need to be care that this is not happening and then everything else you change on the system will be reset after after your session is done but that's basically how it looks like you go into the system and then it's just a very familiar jupyter notebook system so I quickly want to show now um we have for example an example repository which I quickly want to show so I'll just I'll just show it on the website first so what we've done is we build an example repository of lots of different workflows where uh we for example show how to use pipeline systems on here we show how to do diffusion analysis for MRI uh lots of lots of different things and we're currently building that out so if you have a tool that you want in there please go and contribute an example contribute a tool and what we can also do is I can I can easily take this and also run this on our website so for this uh we will just check out this repository so I'll just show that um so just clone our example notebooks then will take a couple y so here example notebooks and then we can just pick an example here so I'll pick structural Imaging and then I'll take the uh I'll take the functional Imaging where is it inro to pre-processing I just want to have a very quick one to show yeah okay so night pipe short so that's a really quick one to show so um what we do is we have a first cell where we set up neurodes this is not needed if you run this in neur desk environment but it's needed outside for example in Google collab and then we output which CPU are running on Just for reability reasons and then you can import any software into this notebook so for example here I install I import FSL and this is a software that we provide so this software will now get streamed into this notebook and then we can run any tool from the software W you can also install any python libraries anything um you need but yeah that was really the streaming that you saw so that this B tool was now streamed into this notebook and now can be used in anything and this the same way we can load pretty much any software and have it also documented what software we're using in which version we're using it and then I'll just uh download some test data and and show that so I'll quickly run this notebook um while this is running I want to show something else because notebooks are bright but sometimes it just needs more it needs a desktop and this is what we can do so we can launch a full virtual desktop in the system which should look familiar if you've used the Linux desktop before but ultimately uh it has a little start button down here where you go into neurodes menu and you see we have lots of different categories right now for functional Imaging electrophysiology diffusion Imaging and then you can start pretty much anything from here with visualization software and you can just start a software it should start it should run and should behave exactly as what you're used to as I said the first startup will always be a little bit slower um because it has to stream the software but then you you can browse brain Imaging data and access everything you want so now I want to show quickly as well um one more one more way of of using that so I said earlier we have this this stup notebook so I can also open this in Google collab so when I open this in Google collab I'll just do the same thing here run all this is now really cool because this is what I mentioned like this you could attach to your paper where you have your whole analysis coded up in there and then run it on on Google collab and it will be behave identically than what you just build on the neur desk instance for example um while this is um see yeah I'll keep this running I just want to show one more way of doing it so uh oh yeah okay but that will not work currently just sharing the screen because I only share the browser but yes so then there's ways of how you can run it locally on your computer uh with a Docker container so you only need Docker installed it can run on high performance Computing clusters pretty much anywhere where you should need it it should run um and let I just go back to our website quickly um yeah your high performance Computing system we have instructions here how to do this and even if you don't want this whole setup around we have ways of just installing a single container and you can use that in Docker or in Singularity so it's really a a very layered approach of this whole system so if you just want everything and no work then just take the top hosted versions and then you can just go down the technology stack and use what you need for your work that's really uh what we've built and let's see if that now finished by now so this this will still take some time um let's see I'll see if this one is finished by now um yeah so basically now this Rand analysis it downloaded data and then maybe I can show that one see uh we also have built in visualizations which is quite nice so we can inte with our notebook this is using a library called me view where we can browse uh and visualize Neuro Imaging data so yeah so the notebook is still running so that's why it's not executing it yet but yeah once it's once it's getting there it will um so yeah that's really that's really what we've built so yeah um while is running we can we can take some questions uh Bradley if you have anything that people might want to ask um I can show that well wait but otherwise that's pretty much what I had prepared yeah that sounds good thank you um I'm a little I'm interested in the containerization aspect of this so this is you know something that you can run kind of on any machine on any kind of computing um paradig like you know if you want to run in the cloud or on your laptop correct so a lot of the people in the group you know they're interested in active inference they're interested in sort of building their own tools and then how maybe to do containerization as part of that strategy yes so can you speak to that a little bit yeah exactly exactly so if you build your own tools um you can just basically build them in here just get them to work first and then once they work I'll quickly show how this would look like then um when you want to containerize it um so for example let's say it's a it's a python tool or a mlab tool mlab we have to compile and then we download it I'll just uh I can show that quickly so for example SPM as an example there so SPM is a brain uh brain imaging software as well and if you're ready to containerize it you can send us this build.sh file where you basically build a container uh based a certain Ubuntu version and you set everything up yourself and here's an example where we download a compile zip file for mlab that then runs in a mlab runtime environment and we have lots of tricks to make this recipe quickly quickly writable so this is actually a higher level language which is called which is developed by neuro doer team and this higher level language makes it easy to build containers because if you if you want to build containers yourself there's actually a lot of optimization that has to go into it and with this higher level abstraction makes it easier to to build these containers but then yeah once you have a software it's pretty easy to run it in here and even if you don't want to containerize it if you got it to work in here and you have the steps to do it it will run on any neurodes system pretty much so that's the uh that's the cool thing so if you have a python library and it's just got three steps to install it you can do that as well you don't have to contr miniz it only if you really want to freeze it in time make it really um robust across different systems then this is the way to go yeah so does this make sense yes yes and then I I would like to ask you know what does the future look like I mean do you have a road map where you're going out five years or so and and if so what are the needs that you need to kind of address in that road map yeah exactly so yes we have lots of ideas how to how to go forward with this it always will depend on funding what we can do in terms of new features one thing we do want is and we're heavily working on this right now is getting into the clinical space more to make it very easy to run the software anywhere so currently we're still need needing some containerization like Docker podman Singularity abum we want to and we we just have a prototype for that where we actually bring everything we need like we use a quo based emulation or virtualization and if the virtualization framework is enabled on a computer it's not even emulated so it's really native speed and with this we can run anywhere on any system really and we have no limitations anymore currently you still need some privileges to start a container on a system and this is really needed for these clinical um scenarios where we want to get into so this is one thing we're building out running these containers on MRI scanner that will be a lot of work in the next months and then uh the other thing is lots of other communities have contacted us so this is now focus on brain Imaging right but active inference might have lots of different tools there and they might want to build their own flavor of this like an active inference neurodes and we have communities like the genomics community microscopy uh dematology they want to do this as well so what we will be doing is we will refactor everything that's not neuroimaging and put that into a new project and then we have spin-off projects like neurodes that uses line technology but then they build the software around it and then Community maintains that one um and builds their own flavor of this this is something we're going to do in the next years um we still need some funding to accomplish that but I'm thinking that is yeah that is absolutely needed to support other communities there and there's also a whole uh underlying thing of what cloud technology we're using right so all of these things don't have to be reinvented if another Community wants to use that so that's that's really our roadmap for the next years making that yeah more General and yeah and specific at same time like more clinical as well yeah well thank you yeah that's good cool okay um yeah not sure so I know some vations came through so I can quickly show that so this is something which is I think really valuable for a lot of brain Imaging people so you can just quickly yeah look at the brain which is really cool in in a browser so you don't need any fancy software there and yeah yeah that's what for example in this case New View makes possible and I think this is really our our fundamental um philosophy with this right we want to just bring the right tools to the right people and make it easy to use them so they can do awesome work that's really what our project is about yeah thank you very much and wonderful people if people have questions after this talk should me an email um and contact us through GitHub or GitHub discussion forums and we're happy to help thank you right thank you yeah all right that's all we have for that um I don't know if we have any questions we want to ask I can try to answer them okay if people have any questions in the live chat they can add but thanks for sharing this and for bringing this important flavor into the mix um where it took me was right where you asked at the end with stripping back to the desk and then allowing the spin-offs and thinking about how we could use that and just how it's like the through line of the education the research the application that's kind of tracing back the community and information supply chain to Applications of active inference so when we come to a symposium like this think about planning a symposium like this think about making a startup or something like that using this in an application setting and we trace it back and then that's where these topics like reproducibility and research Integrity are the foundations application and if there's ambiguity in all these different threads and loose ends like from the availability of the papers things that have been focused on in open science across fields and then these really specific computational issues and then the example with a versioning of a library having these cascading consequences to say nothing of like security issues it's just completely it's like an empirical open science point that was wellmade and so now where do we go from that observation right right yeah I mean i' I've had experience with containerization um you know in the openworm foundation which I also work with uh they they've developed a container for demos and they run their simulations in a series of demos and the idea is to be able to get people to use the demo and get everything to run you don't need any dependencies on your machine you just worry about running the the container and that's it it's been successful I mean it's a little tricky to use and get to work but you know the idea is to make things reproducible so if you see something in a paper or see something that I show you a demo of you can reproduce it for yourself and it's everything's the same so you get the same answer and you can even maybe like modify some of the variables or parameters in that model and see you know work with it on your own and know that that's uh you know that what you're seeing is actual an actual result not just some noise or difference that comes from the computational environment yeah um like practically what are some interesting ways that as we digest all of this year and think about what projects like as a scientific Advisory board or just as an Institute or as part of coalitions what would be interesting how how do you how do you see that popping up across these different research communities yeah I think well for one you could use some of the packages that we've seen in the Symposium you could run them in a container and it would help people if you're running them in different locations like if you're in a research group or if you're in a uh just a casual you know learner who's trying to run an analysis it would equalize the ability to run those tools but also you know we can develop demos um if people are interested where you show you demonstrate active inference and you put it in a container so someone might run a contain you know someone who is learning about active inference could run a container and they could run these simulations and they'd be you know able to run them no matter where they are and it doesn't take them a lot of effort to sort of install dependencies cuz sometimes and when you want to say demonstrate something in a simulation you need to have people run like you know a bunch of things on top of one another and not everyone can either do that or can run them on their own machine successfully so that hinders what we can do in terms of demonstration and Education and Research so that just takes care of that layer and then everything else is like you can worry about like doing the good stuff doing the demos doing the science I as you've had many years of working in open source like what her istics or intuitions do we have in our toolkit when again just cuz something could be open source doesn't mean it necessarily should so how do we add signal not noise or Worse how do we navigate between getting overwhelmed with the technical complexity of trying to take on all of these new research practices and and just like we we could spend all all of our computer time making something reproducible I I I I I don't know exactly where I'm going with that but just how do we guide and just put out especially if we're putting out capacities related to cognitive modeling perception control then is there any kinds of considerations that we should think about before just releasing really effective containers for that yeah so I mean I guess what's in a container is basically the software that you release as open source and of course there has to be a strategy for releasing open source software you just don't dump software out on the web or you just don't like kind of produce a bunch of files and then you know with limited documentation until people have at it and every group has their own strategy and a lot of that is you know to think about the utility of what you're releasing as an open- Source package so sometimes you're interested in the uh commercial value sometimes there proprietary value in releasing something or not sometimes you release you have code but it's not well documented so the utility there is limited and that people can't reuse it so that doesn't necessarily make it good to release um you know so you want to be able to have I think and I I say this as being a full supporter of Open Source activity but to release things that are very Polished in the sense that you know why you want to release it and you have the right support in place and sometimes groups will have the resources to do that and sometimes they won't so if you're a small academic lab then you don't really you can't spend a lot of time you know polishing the software uh in the talk I I gave earlier I think it was yesterday I talked about like kind of these practices of software engineering and that sort of thing and that's something that not every group can do so that means that we maybe need either organizations to kind of have that layer of you know uh making things work well before release them or having meta organizations that can help scientific organizations do these kind of releases and I'm not against like making something open source for like a paper I'm just saying that like in tool building you know you have to have deliberate strategy for it and it's it it pays off if you do yeah that's very wise like your talk in part one and seeing this again and um the prior talk on NGC learn and the Museum of examples and going Beyond just curating links to to hosting more and more and that is what will allow Learners to have access these tools build their intuition about the topic maybe continue on and kind of um converge into the development attractor but even finding friction points and bugs and Reporting it and just putting up a yellow flag or just asking a question a lot of times it feels like people are outside of the development don't realize those visibilities that and experiences that they have into outcome can be in the blind spots or just forgotten by by bandwidth constrained developers and then that is like our research Community with people who are at different stages and using tools differently so it's like how could we not have this yet it's hard to to hold that yeah and it is it it is a Commons for for science which wasn't really part of the pre- internet pre-computer research world and so there's all these forms and functions that we're barely barely even able to describe let alone what has really been tried in different epistemic niches so it's it's super exciting because there's so much experimentation across multiple scales people doing things on their own being their own program manager on through just with friends working with AI and the kinds of things people are asking and the artifacts they're getting to then even up to like the metago type coordinations of organizations who want to have Commons and be particular but also find the interfaces that will help them do things together they can't do separately yeah um do you have any sort of last thoughts or questions and what would be some exciting directions for you and or The Institute in the year to come yeah so I mean uh Daniel asked me to contribute something on this topic and and kind of in the abstract because we do it looks like we do a lot of open source and and things and it's just kind of you know to develop a strategy there to develop a vision there is the thing that we want to get to so um you know I would say that like um you know one thing might be and I think this is very useful is to have a sort of educational component where you can have some you know tools for active inference and to teach people sort of how to you you know we we usually use simulations as an educational tool and making simulations are you know a nice tool because you people can interact with them and so you know and it doesn't have to be like a very high level simulation be a very simple thing that demonstrations of things demonstrations of concept where we might package uh some sample data with a sample simulation and say here build your own sort of active inference model and you know you kind of walk people through how to do it and then eventually they'll start playing with it and they'll say oh this is great and I want to get in more into some of these topics and then that's that's sort of a Gateway into the institute for people who may not feel comfortable with a lot of you know kind of diving right into the tech technical jargon they want to have like a a nice sort of way to ease in and and something that's intuitive and then that helps them um but other things you know like like as Stefan said you know that kind of the thing that they're working on neurodes is for Neuro Imaging but you can use different versions of this for you know different sorts of things and there are also different versions of uh Linux that you can build like your own custom version of so uh in the talk that I gave earlier there's a neurod Debian which has the ability to build these builds where you build them as like a specialized operating system for a certain thing they have neurod Debian which is for Neuroscience but there can be other versions of Debbie in for like you know musicians or data scientists and maybe you know I don't know if it's useful but to have something for active inference with all the tools in included and you just you know download that that rep that drro and you get the tools and they they work thank you these are all great directions a closing note I think it'd be very appropriate in a history Rhymes kind of way that the more generalized cybernetics arose out of neuroimaging and we saw SPM package in one of the presentations and then also neuroimaging and the special constraints like computational resources patient privacy those constraints and that community of neuroimaging also leading the way on the tooling which we can also learn from with more General systems yeah yeah okay thank you for this all right see you thank all right welcome back we are here with John clippinger and we will be discussing active inference biofirm homeostatic agents for Biore Regional and regenerative Finance so thank you John looking forward to this presentation and discussion and more Daniel thank you so much for doing this and hosting this uh it's been a great set of sessions it really has it's been very informative and this is a followup on a a session that we had yesterday on basically um what active inference agents and why they're important and uh and looking at them in an application and in this case in this session we were going to look at active inference agents uh in terms of a concept called biofirm uh as a homeostatic agent for Biore Regional regenerative Finance we're really looking at that the taking the thinking behind active inference agents and applying it to a new notion of the firm that's inherently biomedic and and what I wanted with these series of slides is to sort of set up the conversation and the discussion that we're going to have later with Andrew and Daniel on the providing an indepth look at at biof firms as a homeostatic agent that is capable of interacting with the environment in a way that preserves it and pres preserves itself in the environment um so what I'd like to begin with is sort of this a premise I've been involved in uh looking at regenerative finance and issues around climate change for least the last 10 years and I think there's a growing consensus that we need a new kind of science-based Economy based an enhancing life in nature and quality you really need to to think about what of the economic incentives uh to make a transition to an economy that values the preservation of life and nature rather than the extraction or destruction of it and what we don't need is what we we're now feeling the the consequences of it and that's the commodification of nature in people turning them into objects and assets that can be bought and sold and trade without any discrimination between sort of what kind of asset we're talking about are you talking about something that is is coal as water as life the biosphere they're all assets uh they can all be priced and they all can be traded um and so you just have problem of total fungibility when actually some things are essential for existence and they exist in relationship to other things to be preserved as living things um and then the other problem we're seeing it experiencing it widespread just today sort of the unbounded sort of concentration of of of wealth and power so we have a system that works towards monopolies and the concentration of wealth and privilege and and this of course has created enormous pressures not only sort of on the the ecology but on the whole society itself and you have this notion of maximizing individual utility so everything is about Freedom the individuals in atomic agent independent of the context in which they live and their impact upon others not just other people but their other species the whole environment um and so we have extracted metrics that the negative externality so so now there's you can create quote economic returns but what you're doing is creating a negative externality would' be pollution or it be uh air pollution destruction of species or habitat this is not reflected in the price and then it's left to some other party to a public good party to re clean up the mess but this is they're not able to clean up the mess fast enough than it's being created so we have a sort of a zero some win or take all incentive structure and we have a lack of transparency and accountability so there's no sense of of collective purpose so th this is is really forc people to rethink what an economy should be uh and and what it would be to have biomatic firms an economy something that that actually worked in concept with nature not against nature so first thing is important is not to have any negative externalities this idea that that you you can offset the cost of something is going to be picked up by other parties since their carbon credits and offsets are really an attempt a failed attempt to deal with negative externalities and we we need to get where we need to have nonzero some incentives in other words incentivize the collaboration a cooperation of parties to take Joint outcomes rather than maximizing utility for a single individual and we need to have a new way of recognizing that value is created in in a really scientific incred sense that bio is recognized when value is recognized through a bio capacity and diversity and resilience have that be a notion of value not just to something that maximizes a price Arbitrage but actually creates new kind of value the other important thing I think is is coming into focus is this idea of agency um and this is sort of A New Concept that thinks that ah nature should have agency we should have an agency and we should have agency of of of thinking of nature is being given the capacity to be itself and species to be itself and people to be themselves and not have that be restricted and with that I think we comes the notion of what we call sort of homeostatic fungibility in other words how something is is value of that realized is in relation to keeping other things Al life if you look at the the human body it's like you can't maximize the utility of the heart against the brain or the lung they all exist together so how you allocate resources in the body is really maintaining the balances between those different kinds of organs and that's true of Nature and that's true of organizations what we need also is a sort of scale-free and domainfree metrics that's scientific and adaptable so we really need to have the scientific process be a fundamental organizing principle by how we recognize and and and and and and exchange value um and this means having something that's really is transparent explainable you can't have something that's a black box it's not opaque that you don't know where it comes from you don't know how to trust it you don't know how to derive it it really needs to be something that's transparent and so we've seen people talk about the blockchain and things like that but it's more than just a blockchain it's a whole set of methods being able to see how something is derived and be able to replicate that and then the idea of scalable Learning Systems should learn and they should improve over time markets don't necessar do that they just they actually they they they they have clear a market around a price and then they start again there's no memory there's no learning how to do something better for adaptation as as a people and in the planet we need to learn we need to learn improve upon ourselves and that should be part of our whole economy not something that's tangental to it and finally the whole idea of governance there's governance should not be something a app after the fact or top down it should really be embedded in the whole system of itself and so this been a lot of talk about decentralized governance and decentralized system decentralized autonomous organizations but a lot of that is just talk they're not really fully autonomous and not fully decentralized but how do you really realize that that vision and these these ideas have really sort of were first I think articulated recognized um by The Economist uh elanar a who got a Nobel Prize for her work um on the governance of the econom and she was the first woman Economist to do that and and I'd worked with eler when I was at the Harvard the Berkman Center and looking and and trying to get thinking about a digital common sent it was too early but here are some of the central principles that she had and this is back then when the technology was not very developed but she developed her thinking based upon how indigenous people managed their resources their common resources and that would be their their their their forestry the Fisheries the things that they've done for thousands of years and they're able to keep them in balance and keep their lifestyle in Balance through that and here were some principles she extracted from that so was clearly defined boundaries um and you needed to have a a common po resource where people recognize as a common po resources you have to have it congruent with local conditions and I think be that in other words you have to recognize each little bio region had its own PE uh constraints and and and issues that you needed to recognize and you had to have ways of people could make Collective choices among up Commons how bring people together they couldn't be anonymous different one they really had to be they had to know each other and they be accountable to one another and you had monitoring process you had to monitor it and be able to see where you were and then if something goes wrong you have what she called graduated sanctions which means it was you you the punish was was commensurate with the degree of the violation but you really didn't want to have a highly punitive system that its own negative results and you had various mechanisms for re resolving conflict um and then I think the the the idea of one of the things she her concerns with particularly with indigenous people is that the legitimacy of those organizations were recognized by the larger nation state that that that they had some genuine sovereignty and integrity and finally this whole idea of nest Enterprises so recognizing it wasn't a strict hierarchy there was things that dynamically interacted at different scales and these were ideas that were sort of more metaphors at the time they were aspirational metaphors was very hard to realize in any kind of operational form and she went on to talk about later work and social ecological systems and recognized the biophysical and the social factors how they all work together and V and how you look across different spatial temporal scales and different hierarchies how they're linked um and she was always starting to think about what were the dynamic complex systems that can characterize by adaptation if you're going to have governance but where there really wasn't a mechanism there for for how to realize those ideas now there's another more recent development uh is actually it's called tanu that was started by a friend of mine um name is Jonathan lagard in Africa and they're doing this in Rwanda and they're doing it with all kinds of species are they starting out with the gorillas or one but the idea is is that animals species contribute to the environment and they should be paid for the value that they created so how do you do that you're giving agency not only to the animal but to the to the habitat in this case in Rwanda they want to give not only agency to say gorillas Rwanda gorillas but they also they want to give agency to a rivers in in certain bio regions and by agency the the fact that they these things is staying alive can be valued for their capacity to stay alive and create a habitat that creates a a successful and and resilient environment um and so this is a quote from uh their their literature their website which I recommend you taking a look at and exist with the belie that for these species leave they must actively participate in the economy so how do you create a new economy that includes living things and and and how do you how do you recognize that value another approach is something that's come from regen networks which has been working on this for a number of years and I've worked with them and they're talking about creating ecological institutions and again it's like what is an institution an institution in in the classic sense is separation of people from a set a set of principles like the rule of law you say well here's an institution and it's set up to do these things and people should perform certain roles in an institution um but the problem with institutions is that they're run by people people get corrupted and they institutions don't work in this case what they're trying to do is develop a constitution for they call an ecological subject again is giving agency and recognition to Nature um and then they have a number of of of methods by which to to govern that and put in a ledger create some token system Region's been doing this kind of work for quite a while all part of it a common effort another effort is is something called a bi Regional H this will be my last example before making the transition um but the the the idea of a b Regional Hub is that to pull together different parties in a particular geospatial location and be able to give them a a a set of finance and incentives by which they can act to actually create the bio capacity of that that particular Geo region and so it's really working with core communities that that are committed to actually en enhancing a could be a river a river system it could be the one of there they're looking at in Northeast uh bio region is the Connecticut River how can you bring it back to where it was with abundance when they had full salmon they had was a very rich ecological system but there are other things in terms of land owners and being able to give them the incentives to rewi their property and actually make a living on that and create an economy an exchange of value in such a way that makes it robust and resilient um and so they're working with different kinds of investors and social impact investors in order to make this kind of happen so this is part of a larger collection of people who are trying to do these things now what we're going to say what I and this is a very cred I apologize for the the verbiage here but the we're set and and what you're going to hear in this the subsequent discussion is this idea of a biofirm is a bio which is is a firm that is based upon homeostatic principles and it really I think addresses the many issues both operational and aspirational that you see in the whole biof Finance movement and and and to to to run through this fairly quickly but you're going to find Andrew and Daniel going to talk about these things in detail with examples is that is first principles based it's grinded in scientific method Bas in statistics and markup blankets in other words you drill down to it and you really understand it rigorously as a science you can understand where what it's based upon and why how things work in a principled way it's also computationally tractable at scales and and with free metrics for doing uh the whole kind of reporting uh and and uh that you you have this required environmental reporting they call mrvs um sort of uh and so do you have a method by which you can make statements and claims about the state of the environment that is scales that scales and and and can be used for cross comparisons it's also going to be its open source so the bar form is like a kernel think of it as a Linux kernel for a new kind of organizational structure can then generate a new kind of economy that's Biomet biometrically based another key component is if we're going to scale our capacity to deal with nature it's going to be a lot is going to have to be automated it's there's so many tasks involved in different parts of the world that you have different skill sets how do you how do you automate some very complex things in a in a way that can be trusted and generate metrics that other people are willing to accept this has been a big problem in climate change and this is where the the extraordinary automation of cognitive tasks through through large language models Transformers and agent architecture allows for an automation of many many of the tasks of collecting and analyzing the data and learning from the data this is very significant um and then you can create you have the capacity to have create networks of Agents working together and create a system exchange so that's a Natural Evolution that's happening into edge-based Computing that you can have Computing on the edge can security Edge that's another phenomena and then I think you know the the idea of polycentric governance this is not is made of that but what does that really mean well really means if you look at agents that are self-organizing agents and dynamically relating to one another on the basis of the free energy principle these other principles you have the effect of of of a polycentric governance um and you also have the ability to formally give agency to Nature to different species to different peoples to different kind of competencies so that is the whole nature as to say of of an agentic architecture um and then I think you you have a principal means of interfacing between a biometric economy and the the the current economy that's going to be really important so you can't just live in your own little world you're going to have to Interchange and with the the Fiat economy of the world in such a way that actually protects and and establishes the value of a biometric economy and finally I think it provides a a a principal basis for valuing nature-based currencies and assets that we can justu a little bit in the time to come so that that is that's basically uh my my comments I think that I like to end is said let's make it real that we're actually I think you'll see in the following discussions we're at a point where I think the technology is getting sufficiently mature to actually engage in developing use cases of this and bring it to scale so with that I I'll hand it over to uh Andrew thank you great thanks John um quick mention to Daniel uh John and I had a a really nice conversation prior to this if it's possible uh I might actually start my talk now uh and then yeah perfect I was hoping that would be the case and then that way John and I can also have a little bit of back and forth potentially during the course uh and obviously Daniel you've had so such a big hand in the development so anytime you know yeah would like to chime in please feel free cool um all right so my original plan talk for today was entitled uh let me get rid of this guy uh active inference is a framework for social science-based uh modeling and okay um something happened on the way to the Symposium yeah exactly uh huge updates uh I I want to just give a little bit of context uh for some time now I've uh you know I personally gave a talk a couple of months ago a three point a three and a half hour Workshop basically at an International Conference for computational social sciences uh hold at held at University of Pennsylvania uh July 17th this year and uh what what I was trying to do was basically teach uh computational social scientists that means economists sociologists political scientists and so on who are very Savvy with computers essentially um how we might be able to employ active inference uh in agent-based modeling which is already you know a decades old at this point uh method of Designing uh uh basically simul ations of Agents inside of environments where agents might be you know human beings and an environment might be you know a market system or or some other kind of environment that you know uh humans find themselves in where you can you know you can play out simulations test things and so on I want to introduce that to computational social scientists and I think more broadly what's what's great is that this Symposium now gives me the opportunity to kind of flip it to where I can kind of bring some of what comput ational social science does to other active inference researchers um but to stay on with the biofirm uh I really wanted to again as Daniel said something happened on the way to the Symposium uh I've had these conversations with with John since uh since I did that conference and uh I have a lot of kind of strong intuition that a lot of what he's been describing is is a really wonderful sort of direction for future work in both agent-based modeling but also applications of that kind of in the real world um I mean we're always operating under conditions of you know uncertainty and and trying to figure out kind of what to do in very adaptive ways where our environment is is constantly changing uh from you know implementation of computers social networks uh we have climate change we have uh the typical booms and busts our economic systems and so on so um so I'll I'll here but I just want it to be known like there have been a lot of heavy revisions to this with a lot of development we've made on the biofirm D featuring biofirm uh so again I'm Andrew pase uh I my day job I work as a data analyst scientist engineer always many roles to fill and so uh I I bring a lot of those kind of skills with me to active inference and how I view things like developing um kind of applications uh for specific use cases and things like that I think there can be a lot of Rich uh interdisciplinary dialogue between you know different fields and that includes the kinds of things I do on a day-to-day basis versus uh my interest in active inference which has been going for the past few years now so I've I've been a an intern and I I suppose I'm feeling much more like a like a full-time uh kind of affiliate of the institute at this point I've been teaching the the tech Facebook group um with Daniel co- facilitating that I've been involved in various projects uh and so and then now this kind of partnership with with first principles first and what uh Dr Cliff and jur is trying to accomplish and and developing the biofirm so I just want to put uh you know on this title slide here these are actually links to you know if you would like to get involved with the active inference Institute with uh check out all the kinds of things that are being put out by first principles first and also I just want it to be known like you can come back to these slides as needed as well as any of the other kind of learning materials I've been producing whether it be for the the conference that I did a couple months ago including on the code uh how to actually create active inference uh Agents from scratch uh moving up to multi-agent simulations and so on I also did a a talk with the Institute recently that kind of walks through some of the some of the code so um the the new outline for this talk we're going to touch again on agent based modeling which we already talked about yesterday uh a little bit in h the first Workshop so I'm we're going to kind of skim through that a little bit but I W to I want to put some special attention upon the aspect that uh of Age based modeling referring to what are called micro assumptions about Behavior Point Blank if we're going to simulate humans we have to include certain assumptions uh about how we think humans behave if we're going to simulate them in an environment right otherwise we just we we can create a list of humans they do nothing until we give them some kind of rules or behavioral you know Dynamics or something along those lines so those are essential uh for for setting up agent based modeling and then uh and then we'll shift over into active inference we'll look at the PDP as a kind of Exemplar model talk about some agent multi-agent examples just to give a sense of what kind of work be done in this area many people watching this now who who are already involved with active inference there'll be some aspects of this that are you know might come off as a little repetitive but I just I don't want to I I I want to stick to the the textbooks so to speak right I want to kind of keep this grounded in a sense that you know folks who know about active inference don't know about active inference have some vague interest in active inference in this in between area like can catch on to the material here um next we'll go into this question are llms agents uh there's a lot of exciting work being done with llms uh they're showing incredibly uh you know exciting and surprising uh you know sophisticated reasoning uh skills and capacities for dealing with natural language that said um you know and it's already been said so many times in the other talks given in you know during the Symposium but it's worth repeating like they have limit there are issues and I and I've been doing some work on thinking through how do llms relate to things like cognitive theories what how does an llm think or how how does that work really so that we can really grasp if we want to incorporate LMS into state agent based modeling software applications or otherwise let's let's come to terms with like the reality of what they are before assuming too much about what they are or what they're capable of and furthermore how we might be able to integrate them into kind of more grounded means doing agent based modeling such as working with actual agents the quick answer to our llms agents agents are they agents no that that's the quick answer they they are not uh that's what I will claim and then we'll move on to the biofirm which frankly will take up the majority of the rest of the talk uh from there so uh what is agent-based modeling uh as already stated we have agents more specifically uh you know it's a programmed entity it could be a person a living organism it could be representative of an entire institution uh or otherwise uh and basically programmatically it receives observations from its environment and then it returns actions back into its environment for example an agent could be a person an investor and then uh with the environment uh the the environment could be an artificial or real uh system that elicits observations for agents and then receives the agents actions uh those actions then impact the environment itself in turn for example the environment could be a stock market so we see that there's this reciprocal relationship between the two things an agent or environment's action is an environment or agents's observation so the agent sees a price signal from the environment they make a purchase uh next morning prices have adjusted given that agent maybe other agents purchases you see so there's this kind of back and forth in a loop uh you know certain time intervals such as in time steps or we could imagine them in days like on day one this happened day two this happened an active inference the agent environment's respective Markov blankets which kind of Define what it is to be that entity whether it be the agent or what it means to be that environment including all the the the structure uh the characteristics you know if if it's person then maybe it includes things like their goals or or their um uh what it means to maintain homeostasis so an individual has a you know blood and oxygen flows in their body you know what does it mean to be that entity so from there we simulate them we play out the agent environment Dynamics over time again as I said in time steps some other kind of interval and uh and in active inference we often use the phrase action perception Loop it's like a you do something then you perceive the outcome you elicit your next action you know receive the outcome and then you know arguably this happens at a very rapid Pace in actual human beings right this is goes beyond just time step one or two or looking you we measure this in in in Nan seconds or otherwise so core guidelines setting up agent based models we want to make realistic yet simplest possible agents and environments that has to do with you know uh generalizability if you make something that's so unique to a particular situation it might not generalize well uh you know so if you want to make an agent agent based model of say uh you know agents taking care of a natural environment if you make the agents way too specific to that environment to where they you know if you if you want to set up this agent-based model get insights into what you can do in reality you want to make sure that what you test in the model can actually be carried out in reality it can't be too specific it can't be off course um robustness simply small changes don't dramatically change results uh so you know if you fine-tune something too much such that you know the the the results of the um experiment end up being dramatically different and suddenly once again we have that generalizability problem reproducibility everyone knows reproducibility problems exist in modern science we need to document and publish uh you know our codee make it available to other people so they can understand it recreate the results Etc and finally usefulness which is simply why do this you know if we can't if we can't get any kind of additional useful insights out of setting up these rather complex computer models and and programs um you know why do it so it's it's worth having that in mind you know not not all models are right uh but but some models can be useful right in the stat George Box's words then um purpose so we do agent based modeling in order to predict outcomes we could try and extrapolate from the present uh we can entertain different kinds of hypothetical scenarios design new policies and and have them play out over the course of the simulation to see what the outcome is and then we can also look at Collective outcomes of aggregated individual behaviors if all uh you know stock investors in my simulation All Began investing in this particular stock like what happens by the next day do we end up with some kind of bubble and then bust kind of situation like what what occurs you know that that kind of thinking and then over the past couple decades and this is really what I want to highlight a couple months ago and what I would like to highlight now again and I think it's especially relevant for anyone wanting to do agent based modeling and especially folks who are interested in active inference there's been a cognitive turn uh you know a lot of these traditional a based models you know they're very you know very interesting a lot of thought has has has been put into them but their agents tend to be very simplistic um in part because you know 40 years ago we were working with uh you know much more at this point archaic uh computers and computer architectures and and devices that simply could not run uh simulations that involve many interacting agents who all have highly sophisticated you know Behavior furthermore it's called a cognitive turn not a complexity turn but a cognitive turn in the sense that we're looking at actually more closely approximating human behavior so that means actually looking at other fields ranging from you know Neuroscience cognition psychology social sciences that means looking at uh you know cybernetics uh biology you know trying to open up the the discourse and the dialogues over how can we more better approximate human behavior so there's some common characteristics for these calls to cognitive agents that includes like having inter you know agents have beliefs um they update their beliefs based upon what they see they have to infer what to do in a sense that they're more autonomous they they kind of make their own decisions traditional agent based modeling it's like well if if if if the agent sees X then they do y otherwise they do Z and these highly simplistic you know programmatic rules that you know the program you know I program them that way so I already know that they're going to do that uh whenever it comes to agent based modeling with cognitive agents I don't necessarily know exactly what they're going to do they kind of make their own decisions right they have the architecture for that and then also reinforcement learning which is kind of a you know adjacent or subfield uh in some ways of like machine learning data science you know a lot of these kind of state-of-the-art but still rather standardized at this point ways of doing modern uh data uh you know processing and analytics and and predictions and forecasting uh reinforcement learning is the way of doing that where you actually have an agent all right so um so that's been kind of one of the main ways people are trying to to continue this cognitive turn and agent based modeling now micro assumptions there a lot of information for a single slide so you know it'll be over soon but I I really just wanted to give um you know kind of a picture of there there is so much research that's been done so many more kind of dialogues between fields that that can occur uh here now um like when thinking about how to make cognitive agents what makes sense you know what what what's realistic but what also balances complexity and accuracy keeps things simple enough to where the agent-based model can actually be useful so there's been a lot of criticism at this point centuries criticism of this kind of neoclassical tradition classical than neoclassical tradition Adam smithian kind of tradition uh that tends to paint humans as utility maximizers you know it's often criticized because it's too simplistic it reduces people to being uh greedy or immoral you know as if they're just going after you know profit without any concern for uh any other you know condition or needs or wants or you know um so is kind of selfishness right without any form of Al altruism or sense of community um some people point to Economist Lionel Robbins you know kind of a classic in the field uh who claim that unlimited wants scarce resources that is the fundamental economic problem yet this assumption I think this gets overlooked I also criticize the utility maximizer assumption but I I want people to be aware that this this leads to a a really useful tool which is for agent based modeling you can severely reduce computational costs for putting together agent based models if you just have a bunch of agents who are trying to act to maximize a singular value there's not a lot of complexity there you can you can run those equations quite quickly right and then that also assumes that the main thing in your environment is the singular value that is utility or a standin might be uh you know money or or some general notion of liquidity and so since then we've had so many other ways of looking at Human Behavior Uh behavioral economics and finance psychology cognitive science social theory you know humans aren't just trying to maximize utility they're they're they're speculative they don't have fixed preferences they actually look at one another's decision makings you know this goes back to you know John Mayor KES we had h himman p Minsky who looked at uh you know Market crashes we have Robert J Schiller who's alive today in writing on narrative economics the idea that you know ideas have an impact on how we uh interact with our economy and how how we behave um there's economan thinking fast and slow like sometimes we just rely upon horis sixs or maybe we do it a lot of the time you know it's quicker it's it just um it it mitigates having to put a lot of thought into things where we just kind of want to get something done even though it's not necessarily maximizing utility um there's fance sunstein nudge it's like sometimes yeah you do just go with the default option some people whenever a website asks them like do you accept all the cookies they will click yes quickly just because they want the thing off their screen right um so and then we have social emulation am meis convergence ideology all these theories um you know what is what is money it brings up the question of what is utility what is money currency actually isn't just a means of exchange it's not just you can be a store of value as its own value changes to to you know TR try and do kind of economic transfers to make sure that you get higher value out of whatever form of currency you're holding uh it's also a social institution though it's much broader right so be having access to money as many of us know allows you to be uh first you have access to a variety of Commodities assuming you're in a market economy where many things are fungible and tradeable in that way and then we also have social belonging it's very difficult to maintain yourself in a society where you have no money right so that's a it's a whole dimension that that is lost we just look at something like maximizing utility uh you know I've put a mention of Lou alus which is a much more kind of critical theory approach but it looks that the impact of state law uh you know different forms of coercion upon people to to act and behave in particular ways that go beyond just their ability to to try and maximize their own profit the pragmatism people update their beliefs they're forward-looking they don't maintain the same preferences and ideas and rules uh you know throughout history you know they change they have their own circumstances and so there's a kind of pragmatic uh approach to life Carl pan economic activity is embedded in society uh meaning that a lot of the AC economic activities that we have you know they themselves are predicated upon our ability to trust one another work with another and all these other kind of things where whenever that trust is harmed it can dramatically change how that economic activity plays out and then we have Neuroscience biology physics and that you can see how we're moving more in the active inference Direction now but you know that we have these are predominant theories now in Neuroscience the idea of predictive processing that you know that one comes to a situation with their prior beliefs and they they they can be updated in the sense of the the Basi and brain hpis a kind of way of mathematizing that that process we also have and this is quite old now but it's still still stands as a rule like heian learning or associative learning you know so someone who Associates money with being something bad very well might not you know have such a drive for money versus relative to other things um and then homeostasis allostasis allostasis being planning for homeostasis or homeostasis in the future these are obvious things for us that we deal with on the day-to-day we need to fulfill our needs uh physiological needs neurobiological needs uh unconscious processes in the brain you know that goes beyond being able to say that every single thing that we do is necessarily conscious or deliberate right I I'm not actively trying to think about how to regulate my blood or oxygen flow right now uh we also take preventative actions you know we do things that might be a cost right now but they change you know the impact of whatever happens in the future for us uh we are for looking in in various ways and we do often rely upon habits as needed uh to fulfill our needs habits allow us to do things without having to put in too much expenditure mental expenditure you know metabolic expenditure to ensure that our needs are going to continuously be fulfilled and cybernetics embedded cognition uh you know PE this is the idea that behavior and even intelligence arise from interactions between brain body and environment which kind of overcomes some of these uh ideas you classical ideas of a cognition mainly being outside in where you just kind of look at what's around you and then you you take that in as a passive Observer um instead we're actually always in communication with what's around us a mortal computation in the sense of we can move in the direction of you know if we want to do something like cognitive modeling then we're looking more at uh you know actually trying to mirror these kinds of processes they're much more realistic C collaboration imitation computational Neuroscience Psychiatry and modeling modeling internal Dynamics in form of you know time series neuroimaging data uh you know applying different kinds of techniques like causal inference to see how one factor impacts another one variable impacts another um actually trying to find correlations between internal neuronal Dynamics with actual uh behaviors that occur uh you know so so for example sticking you know kind of electrophysiological like electrodes on you know a mouse and a in know tze to see like okay what kinds of neuronal Dynamics correlate with you know the rat choosing to go left to find the reward uh you know find the cheese or otherwise and then and then taking that and trying to move in the direction of you know creating an agent who who basically is fit to the behavior of that that mouse or rat in the t ma right and that's a classic kind of example that's looked at in active inference and we also have the hamiltonian principle of least action much more in the realm of physics um and and it's it deserves a whole you know slide or 10 you know on its own but the the main idea figuratively is you know the idea of something occurring uh based on following whatever the easiest path is from one state to to another so a lot of these things the idea of you know social homeostasis uh you know relying upon horis sixs versus planning and forward looking versus dealing with whatever's going on in the moment just because you need to figure out what to do now even though it's not something you plan for all these kind of aspects start to come together I argue an active inference uh so it's kind of like you know is this should we be considering something like com economic is Revis you know again liquidity is the means for all Commodities at least in market economy at least assum assuming fungibility of things that liquidity can get you many many things in a certain way I almost want to dial it back on saying it's immoral to want liquidity uh it allows you to fulfill your needs again in a market economy by seeking out Necessities for your own homeostasis or allostasis you know physiological relief or otherwise even then even as your bank account increases that doesn't necessarily mean you're going to have your social needs met kinship or familial needs met physical health uh you know if if a doctor tells me after I've suffered some kind of injury that I actually need to do stretches now in order to make sure that my arm will move going forward in the way that I need it to it doesn't matter how large my bank account is I'm still going to have to do those stretches right um so there this a simplistic example but all of this is to say uh you know for agent based modeling rather than thinking of utility especially utility in some vague abstract sense of you know is it money is it liquidity is it something else there could be a shift to a kind of alternative uh you know what is essentially a virtually textually Universal value computed by the brain which can Encompass not just to drive to liquidity but other aspects and means of Life simple as possible yet no simpler and then it itself becomes a simplified recortable rip metric for measuring one's capacity for current and future survival uh and so that you know enter the free energy principle which is one of the central principles of active inference uh here we're minimizing quantifiable uncertainty uh you know that by minimizing this it allows us to achieve our goals fulfill our needs we learn we actually change our minds you know we learn new information we are not fixed beings who you know are are are kind of one-dimensional we update our beliefs and uh free energy is a proxy for uh uncertainty or surprise you know it's it's computed quite easily uh you know because it's a it's a variational quantity there's a approximate computation which you know in in physical living beings that keeps our metabolic costs low uh and and for agent-based modeling that also means keeping computational costs low and so it still kind of fits you know that the the the usefulness of what utility was for agent based modeling I think that free energy can instead be the new kind of metric that we're looking at um and you know I included this kind of a you know fun fact that there was a nice talk given not too long ago with AC of inference uh Institute uh you know some folks who do neuromorphic Computing just reminded us like despite how many calories their bodies use on the dayto day the brain only requires roughly four bananas worth of calories despite the fact that it makes 10 to the 15th uh calculations per second it's just a reminder of like this this variational aspect fact than being able to do quicker approximations and then um you know we talk a lot about free energy but I just I really you know in the in the kind of funniest kind of way I just uh have always been fascinated by this equation uh this is you know textbook equation uh from active inference but you know so G is our expected free energy related to deliberative planning goal attainment information seeking conditioned on our homeostatic preferences right so we plan for things take care of ourselves to to accomplish things we need to F is variational free energy uh another quantity Where We balance the complexity and accuracy of how do we deal with the moment then we can think of this as like you know say you're camping in the woods and suddenly there's a bear it's like well you didn't plan for that uh and you didn't expect that and coming up against a potentially dangerous bear is probably very much against your preferences for maintaining your homeostasis in this case your sense of physical safety so you have to to deal with that and then e uh you know those are your habits your horis so so what

I'm saying is that I just I I'm particularly appreciate that we we have a quantifiable way you know here of trying to decide what uh you know what it is that determines someone's actual Behavior you know that the posterior beliefs High meaning what policy or action should I take right uh all of that is kind of all all these different considerations habits deliberative planning momentto moment planning all these things are are involved in that so it's just it's kind of it's it's an attempt at from my perspective like we're actually combining many of these inter disciplinary perspectives into a single equation there and so active inference I think I I I appreciate his attempt to try and do this kind of mapping of you know sub you know neurobiological substrates to actually interpretable cognitive theory regarding what is a habit how does that you know what what composes that in the brain what are goals what is planning what is action and then from there moving on to uh you know these kind of Exemplar uh initial active inference uh models which themselves are highly relatable to to reinforcement learning so it's all to make the case that active inference I think it draws connections and kind of builds Bridges between all of these U you know these other disciplin in a way that that is very rich and it's not so uh exclusive right it doesn't it doesn't stop conversations it starts them so and then I also wanted to show like here is just a quick list of some multi-agent simulations with active inference that I think you know heavily relates it is Agent based modeling you can open the the textbook from a couple years ago agent based modeling is there you know it's this isn't foreign to to active inference but just I hope those in the agent-based modeling Community can start to see how strong these relationships are with active inference so um epistemic communities under active inference is this uh paper was um you know recreating agents who hold their own opinions uh about things and they're they're in social networks so you can study how they share beliefs with one another and yet given you know a lot of these kind of formal principles and active inference seeing how that plays out we can actually see what the kinds of dynamics that we start to see on Twitter and other social networks where you know opinion polarization begins to happen depending on uh how you know how the network is set up in the first place so um you know it's just it's very interesting we also have active INF for ant this is much much more kind of a biological or Zoological example uh you know where there ants in a colony who who uh have to adjust to changing context such as changing food location and they leave one another pheromones so they can um actually follow one another's Trails so to speak to find the food and come back um and then this is very recent but uh we had another talk earlier today but the um this paper with Connor Hines who is one of the developers of pmdp which we'll be seeing shortly uh he and you know frisen and others are are developing further tools for doing active inference and this kind of gets at the question of like nestedness and and looking at how um you know whenever you have a bunch of individual agents what where are the cases where their Collective Behavior actually you could fit that behavior back to the individual parameters that each of the individual agents hold um and then again with social science like looking at agents who have distinct roles uh such as leader and follower where or it could be you know you could apply this logic to many other situations the idea is just looking at how agents who have their own distinct uh you know uh uh available actions and beliefs nonetheless uh you know whenever they're kind they can kind of synchronize in their capacity to fulfill uh you know a goal that they collectively have um you know so it kind of speaks to the idea of having heterogeneous agents with heterogenous you know concerns but nonetheless with an umbrella Collective concern they're able to accomplish their goal um this next one uh rather long title active inference framework for Collective parallel problem solving on ink Landscapes this is uh you know I gave this at the tutorial a couple months ago um I just recreated uh so I'm this was the point I was trying to make you can recreate a lot of these old traditional agent-based models you know using active inference agents that's what I'm claiming so so I'm not trying to say oh all of these traditional models are obsolete uh not at all I think there's so many fasinating design principles I would love one day to recreate the uh the Santa Fe institute's artificial stock market but with active inference agents right we can do that so um so here I just recreated one where you have agents who are collectively trying to solve a problem represented on an Inkay landscape meaning that they're trying to solve a very complex problem where there are a lot of interdependencies in different subparts of the problem you know it's like uh trying to uh trying to design a car it's like well you can have a great car but if you remove the wheels it's no longer going to really be the kind of car that you need right or you can remove the steering wheel right you you need to have a certain combination of pieces there to have a proper solution so agents collectively do that and you can study like well how well do they do that do they get stuck you know with a suboptimal solution uh and then furthermore you can look at metrics like oh if we run this in different ways uh what is the connectedness needed uh to for them to reach the optimal uh um solution or otherwise so and then finally this is not an active inference uh paper um but this is this is just a a reinforcement learning uh you know based problem but I I included it here because I just want to make the point that this is this is focused on industry right this is focused on multi-agent settings where you can have them kind of work in M manufacturing firm that involves like making a lot of kind of Rapid decisions in order to manufacture particular products how you could have agents kind of synchronized in a way that they can accomplish you know that goal and uh again similarly with with traditional agent based modeling where you could you know swap them out with active inference agents so to speak might be able to do this you know take that same approach with looking at how people are using reinforcement learning today I think it lead to very interesting results so I just again just trying to bridge these kind of uh you know dialogues between people so now we'll move to LMS as agents um this I just think it really needs to be uh addressed um so llms have been uh increasingly integrated into a wide array of potential use cases uh you know I I included some resources here like there there just uh you know dozens on dozens of of research papers that are being published uh that try to use llms in these kind of agentic ways where it's as if an llm is an agent who you know supposedly makes decisions autonomously and so on um again they're showing more and more sophisticated reasoning capacities um they're they're they're having uh kind of longer uh context windows and output Windows meaning you can feed them even more information at one time they can release even more or output even more information at one time uh you can also upload your own uh text files documents write your own prompts for them to respond to so in that qu in that case if an llm uh really is an agent then why not just skip all of this stuff with reinforcement learning with active inference why not go straight to using LMS as agents of course I'm gonna argue like we should not do that that be um I mean it would it would be IR responsible at this current moment in time in terms of llm development I I would I would argue that that would be kind of responsible unless you're just kind of making you know some kind of you know entertaining or or fun app that uses llms or something but for for something that that that calls for you know much more discretion I would not do that so what what are the biases here um uh llms are trained on large corpuses of existing text we all know this uh what that does is it kind of sediments a series of cultural scientific explanatory rhetorical Prejudice information figuratively the snapshot of the retrievable Web scrapable Internet other digitized human and AI produced texts to which uh the lm's internal parameters which are you know blackbox parameters now we have llms being trained to where they have seven plus billion parameters on the inside um you know and then consequently the their outputs are just kind of reflective of this so so a lot of these uh you know prees prejudices have been addressed you know very obvious in a certain way very unfortunate things that we saw with initial llms were like racist you know or or sexist or otherwise kind of outputs coming from them based upon the materials they were trained on people are trying to find different ways of of training them differently to to get them to elicit the responses they uh that we want that's the thing what's crucial when working with LMS is the prompt itself so I you know I I want to think you know just put yourself in the chair you know facing a computer and you're using an llm what's really going on you as the user you prompt the llm after it's already been trained you have your own purpose or goal in mind that the llm does not hold for itself quick mention I'm aware that I'm doing some slideshow karaoke I just for anyone who's like but uh just want you to know there's a lot of information I want to make sure I cover it thoroughly uh so so the user must provide the context for this goal uh or purpose and their prompts in the hope the llm will catch on less figuratively they hope that the llm will be able to predict the series of words and sentences which correspond in the user's final opinion to the answer the user needs or wants right and so after training the LM it can't actually learn what the user needs it's it's always in the user's opinion and we'll get more into that again it's just making the point The Prompt itself is incredibly important this is kind of a a simplifi uh ever so slightly exaggerated uh you know illustration of the kinds of things I'm talking about so if I go to an LM and say hi my name is Andrew it says hi Andrew it now knows my name how can I help you today uh I I ask it help me prepare you know I have a certain goal in mind I tell it help me prepare an initial draft for a proor of business model you know for an idea I have uh you know here are the following details that should go into that that business mod uh model then it produces it for me and today it can even release it as a text file and so I could click on it download it great now what happens whenever our conversation history is deleted what happens whenever we exceed the context window that is repeatedly being sent back to the llm every time I prompt it um we have context loss so whenever that happens the llm you know I say okay now please help me revise the file please provide the file you mentioned it does not recall anything that came prior right what is my name your name is me which is is a slightly exaggerated example but it just really wanted to make the point like it will use its reasoning where it's capable right and and and confabulation uh is is a kind of a technical word LMS hallucinate in the sense that they can they can kind of reason things in a way that they end up coming up with false information you know um we have to be careful about that right we have to be very cautious about a lot of these characteristics of LMS they effectively I mean I would highly relate it to someone who has interr Amnesia right it's where where you know someone cannot actually recall what has happened uh in the recent past and nothing you tell it actually gets committed to long-term memory and so I've kind of crudely used the phrase like the LMS are kind of like Talking Heads uh you have to repeatedly tell everything and every prompt and what's going on when you interact with chat GPT is that a lot of this conversational history is being stored in the site in some kind of cache and being resent and you're not seeing that process because they want it to be a nice user interface to where you can more easily use it so recent agentic Frameworks and workflows abound in recent Recent research get the issues presented by limited context Windows confabulation Etc still abound elens with these conversational histories have been considered to be age in various cases yet in order to try and overcome some of the issues that I pointed out there are a variety of Frameworks that are being developed you know so one idea is an actor critic framework you could have two or more agents you give them different role prompts you tell one of them oh you're a creative you know scientist and the other one you say oh you're you know you're a harsh critic and and expert in literature on uh this scientific topic and then can have the first agent try and innovate and come up with something and then the the second agent critiques the former agent you know so you have this kind of way of like oh in order to better ground the information that one of the so-called agents is is creating you can um you can use another agent to kind of critique that make it more uh yeah grounded to reality um another uh way of doing this is building up databases of retrieved documents that is you could you could it's it's an attempt to overcome the limits of context windows so what you can do is you can repeatedly query an llm and any true information as well as any information you provide yourself through your own documents your own promps uh you know any kind of formal documents you want to submit you could kind of collect these in some kind of organized database and you could query that database in a way that repeatedly send into the context window here's only the information that needs to be sent right now based on everything I have and then that database kind of acts as your ground truth um that that's a means of doing it it's it's rather sophisticated there are different ways of doing that we'll get to that soon and finally there's a thinking or reflecting framework I'm just spending a second on that it's where you have an llm uh come up with things but then have it revise its previous responses U very similar to an actor critic Frameworks it's just you're viewing it as one one so-called agent doing this as opposed to two um in all cases the prompts are still incredibly important so what's in a prompt well uh there are many people who have now you know uh coined the phrase prompt engineering which is at this point has kind of become a field or even art in and of itself this is a recent recent systematic survey prompt engineering uh that that covers literally dozens of techniques uh for anyone who's you know interested in that um there are many ways of doing this and furthermore some other problems include uh difficulties handling numerical or symbolic problems so intuitively you know if you read 2 plus 2al 4 is a string as opposed to being able to recognize oh these are two numerical values uh that need to be added be an addition rule to compute a further value you know it might not be tokenized properly um and so that the LM doesn't actually know how to deal with that despite the fact that to you it's very obviously just a mathematical statement um so llms are getting better at dealing with math much better actually but they still risk this kind of in Precision so if we want to actually incorporate llms into software applications especially those that have Stakes right those that have real world consequences where you're actually employing them in any kind of like in depth industry related setting in relation to what John was saying in any kind of ecological setting uh you know situation where you'd want to really carefully take care of things uh verify the information and so on uh you need everything to be much more precise and trackable than uh whenever you whenever you uh use llms in your application you could have your application such as you know if you're developing this in python or Julia or otherwise you could have that uh code actually handle the math which is something we typically already do um as well as any kind of like if you're able to hardcode any kind of rules that you can extract out of the text and actually consider those to be kind of you know programmable and then you can have them in in your program and and have that logic be carried out outside of the llm and you can just have the llm handle the natural language aspects um so kind of my proposal and this I don't want to spend too much time on this because we'll see some diagrams that actually represent the ideas here but uh you know I argue that that we should be thinking about things like pre-structured crops with what I'll call empty spaces uh for anyone who uses python it's very simple you just create a bunch of preconstructed prompts that are structured in a way that you can basically Supply uh The Prompt here you're you don't have to do this word for word but you're effectively saying here is what I need here is how I need it how I need your output to be structured here are my current you know thoughts or beliefs of what's going on give me what I need uh but then make it an FST string to where you can have kind of these empty spaces in between the prompt uh to where at any given moment in time say in an agent-based model you can send the most recent updated information through the f-string literal so it's like you're you have a really nice prompt that really gets that what you want the llm to send back to you but if you want to have this run algorithmically uh you know in a way that you let it run autonomously it needs to be that the information in that prompt needs to be repeatedly updated again like like reminding the the person with anterograde amnesia what's going on now then an hour later what's going on then so on um bringing it back to active emerance agents we know they change their minds through perception they act to change the world they actually have capacity for learning including online learning learning in real time you know updating their internal model parameters which might be liken into synaptic learning or modulation in the brain uh you know updating their beliefs about the relationships between observations and states the efficacy of their own actions you know a lot of these other kind of cognitive mechanisms that llms lack so uh to bring it back llms if they were like agents I would say that they're like agents who cannot act nor learn they only observe and infer uh there's been one argument that they do act it's just they they don't quite U you know that because they don't learn anything it doesn't quite matter and they can't really observe that outcome of their their action it's like their action is they output you know words and sentences and and that doesn't actually come back and impact them necessarily in any kind of tangible long-term way um they are typically trained once very expensive process some people have tried to specialize llms by only training them predominantly on medical text to make a kind of a medical llm or legal texts for legal llm you know but still we're stuck in this like lack of online learning or or or something like clear preferences from a neurobiological perspective LMS have very broad yet immutable long-term memory and uh you know they're trained on a lot of information and then they have a highly variable short-term memory their context window which if anything happens to that and it's gone then they've lost you know all context um and so I argue that we should think of uh llms uh you know as as a a language processing tool I think they're very effective at that they're they they absolutely outperform Us in terms of being able to feed them many many texts and then you know quizzing them on those texts just in a plain as day they they far out you know they exceed us and this this part of why they're so popular now and being considered for software applications then from this vantage point we can Envision augmenting agents who learn and act such as active inference agents with something like the capacity for natural language is how I'm I'm currently phrasing it so this is kind of my diagram uh diagrammatic you know attempt at formalizing this for for anyone um these are a lot of the design principles that are going into our biofirm that we're developing right um I I started developing this whenever I considered how could I have an active inference agent write a scientific paper by prompting an llm like on its own right so so each action of the agent was you know uh prompt the llm to find more literature and citations uh another discret action is uh you know write or revise your abstract you know and these other aspects of writing a paper and seeing if it can kind of you know choose for itself what to what to write rewrite update so on and how would that work and so a lot of that thinking has kind of gone into this project um in this case what you would have is you know for for a user of the biofirm they would be able to kind of uh you know submit their own documents PDFs text this is kind of the grounding truth representative of you know the users own preferences their goals what it is they want to accomplish that gets input into structured prompts you know F string literals like the the empty spaces that way it'll it'll basically take your inputed information and restructure it in a way that's needed to get the kind of answer uh and the kind of response that you need right so the documents are sent to the structured prompt uh which is then itself sent to the llm which processes it and returns your structured response structured responses uh you know allows encoding for agent observations meaning you could in a very simplistic example you could ask an LM is this a good idea yes or no and that can be encoded as a zero or a one and the zero or one can then be you know read up as an observation for a particular observation modality that that an agent a PDP is equipped with right that that's a that's very kind of Elementary approach uh but it's I hope it maybe explains the idea for for programmers who who are interested in these kinds of things um and so uh you know that observation is then taken in and then the agent can infer States per usual What It Takes the lm's you know feedback into account it's decisionmaking and then we have expected free energy meaning you know um the free energy it computes condition on its preferences right so it's like saying oh the agent can take in an lm's you know natural language based feedback uh to the users initial preferences as elicited and now natural language and the agent can use that feedback to make decisions in order to follow through on its own preferences or goals right so you have this kind of reciprocal relationship between um the the user and via the llm by curiously to the the agent and then back again as you can uh take the agent's current internal beliefs you know it elicited this policy it it now believes this it previously observed this those can be put back into a structured prompt uh which could then be sent to an llm for a new structured response and that might you know that might be something that involves like reporting like the agent is you know in natural language reporting its beliefs about what's going on and then that can be useful to the user and then be sent to uh what we'll for now call A A Knowledge Graph apologies I just now realize that this is being covered um yeah and so then this knowledge graph and we'll get more into that I was very happy to hear some other people making references to to knowledge graphs during the Symposium um you know it can be done in different ways but a general sense of a knowledge graphs that this is kind of like your database of of knowledge that you're gathering over time in terms of you know natural language based knowledge so it's kind of like you're storing the context of the moment you're kind of storing the truth you're stowing the history of what's going on in terms of these agent llm interactions uh you know and then that way the user has the means of reviewing for themsel in everyday terms what has played out during the course of simulation so that's kind of the design principle and then the knowledge graph this is a way of kind of overcoming The Limited context window issue so in a general sense uh and I'll I'll mention I'm you know a lot of my thinking about the knowledge graph comes out of this really like very recent exciting work uh being done by Marcus J buer uh who works at MIT right now um you know he's been trying to find ways of of using llms to to you know um basically go over many texts in order to find patterns in the text that can then be you know kind of put together almost in this like a sharian like Innovation Way of like uh just recombining materials that already exist in text but maybe humans themselves have not yet made those connection themselves of oh uh we know all of these many different things uh if we look at it this way that actually leads to a new innovation that we could come up with you know uh we we just didn't make that connection before but we did now it's like oh you can have LNS who are highly you know sophisticated whenever it comes to to going over large amounts of tax having them make those kinds of discoveries um so anyway that that's the kind of where this is coming out of um and then uh sources for updating the knowledge in the graph may include user documents up-to-date environment or resource reports agent beliefs domain knowledge regarding to your use regarding your use case llm responses to structured queries we kind of looked at that before um business model as a shared narrative it doesn't have to be a business model but you know it could be you have many in the context of the biofirm which we'll get into shortly um you know the idea is agents are working for a firm so they have this kind of they need to have this shared narrative between them that kind of captures all the crucial core elements uh describing what are the operations of the firm what are the guidelines what are things that they need to attend to uh you know regulatory or legal standards um what are what are the their current financial statements what resources do they currently have available uh what are what are what are the perspectives of all their different stakeholders you know being able to synthesize that information into you know kind of a singular document that's that's short enough that it but but nonetheless uh Rich enough that can be repeatedly sent back to the llm to remind the llm this is what we care about these are our constraints these are our Central concerns right now or otherwise so anyway uh I'm getting a little ahead of myself but it just that's how we can start to develop a knowledge craft overcoming The Limited context window by having a graphical structure to the knowledge you can keep it nicely organized and uh as we know with graph structures with having nodes and edges you can have along the the the edges can kind of be the descriptors that relate you know multiple Concepts as nodes right so you can kind of Traverse this knowledge graph to make connections between all the entities within it uh these are the visuals here actually Daniel created these based on some particular LM uh structured prompts that that he ran and then kind of analyze um and so the these kind of principles are going into it so by using pre-structured prompts and quering our knowledge graph um you know we can repeatedly kind of communicate what is the situ situational context of what's going on uh in the moment as well as here's the knowledge that we've accumulated so far that can help with problem solving during the moment this allows us to filter down only to the needed information and synthesize it reducing the final amount of information uh in this case tokens which also if you're using the open AI API to use the most uh recent like state-of-the-art models where we're usually spending at least a few pennies per llm uh uh prompt or well it depends but the the point is by reducing the amount of information flow needed you can keep costs uh significantly lower as well and then also um kind of uh quickly overcome that issue potentially exceeding the the limit of the context window and I you know I'm I'm interested in finding different ways of describing this I'm a particularly a large fan of Yori Baki neuroscientist NYU who does a lot of work on on neurobiology and and has even worked with friston at a point in the past but uh uh so I I included him to try and give a metaphor of this is this conceptually can be likened to theories of episodic memory in the brain uh for example this this uh one Theory uh that the hippocampus in the brain is something like a pointer so whenever you try to recall you know an episode in in your life you know that involved uh you know certain many episodes like in order to tell a story of what happened yesterday starting from the morning and then what did you do uh you know at noon and what did you do in the afternoon and so on like the hippocampus is a kind of pointer that that kind of kind of or almost like a librarian that goes through through uh your long-term memory to to kind of pull out sequentially what's needed in order for you to tell that story or narrative in the moment um or from a kind of more industrial or a modern data analytics example you could think of this as querying a database SQL or mongod DB meaning could be a structured or unstructured database where a report is synthesized via joining inquiring only the necessary tables or cataloges of blogged information needed for the report so again it's graphical structure that allows us to kind of more quickly get the information we need without having to pull too much more than what we need so now this is example of I'm I'm just currently phrasing this is like agent foraging and knowledge updating so it could be uh starting from the top left we have our homeostatic agents uh John made a reference to them earlier um the sense is that these are agents who have their own preferences for you know maintaining their own home stasis they might be trying to maintain the homeostasis of their environment in the case of an agent who designed to take care of an environment which is what we'll see in the biofirm uh they'll have their own current beliefs they're being updated over time uh you know we saw kind of an image of the the inner dynamics of what that looks like for a PDP earlier they might have their own goals which again in this case would probably align with the idea of maintaining homeostasis you know based on all of those factors uh that might lead to you know some pre-structured prompt leading to something more specific so in this case this is an agent who um wants to take care of an environment uh and this is an agent who specifically is looking at how to take care of the soil on a a natural Farm uh like a land preserver or Farm uh and so they need more information about the soil structure so they query they prompt the llm water causal determinance of soil structure and Farms you know this an initial this is this is whenever you're first making the knowledge graph you would want to have this uh this knowledge at the very beginning of a simulation right so that it's kind of building out your knowledge graph to start so you could ask it this you could tell it to site your sources or you could integrate some kind of API uh such as semantic scholar or other apis try and check if it's actually pulling from real citations again we have that risk of confabulation making up seemingly true information uh and and so we want to uh just kind of keep a check on that so the llm tell you a lot of natural language based information about soil structure the determinance different kinds of manage P management practices that could be used to take care of it uh different kinds of conditions and relationships between the soil structure and other aspects of the soil what's around it all these different things and then from there we could kind of go back to the knowledge graph store this deconstruct this information you know ideally this would be we'd ask the llm to structure its output in a particular way and then we could send those different aspects of its response into different areas of our knowledge graph um so soil structure determinance management conditions and then also track the citations the knowledge graph again it's almost like a threedimensional structure in that you could add the time Dimension that is you could time stamp when all this information is collected uh as well you know so it depends on how sophisticated you want to make your your knowledge graph this is the last diagram I have related to this concept but I just want to include it this is what I mean by um a structured prompt with empty spaces right so we have that same agent from before um now we're sending a structured prompt I believe so this is pre-structured right these these these curly brackets kind of denot like these are one of the so-called empty spaces I believe the whatever your current environment variables are are currently here's what the agent beliefs are but our goal is here's our current business model and our current constraints are here are current constraints what Solutions exist to address this this is incredibly short and simplistic just for the sake of illustration but the the idea is you could input you could you could keep these environment variables stored in you know your local environment you know python is storing and processing them um you know the the agent beliefs are being uh extracted from the agent you could Reen code those with natural language you know you could you could you could include information like oh this agent's belief is representative of the quality of its environment so then this gets read as a sentence that's something like the agent believes with a you know 02 probability or very low probability that the current state of is environment is good you know you could you could re encode that so this is really a conceptual kind of move to try and find ways Bridging the Gap between qualitative and quantitative information uh at no point in I saying that all of this is the absolute best way to do this uh what I am saying is that this feels a little bit like Frontier work uh the llms are bringing up the question of how can we properly integrate integrate them in the software applications and I think these are the next steps in terms of our thinking of of how maybe we should be approaching this so and then you can see again uh for the rest of the the diagram we still have this communication we can continue updating our knowledge graph and also pull information back out of the knowledge graph to send into the structur prompt Um this can all be sent to the llm uh you know and and so the llm could do you could you could request it to return your response in multiple ways one way is you could tell it explain to me a natural language uh you know what Solutions exist to address this and then also ask it in the same move uh you know also give me uh you know these encodings of of of your answer so so could be that that the llm response some kind of natural language explanation of you know how to how to handle this problem but additionally uh here's structured encoding you requested to noted as follows uh a positive one approved recommendation to carry out available action six might be the case that this uh your your agent has an observation modality meaning it has like a sensory receptor uh for the llm where a one corresponds to like an observation level of like a yes or an approval or you know go for it basically and then and then six is kind of like a second observation modality where the Lin tells you which one of your available actions you should take so this is just an attempt and trying to find ways to modify an agent's architecture such that it has a kind of receptivity to uh llm outputs uh in a way that could go from natural language to actual directives over what to do given the current context and all the information you've asked of uh that you've provided to the LM uh in the structure prompt Now we move to the biofirm um so the biofirm like this is a lot of the principles I've described th far uh kind of play into this um again a lot of work has been done on this by by by Daniel fredman here in the the talk and um and then a a lot of initial kind of coding work was done by myself um and then we've of course been under Guidance with Partnership of John clippinger as well as David Lovejoy with uh first principles first uh I really uh like a lot of kind of the media and writing and and uh podcasts that they've been doing and so um I I would strongly suggest that checking them out uh and uh it's been a pleasure working with them so far and uh so with the biof firm we actually we have a significant amount of code largely built out uh you know and kind of organized by Daniel uh with the under the active inference institute's own GitHub repository currently in a sub project biofirm um we also have a whole another library that we'll look at soon uh but for now this is kind of the foundational the starting point of the biofirm tool highlights right now we're building active inference agents as a you using PDP which is you know essentially has been has already been referenced earlier in the Symposium a kind of Port of uh of SPM the the mat lab Library developed by you know friston and others um and so uh you know it's been a really good good library to to kind of work with and quickly start trying to implement some of our ideas the open AI API um many of you who have worked with LMS and and and tried to integrate them into code like are already familiar with this um but basically it's a way of quickly being able to prompt within your your coding environment uh a wide array of llms uh including the most recent ones and you know the new chat GPT or llama or otherwise models claw on it uh and en cursor which you know this is not necessarily essential to developing the biofirm but uh it's just uh Daniel has kind of opened my eyes so to speak to to the wonders of of kind of integrating for developers um being able to have uh an IDE where you're editing code and then you can use an llm to help you write that code and obviously you always want to go back and look and make sure that it wrote the code that you actually intended but uh it it can it can you can dramatically reduce the number of times you need to hop into say stack Overflow for example if that makes sense to to programers so it's great for productivity uh and then also some major things Daniel put out these um these uh live streams that cover a lot of the development of the biofirm so far um both of them are huge I mean we're looking at a couple hours or so of of material and we have close direct code walk through today originally ideally I might have gone through some kind of code walk through um one I don't think we're quite there yet in terms of being able to to um we can demonstrate the principles but in order to have something that's kind of like within the context of okay we're doing you know the active Ence Symposium uh and and we kind of need to keep things streamlined we're not quite there yet but you can get a very big picture of what's going on in that code and and get an actual code walk through by having a look at these live streams that that Daniel put out so the biofirm excuse me um uh essentially in a biofirm agents are working in a firm is like setting the scene uh to maintain the homeostasis of an environment be a Farm Natural land preserve and it could be it could be many things but for now it's I'm trying to kind of constrain uh to kind of an example um agents prefer both in a colloquial sense they want this but also in a sort of basy and prior computational sense to observe themselves maintaining their environment home stasis their own Survival and health in this sense are directly linked with environmental uh survival and health mapping internal states to inferred external States meaning their own internal beliefs are you know they're trying to model their own beliefs uh directly as you know you know kind of these these proxies for what's going on outside of them in the environment so they're they're very directly trying to map themselves to to their environment all of that being conditioned on preferences again what do they prefer homeostasis right so they they're trying to model their environment internally for themselves while simultaneously trying to keep that environment within some kind of homeostatic range we'll get to a more concrete example what that means um but for we can just think of it as like um you know that an agent's own homeostasis involves realizing its preferences relatable to how humans act to maintain their body temperature oxygen and blood flow since it's safety security whether it be physical or or economic um so in this case minimizing their own uncertainty condition on preferences maintaining homeostasis is equivalent to M minimizing uncertainty about if the agent is successfully keeping the environment in a homeostatic state kind of a one sentence you know concise way of looking at that they're minimizing their certainty about if they're able to successfully maintain the environment of homeostatic State uh from there we uh you know ideally what we would do is move on to building a knowledge base for example a kind of Knowledge Graph uh which itself would in an initial setup phase would involve a you User submitted documents uh so for our example case we actually did cmit both a real document uh that John supplied uh related to uh this this Farmland uh kind of natural Preserve that he himself has taken care of of course it's been redacted um but uh the point is that this gives us a kind of real world example that we are quite familiar with uh containing a document that we can use you know for an initial testing use case um and then um you can also do other things you can send in specific kinds of like structured queries uh you know asking for more domain knowledge or information about say how do we take care of a farmer otherwise and we have all of that then we follow that up by yeah structured prompting another idea we have that I won't go too much into right now but one idea is that you could actually extract out of natural language potentially like syllogistic reasoning based rules uh that there's a you know researcher axle constant who's done some work uh in the legal domain and looking at uh kind of basian uh probability and graphs and relating that to syllogistic reasoning I think that would be a really nice way of concisely extracting natural language rules turning them into something kind of like symbolic you know uh objects in Python that can be used as formula for for doing different kinds of reasoning that that increases precision and reduces the need to make so many llm calls having the LM do that but it's a thought um anyway in any case all this all the whole development of the knowledge based Knowledge Graph is motivated by agent preferences and beliefs um so social science I don't want to get too far away from that um you know I've been talking about a lot of design principles but I think the biofirm itself as currently it is an open source uh you know sort of resour it's still heavily under development I'll get back to you know the aspects of what's going on with that a little bit later um but but currently there's a lot of De developing development being made and is being written in a way that we ideally in the future would like this to be adaptable to different circumstances it might be the case this could be quite useful actually for continuing social scientific research the form of an agent based model um you know you could consider that these might not necessarily be agents in a firm you they might not have a a business model they might have something else they might have you know a constitution uh you know that that links all of them or otherwise um for now I'll continue working U focusing on the biofirm but I just want that to be born in mind that I don't want to present this as if it's a highly specific use case that you know doesn't go that doesn't generalize you know allow for generalizing Beyond itself um but for now we'll stick with with uh focusing on it um so social science we have have agents who are collaborating uh in coalitions of agents and networks of Agents uh we have uh you know attempting to say that we can augment agents with with natural language State belief reporting uh with consequent development of a cultural uh and this is Dr Clipper's view you know if we have agents who are motivated to take care of land and then they collect uh knowledge that that corresponds to that and then they're using and communicating using this knowledge you know that then this kind of becomes something like a culture so we're looking at something like a kind of cultural grammar or or set of scripts that are being developed by via these agents taking care of uh you know the the the the land around them and and we have shared narratives and then on top of that we also you know in a realistic scenario we would have agents who are doing economic resource management and potentially trying to uh you know generate some kind of value in some kind of way basically replenishing or regenerating sources available to themselves or otherwise and then on value generation and this comes back to the very start where I refer to utility as being kind of like what was previously the pen penultimate sort of uh uh incentive for for agents in traditional agent-based modeling now if we shift that to where now we're looking at expected free energy as the thing to be minim I um you know and in correspondence with kind of the principles of active evence then uh what we're doing is that we're looking at this is you know like the approximation of uncertainty condition on preferences uh agents are trying to minimize that value where there uh with value generation a whole another thing we could do is look at each of the agents marginal contributions to the minimization of this quantifiable metric uh that is uh expected free energy uh for the whole firm like look at their marginal contributions to minimizing the uncertainty of the firm and then that itself becomes a kind of value you know is we can accumulate that view that is kind of like uncertainty reduction is what generates value so that crucially this accumulation it's not some arbitrary floating value who whose own value you know changes based upon kind of Market speculation Dynamics or otherwise instead the momentto moment Val generation car that that that occurs during the course of a simulation of a biofirm it reflects the agents actual learning and capacity for taking care of their environment as proxied by expected energy and then the the kind of broader multi-agent setting and Future Vision for the biofirm their cultural aspects again as I mentioned earlier this kind of an Ever evolving accumulation of shared knowledge again we could have time stamps that also means that we might need correction of that knowledge over time um we'll talk about that more shortly um uh anyway the the evolution of the knowledge graph it's motivated by mirrors agents motivated to accomplish Collective goals for homeostasis from various perspectives um you know you might have agents who are specialized in one way agents in another you might have multiple stakeholder uh perspectives that are being fed into that initial Knowledge Graph or or along the way during the course of the simulation ethical uh ideally would um it would exhibit these kinds of principles like by accumulating different perspectives between agents between users stakeholders uh you know in the case of Biore Regional modeling where where there can be very very many people potentially involved in in looking at uh or being affected by how an environment is uh is being is being cared for taken care of or managed or governed you know we being able to actually submit these these potentially conflicting or at least different uh perspectives from various stakeholders those could be submitted to the program could also include Financial regulatory or sustainability related knowledge um you know that can allow for comparing and synthesizing various perspectives again in some some kind of after critic framework you know is I mentioned earlier like a lot of methods have been used to try and overcome the issues of llms and so we be remiss not to to uh employ those in various ways where appropriate that way we can better ensure that the information that's being used and processed and being fed into agent decision making and and so on uh taking care of an actual environment uh we can make sure that it's much more accurate coherent also identify faulty or contentious information if the Agents come up with a very aggressive policy you know prior to its implementation could be critiqued and revised from an ethical regulatory or legal standpoint and all this is collective uh you know we can kind of synthesize the information I already touched on this earlier resynthesizing core information or primary information in the knowledge graph and being some kind of summary business mod Constitution set of protocols protocols or missions that the agents have um and then uh sustainability Resource Management um you know making sure that agents are aware of available resources uh combined with quantitative analysis you could you know if you're looking at a real world setting you'd be looking at things like price information forecasting supplies needed for a particular policy might have qualitative reasoning involved such as why like which supplies are needed for a task what purposes uh within what guidelines um that's something I've been thinking about as well for a related Institute project uh called farmworks uh we might talk about that later we'll see um and then heterogenity a agents might be preconfigured or learn they necessarily will learn over time assuming we we Implement learning and their architecture more than likely well it depends on how and for what purpose and then they might even you know evolve so to speak to specializing different skills knowledge uses allowing for agents to collaborate under broad shareed goals while working on subtasks there's some nice um we were joined by m yesterday um during the first Workshop um so has been involved in a lot of this previous work and then um you know another reference to kind of the as well the industrial example that I gave earlier to how how we might think about some of these things um basic form of a homeostatic agent um this is uh just take a second homeostatic agent this is the general just kind of quick natural Language summary of its its internal architecture it's it's incredibly simplistic this is sort of like a sort of a kernel of what is the initial formation of a homeostatic agent prior to any uh additional you know modification or or changes in its architecture that you think might be required for the situation at hand for now let's say there's a single environmental variable uh what we did and um you I'll talk more about it later but the idea is that we have an environment that we've an artificial environment we generated we generated it based on the documents I mentioned earlier including John's Farmland document as well as particular prompts submitted to the llm to kind of compose what are the initial makings of our our knowledge graph and it came up with soil structure as being one of the essential uh variables that should be included in included in our environment and then we have an agent here who's kind of mapping itself trying to map itself to this quantitative value uh it is a question uh right now of you know I I can imagine those who do active inference like the question might be why use a PDP why would you not use you know a continuous time continuous State space model uh for for now we're going with pomdps uh in part because uh we're assuming that we'll be using agents to model highly complex environments where it's not so simple as to kind of follow uh simple up or down uh scale or say that the the the variable is necessarily being defined by you know some kind of gosan uh distribution and being sampled from it um it might be the case that this is much more complex there might be changing conditions or otherwise in any case for initial testing environments is also a rather quick thing to two I will admit so um but for now uh the idea is that you'd have a PDP who can view the data over time at each time step uh Daniel ran this one for uh actually I don't remember if I ran it he probably did uh over 10,000 times steps um and so at each time step uh the agent observes if the current value is above or below some homeostatic range which the agent prefers it fly in right and so it infers the H state of the variable is it actually if I observe low is it actually low uh if it's in its homeostatic range is it actually in its homeostatic range or is it act in reality is it really high or otherwise you know and uh the another question as to whether this is necessarily a partially observable environment since this do seem to be reflective of real data but again this allows for the space that there might be very complex relationships between soil structure and let's say uh um uh moisture levels in the soil or or micro bioactivity within the soil or other kinds of things that that that really matter here to where it's much more complex than just being able to say oh just because this went low that means I need to try and bring it back up as much as possible really fast right so uh there might be more nuance and having more sophisticated architectures might be able to to kind of parse through that problem including increasing uh inference Horizons or or including more other variables that it tries map um and in actions the agent can increase maintain or decrease the variable uh right so again very simple a little more details later on on how that can how that can work but just say straightforward it's kind of like the agent is trying to balance out this metric um given that a lot of it all of fluctuations uh aside from the agent actions impacting it are coming from its relationship with all the other Environmental Ables in its ecosystem so it's kind of like the agent is trying to map its internal understanding of what's going on with the environment's behavior and then the agent of course prefers to see this within its homeostatic range and so it will choose actions that it believes will help this variable to get back into its homeostatic range so from there we can view this like as a whole ecosystem of variables so this this kind of defines our environment that we're looking at right in agent based modeling setting we have our homeostatic agents and then we have our entire environment composed of all these environmental variables and over in a real world setting whether you're running a firm a farm or otherwise you know each of these variables could be mapped to you know an iot device uh they could be mapped to you know some kind of uh uh logged uh observations you know that your team makes as so what's going on with the farm right now um I guess I'll bring up Farm works again briefly we've been looking at a lot of farm data lately that's been collected by farmers and getting some inspiration on how they go about doing those things but I will say that this uh this environment that we have here is entirely artificially generated based upon inspiration from text um so if we think about this as something like agents are trying to solve a non-equilibrium steady state uh what does that mean I I'll try and summarize by saying that we're trying to maintain the structure and properties of the farm what is it that makes a farm a farm well if your soil moisture Falls dramatically too low and then suddenly all the organic matter and it dies and then many of these other things start to fall apart well now we're not quite looking at at at at homeostatic uh soil anymore right so we kind of have like this this natural decay or dilapidation of of the system the ecosystem and so agents are having to repeat like continuously uh you know kind of actively maintain uh through control inputs meaning through their actions uh they're trying to keep these variables within homeostasis and these it's worth noting these these Target ranges homeostatic ranges might change themselves over time right due to uh you know we can imagine um the the target soil temperature range might change to to adjust for you know seasonal crop change um it also might also account for certain kinds of like exogenous or external shocks to the system the impact of climate change there might be a natural disaster that occurs or otherwise that itself can really change the Dynamics required to attempt to maintain the ecos system uh as it is or as it was or if it needs to kind of change in order to continue existing given uh what kind of shock occurred so to maintain these desired homeostatic States the agent must uh continuously work against yeah these these natural tendencies towards dilapidation States I've already mentioned this many times these these states these envir uh variables may have complex interdependent relationships may call for agents with forward-looking inference Horizons um you know of course if we over maintain the soil uh you know put too many s in it or or do other things that might still be okay for the crops it might harm local biodiversity uh you know local animals accidentally cons you know consuming different kinds of uh um um you know materials that were that were used to take care of the soil and suddenly they die that harms local biodiversity and yet uh biodiversity might also include uh pests in the environment that themselves then harm the soil or crop growth so it's just um being able to to kind of adjust to all of these highly complex variables and crucially in reality we might only be able to impact some variables second some variables and not others thus agents might specialize in handling one particular controllable variable and study its relationship with the other variables leading to actionable predictive planning another kind of everyday explanation of this is like um you know I I I mentioned I'm a data analyst I work in education and so um you know I'm I'm fre ly looking at a lot of data regarding um um students and kind of our institution and so on I'm based in Chicago uh some things that we can change our bus routes for our students to make sure that they can get to school safely uh something we cannot control is the amount of crime in Chicago which um anyone in the US knows is a that would be very difficult to do right so so so we're looking at there are all kinds of factors in our environment that we cannot control and then there's what we can and then trying to find how can we control what we can control in ways that kind of can balance out with all these things that we cannot control right so that it's very crucial to be able to identify and then measure analyze these kinds of relationships whenever we're trying to design any kind of concrete actionable uh uh policy right or intervention that you might want to make and then uh John referenced the commons earlier right and Elanor ostrom's work um homeostatic agents solving non-equilibrium study States the ever ever solving the commons so so I'm I'm trying you know I I was thinking about a lot of these principles and ostram was required reading for me as someone who coming you know from a social sciences background and so I this part of why I find this project so interesting and why I find it fascinating to try to realize a lot of these things um in principles involved so we we have local rules conditions adaptation I mean the agents necessarily have to adapt to the environment around them if you provide some kind of Kernel agent and and give it some space to figure out what it is it needs to do it's necessarily going to have to adapt right so so we have that attention towards the local um recognizing uncertainty and adaptation you know agents will have to take care of homeostasis manage resources through learning uh to to change conditions and broader unobserved uncontrollable environment in the sense of um you know there are many things going on that the agents cannot directly observe um there are other things that will occur that were unexpected you know so and then also there might be relationships within the variables themselves that's you know for example there might be some kind of critical threshold that's hit by one variable that then has some kind of domino or Cascade effect on the others you know so that that's why I include both the phrases endogenous or exogenous shocks um you know being able to to watch out for those kinds of Dynamics then governing the commons you know that's sort of that's sort of the benefit uh of you know if if it can be done right if it can be done successfully integrating llms in you know a very in a very thoroughly considered way that allows for you know accurate information uh uh uh accumulation and sharing and then using it for implement policies um you know that that's the aspect that kind of keeps things grounded and then and then gives this kind of natural language based way of dealing with an environment you know this this kind of just repeats many things I mentioned before but we're synthesizing a lot of info from various stakeholders uh various domains you know that are needed to handle the problem repeatedly grounding agents at common goals maintaining the environment against perturbation capture maintaining its Markov blanket and then nested collaboration potentially so just as users can practice oversight by running these models um agents themselves you know you might have agents who are at another layer so to speak who who maintain their own oversight over other agents or coalitions of agents and those uh those kind of hierarchical Agents might themselves act as something like an observer role um they might recognize a broader issue going on on uh in The Firm that isn't being accounted for by specialized agents and so you can have those kinds of kind of hierarchical uh uh specialized uh structural roles um that that allows for this kind of nested uh nestedness and how the environment is maintained um you know and and then it's a very interesting idea that I was going to bring up later but in the context of like looking at Biore Regional hubs or something along that line I mean if you were actually employing something like a biofirm to a real world scenario then I don't see any reason why you couldn't have some way of interfacing between different biof firms right because if they're actually playing out in reality and building their own knowledge graphs you know say in real time in s some kind of way then then you could very much have that kind of interface and you could have you know structured prompts pulling from the knowledge graph of one biofirm you know sending that information to another biofirm and back uh you know so so that's you know I I don't want to get ahead of ourselves but because the vast majority of the actual concrete work that's been done on the bio firm has only been over the course of frankly the past couple of weeks I mean longer than that for sure in terms of of of John developing the ideas conversations we've had over the past couple months but whenever it comes to developing this stuff and actually realizing it it's it's still quite new so it's very exciting to to report biofirm for users let me just see where we at okay um I'm gonna have yeah yeah I'll speed this up a little bit um biofirm for users yeah uh one you know for for for for a user's perspective first you submit your documents as I've mentioned many times uh you know your own PDFs real world information uh you know if if you want to submit textbooks you know regarding uh agricultural techniques and methods for taking care of farms you know including formula that describe relationships or something you could have from there the direction is to have there are two intended kind of potential paths you can take right now all the testing is being done with the first path but the ideal is to move it in the direction where you could have both available first one is uh for being able to explore counterfactuals so we generate synthetic data using domain knowledge or historical data that's what we're already doing and then second direct application that would be where you're actually able to send in data and have it being streamed and have this play out real time is is very much you know kind of in the future but that it's sort of on the roadmap so to speak um and then once you followed one of those paths uh you could generate an environment for your agents and then also you could construct applicable agents meaning agents to your design or ones that kind of fit the situation or be mapped particular environmental variables as we saw with the agent who takes care of the soil structure earlier using agent maker uh which is kind of a entire addition that that that d manual kind of Zoom through uh being able to incorporate included there's more information on that uh the second live stream that was released um from there you would uh simulate uh your agent based model meaning there would be an initial uh setup phase that takes in your initial documents and so on constructs the agents constructs the artificial environment then runs your action perception Loop it keeps things moving data and Ln responses are encoded as observations for agents agents take these observations take care of their environment so we already know homeostasis for the agent based on how it setup is kind of hardcoded into its design and maap to the home homeostasis of its environment and finally there would be obviously we would want this analytics so having some kind of analytic Suite that allows you to derive insights from different kinds of metrics that are being recorded over time there might be different kinds of like llm summary reports and recommendations that actually extracts the metrics and you know through a pre-structured prompt as I've said a million times during the course of this you know using prur prompt can input a lot of these results and then and then have an llm kind of put it into a natural language like just explain for you in everyday words here's what happened here's what's going on here what the agents believe and so on uh Biore Regional modeling I want to look at this really quickly this is uh a lot of Daniels work on uh I mean this is just an initial step but what's happened is that Daniel sent particular kinds of structured promps you find more information on this in the first live stream uh but basically ask about information uh in very particular ways um to llms and did it programmatically was able to extract incredible amount of information about these different regions different counties and uh he started with California and then moved to to New England and then it's expanded from there and so we have like Maine and Massachusetts so for example uh Massachusetts uh Hampshire County like you can uh he sent in like you know generate some kind of uh Supply strategist statement um we have copies of these this is a Json form meaning it's kind of a set up in a way that can be read by the program then here's just a more readable legible you know every day for a user report it's like here is you know your supply chain strategist report on the ecosystem of that County uh references to what's going on in terms of its biodiversity how climate change is impacting it how land is use habitat fragmentation water quality so just a lot of the different things that are going on specific to each of these counties you can do that from an art a market analyst perspective is as well try and consider different kind of opportunities what the current economic landscape what kind of regulatory environment compliance requirements or their state federal local of course these are currently very short descriptions uh ideally over time you would build out your knowledge graph your your knowledge base You' actually you know kind of zoom into each of these send more follow-up prompts like tell me more about this you know site you're site the information you're pulling from um and then you could even consolidate that so this is this kind of a Consolidated uh report you know very long for all of these different Market analyst perspective and ecological perspective supply chain strategist and just kind of finding ways to to synthesize a lot of um information in one move this was a business case and I just wanted to show quickly like here are you know ideas of you know here you know defining out potential revenue streams risk R assessment mitigation strategies you know so it's very five-year Financial projections of course we could pull this kind of logic back into the program have local computations done that actually comes up with much more uh uh uh uh uh grounded realistic and precise information based on available resources that you have or that that at least artificially the agents have if you're doing more of an artificial setting um so yeah it's just it's making all these linkages so um you know that that becomes sort of the basis the the starting point for for our um for the biofirm so we we then you know again we generated this artificial environment all we did was send in user documents structured prompts through an llm you a structured response um let me quickly yeah um this allowed programming our environment uh it's it's Dynamics configuration generating descriptions for each of the variables they get stored in this nice Json structured file ecosystem config you know so I mean we it you you saw previously like I was looking in a GitHub repository we like we were really building out the code in that in that base um and so anyway that sticking back to the artificial environment here this allows us to pull out yeah what are our variables we proxies for variables such indices like it can be quite difficult to measure biodiversity so we make an index for that there might be other ways of doing that there might be more specific ways of doing that that fits your own use case or otherwise we could you know tailor that um domain inspired quantitative relationships between the variables you can even you know with more manual control you can set the noise levels for each of these like if you assume that there might be some noisy process you know uh from a more mathematical or statistical perspective how what which of these are actually controllable or not for those which are controllable how much are they controllable can you dramatically change biodiversity in one move probably not um or you could and I don't want to follow that thought further biofirm development and performance benchmarking uh this is another crucial part of the biofirm development being able to actually Benchmark how our active inference agents uh uh perform relative to you know some kind of benchmark basine such having a more reactive or random sort of thermostat style like reactive um control so basically on the left we don't have active inference agents it's just trying Instead This is a a more basic uh uh control strategy that kind of just reacts to the data it's seeing uh and of course the DAT is getting very out of hand I'm not saying it's not getting out of hand with the active inference agent but much more so that's what we're seeing from initial test case and this active inference agent on the right is still that basic kernel homeostatic agent right it is keeping its Target within range you know 21% of the time relative to the random strategy which is only seven um you know the next steps would be to actually fine-tune and modify uh the agents to better match the situation so far it's it's it's very simplistic it's only looking at one variable only tracking one variable has it's currently not taking in any information from an llm it's currently not taking in any information about the other environmental variables um this was really interesting to see though that it already has this capacity for outperforming the the reactive model now we've had some instances where the active inference agent doesn't uh you know it maybe matches the reactive model so it's just you know it's prime time to to step up the agent development game you know so that this was just step one it's like can we can we build it you know we yes we can so now let's improve it um analytic Suite we can look at you know the the variables themselves their correlations with one another we track the metrics about the environment over time then compare how the environment behaved in different different simulation settings um you know Allah that's what we want to do with agent-based modeling in the first place typically you want to be able to run simulations look at the outcomes based on different circumstan and then we can compare you know how this went using different agents um and uh you know uh we could use llm integration to to to translate this into like a a more readable report for users um this is a whole another aspect I I really wanted to emphasize it I I've made this kind of as an additional module so it can be edited kind of after the fact like outside of the main program uh which also means that you know you could you can modify this fit as you want but basically the idea is you try and measure the surprise or uncertainty or expected free energy or otherwise of the environment here um and and um basically look at how the actual variable diverges from where you want it to be you know the difference uh you know it's of course it's mathematical and strong from active inference and other areas basing in statistics and mechanics but um the idea is you can measure how far away it is from where you wanted it to be in terms of surprisal uh you know actually capturing kind of this more rich or nuanced information about how that variable moves versus where you want it to be um and then you can kind of flip that say oh well in a non-equilibrium steady state system where more usually than not the system is going to have a tendency towards generate uh generating more and more surprisal that is the the the complex um relationships between the variables is going to lead to things typically going off course Very quickly if you don't touch them um then in that case any move at reducing surprise meaning any move at combating that natural tendency towards kind of like chaos or entropy uh you know can be viewed as a form of generating value so this is a kind of additional metric we could add saying that there's a kind of value added to having these agents uh minimizing the surprise meaning minimizing the free energy meaning minimizing the distance between homeostasis and where the data is now uh right so so we could see that kind of accumulation over time and again it's modularized meaning for a user they could potentially experiment as they like they could change how this value added is computed I think that'd be very interested interesting for people who are interested in economic simulations or or doing this you know differently or want to see like how how does this work you know um then also shle value analysis we have shly values so um in the course of you know these simulations we have agents trying to solve or take care of their environment and so in the case of U shapley value computation what it's doing is that we can compare how do certain coalitions of Agents run through the simulation compared to one another uh in terms of U you know these different metrics and so you can see like oh agent all five agents together uh you know did did better at maintaining their environment homeostasis uh they they did better than only having um you know these this other particular Coalition of Agents but they actually did worse altogether versus when you only had these two agents working together right so it's it's a very kind of sophisticated way at looking at how agents are actually solving the problems and actually carrying out actions that they need to carry out um and we can do this you know for different metrics and we can view that as you know that that can be another way of saying this is how value is being generated from by these agents right so being able to make those kinds of you know those kinds of analytics then uh final yeah these are the last couple slides GNN and agent maker um this is very cold it's kind of like uh you know did so much rapid development of the biofirm over the past couple weeks that this just kind of like hit me I was like oh I don't I didn't really think about that but Daniel has incorporated along with his co-authored work with Yak smel in the past um they've worked on what's called General notation notation and and and it's a it's a way of kind of translating active inference uh active inference agents meaning models or architectures uh between different programming languages including the to being able to compose them using natural language uh and then also being able to visualize them graphically like that's you know it's not quite there yet the kind of makings for um being able to have this kind of translatability like uh you know the ideal would be being able to write to an llm like hey make an agent for me that can do this and this and then by giving the LM certain kinds of contacts it can fill in the matrices and and so on as you need it of course we're still a distance from that but it's kind of building the framework towards moving towards that and that way you can make agents who are much more tailored to the situation and furthermore ideally you'd be able to uh translate them between programming languages meaning you know being able to switch if your applications in Python or Julia or otherwise um being able to have that done for you and then uh Daniel has basically forked uh the PDP repository and then built agent maker on top of it so right now it's kind of tailored to PDP um but it could be used in other ways as well and so we already have some next steps on the kinds of Agents we want to make for me it's agents who should be able to observe the whole environment but only be able to control their own respective part to me that's what would make the idea of like a having a shared common goal amongst heterogeneous specialized agents like that's what would ize that so that's kind of the broader view of the biofirm you start with inputting your documents your texts your PDFs uh data for fitting not act um excuse me that's optional right now uh it would be more ideal for a real world setting but for now just generating an artificial environment just the text is fine we've been able to accomplish that far and then from there you could fit agents to the situation or at least maybe customize them s using agent maker you have the the start of the your knowledge graph or base being developed and you have that uh ecosystem all of its Dynamics and configuration kind of Spun up then after that you just run the simulation uh looking at you know analytics uh we want to move towards having an interface in future having dashboards we've already seen a lot of these um graphs earlier so it's you know and then we'd also want to have llns that could actually report the results and this is basically towards an adaptable kernel of a biofirm this is kind of the bigger picture we have different kinds of design considerations uh couple highlights we what we're looking at things like implementing potentially amortized inference we kind of accelerating the inference process um using continuous State space predictions and models you know and then considering you know we we have an RX and fur group at The Institute so we'd like to potentially move towards moving using a more reactive framework using RX and fur the Julia Library as well as kind of complimentary uh library for environments um the ARs environments Library plenty of different ways you might be able to use different kinds of apis or Integrations with other tools um locally storing LMS to mitigate the issue of having to necessarily like uh use the open API uh open AI llm API um so and then obviously integrating you know key aspects are like the resource management right and that that's that's actually I would argue is going to be one of the easier parts of incorporating because it's like just as we can currently read real data or gener generate artificial data and have the agents manage that like soil structure we could do that similarly for different kinds of like U you know Financial resource uh values or managing an account something along those lines right so bringing it back biofirm development testing this just a slide I showed earlier but it's it's just a you know reinforce here some of the tools we use some things we want to do in the future some related Institute projects related to you know we're working on projects that relate to mitigating future existential risk we've applied to a grant with Forest site there's the fli grant project farmworks ARX and fur group there's a a new economic simulations Related Group some people are interested in looking at things like you know simulating currencies and and so on um in in agent based modeling then those live stream updates um my computers having uh repeatedly deciding to cover things though and that's it so uh we we went much further on time intended there was a lot to there was a lot to cover um really appreciate every everyone's attention though um and you know in these 11 minutes we have left you know if you guys want to give anything a shot please go for it voice is going out no take take some yeah anyone in the live chat can write a question John where where does this take Andor what will you plan to do tomorrow well I think what we tried to lay the foundation here in a very thorough way is uh what is a biofirm and uh and what how is able to address uh many of the issues that you see and in people trying to work out a bio uh bio Financial the Biore Regional economics and create the proper incentives and so um I think what we're trying to do here is really show that there's a number when I started to say here are some of the criteria that we're looking for in particular building with with eler ostram and the whole idea of how to govern a commment is a profound issue that seem to be intractable to a lot of people um I think this in my my view this provides a framework for doing a scalable framework for doing it a way involving people a way of applying scientific methods and creating fin Financial incentives and automating a lot of the the data collection analysis task and how you can agents can build up their own competencies of over the period of time so there's a lot here there's a lot to digust um and it does recommend it does provide a new metric of value that's based upon an and a you're not you're not establishing value the classic economics that our price Arbitrage or information asymmetry Arbitrage you're actually saying there's such a thing as a free energy princip a scientific process and we can show how the value is created and then how to allocate it to those people or those agents that are creating it so this is really sort of foundational so we want to take it in the next steps um and what we're going to be doing tomorrow is is um actually there's a group of people who are experts in the sense that they work with uh how to take multi-agent architectures and apply them to in some cases the to the Enterprise from Del that's doing that another grp uh Simon uh uh Torrance is doing it in terms of AI looking at framework for agenda architectures uh there are others are looking at it applying to the bio uh Finance bio Regional Finance issues um so we have that Susanna Choy we also have uh uh we have Matt who's head of sustainable finance and metabolic and then there's John Havens who's been involved with itle e establishing ethical Frameworks for application of geospatial models in AI so it's quite a combination of people but what we want them to to look at is what's the impact of this when you're having fully autonomous biomedic based agents uh and what happens when you can apply them not just to the biometric to the Biore Regional Finance but also the new notion of a firm um and I think there's is a real profound issue here what happens when you really have autonomous agents and you create institutions we create institutions that are supposed to be above people but the problem is people populate institutions and they come captured but what happens if we start to put these agents and build institutions that are that are function this way that are verifiable this way and what does that happen to what we think about employment jobs and I mean there's some really substantive issues here um and how do you create this as a a public resource a public good that doesn't get captured by particularly small interests so it's a broad spectrum of questions we're putting out there um that I don't think that the public in general fully appreciates the the the capacity and the power of an active INF free energy Principle as applied to economic and and societal issues so that's we're going to try to tackle those issues and get people talking about it and this is this is a beginning but we want to take it next steps and really get reference get actual reference Technologies out there that we can actually show and demonstrate and may be able to replicate that's that's the the goal yeah and very helpful I gotta say you know Daniel kudos to you from all along I mean both you guys and Daniel just you've done remarkable work and and and accelerating this process and we're very very grateful for that and what active inferences institute there's other things that the other sessions I saw here that seem to be very relevant to what we're doing so I'm really looking forward to collaborating I'm I'm quite fascinated with with thought Forge with Matt Brown is doing and doing you homeostatic networks and it goes back to Ashby and the law of re with a variety I think all that's applicable to this too so there's this wonderful synthesis that you've enabled you you really have I and and through a singular effort so again extremely grateful for this thank you it's been awesome and and uh a lot more I can say but what what what can we're gonna we're gonna ask more of you you're I'm not done yet you're not done no no we we're no the the part that resonated with me in the presentation over the last two days of biofirm and how Andrew laid out all these people pieces is like on one hand we have that researcher lens again about similarities and differences and about treating it like it's an engineering problem and then I think John you've brought in the deeper history of Commons and cybernetics and these more open-ended problems about how do we connect any technical capacity to the embodiment and the co- enactment of systems that are livable and regenerative and that's a little bit of a different kind of topic that's one though that that uh mass convening and awareness can can fertilize and make real in a way where the technical understanding it's like you can skim the textbook you can read it for 20 years it's like it is just going to be the textbook it's just going to be H something to jump off of and take into another way it's not like some of these questions about how different regions will work it's not it's not in the 2022 textbook so that's applying active inference is taking the high road and the low road and variational free energy expected free energy like all of those formalisms and then like surprise surprise applying them to what matters and figuring out that project and operational part and there's just like there's so many pieces was just like thinking of the yarn connecting the the the needs to what we do and don't have at different levels of Readiness that takes us back to the tech tree and all these other ways that we've discussed coordinating around that balanced way to think about Regional and what is Federated what is transferable how do we deal with all of these aspects and it's um been a great journey over the last few months also well thank you I I what I what I find so reassuring is that and and you you see applying the method and then it starts to explain and you keep applying it and so it it it doesn't it doesn't Fade Out it doesn't you're not you're not get o h it doesn't become OD Haw and so I feel like we're doing something really principle that is extensible and when I come from the the the bof finance community and see all the things that they've been trying to do are in the Commons Community say actually there's a wonderful fit here and getting these two worlds together is is is is is really important it's a challenge too um but i i i people want to do that and and they're looking for something like this and so I really hope that we going to be able to create that marriage as it were to add one comment on that I think when you opened today and said well we could make things fungible we can connect but that is also part of replicating if not elaborating the problem which is the fungibility of drinking water here and this calorie here and this unique species here this facilitating that fungibility is facilitating the capture of the underlying resource so even if it were somehow trivial to make like assertions about impact monetized and the whole hypers certs and impact certification even if that were trivial and we can just assume that that we have full observability or something like that it could even accelerate Financial cap Capital based Cascades and so what what can be an escape grounding to that and that's where seeing like whoa the models on through the Paradigm of incentives are maximalist and reward oriented can they be homeostatic or allostatic oriented to satisfy and be on paths that bound surprise rather than try to maximize but then as soon as we look around the world there's things that don't make sense we're not trying to maximize or minimize like the reservoir but rather within a range and so there's so many things where we can just connect first principles of intelligence and research questions to just like the calm subtle sometimes ways that that the technology because to to to move too fast also would would again replicate fallacies right okay well good luck with the session tomorrow okay and and I I hope many people come join and and engage in these Fun open source meaningful projects this is just the beginning it's just a checkpoint and and uh it's a strong one and and I'm very pleased with it yeah cool all right thank you fellas see you later all right welcome back everyone my name is Devon Bay um I'll be presenting uh accounting for multiscalar and conscious psychobiological phenomena in the active Paradigm so for an overview I'm going to be going over the marov blanket trick and then critiques of the Markov blanket and address how some of the previous work of Chris fields and friston partially addresses these critiques um before finally going over uh problems of Consciousness new approaches and theories on Consciousness and how these fit within the Fe and active inference and then provide some ways to update the framework uh based on this information I'm also going to uh take an epistemological perspective on modeling and I'll focus on how both the conceptual mathematical formalizations of superposition and entanglement can maybe alleviate some of the problems that are identified in the critiques so um the Markov blanket since we are on uh limited time here um I will assume that most of the people here are already familiar somewhat with the marov blanket or active inference framework um but I'll give a brief overview of the marov blanket focusing on some of its mathematical Frameworks as well as some of the Assumption assumptions implicit within the framework so um for the mathematical overview given a state s of a model it's marov blanket is the set of sufficient States within the network needed to infer that state and it essentially partitions autonomous and external states and uh is helpful in identifying which variables are conditionally independent from each other and expresses The Joint probability between external and internal States um as mediated by blanket States so uh the assumptions here is that there's sparse coupling between the different states in the mark of uh in yeah in in the uh in the overall system and that there is uh somewhat of a delineation between external and Inter internal States and uh that these external internal States can be properly partitioned via the marov blanket so uh the implications here the marov blanket is metap physically distinct from Pearl blankets uh at least in the context of active inference or free energy principle I think this was uh raised by um uh by uh boomberg or and and his colleagues um in the uh in the Markov uh in the emperor's new Markov blankets paper um and so this essentially means that they are um they are uh they differentiate themselves from Pearl blankets by nature of their application and the frisen blanket Itself by Nature uh takes on assumptions like the existence of external internal sensory and active States um and so this is the the main sort of differentiation also uh the firsten blanket is not providing a metaphysics um at least uh at least in my interpretation and so uh so we assume here the existence of distinct states which may be modeled um and this means that there are distinct states which are properly separable and that there are either epistemological or ontological grounds for doing so and so the main problem here I think is that there is not enough justification to argue that the the marov blanket partitioning is an epistemological or um you know in in a more broader sense an ontological um boundary and that is uh you know causally um correct or is it causally captures is able to causally cap capture causal relationships between variables um because the existence of sensory um of sensory active in internal States um and external States is something which can be disputed and which I'll discuss later on in this talk and so uh another implication of the Marco blank is is that it allows the modeling of recursive structures and Dynamics by use of the re renormalization group for example which um apparently allows to marov blanket to capture multiscalar phenomenon um and finally the question of whether the marov blanket aderes to the beian a Razer um it would simply depend on the choice of parameter for the specific model and does not inherently implicate the marup blanket so um critiques of the marov blanket like I mentioned earlier the ontological critique of the marov blanket is leveraged by both brunberg and Raja in their respective papers although as we've seen in responses to these papers frankly it depends on the ontological interpretation of active inference and uh the free energy principle and if as Keever in his response to bromberg's paper claims the Markov blanket should not be interpreted literally then this problem partially goes away however the epistemological status of the marov blanket is still under question as raised in a part of raja's paper in so far as the explanatory power of the models under some pressure so chiefly there's a broader philosophical question about the purpose of models and this is not necessarily unique to Fe or active inference or the Markov blanket but nevertheless since it affects all models these are issues are raised um are raised in in in uh in in the context of the markup blanket as well uh next we also see that there is a mathematical uh problem with the selection of markco blanket States um because we see that the uh for example uh Dynamic causal modeling in neuroscience does not demonstrate that uh that models provide scientists with a plausible account of the behavior of real world biological systems since successful applications of of caal modeling does not give us any reason to believe that neuros systems plausibly for example minimize free energy or that they are arotic active inference systems and so uh this means that even even though the model may be applied successfully and may predict some uh Behavior successfully it doesn't necessarily mean that the model is number one an accurate representation of what's ongoing with uh you know the internal dynamics of the system and number two that it would therefore be able to capture accurately all the internal workings of this of the system um and so uh so yes on this mathematical side as well we also see that uran Elon published in 2023 um a paper on the constrain disorder principle and how that would uh possibly dictate neuronal function and so Elon points out that despite the stochasticity exhibited by the neur uh nervous systems operation there's not necessarily a high level of control since Randomness at the lower levels never stops and thus it dictates a paradigm of reaction not action by organisms um he uses the free energy principle but does not use it to justify a form of active inference which would be minimizing uncertainty but rather argues that the constraint disorder principle um uh is is more preferable where a process of balancing disorder and order is more inherent to systems operation in highly Dynamic random environments so whether all environments are highly Dynamic and random and at what scales this Randomness can be asymptotically ignored becomes uh or becomes irrelevant is still for question however the fact that these boundaries are not clear and in fact that they arguably cannot be made clear for example via the um existence of Maxwell's demon or limits on computability Etc should point us in the direction of using better models and better parameters for these models so what can this tell us about the free energy principle in the marov blanket first it points to broader problems with the free energy principle but these still carry over um implications uh the sorry these still carry implications for the ways in which the free energy principle epistemologically treats the Markov blanket mainly to delineate uh a cross in between scales and uh causal uh relationships in the later portion of this talk I'll explain why entanglement and superposition are suitable replacements for the Markov blanket when they actually offer a greater flexibility without constraining the mathematical and philosophical operations and notably Chris Field's paper on uh the FP for generic Quantum systems makes implicit use of the same some of the same affordances offered by superposition and entanglement but takes a very specific interpretation of quantum systems and I think limits the flexibility of of such modeling tools by adhering to the holographic principle um so uh now onto the uh epistemological uh status of models which I think is important to raise here because the because the question of whether the marov blanket can accurately Encompass multi-directional uh what I'll call horizontal Pathways vertical and horizontal Pathways even without the presence of clear attractor States is still under question as is the more General Assumption of ergodicity within the FP framework and so um as experimental evidence from physics shows um systems in general not simply at macro or micro states are afflicted by complex Dynamics such as replica symmetry breaking which fall into a sort of gray area between ergodicity and non-ergodicity and thus render the determination of of ergodicity versus non-ergodicity much more complic ated even with suggestions such as local ergodicity as outlined by ramstead in uh in the 2018 article so um I think it's important to outline the epistemological status of models because um it's important for clarifying whether the marov blanket itself is problematic within the uh the FP framework and whether it can potentially be reworked so from a beijan perspective models should want to to minimize both epistemic and alator uncertainty where uh epistemic uncertainty is the reducible part of the total uncertainty and alator uncertainty is basically the irreducible part active inference models therefore should minimize uh both epistemic and alator uncertainty um and represent possible model configurations and parameters via prior hypothesis space where the choice of model is informed both by where we believe the quotequote truth truth is located or at least the uh the most truthful uh model is located as well as the model constraints so here we see um in green is the ideal situation where we have an ideal selection of model with some with an ideal convergence to some truth X um and then uh the purple shows an example here where uh our prior hypothesis does not accurately Encompass the true model and hence by extension the true model is not included in the posterior finally the pink shows where the posterior does not converge to the true model um and so this is illustrating how problems can arise in these two areas first the hypothesis base does not properly Encompass the true model X which is the purple or blue example and then second the posterior space does not properly converge to the true Model X which is our pink uh Pink example here and so constraints on X uh depend on if you're a realist or an anti-realist so for example if you believe that there is some ground truth that is universal and um unchanging and it's you know physically uh accessible then of course uh the true model would be a a real true model but uh if you're more of an anti-realist then it would depend on uh you know your your hypothesis B and you know of course whether you believe that troops can be can be accurately represented or even be accessed um and that they're Universal or whatever so yeah is now the question is is the Markov blanket the proper choice of parameter for the free energy principle and active inference so the purpose of the marov blanket is to provide a statistical boundary which delineates a system from its environment it ensures that variables are conditionally independent um of that certain variables are conditionally independent of other variables in the network and it formalizes the belief or the prior that the system and its environment are separate either ontologically epistemologically physically like literally or even causally right so um Columbo and palasio actually indict the marov blanket and Fe active inference model by saying that it achieves maximal generality for minimal biological plausibility because these boundaries are can be drawn sort of arbitrarily and uh even the existence of cellular um cellular membranes for example does not necessarily constrain causal relations between or across scales so uh notably uh Noble um argues that organisms are open systems that operate far from equilibrium and so again drawing from Noble's arguments uh Elon in the 2023 paper argues that organisms develop numerous emergent boundaries between levels which are not present at the Single Cell level so even though they may have distinct functional purposes boundaries like the skin intestinal tract or even vastus medialis depend on exchanging matter and energy through cellular membranes and so boundary is thus dynamic interactive not passive and cannot always be defined anatomically so for example electrical gradiance and onon concentration gradients are actively generated across cell membranes where interactions across boundaries determine an organisms functions uh not just parts of not just the the composition of the boundary or uh or the genome so of course one way this has been explored in the free energy principle in active inference is via uh the sort of of recursive Markov blanket but uh we see that this sort of recursive modeling is not able to properly capture um multiscalar relationships and interactions uh across and between boundaries where essentially we would have uh you know um marov blankets that are um that are causally implicated in in more than one uh in more than one way that that can sorry in more than one way that can be described by simply the repertoire of active external um internal States uh uh and sensory States and so this is sort of where the case for entanglement comes into play because uh entanglement is uh is has been argued for as a way of causally representing relationships between variables in uh in biology so first of all we see that organizational boundaries of living systems are open and flexible such that these boundaries need not be coextensive with the organism's bodily boundaries number one so again I guess this is just to drive home the point uh that the mark of blanket should not be taken literally um but we also see that biology is inherently Quantum so uh so these um so these uh these processes of uh cellular differentiation for example uh are under are fundamentally underlied by or sorry un fundamentally you know determined by or perhaps uh sort of controlled by uh Quantum processes that dictate you know um what chemical bonds the cell is able to break for example or or how certain concentrations of of ions for example flow across um across the cellular membrane and the permeability of the membrane as well can also exhibit Quantum effects so as you know the field of quantum biology grows I think we will increasingly see that the quantum levels is extremely important for dictating how uh organisms evolve and especially if you know um um Quantum Consciousness theories continue to gain support uh we could see that uh Quantum processes may also be implicated in Consciousness and so causal entanglement addresses the sort of asymmetry um between what were previously considered linear processes um and um the sort of asymmetry across and between scales so uh I basically argue that causal entanglement is is a counterintuitive form of aams Razor because when we have entangled variables we don't need to always explicitly represent the dependencies of these variables um with each other so uh that means that we are we can you know in an infinite dimensional hbert space or a finite dimensional hbert space represent the variables uh in a way that does not necessarily dictate uh specific boundaries or um or constraints on these variables which may impede the cause of modeling of uh certain phenomena so as the scale and complexity of the model increases representing relationships via entanglement becomes much more peeling and of course uh you know Quantum Computing also offers ways to um to speed up um uh the the computation of of machine learning models like act like an active inference so uh vei actually puts forward a case for causal entanglement using DNA and essentially we see that uh that entanglement can serve as a form of um uh a form of of sort of outlining the the total the sorry the different causal dependencies in a system without necessarily ly explicating them without putting too much emphasis on the uh the specific properties of the causal relationships if we know that causal certain things are causally implement or sorry are causally implicated then we can take the sort of rough approach of assuming that they might be uh you know entangled to some degree um and so basically entanglement can serve as a method for representing murky dependence um without necessarily misrepresenting causal Relationships by conflating correlation with causation and so the um the in the paper vei uses chromatin structures during the interface period of mitosis to show how and basically show how it can be characterized in terms of interactions between entities at uh three different levels of biological organization so essentially the the parts or angles between the linkers the uh in the chromatin and um then uh the the chromatin structure or the topological invariant structure and then finally the the the feedback loop between how the whole which is the the overall chromatin structure depends on context properties in the feedback loop uh for example via histone deacylation or histone methylation and so essentially uh we see that they argue that DNA is a total both a total and direct cause when being manipulated where it is a total cause because DNA is is implicated in its in its own manipulation and it's also a direct cause because um it's uh it's an inter prevention on DNA will change uh will change the the you know this the functional or structural properties of the DNA um with with without changing necessarily um additional uh non lo non-local you know properties of whatever substrate surrounds the DNA and so specificity still allows linear models to capture causal relationships in terms of you know isolating specific variables but when we increase the scale and complexity of these uh of the model for example or what we include in the model it becomes increasingly more difficult to separate out and to sort of distinguish okay where exactly does the specificity lie and so uh this takes me to this differentiation between repertoire and connectedness when it comes to causal specificity so repertoire is basically how many possibilities exist on each side the cause side and the effect side and we can essentially in in a modeling context take this to mean the number of different values that an effect variable and a cause variable can take meanwhile the connectedness is how many or how the possibilities on both sides are connected in particular and relevant ways so we can see uh again in the modeling C test that connecting connectedness is a condition for a given function connecting the values of cause and effect variables and so repertoire does not entail connectedness and connectedness does not entail repertoire which is important to note um because essentially we see that um that uh repertoire and but by splitting causal specificity into these two independent properties we may not be able to compare causal relations from a single point of view and therefore uh you know causal entanglement will may allow for the comparison of both without necessarily giving up you know one one for the other and uh and of course uh Ru notes that you need both repertoire and connectedness in order to come to a complete causal understanding of you know whatever processes are implicated or you're examining within the system and so the the form of entanglement between between uh you know repertoire and connectedness um would would allow you to express uh both of them in the system simultaneously and potentially uh you know um extract results from uh from either of them and so uh what are the practical constraints and limitations on this uh this approach well first of all we should uh view we might view it as a case similar to Quantum versus classical Computing where as the complexity class of the problem increases Quantum Computing becomes more appealing so utilizing entanglement may not be computationally feasible right now given Hardware constraints and um modeling using the entanglement May aisc uh the extraction of direct causal relationships but um as Noble has pointed out and this previous researchers have pointed out the the extraction of direct cause of relationships is a very difficult task and epistemologically it's not clear whether this is even possible once the system surpasses a uh given uh given you know point of complexity where it's simp where the dependencies between uh and across scales simply become too large to um to properly uh identify you know direct causal relationship sh and so um essentially uh this this entanglement approach may also become computationally invasible when uh we have lots of uh you know dense variables that are not sparsely coupled but I think because uh you know the Markov blanket also you know inherently relies on the sparse coupling and sparse coupling and neuronal Dynamics has been shown uh entanglement can be taken as an as a as a sort of a placement for the Markov blanket um and so I'll just uh get into some previous work involving the free energy principle in quantum mechanics so Fields uh Chris Fields integrated Quantum information Theory with the free energy principle and proposed that physical interaction is a form of information exchange uh where the holographic principle can essentially be used to formalize boundaries between systems uh additionally warier investigated the potential for tensor networks to represent um gener generative active inference models so tensor networks of course can offer computational benefits When run on quantum computers and the tensor Network itself was originally um designed to be used uh in um in quantum mechanics um so specifically for Fields approach uh with the the quantum reference frames we see it's a different way of approaching the Markov blanket um which still formalizes boundaries using the holographic principle so it's still assuming that we can properly distinguish the boundaries between uh you know organisms and their environments and that these boundaries are both causally uh necessary right so that there's some causal impact of the boundary uh that cannot be ignored and that the boundary itself uh is you know properly distinguishable um so now I'll just I'll go into uh some theories of sorry the overview of Consciousness and how it's implicated in uh both FP and active inference and how it fits into the proposed uh framework for um for for uh for utilizing entanglement so uh the minimal causal mechanisms of Consciousness we see that there are regions in the brain that are implicated in Consciousness where you know if there uh if these regions of the brain are damaged or not functioning properly this leads to changes in the overall phenomenal and conscious uh perception of individuals so we also see that non-equilibrium macroscopic brain Dynamics are a robust signature of Consciousness so at some level consciousness uh can you know statistically and um uh and physically be proven to be the sort of higher order uh um process or or dynamic that is going on in the brain and so uh Consciousness and the marov blanket so how does Consciousness interact with the marov blanket and the free energy principle does the link between phenomenal perception judgment awareness observation Etc disrupt the epistemological application of the marov blanket and do competing theories of Consciousness pose problems for active inference so uh on the first question I think that when it comes to the Markov blanket and for energy principle Consciousness pres you know as I guess with all neuroscientific theories uh is the sort of uh throws a renion things when we are trying to uh uh consider how you know conscious States or different levels of conscious States may not be easily or neatly put into boxes of either internal or sensory States because again you're assuming that the organism has either the organism itself has some capacity to differentiate between these states and that the states themselves can be placed into this sort of uh into these these sort of categories for representation by the marov blanket so I think this is especially apparent when we are discussing phenomenal perception judgment awareness uh observation Etc because um these types of phenomena are very very broad ranging in terms of their uh phenomenal uh activity and their corresponding correlation sorry their correspondence to underlying neuronal Dynamics um and of course the research there is still ongoing but uh based on you know whatever what research there is into U awareness for example attention or observation um there there there seems to be um varying sort of uh degrees of complexity and uh also degrees of of causality when it comes to you know which areas of the brain are implicated in minimal self-awareness and to what extent these boundaries uh initially observed can be modified um or you know modified either internally by by the the person The Observer Modified by some external uh external causal Force for example uh like anesthetics and so the competing theories of Consciousness sorry do competing theories of Consciousness pose problems for active inference and I would say depends on the mechanism of Consciousness right so it depends on how these theories lay out Consciousness because they may or may not implicate active inference for example uh theories which are more volitional on the on the more volitional side that attribute um you know Consciousness to volitional activity which in itself is uh may be distinct from you know uh free energy minimizing um or uncertainty minimizing activities then we would see that uh that yes these competing theories of Consciousness do or would PR problems for uh for active inference and another Dynamic that I want to explore here is the unconscious right so experimental evidence supports a sort of minimal unconscious where we can broadly distinguish between type one and type two thinking so type one thinking is UNC ious processing and type two thinking is cons conscious processing and um researchers have sort of found that there is a form of dynamic unconscious which sits between the sort of conscious processes that are you know directly under observation or awareness and then those perhaps autonomic processes for example where uh they're they're entirely uh unconscious and so the dynamic unconscious Theory essentially posits that the uh The Observer or the the conscious agent themselves has some capacity to uh to guide their awareness to processes that were previously um previously unconscious so uh you can essentially make uh previously unconscious processes conscious and again the implications for the mark of blankets uh neat delineation between sensory active and internal States um I would say this this sort of frustrates uh the the um the thinking behind Okay the thing behind trying to separate out uh the uh these um these these different states right and so theories of the brain should incorporate a transition from unconscious to conscious awareness um and of course this is extremely difficult because we don't have really a solid understanding of what would what Consciousness even is what are the minimal phenomenal sort of uh requisites of Consciousness and how sort of expansive the operations of the unconscious are and whether those unconscious operations are therefore implicated in conscious processing as well for example with uh underlying cognitive biases so uh what are the implications for the free energy principle so both conscious and unconscious processes should be modeled however the distinction between uh sensory and internal states does not do uh this uh these conscious and unconscious distinction I think it doesn't do it justice so why is that the case well first conscious processes are sometimes unconscious processes are processes that may inherently be uh be sort of semiconscious for example when someone is uh is scratching their their leg for example right the the distinction between uh conscious and unconscious processes is one that is fundamentally Dynamic and therefore um we don't know whether there even is a boundary between the awareness of scratching leg and and not scratching leg and especially if that boundary is uh is sort of um is highly Dynamic for example Le uh when someone is slipping in and out of Consciousness post anesthesia um and so we see that unconscious and conscious processes could influence active inference in ways that are uh particularly unique to them so for example unconscious processes uh could be represented as uh as you know as priors within the model um but that doesn't always necessarily have to be the case because these unconscious processes could also be interacting as um as as internal States for example or they could be uh they could be present even as active States once they are made uh made conscious um and so the mechanisms of Consciousness uh various uh theories which have been proposed could provide a mathematical basis for expanding active inference and the free energy Principle as well so I think this is very important to note because uh theories of Consciousness um sorry novel theories of Consciousness that incorporate you know more Quantum information side or that correspond to um you know the electrical sorry the the electrical um the electrical correlates of Consciousness for example um are extremely important for uh for sorry May provide mathematical Frameworks for updating um active inference or you know potentially May provide a useful uh mechanisms um for modeling specific conscious processes that could then be incorporated into uh the framework of active inference um and so see and so yeah so uh I think I'm missing a few slides here but uh basically the the the overall theories and most prising theories of Consciousness um uh that that are that are up coming are you know the sort of um the the electrophysiological and um electrical the electrical field uh sort of theories of Consciousness which posit that Consciousness is actually a result of the electrical Dynamics in the brain which of course uh you know would need to be a case as as for for any minimally causal um sort of uh representation of Consciousness but the presence of electrical fields in the implication of these fields in Consciousness is uh you know a particularly new Avenue of research and we see that for example firing patterns in their brain have produced uh quasi Crystal representations and and can be represented as well um in in in the Taurus uh for example so these uh higher dimensional uh uh Dynamics within the brain correlating to Consciousness or um or these sort of UNC ious processes will be extremely important in um in determining you know what processes are relevant for Consciousness and how these might be implicated in uh in inference for example or uh thought patterns you know traversing uh whatever sort of dynamic landscape that exists in the brain which again may be determined by these higher order or uh these higher magnitude or dimensional structures so um overall we see that investigating the mechanisms of Consciousness I think is fundamentally necessary for uh for FN active inference because these uh structure um or have the potential to structure uh uh thought and the ways in which uh certain uh you know energy landscapes for example in the brain are uh are traversed or are um are fundamentally interact with each other and so uh the areas for future work here um would be to unpack the philosophical dimensions of active inference which would include implicit assumptions about intelligence sapience and sentience so for example you could utilize Den Jean's conception of intelligence which um which argues for a a a a uh sorry a relationship between the model and the modeling so and sorry the model and the modeler um as a as a direct form of intelligence there or you could utilize Simon Don's theory of individuation as a more productive method for explaining uh the boundaries um and cellular differentiation which I've critiqued previously uh you could also utilize Quantum Computing to take advantage of quantum specific algorithms and uh to investigate uh Quantum biology um um with an increasing uh uh with increasing scrutiny for example and and discover ways in which Quantum biology is implicated uh in you know processes of Consciousness active inference and in the brain and so essentially the the the overall Theory here when it comes to when it comes to uh when it comes to um to to active inference and uh and uh multiscalar phenomena and uh and Consciousness is that we can essentially um if we take that the boundaries between the the system and these systems are not actually uh ontological or physically causal causally implicated um and we utilize the sort of uh causal entanglement to represent relationships between uh you know different parts of the system uh Consciousness can be taken as a form of observation along uh specific uh you know um causal trajectories within the system that would essentially collapse the superposition of these entangled states which would produce you know specific uh outcomes and so that's uh very Broad sort of method for integrating um Consciousness with you know Quantum phenomenon and entanglement but I think that there are many uh many many other sorry different there are many there are different methods of integrating quantum mechanics with uh with the Paradigm of active inference for example that don't just that are that are distinct from the holographic principle and do not rely on the explicit representation and encoding of information in uh in boundaries for example um and when it comes to for example these unconscious processes as I've outlined in the abstract these unconscious processes could be represented more so uh by you know the minimization of uncertainty um uh via you know the free energy Principle as always ongoing processes similar to Elon argument whereas conscious and more volitional um uh actions would be represented by sort of observation on this Quantum system where we are uh where we are collapsing the superposition of specific entangled States and so I think that in this in in that way entanglement offers two sort of uh sort of benefits to the ACT inference Paradigm the first is that it allows for some flexibility when we are talking about uh different variables and causal connections between and across scales because we can entangle two variables that are causally distinct um sorry that are that are uh scale that are that are distinct across scales for example but which may be implicated in some uh feedback loop for example um and this entanglement can then uh represent the causal dependencies between these variables on the other hand um if Consciousness you know is a sort of quantum process then it would allow uh it would allow for the integration of whatever Quantum processes that are ongoing in in whatever you know Quantum Consciousness uh um so if if you believe that Consciousness collapses the wave function for example then uh then incorporating entanglement would be a mathematically feasible um technique for incorporating this uh this form of volition where essentially the uh the conscious agent is collapsing a specific wave function and that leads to uh you know that allows them to perform uh active inference for example by minimizing uh specific uh probability densities so um yeah that's overall uh the the scope of my talk today um and yeah I hope uh everyone uh enjoyed that and uh if you have any questions please let me know thank you Devin great times so I'll ask one question you'll give the closing word for the live streams for the whole Symposium so you're the only undergrad presenter what would you say excited you to get into the game and what do you say to younger Learners to come um I am mainly interested in uh um the sort of mathematical and philosophical um uh I guess um ideas behind uh Act of inference and um I was I'm particularly um uh happy to see that the active inference ecosystem has you know thriving uh philosophical and uh and mathematical um Community um so when it comes to uh um younger uh um uh participants who may be interested in active infint I would say uh you know keep uh pursuing um what you're interested in and um if you have any particular areas of research or ideas you should always um be be looking to share them I think and uh be looking to expand your horizons so yeah cool well looking forward to hearing more from you and from others so thanks again Devon for bringing so much together in this last Capstone talk thank you so much for inviting me to uh give the talk and apologies uh I was a little bit short on time to uh to put together the slides so uh this was a lot uh messier than I would have liked unfortunately but uh yeah real life is like that all right thanks Devon peace thank you so much e